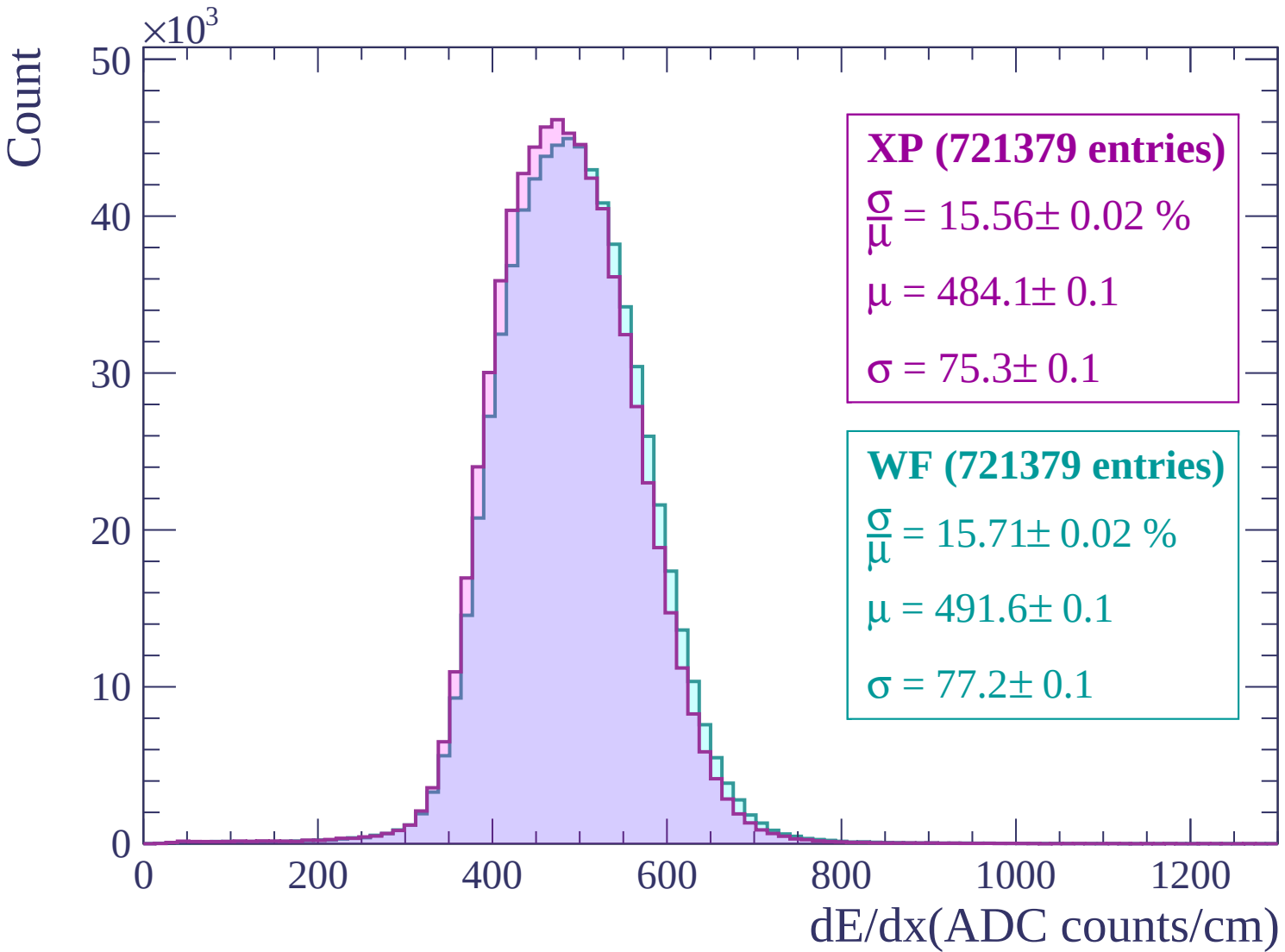
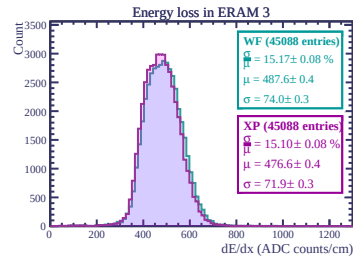
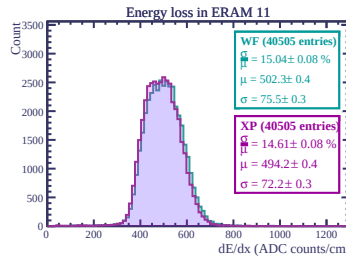
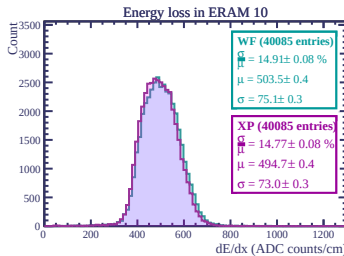
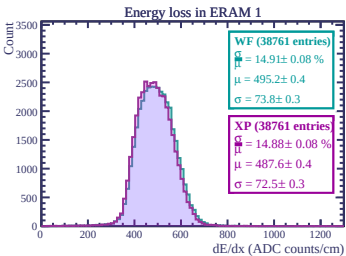
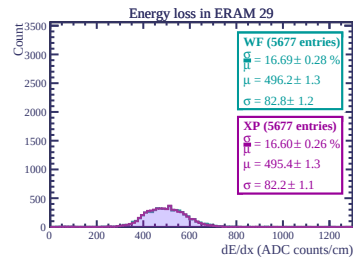
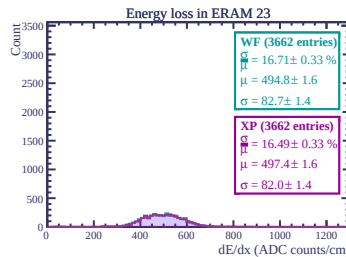
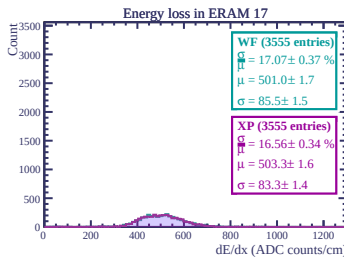
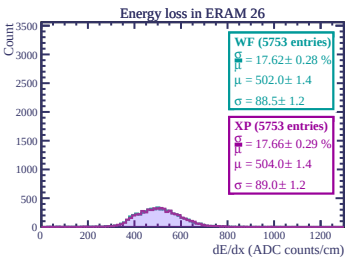
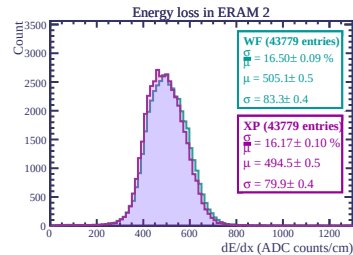
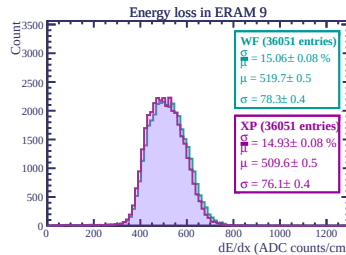
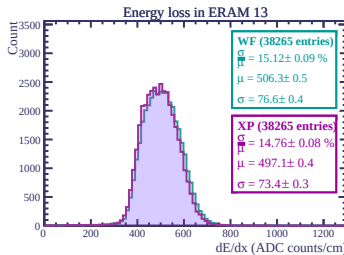
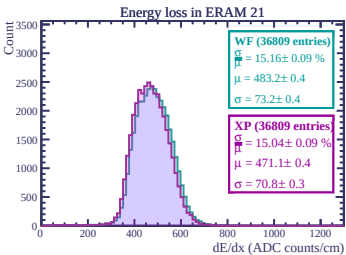
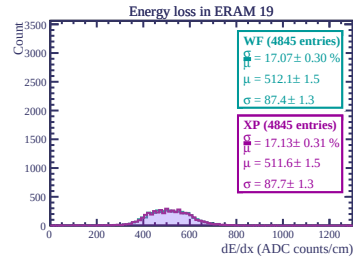
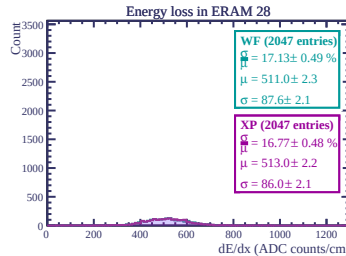
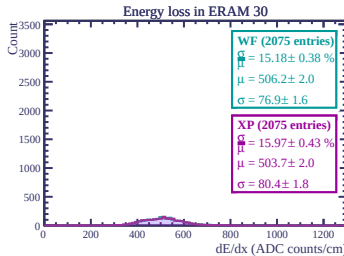
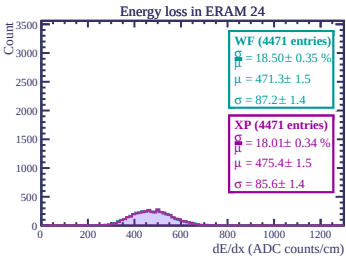
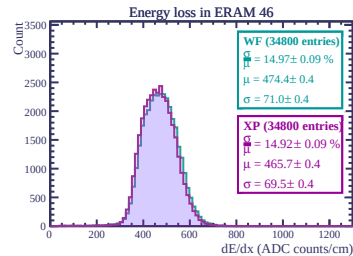
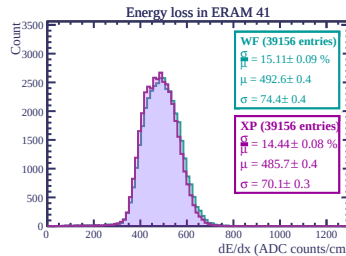
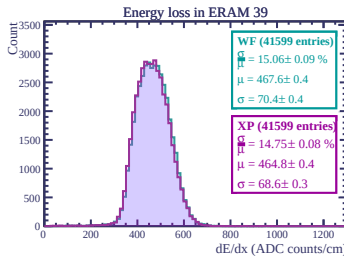
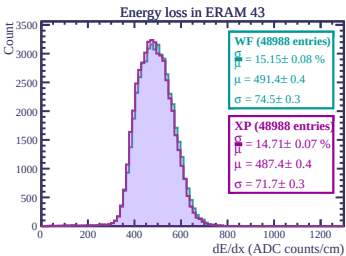
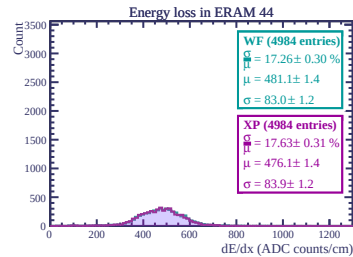
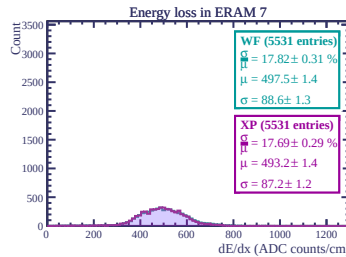
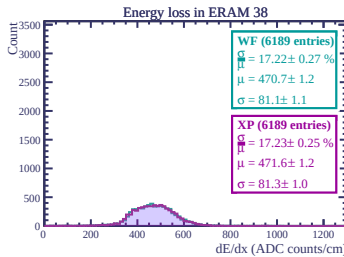
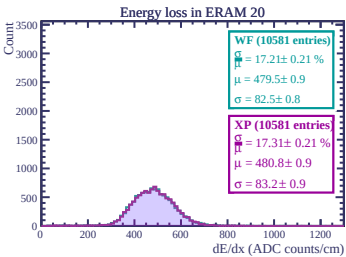
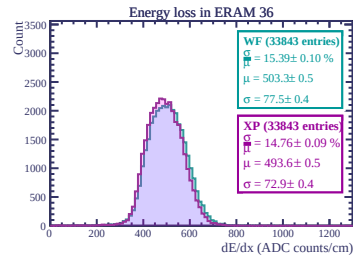
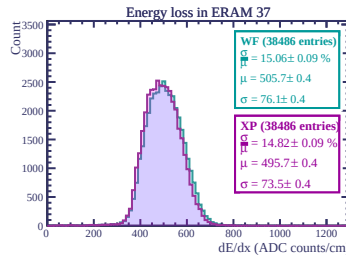
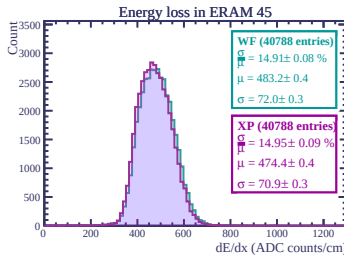
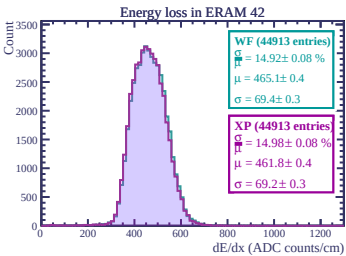
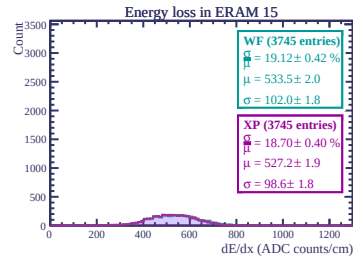
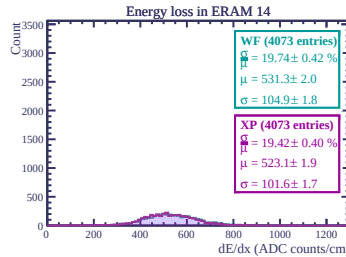
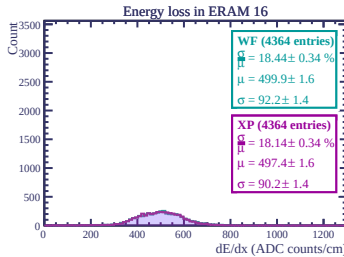
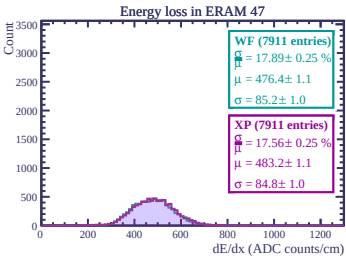
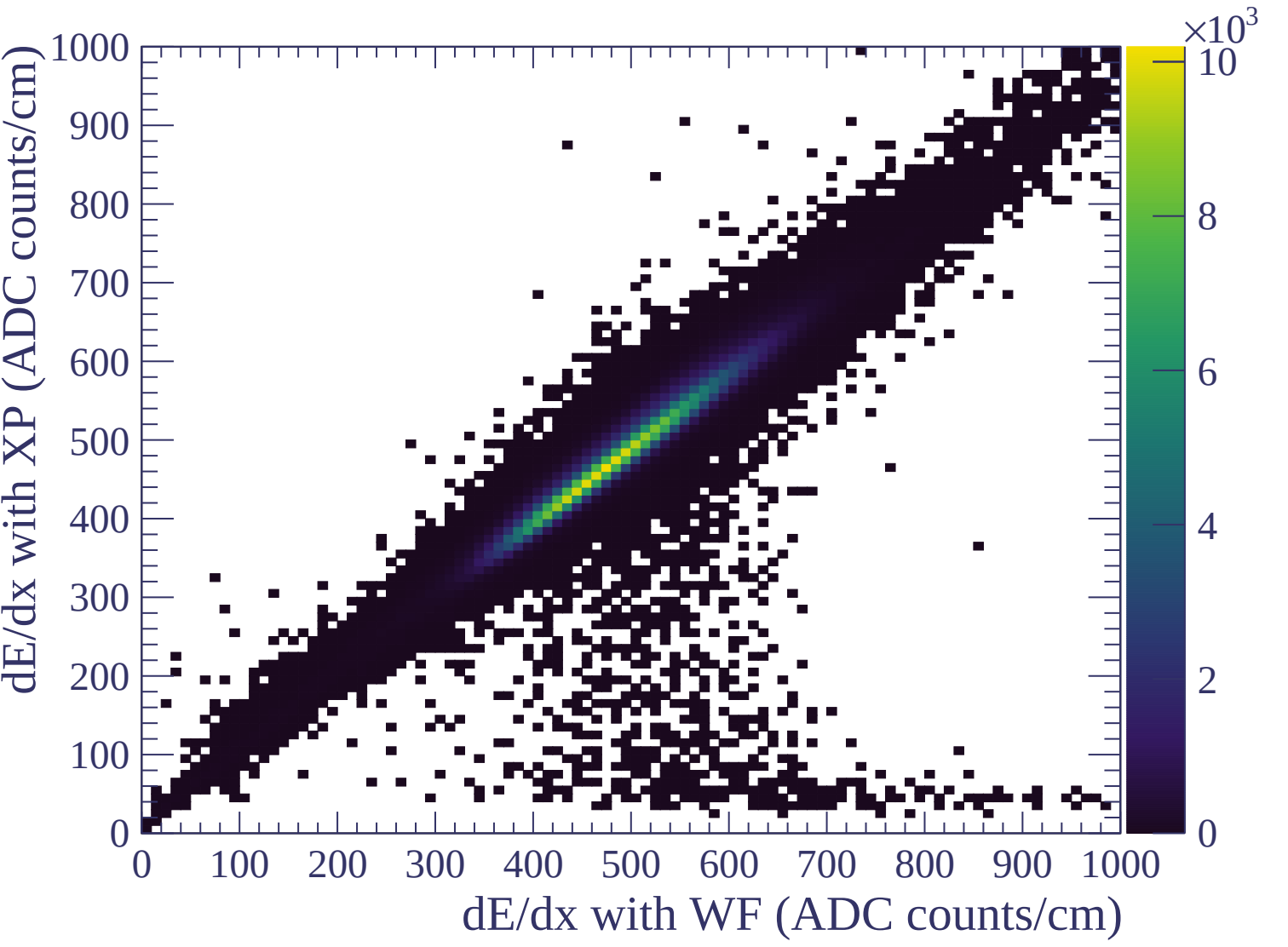


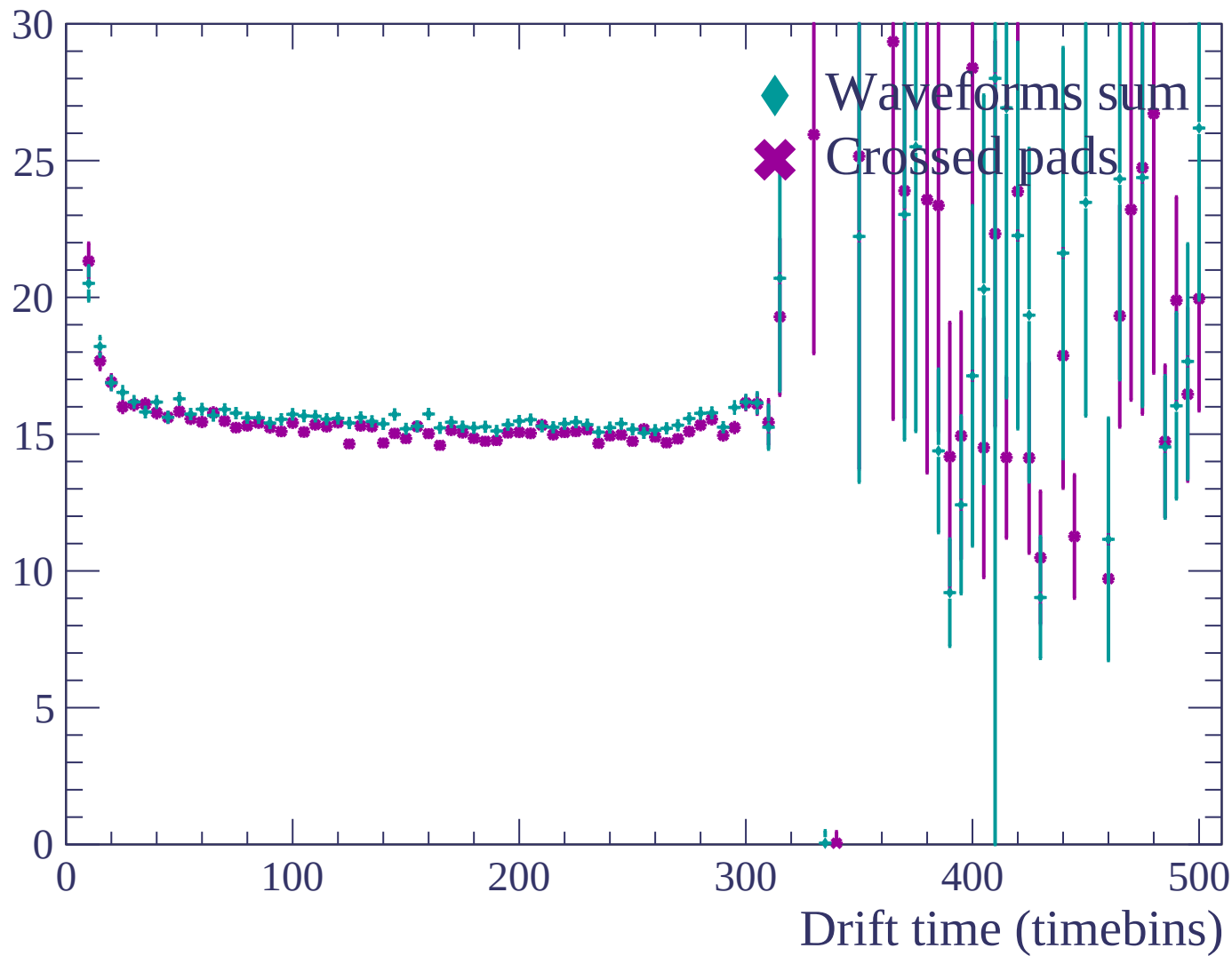


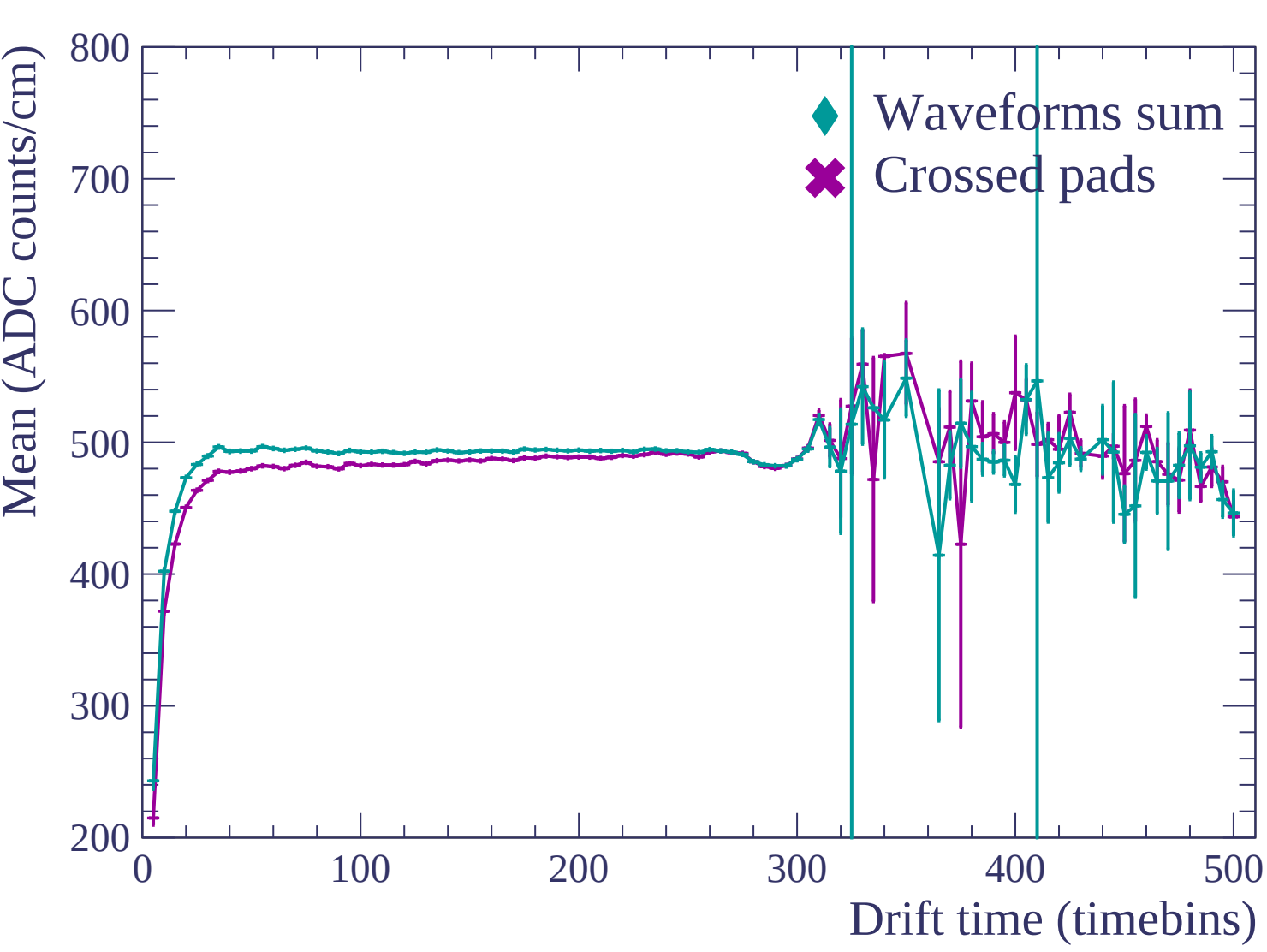


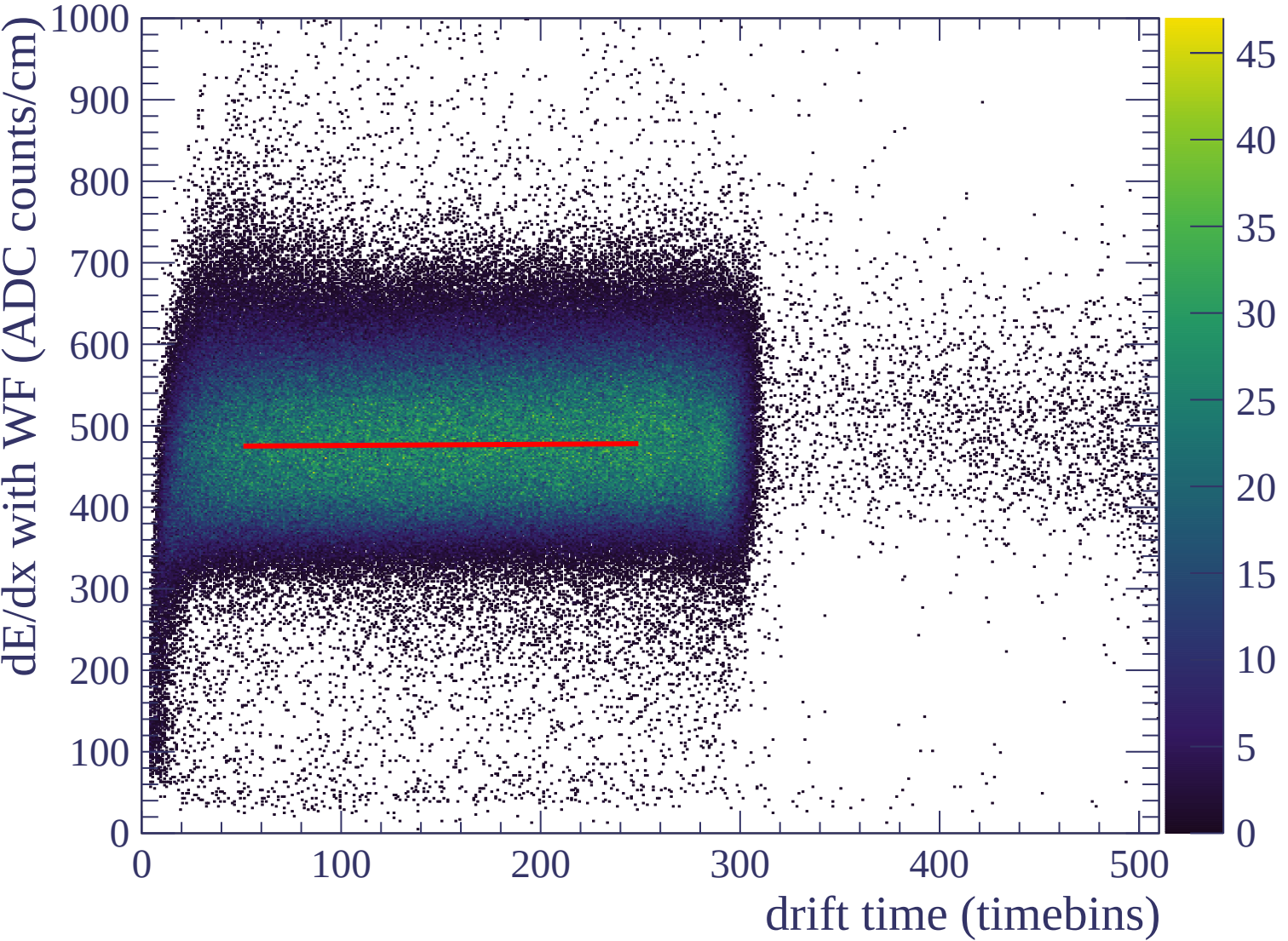


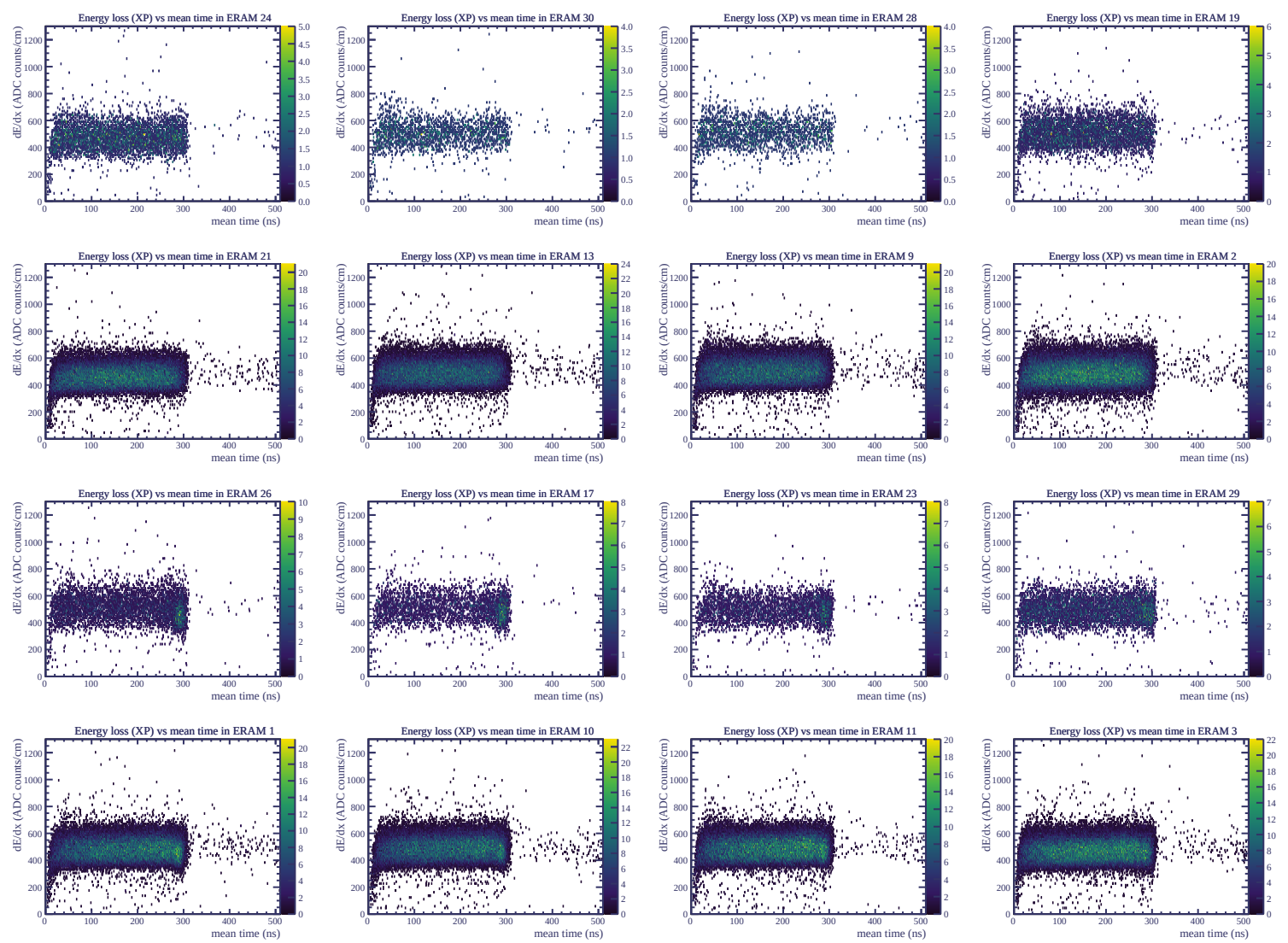


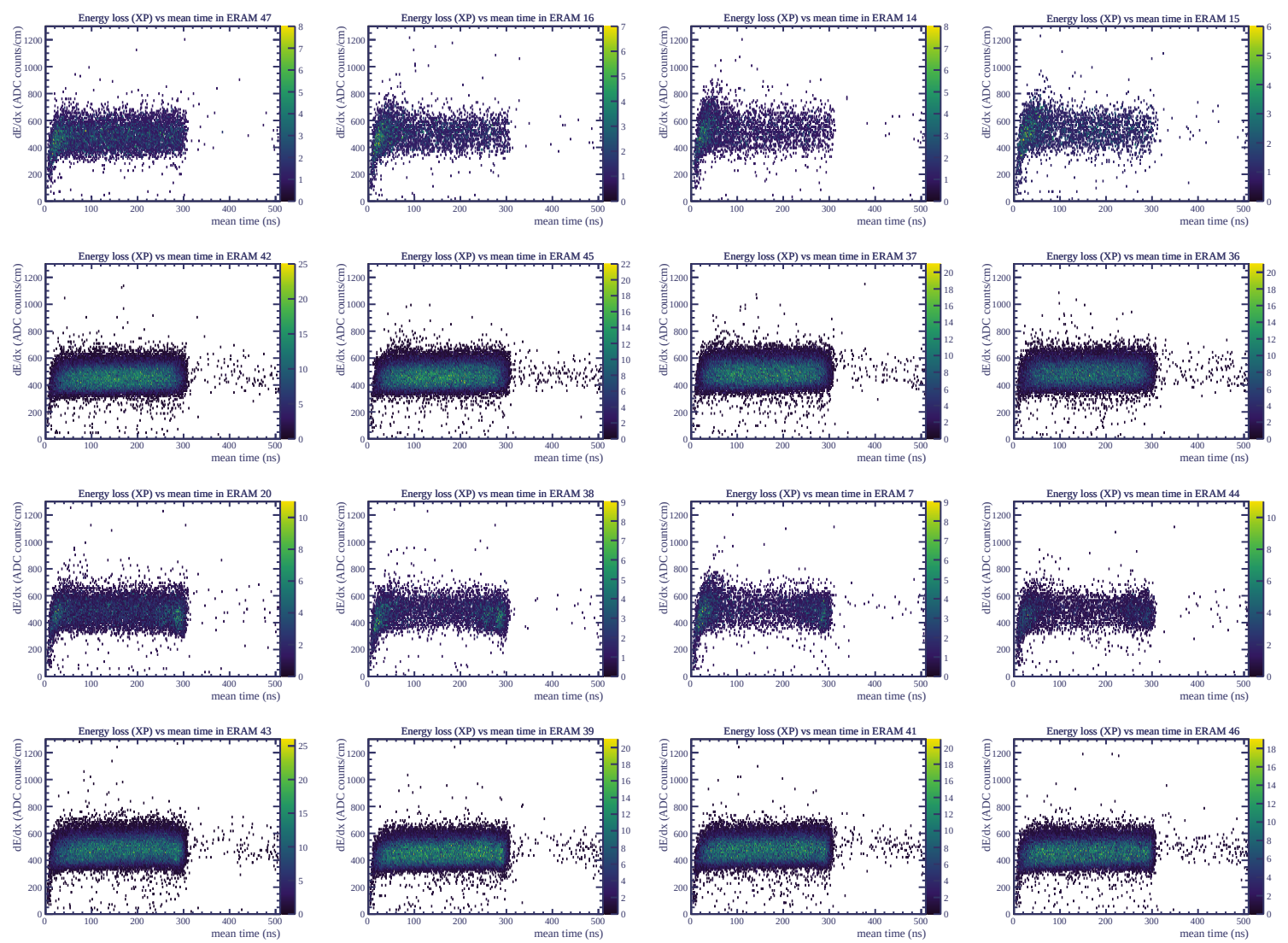
Resolution (%)

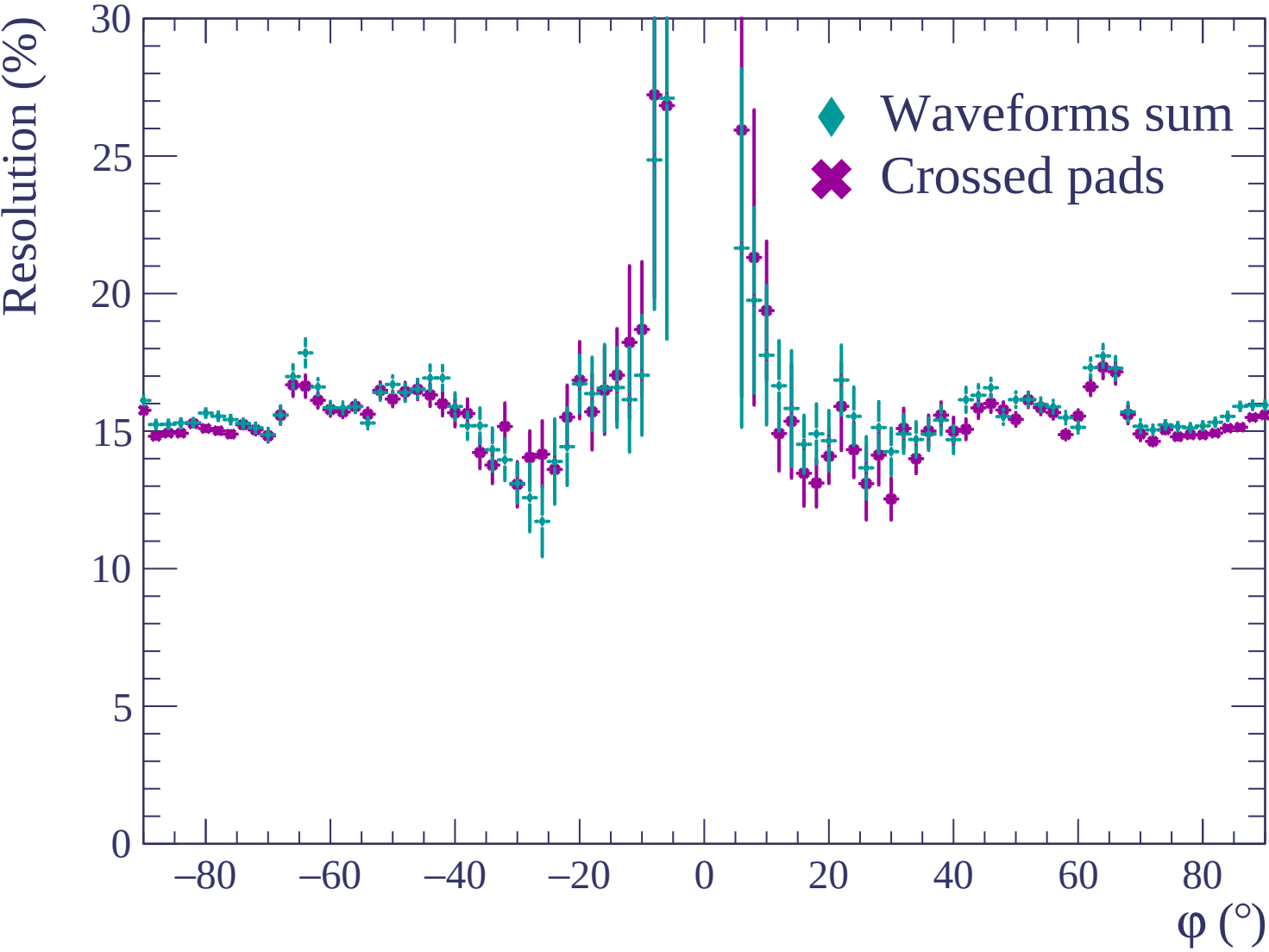


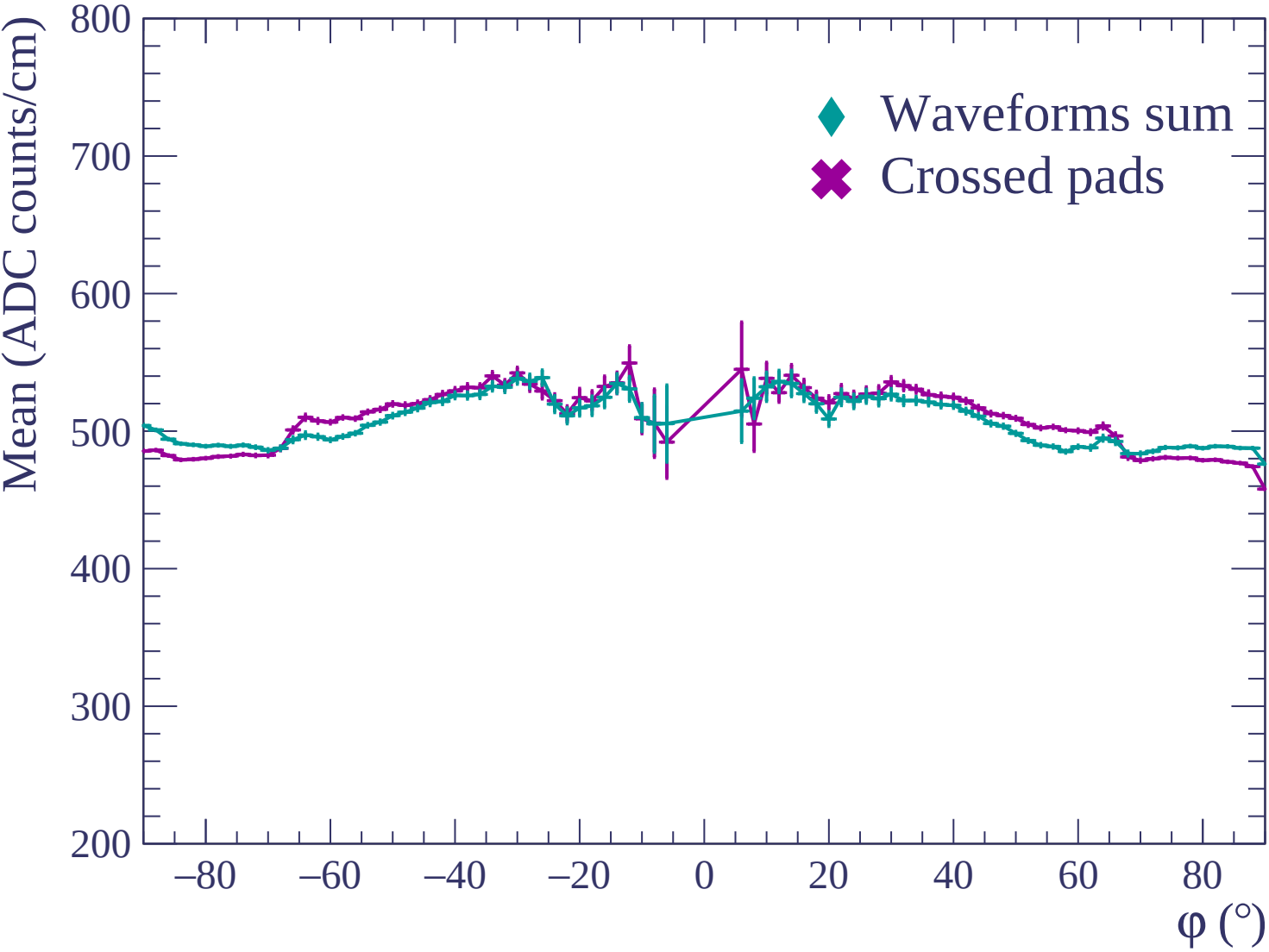


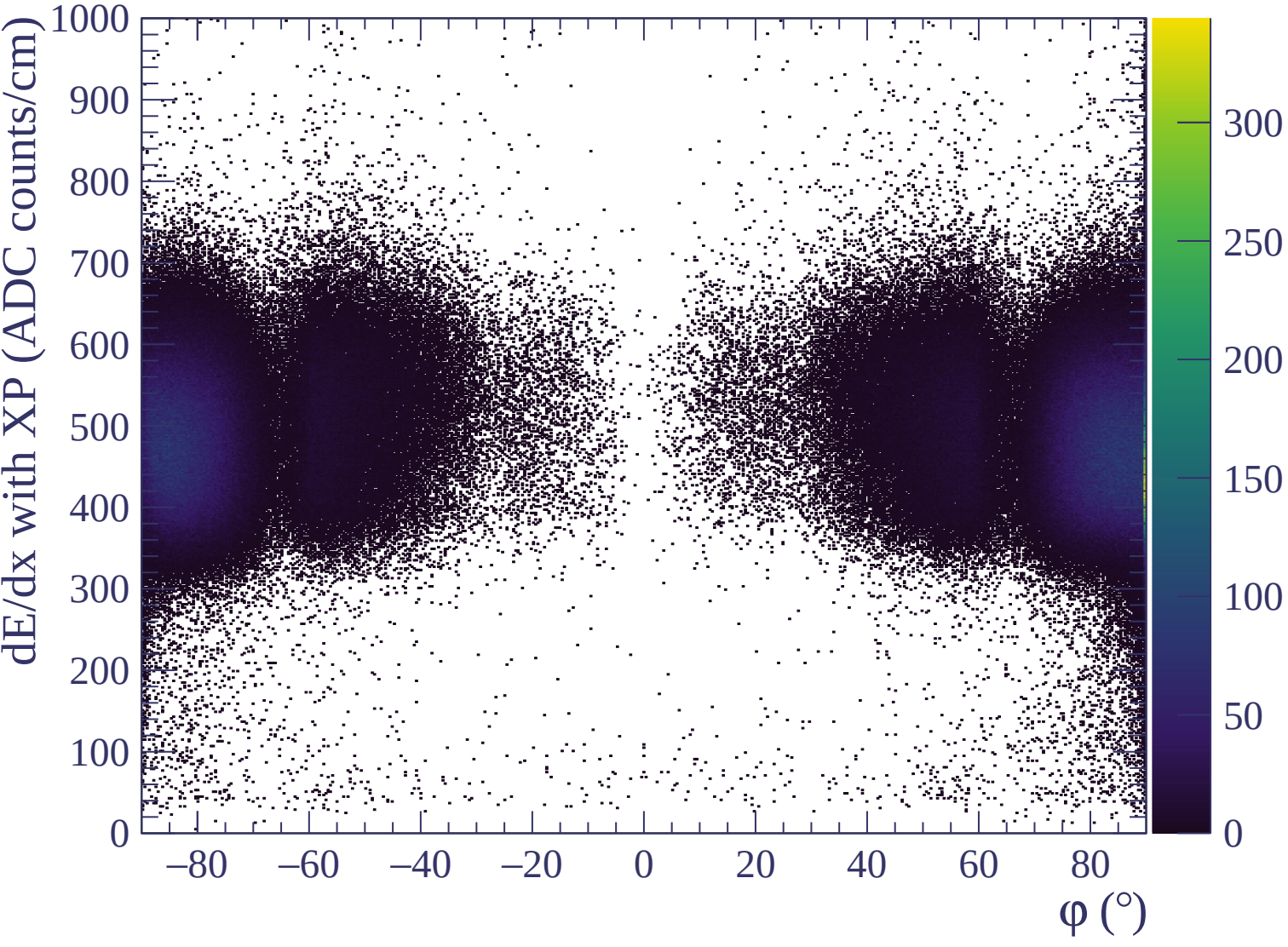


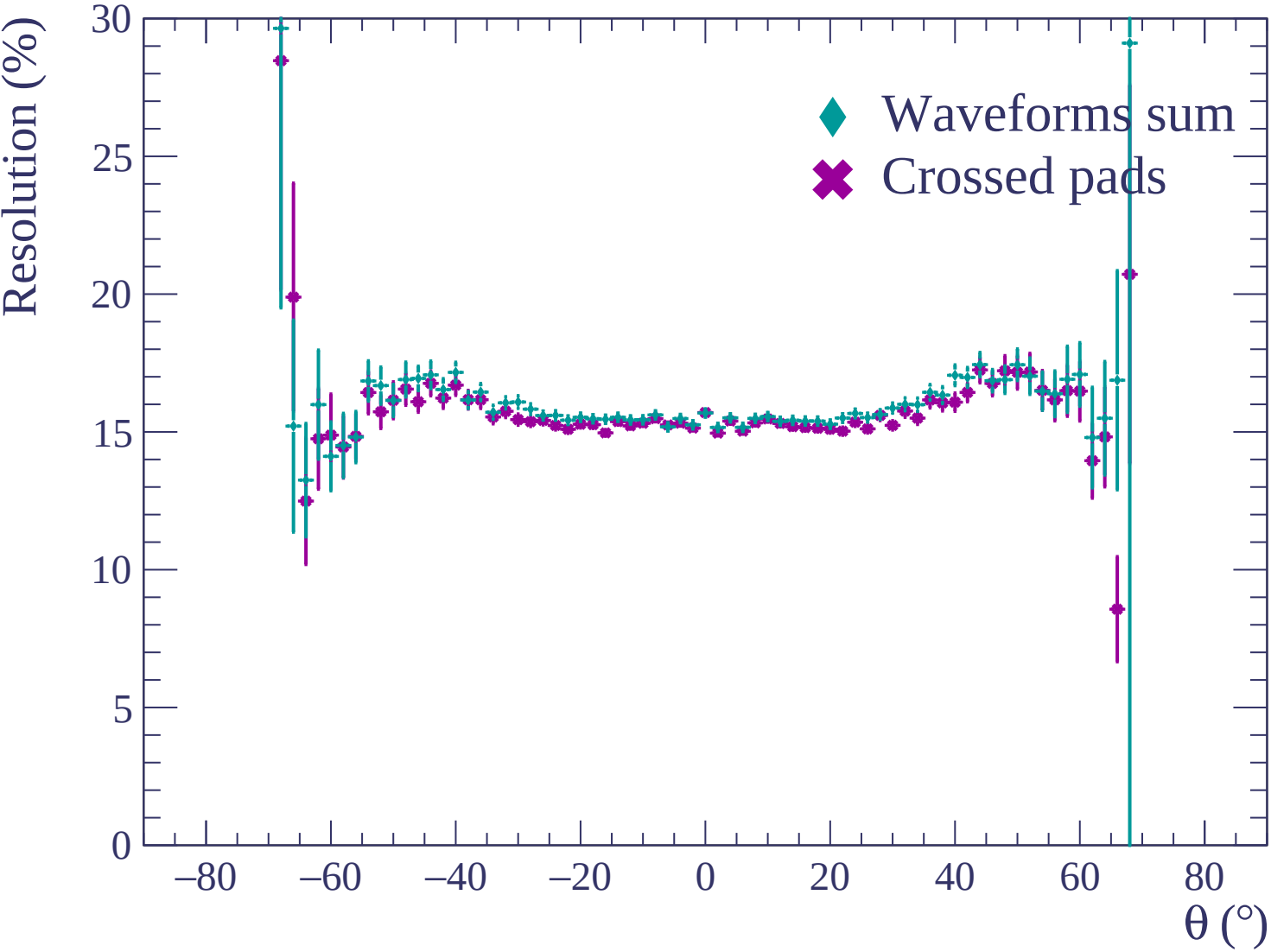


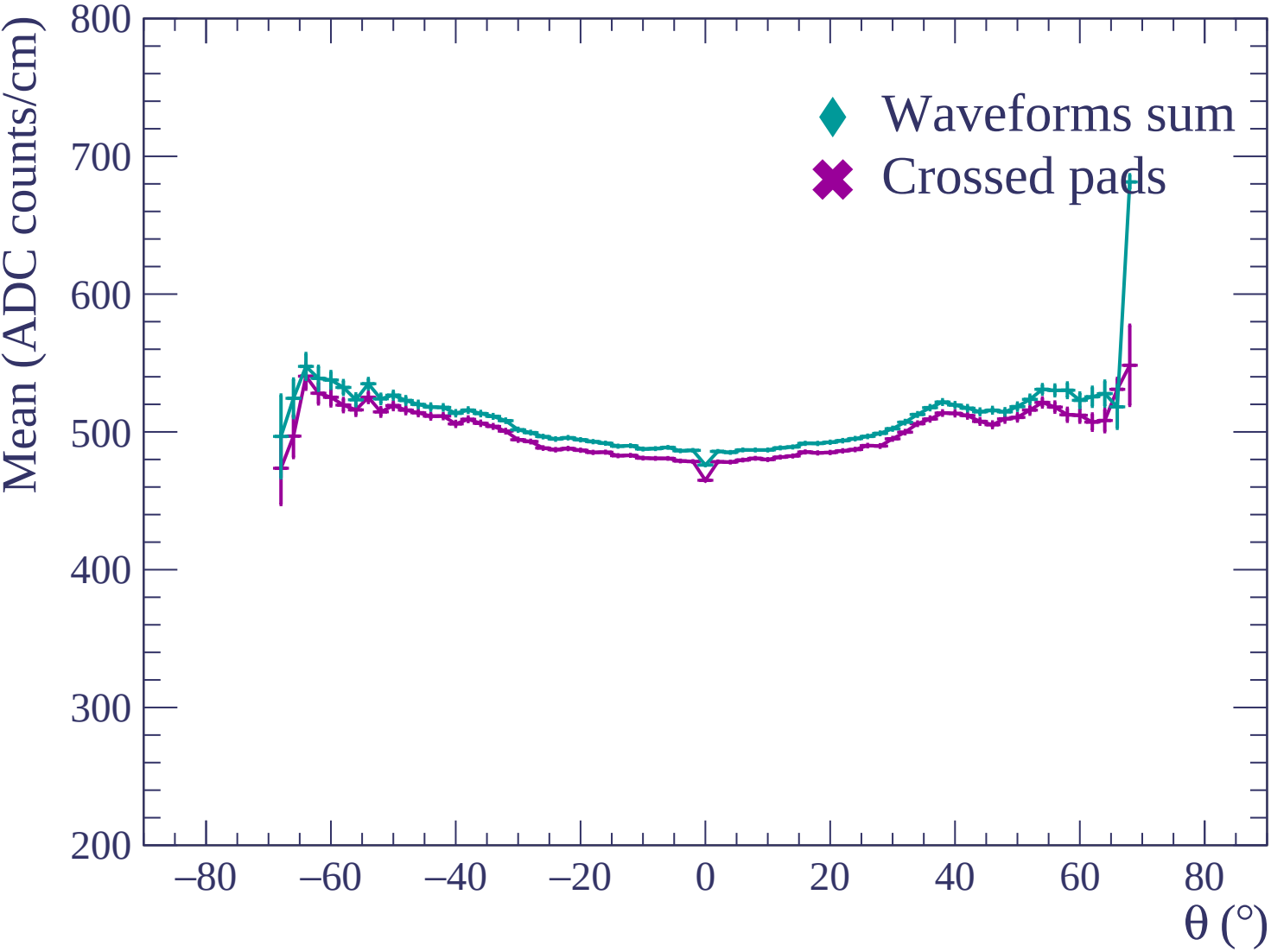


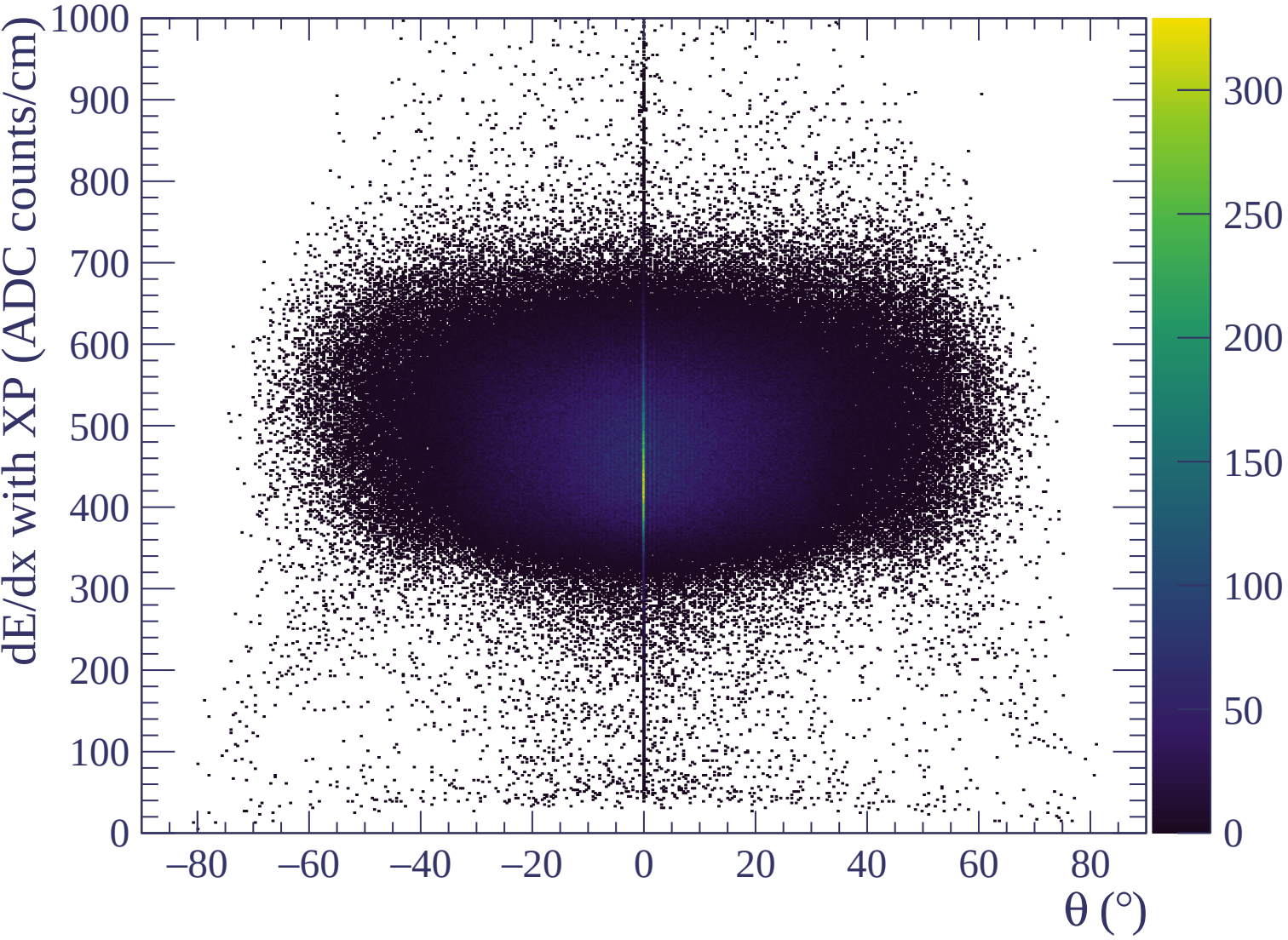


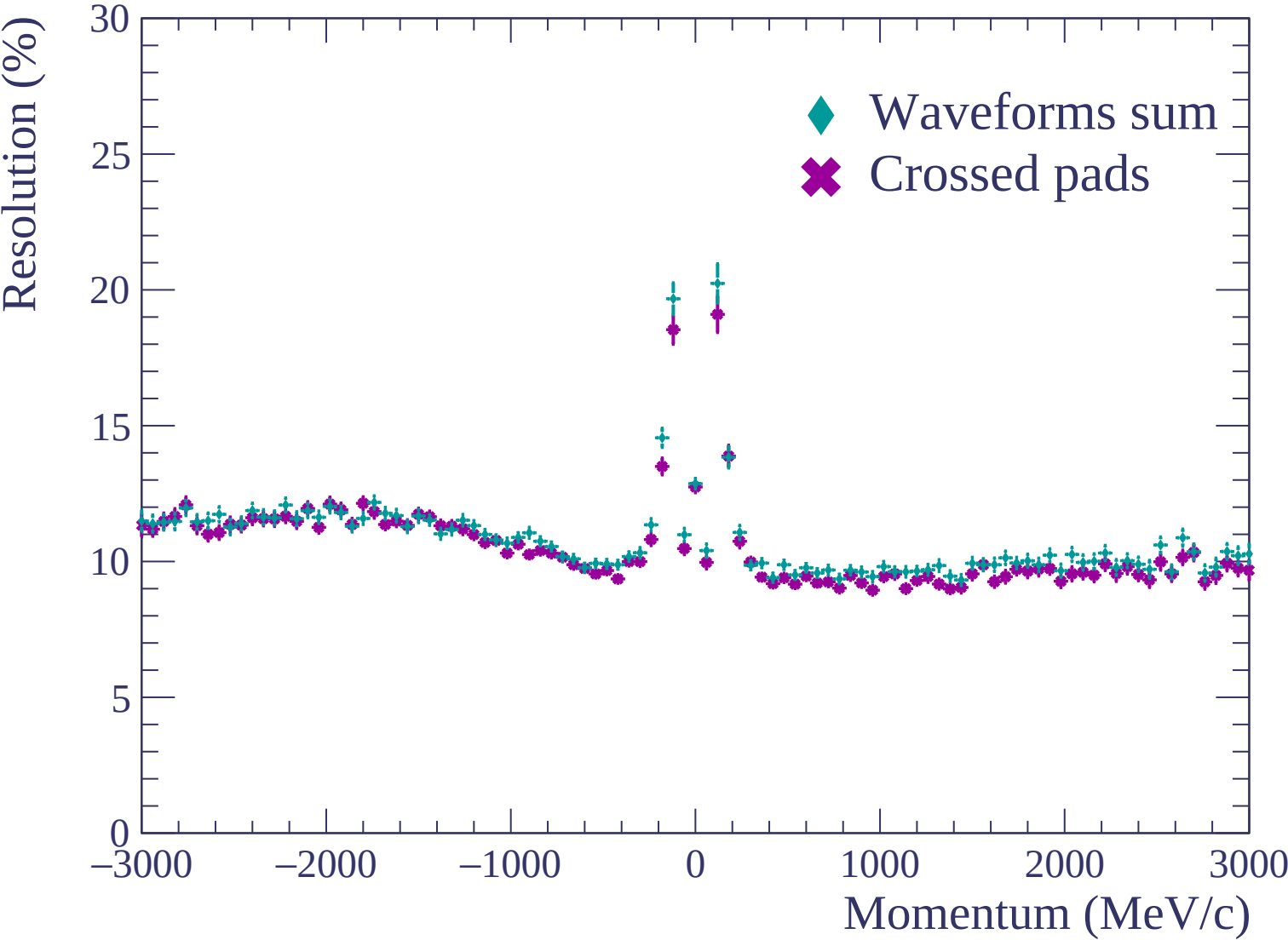


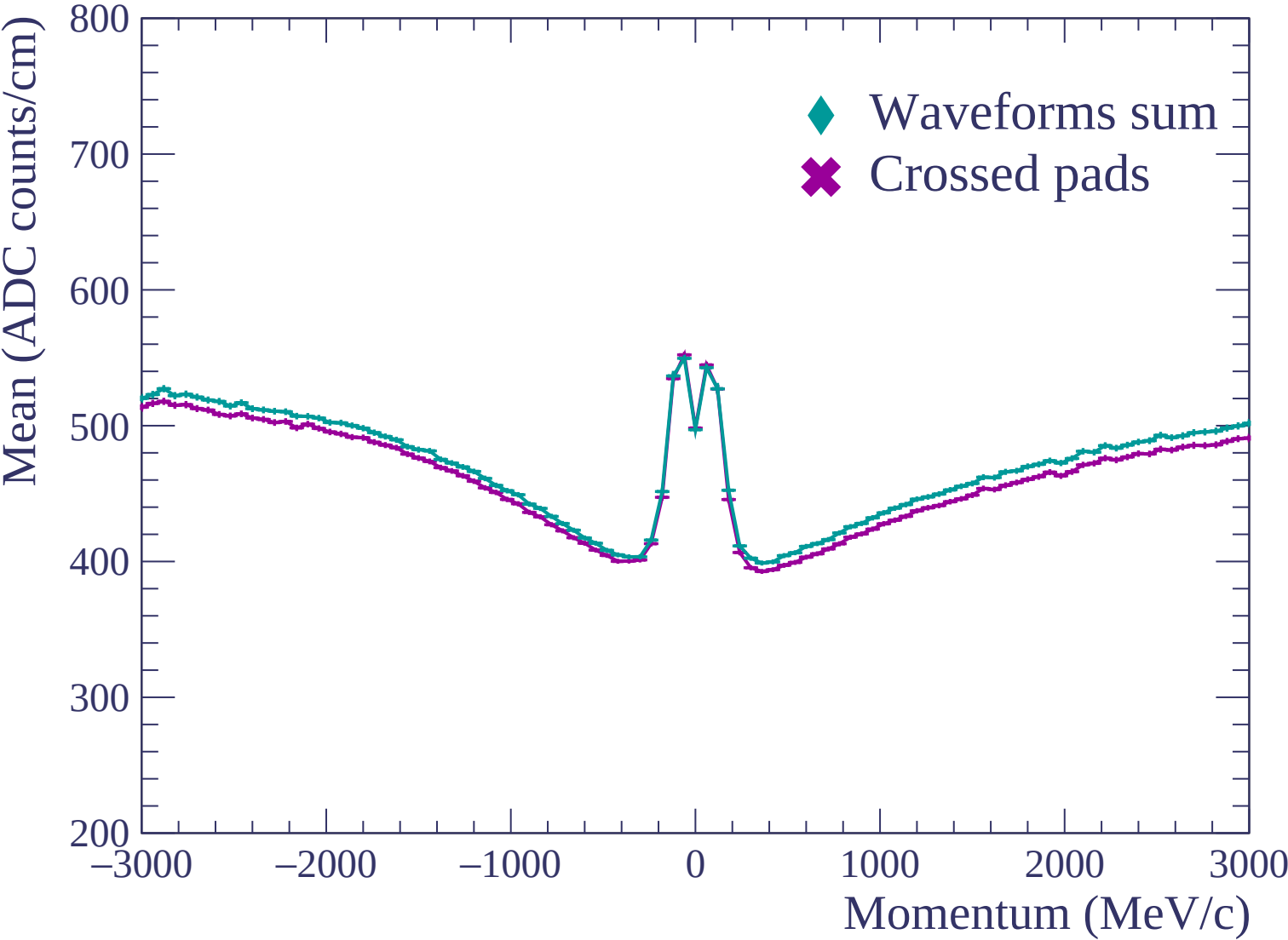


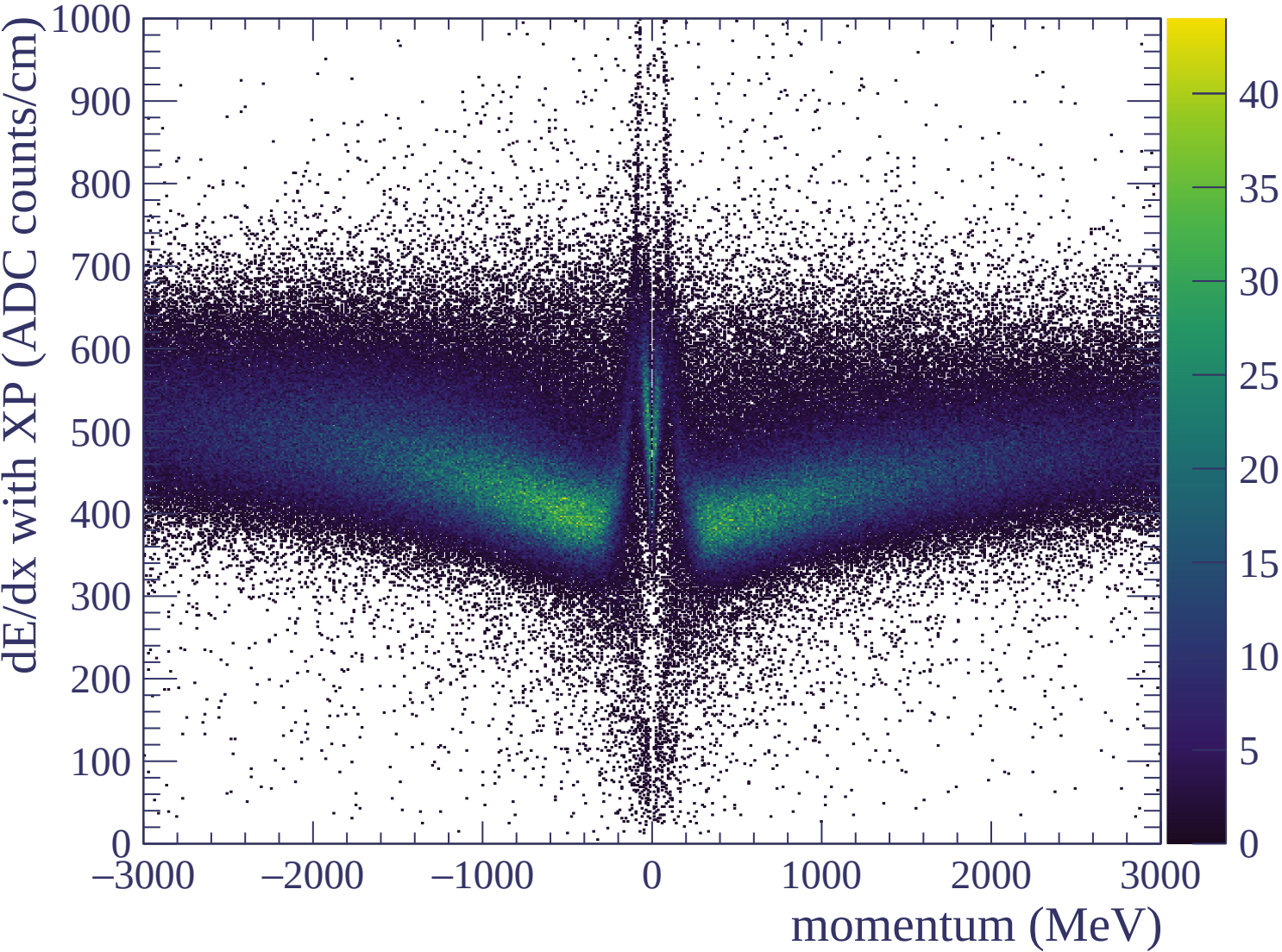


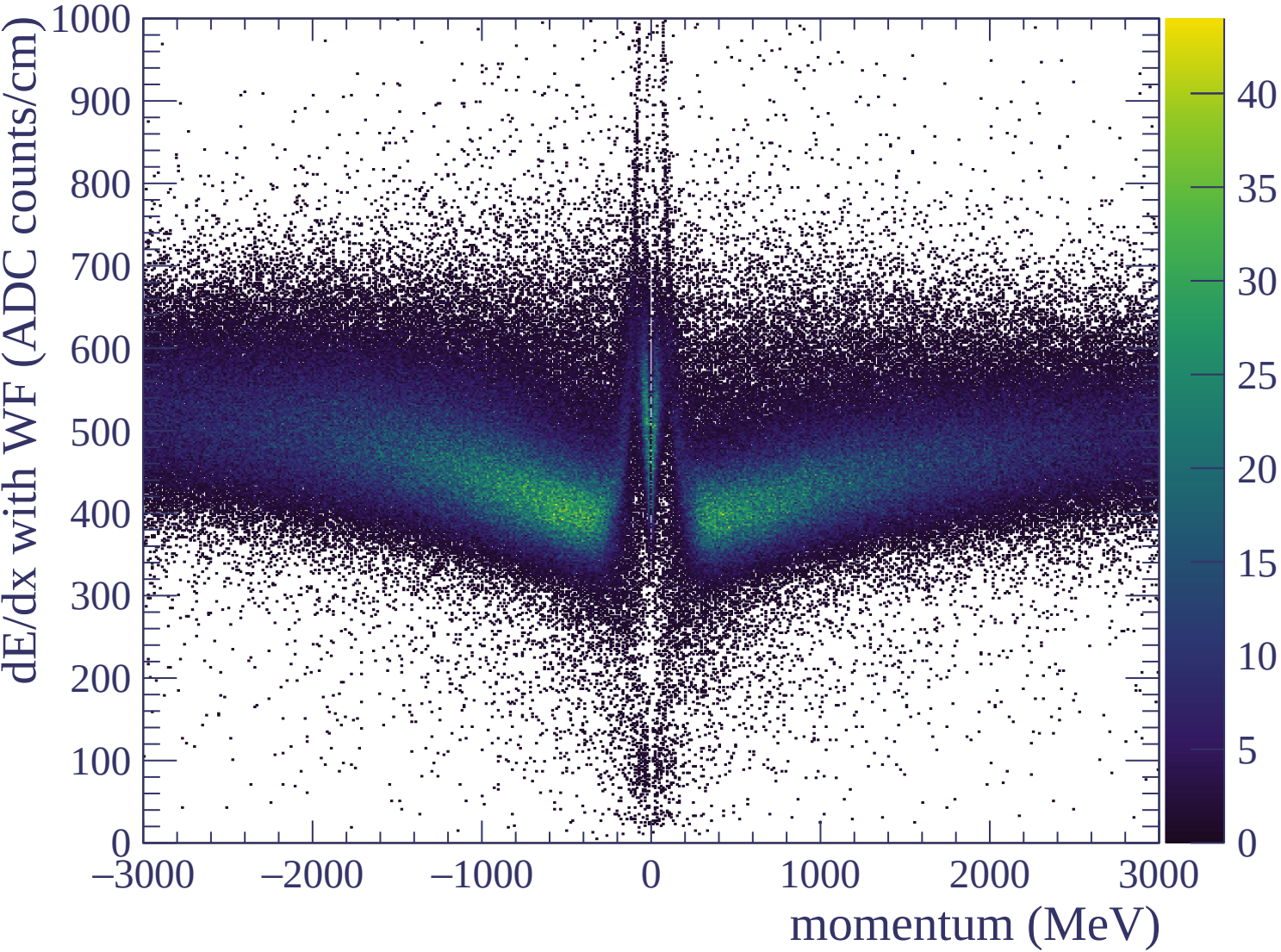


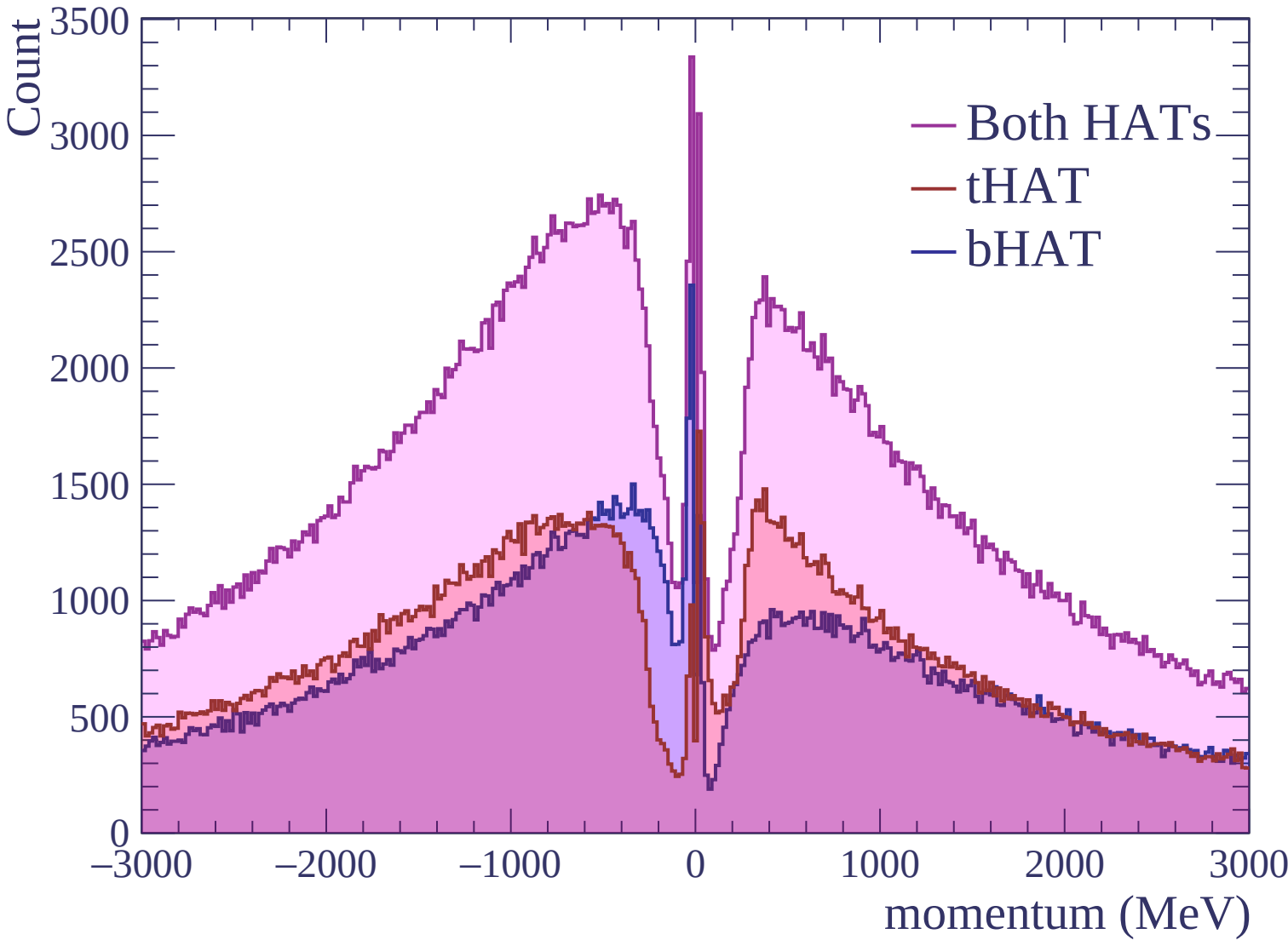


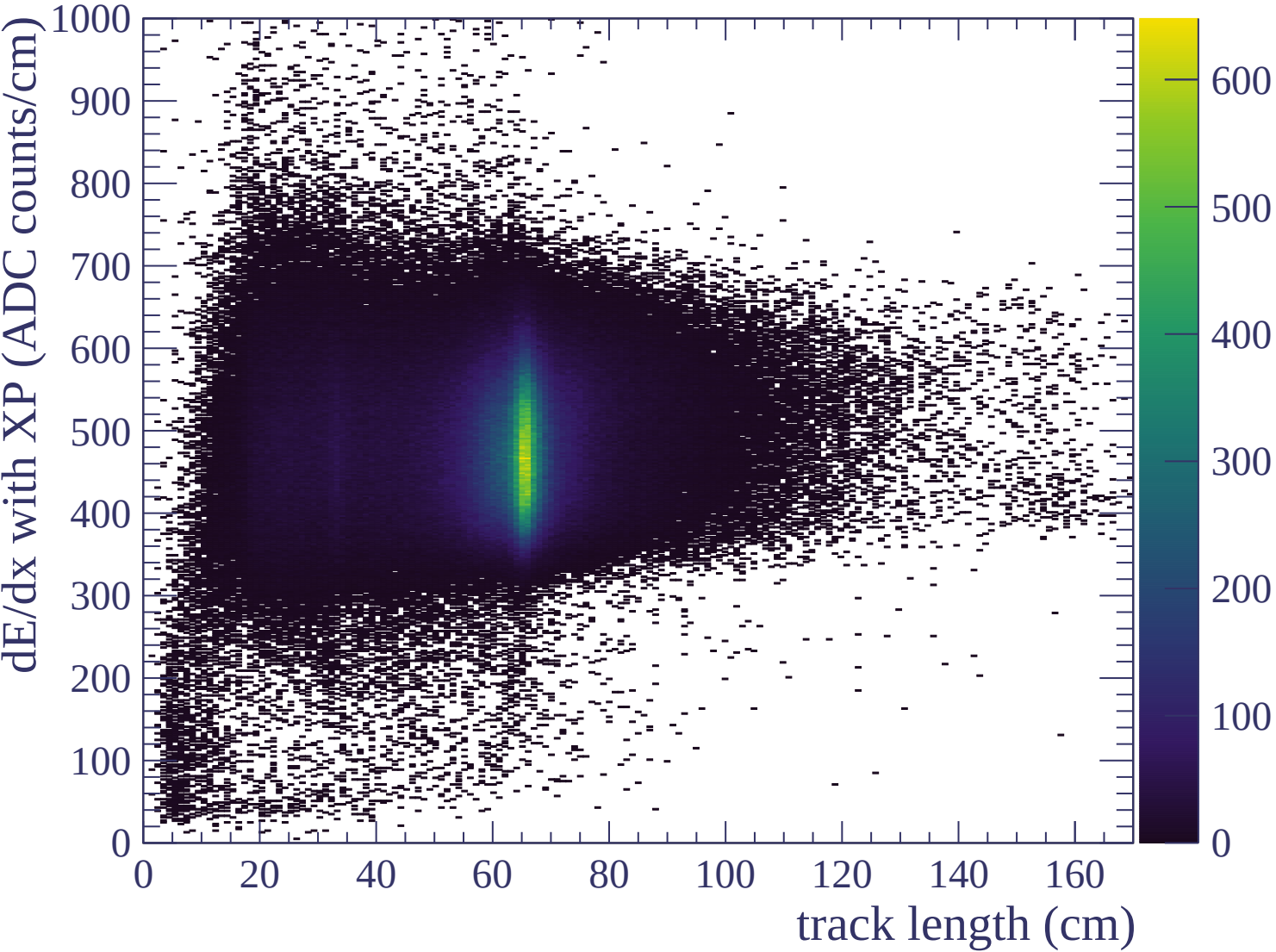


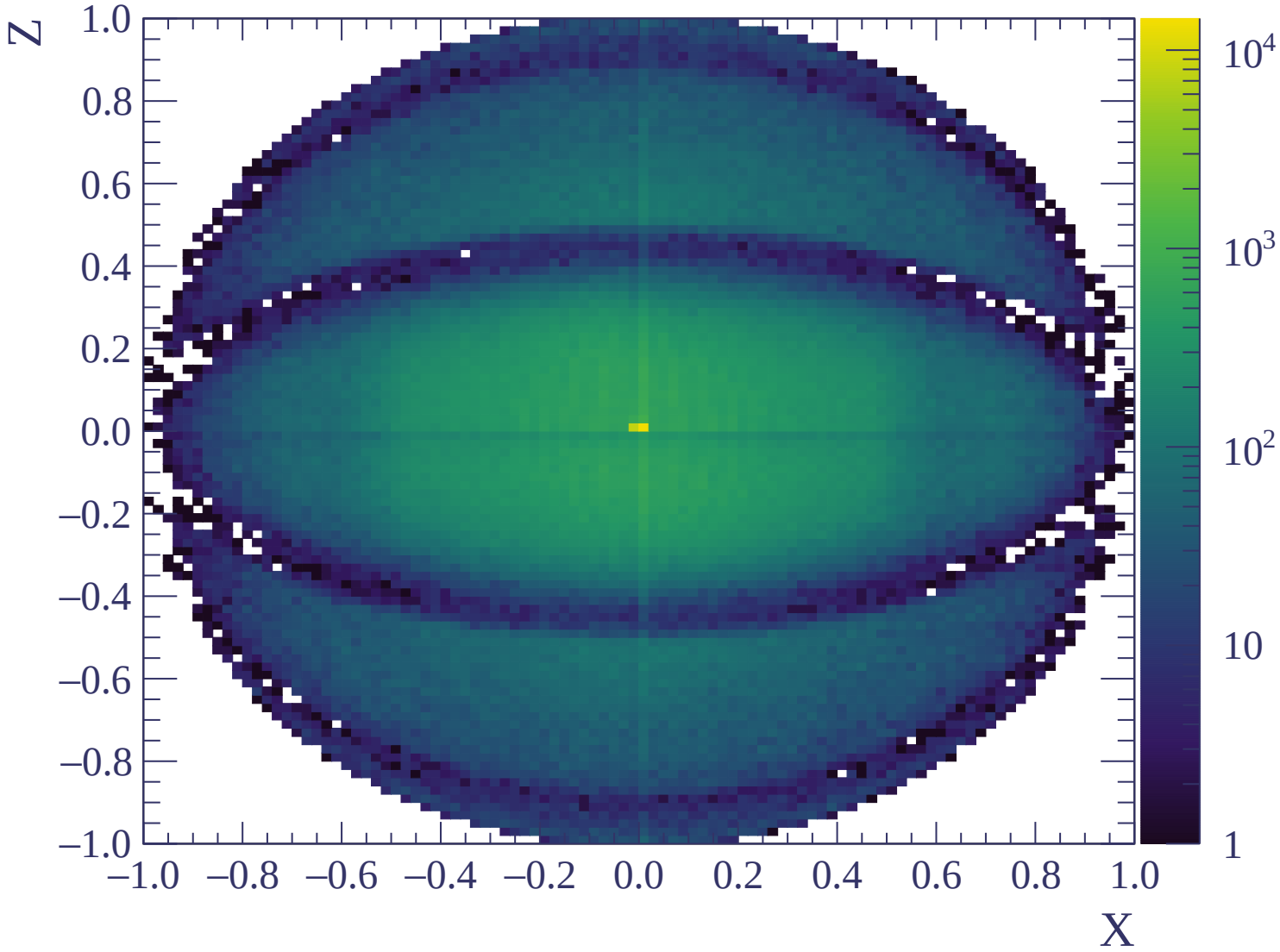


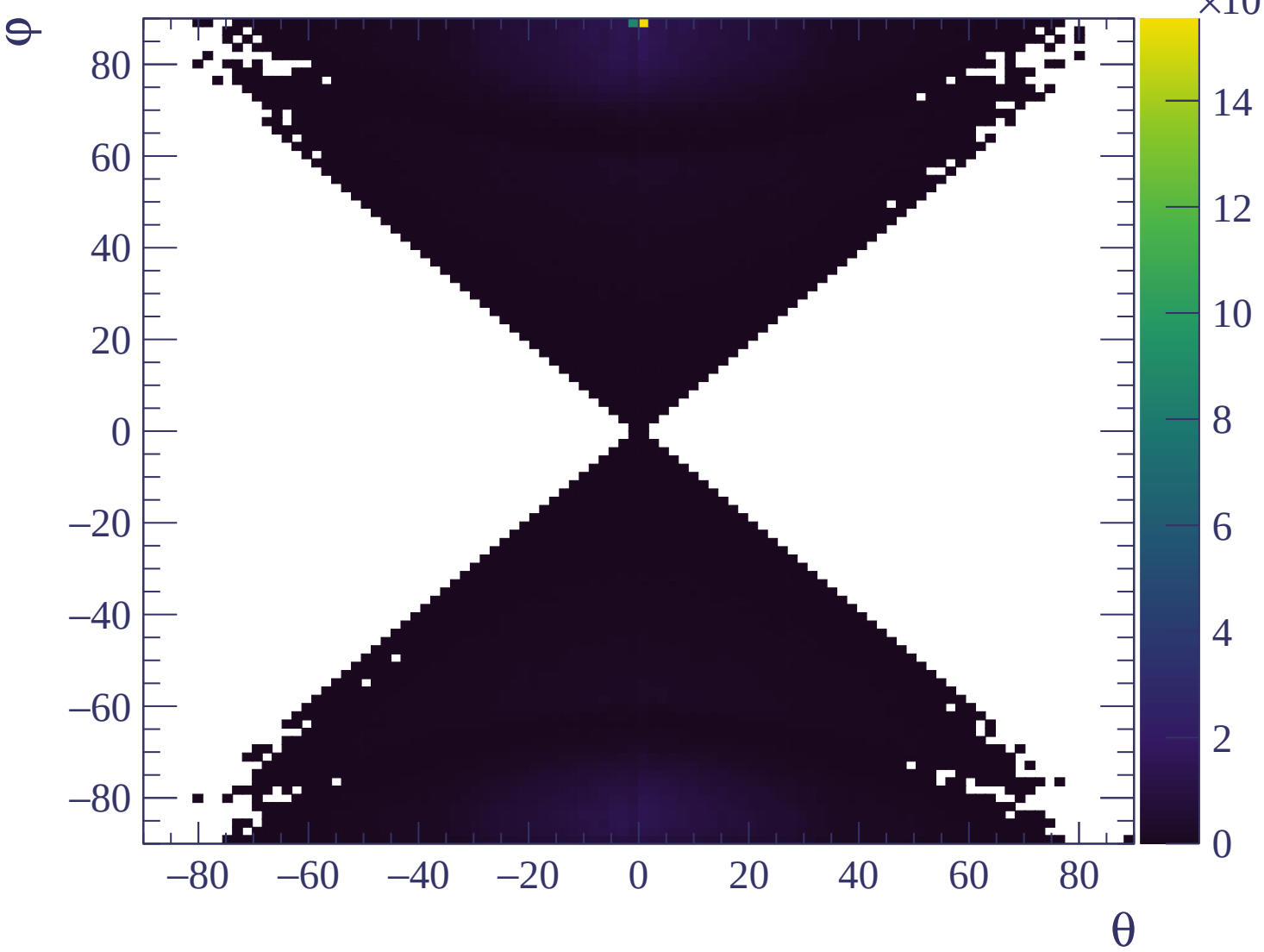


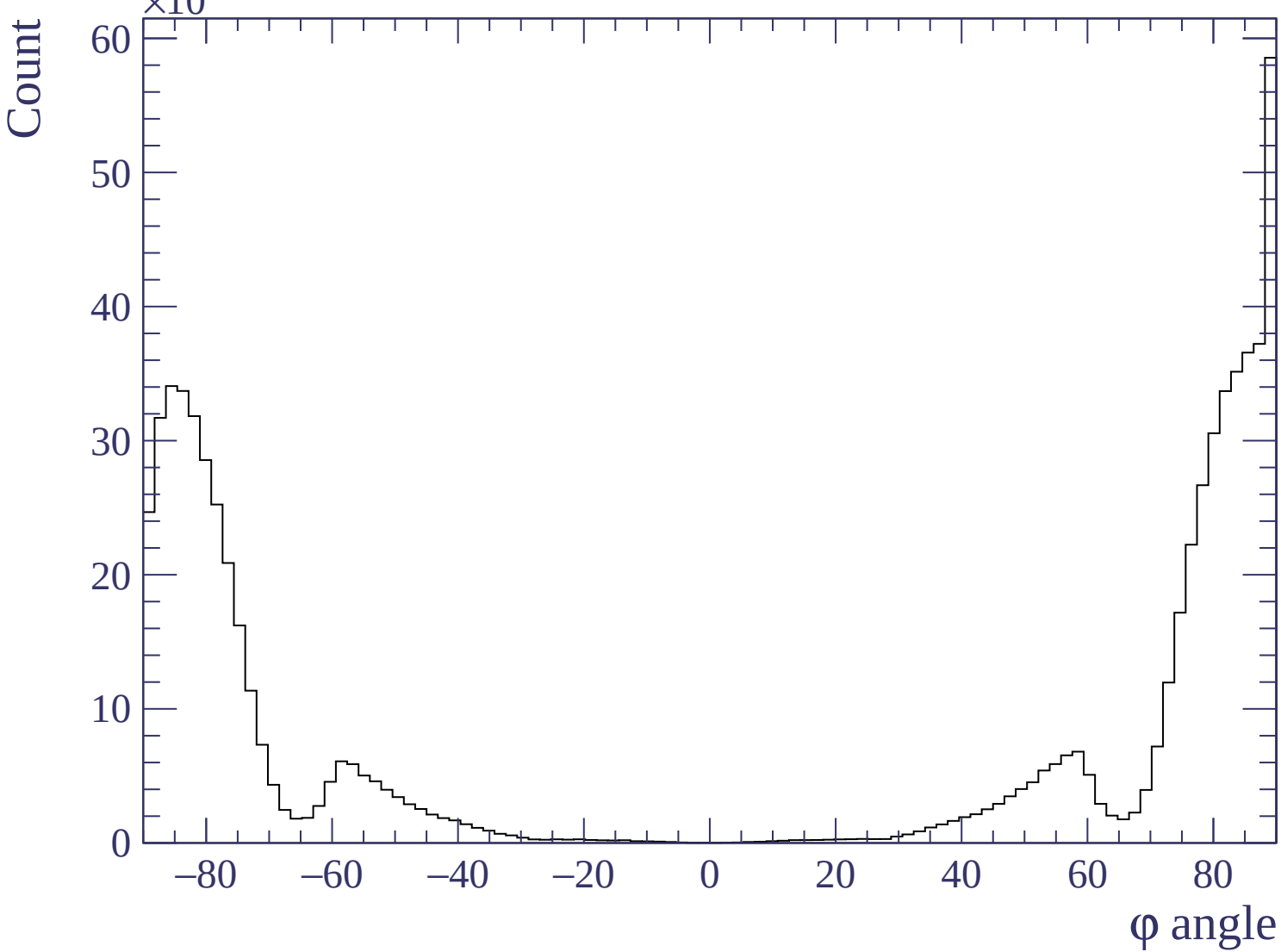


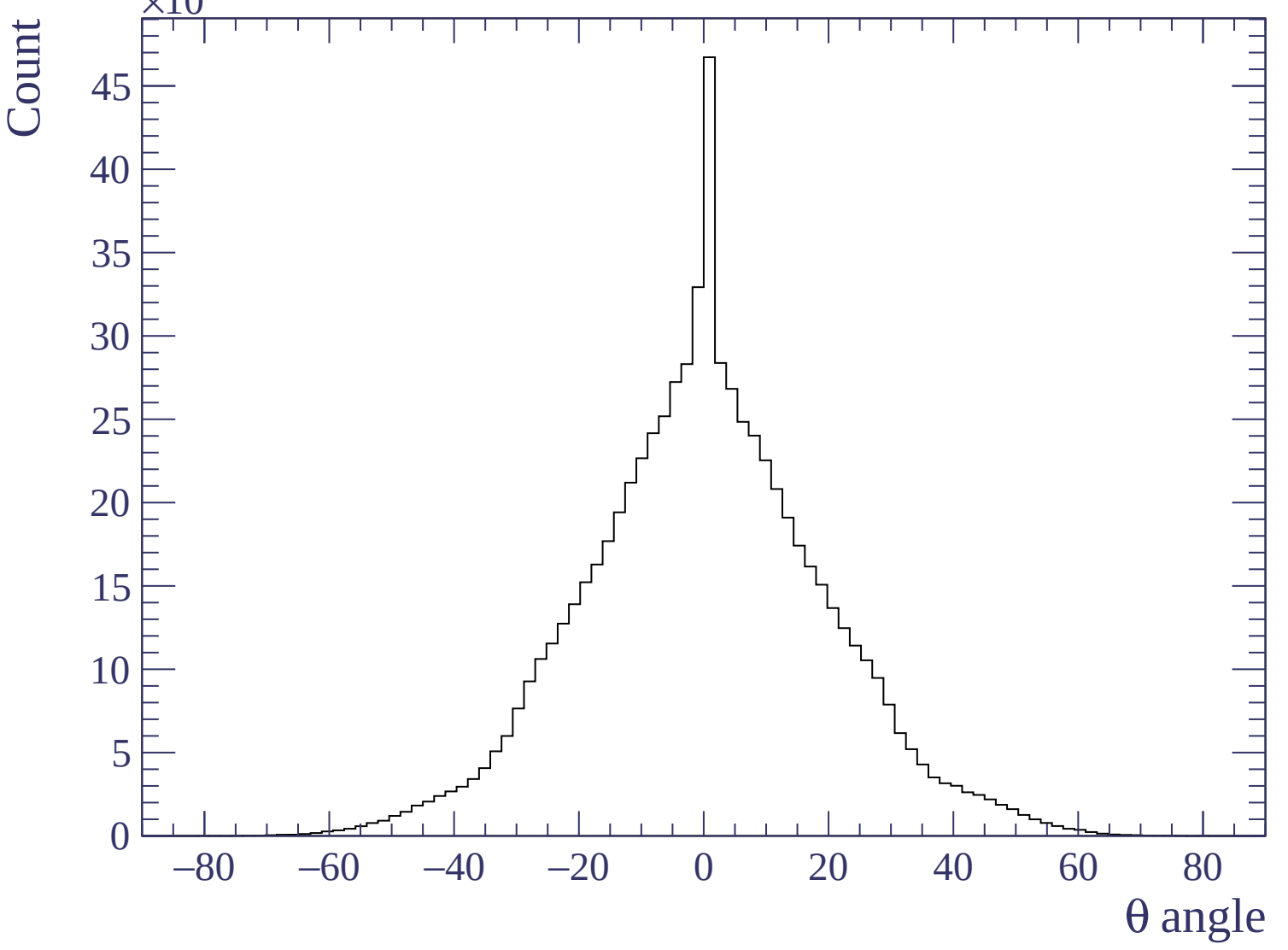


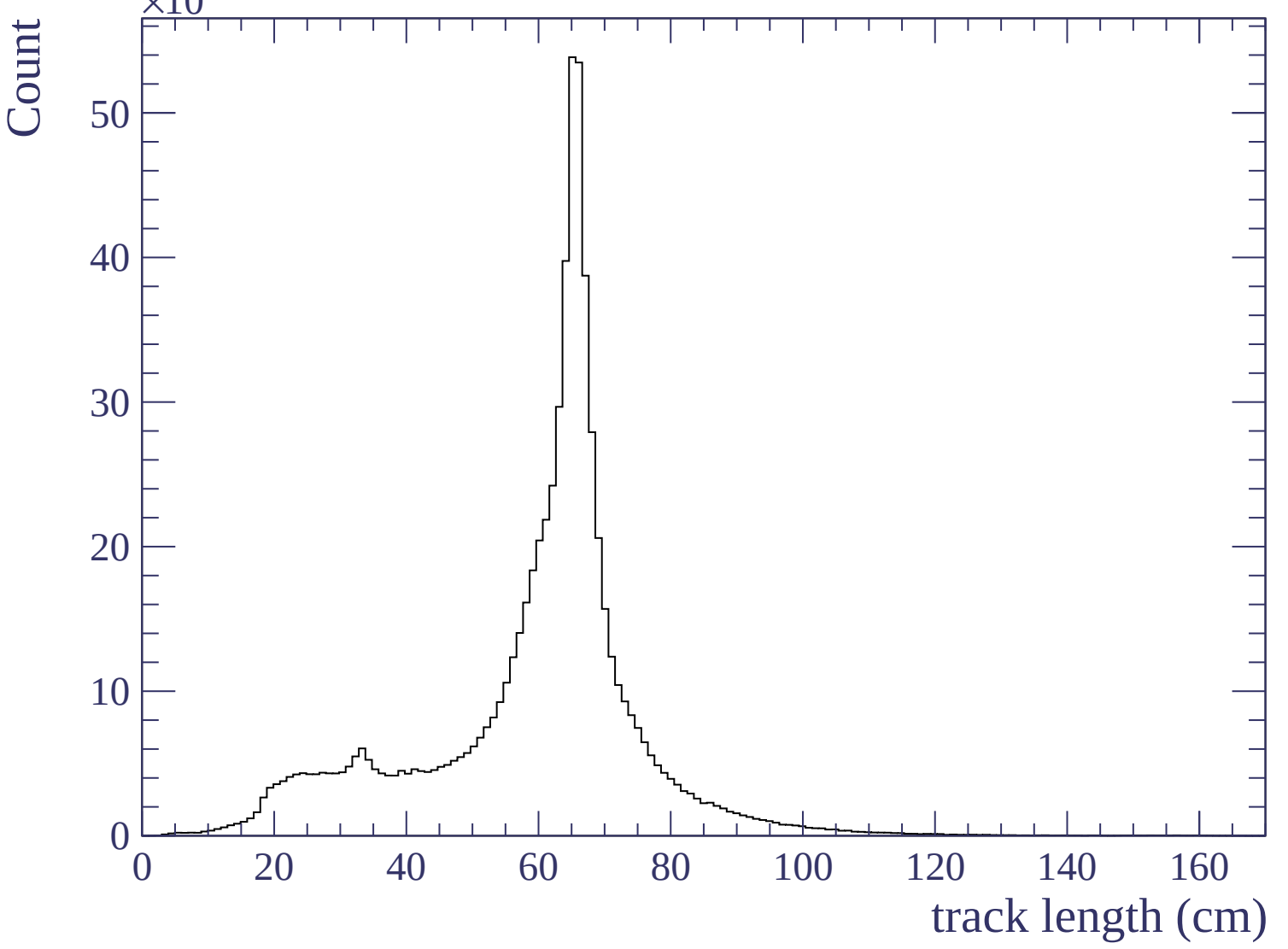


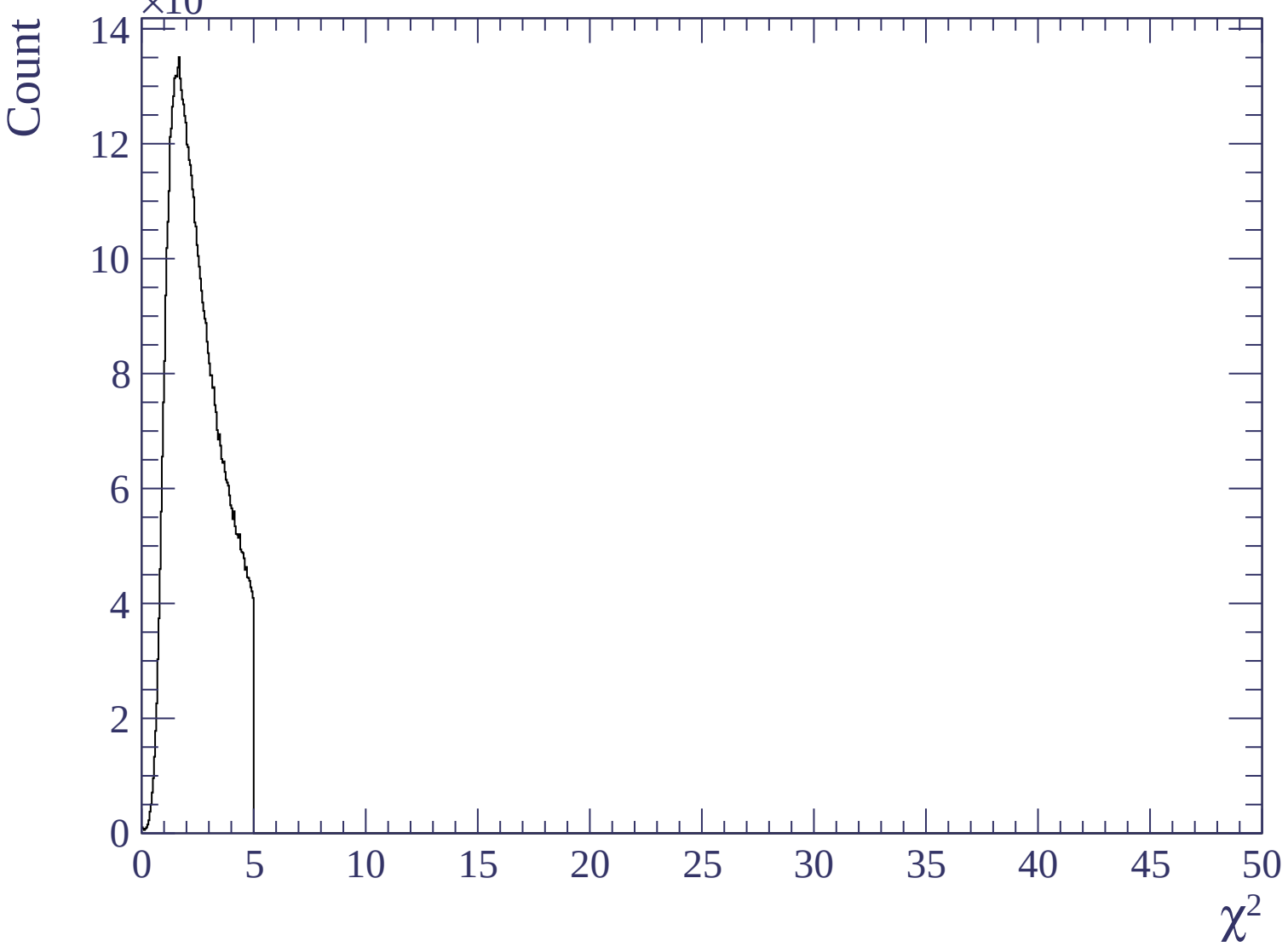


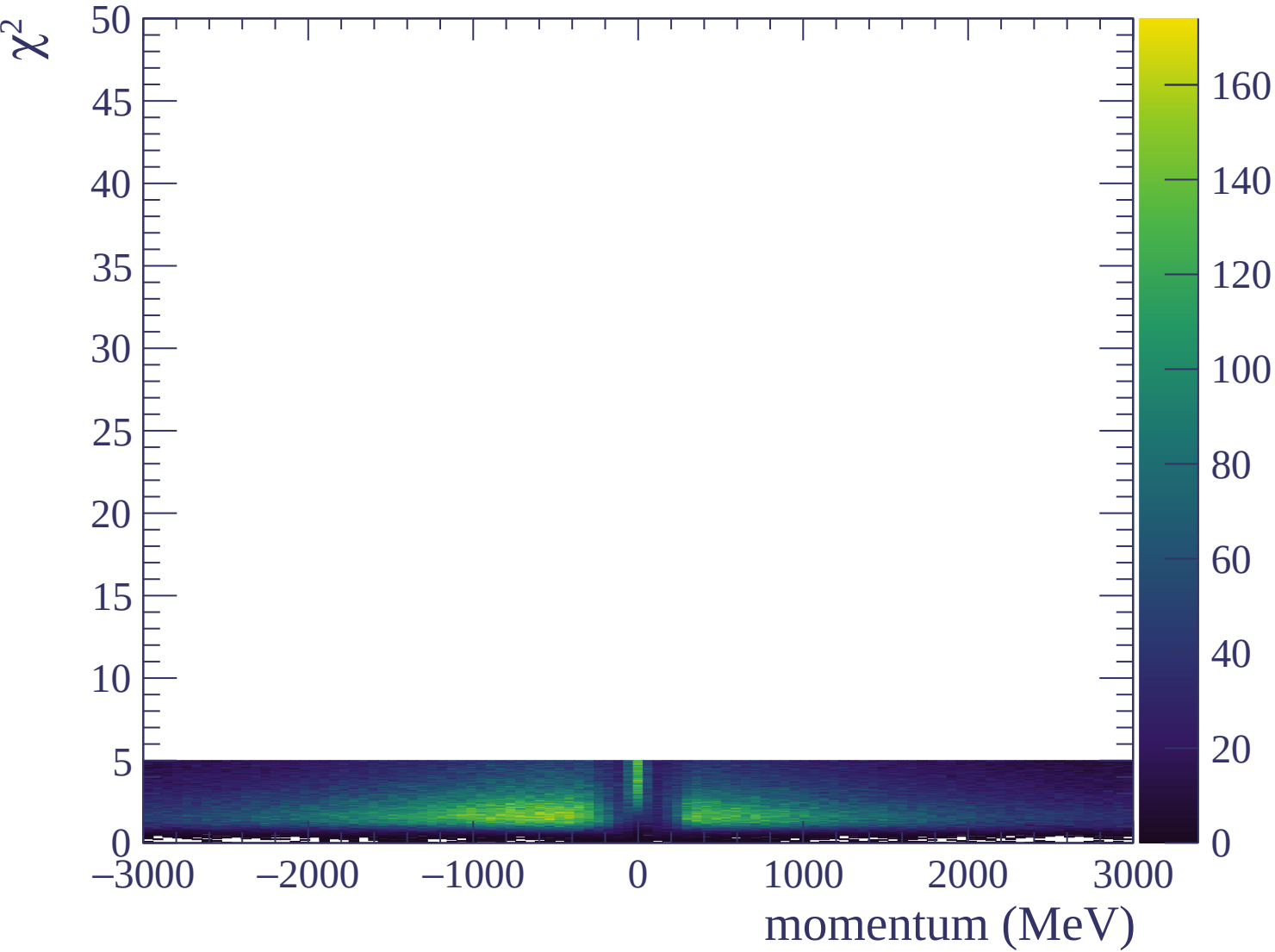


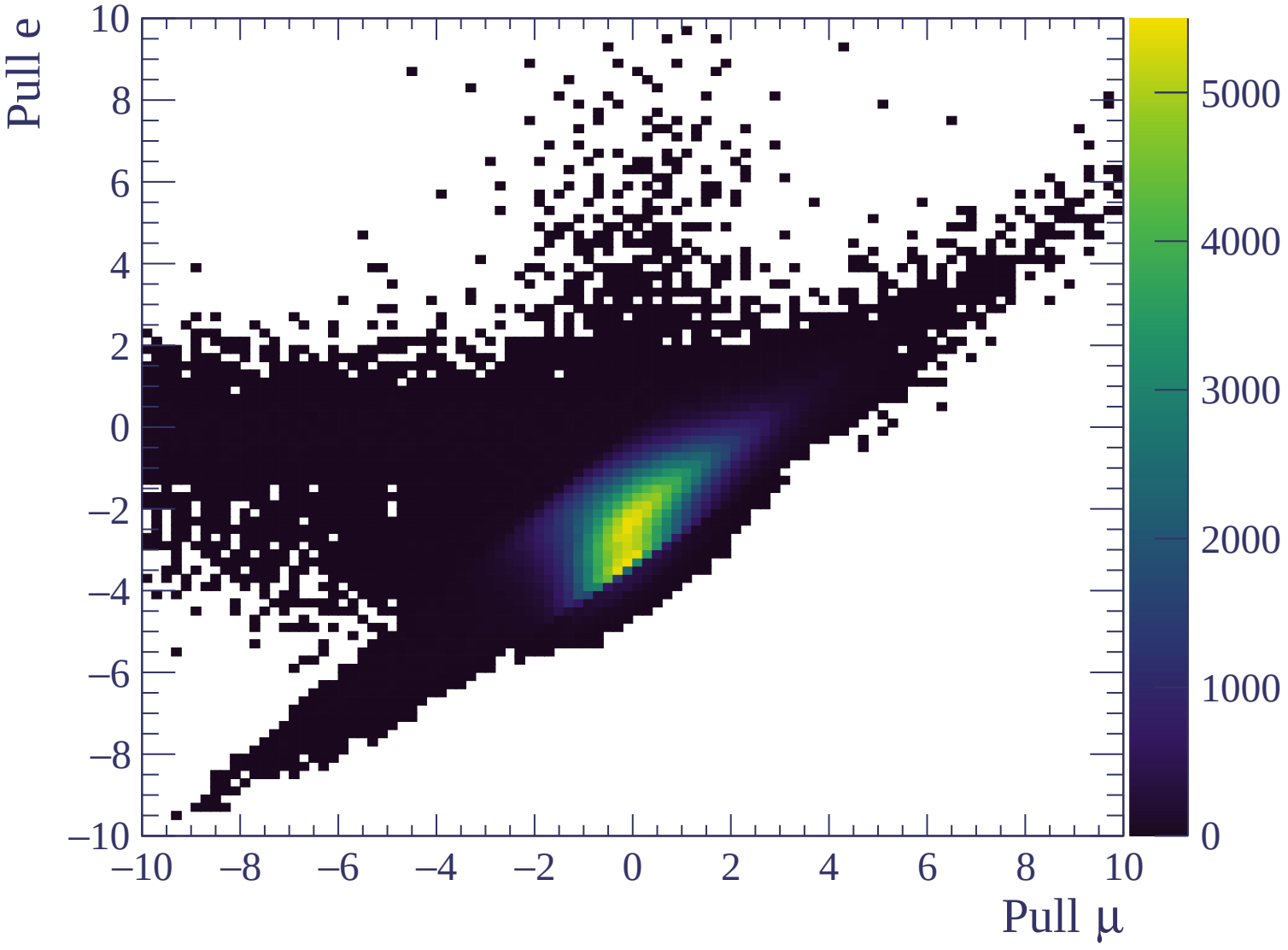


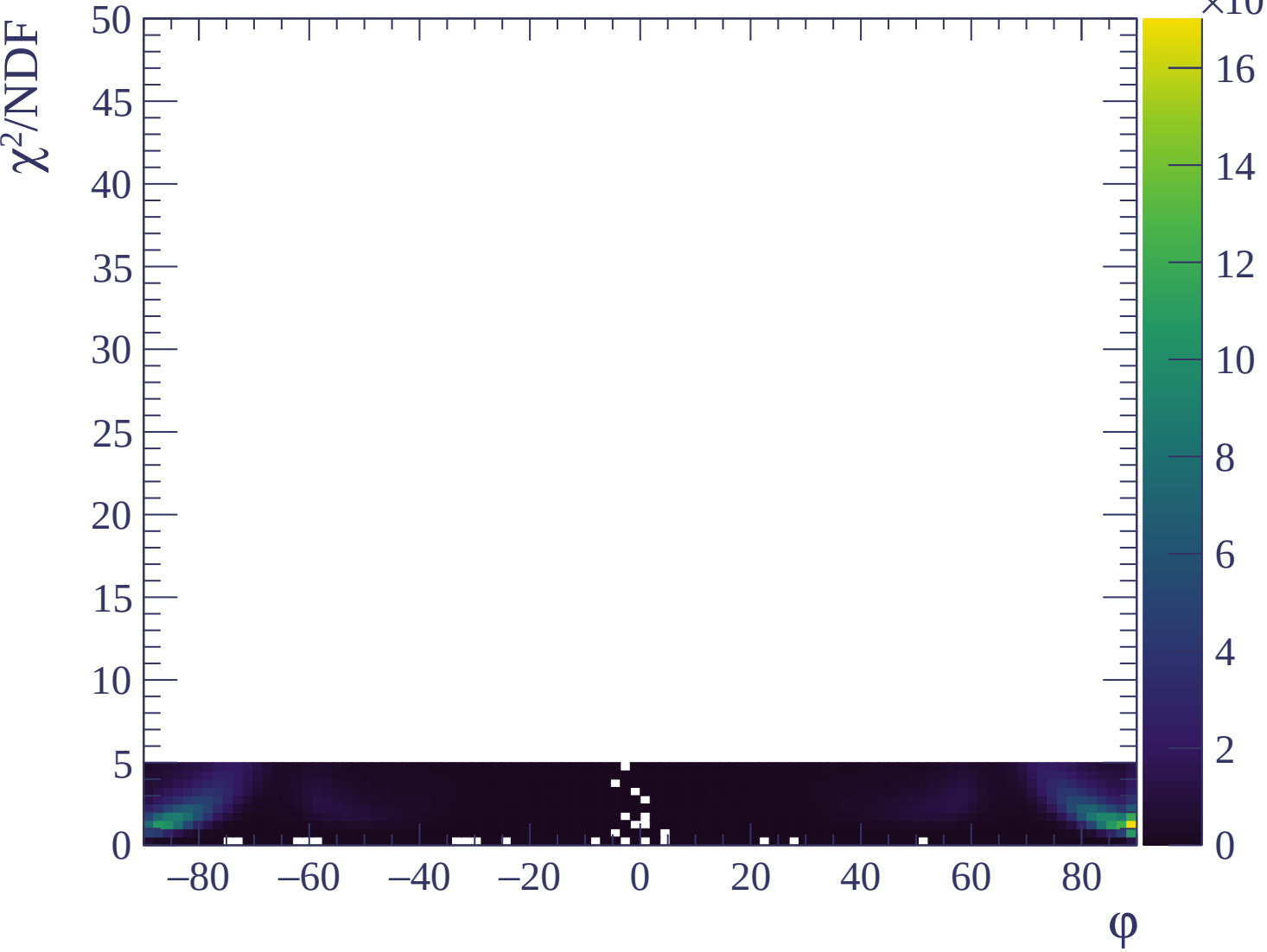


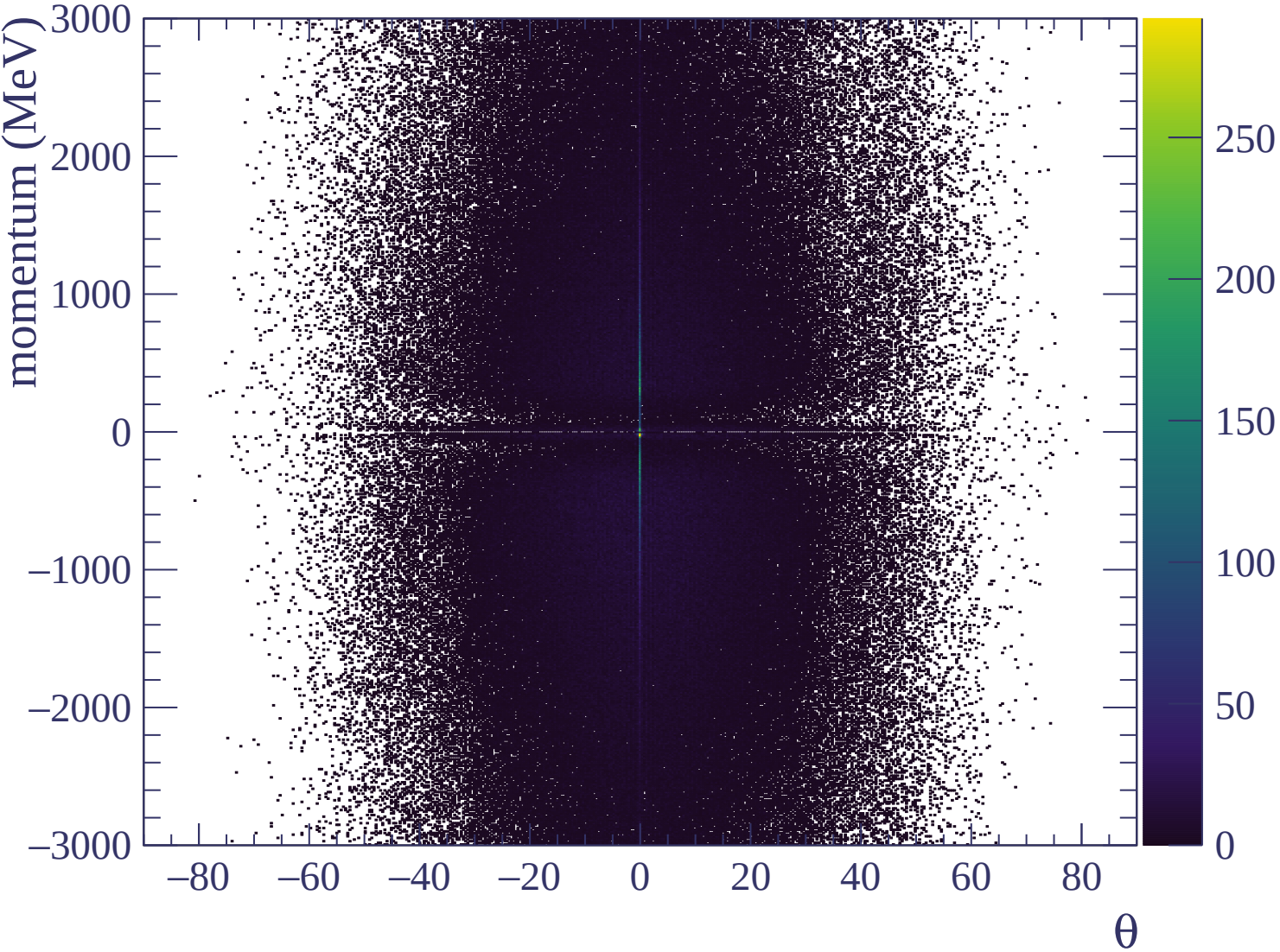


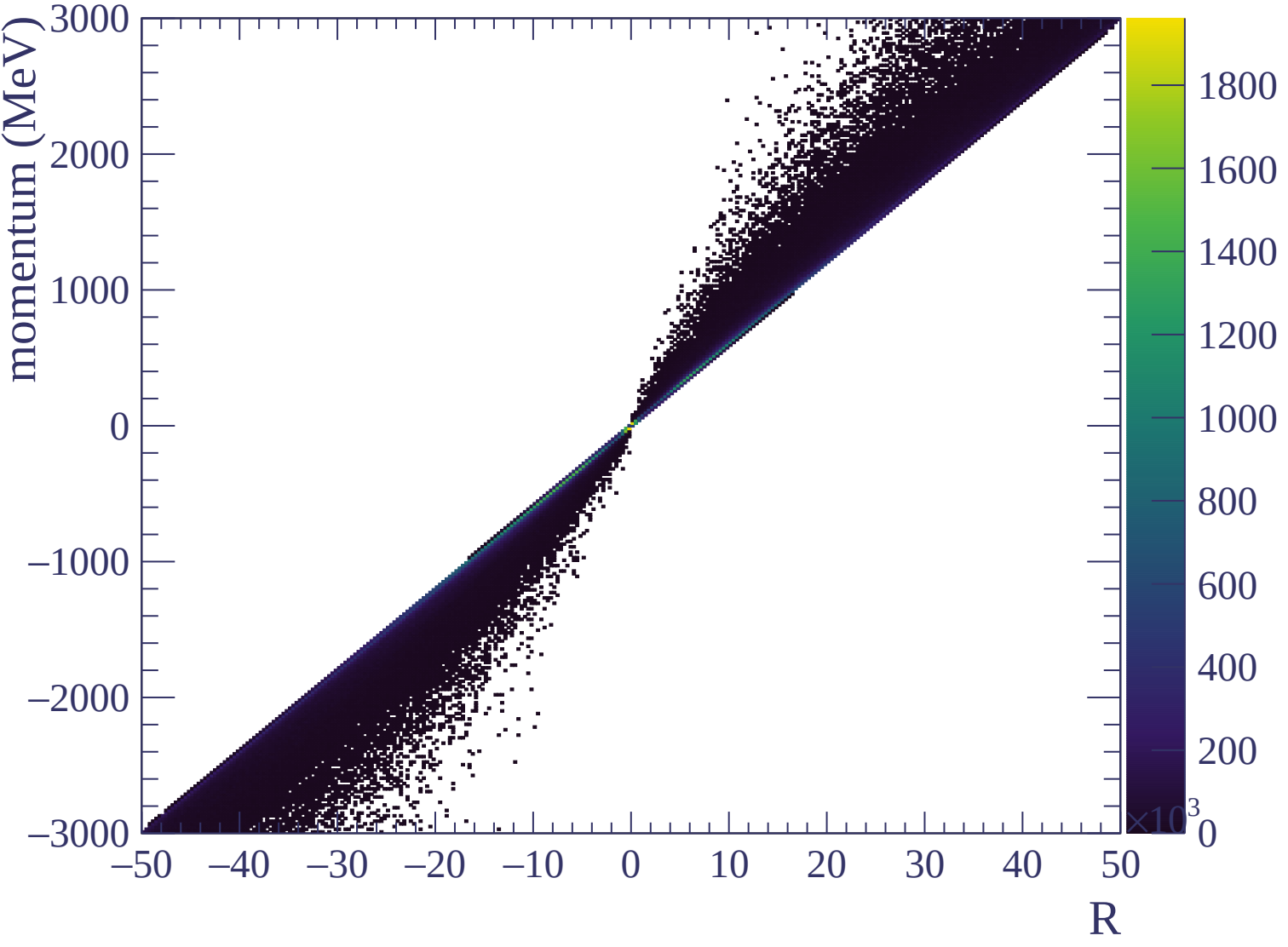


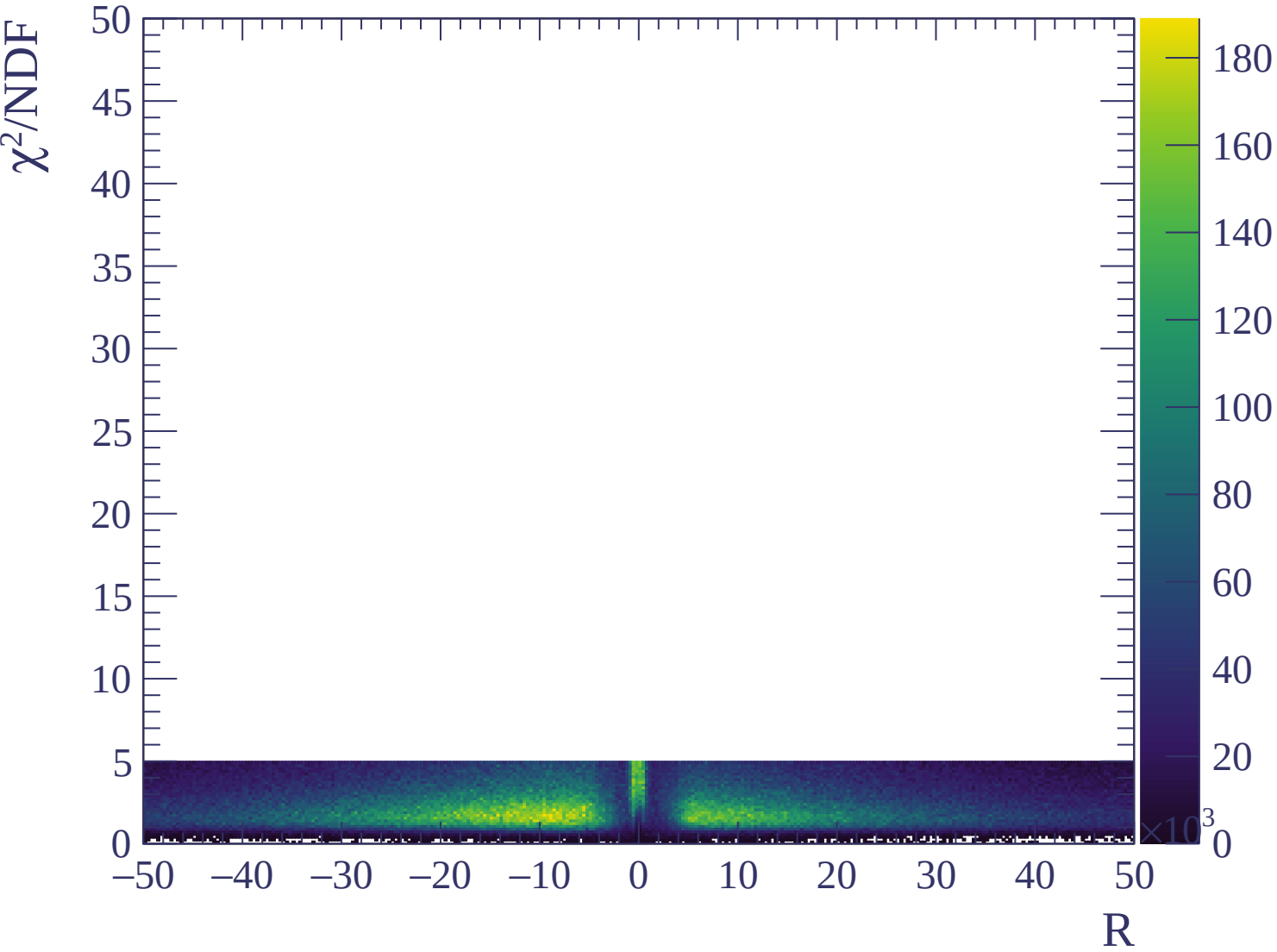


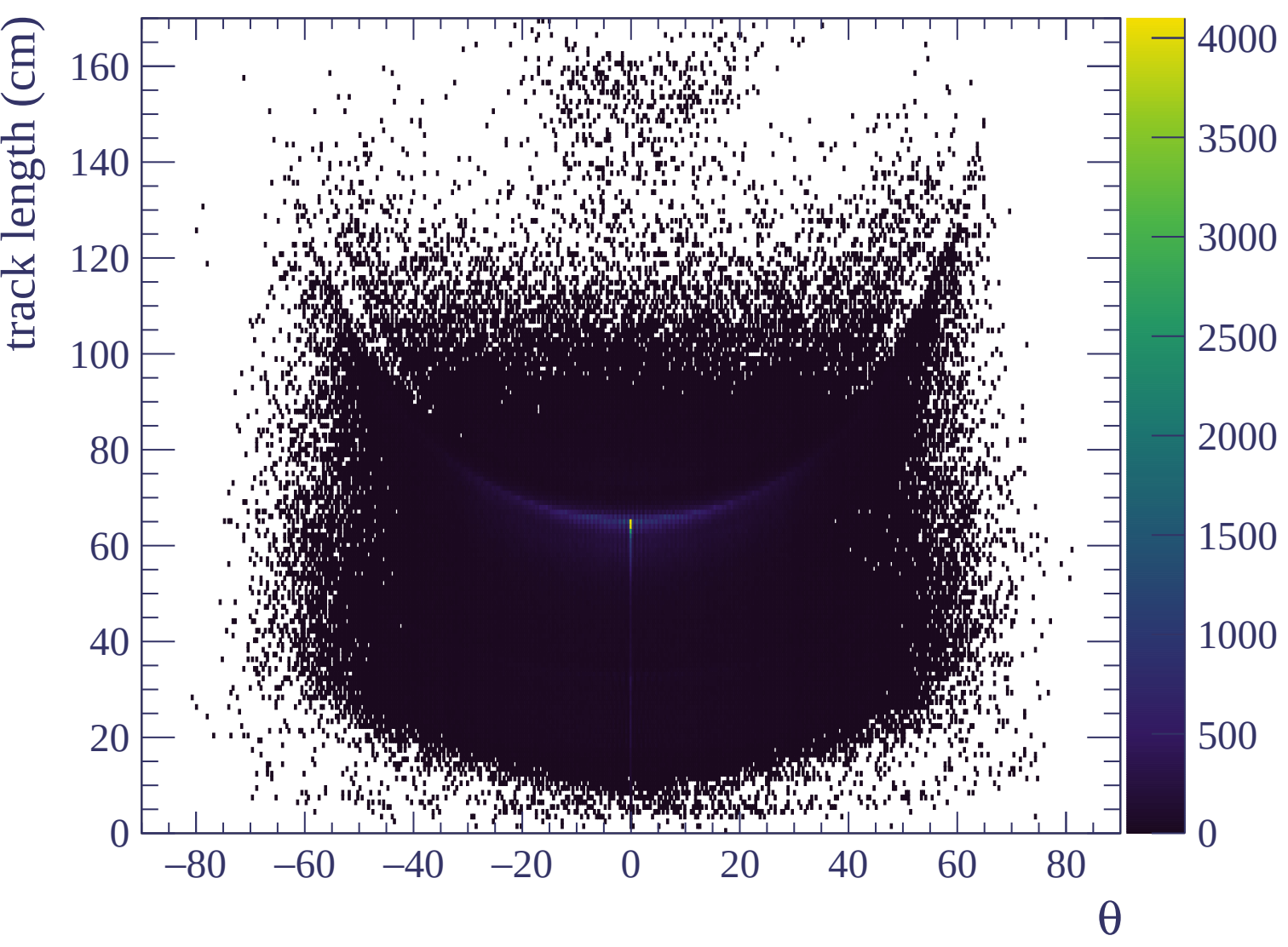






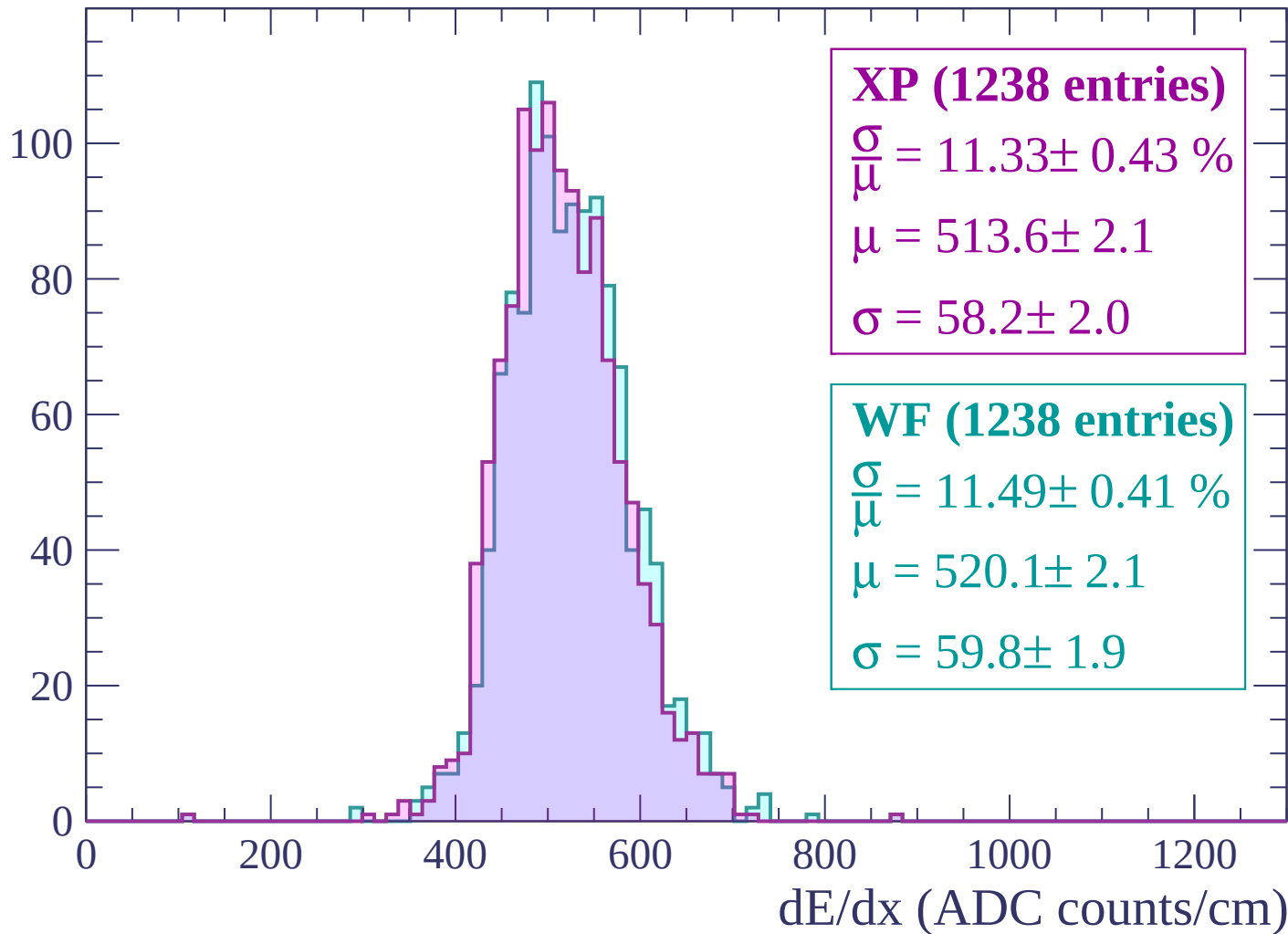






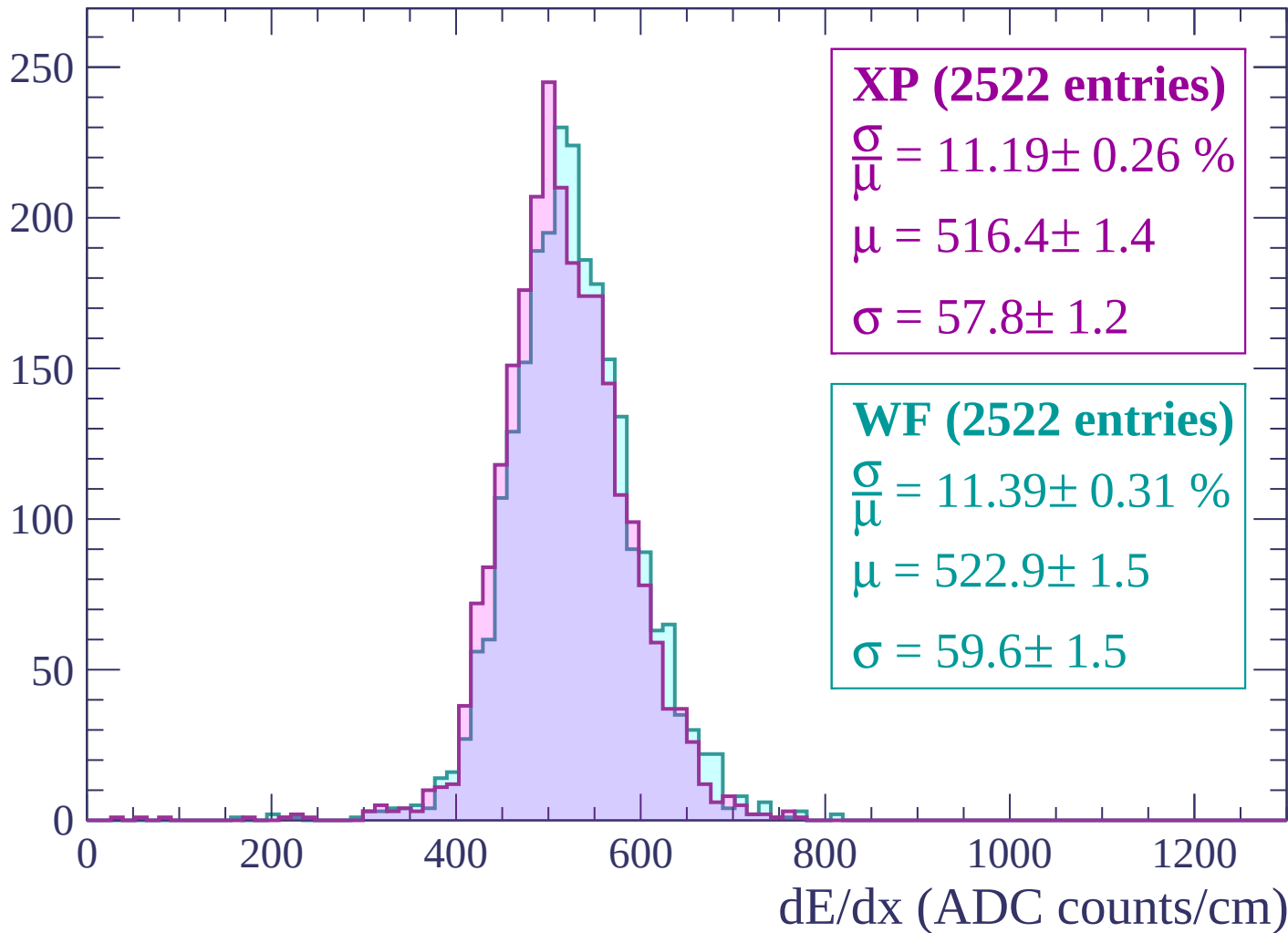
Energy loss | $-3000 < p < -2940$

Count



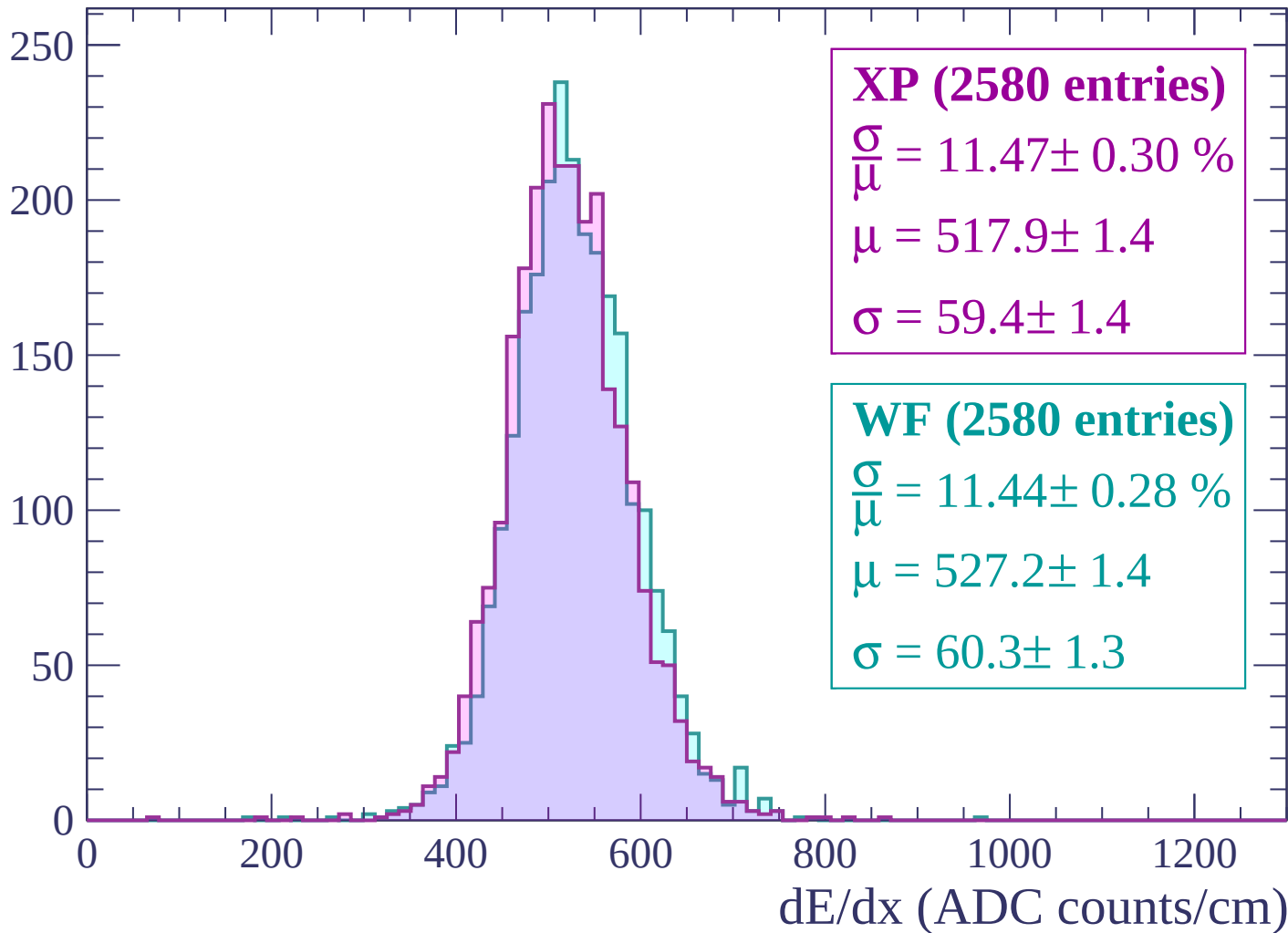
Energy loss | -2940 < p < -2880

Count



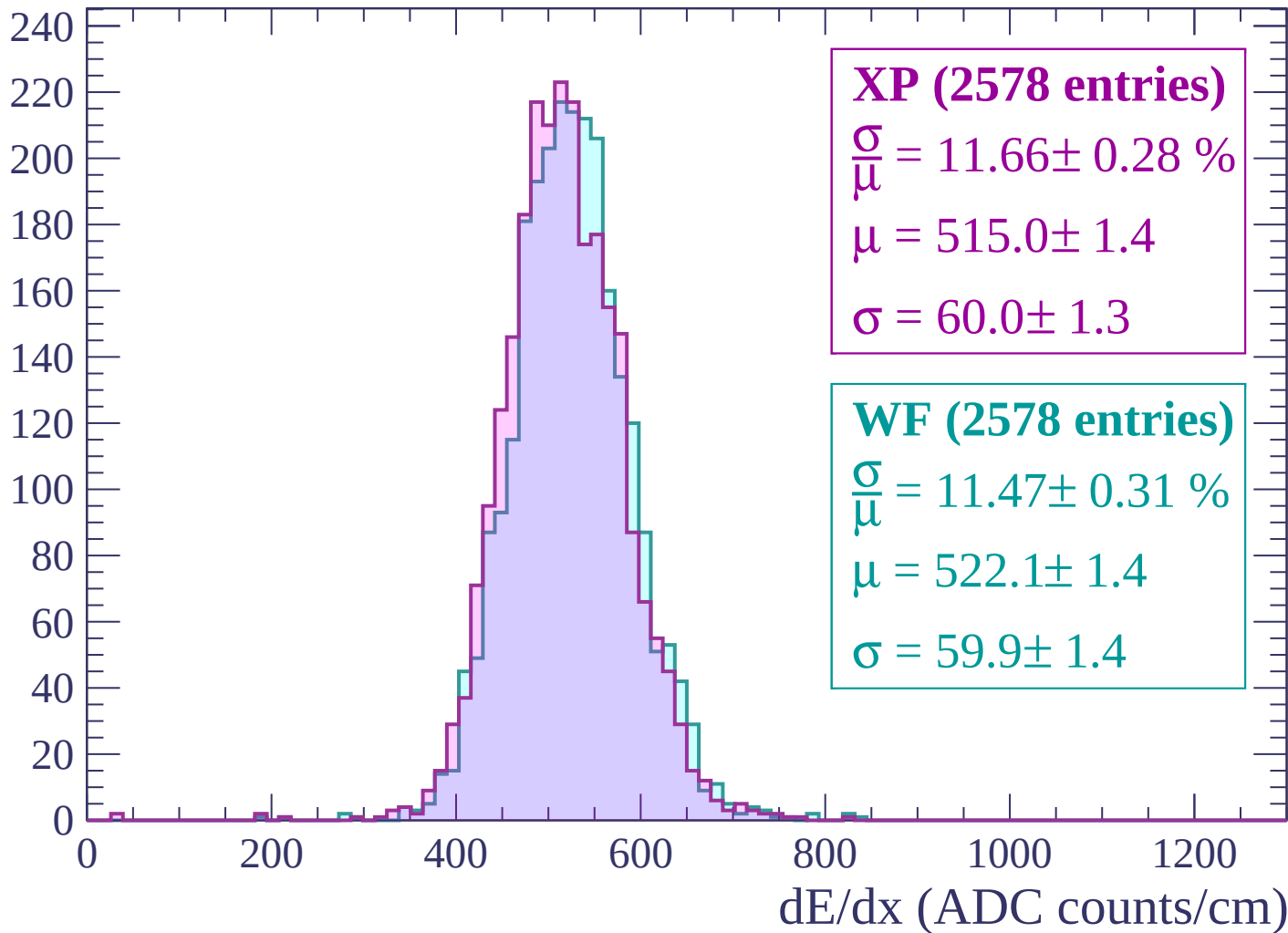
Energy loss | $-2880 < p < -2820$

Count



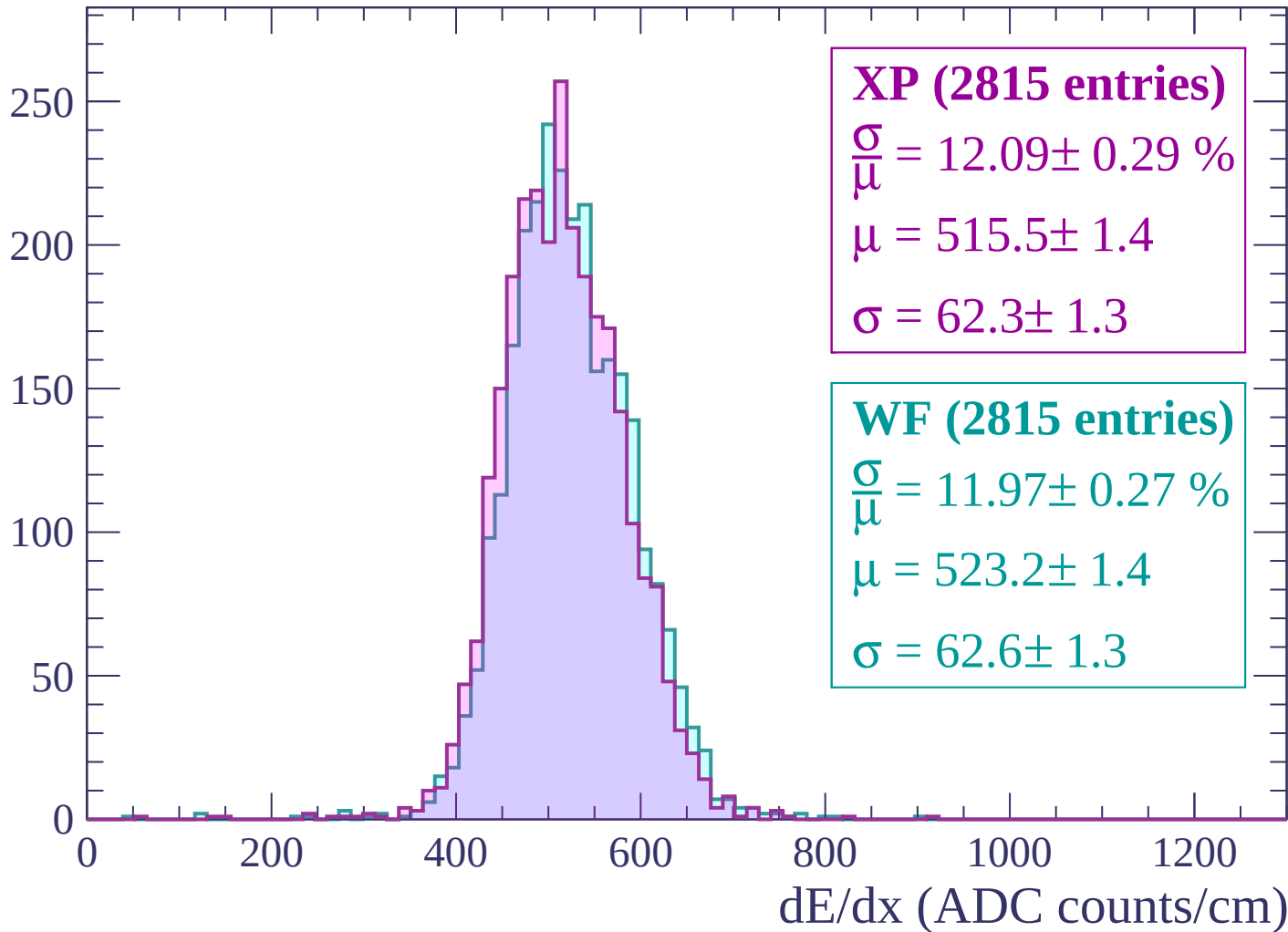
Energy loss | $-2820 < p < -2760$

Count



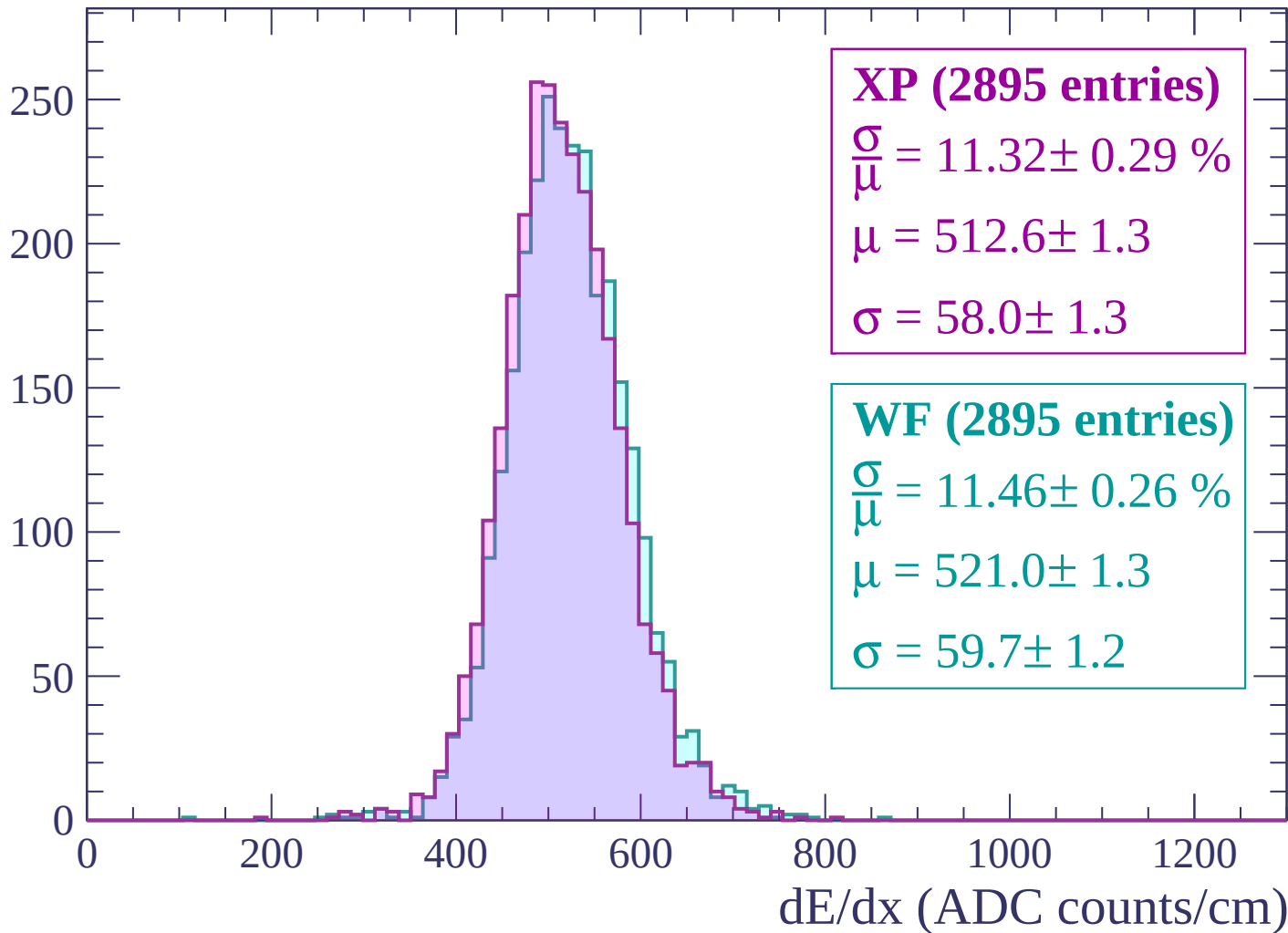
Energy loss | $-2760 < p < -2700$

Count



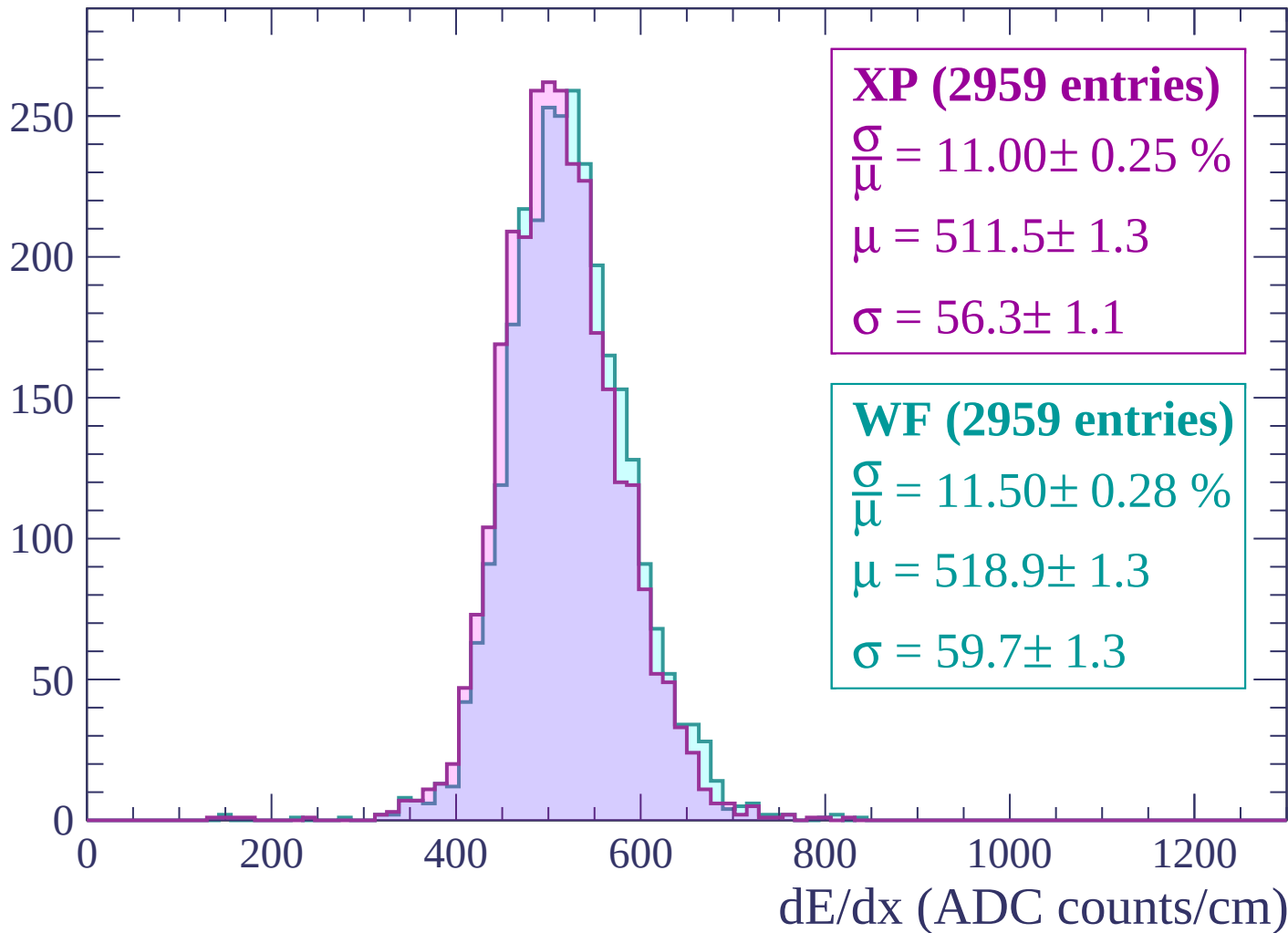
Energy loss | -2700 < p < -2640

Count



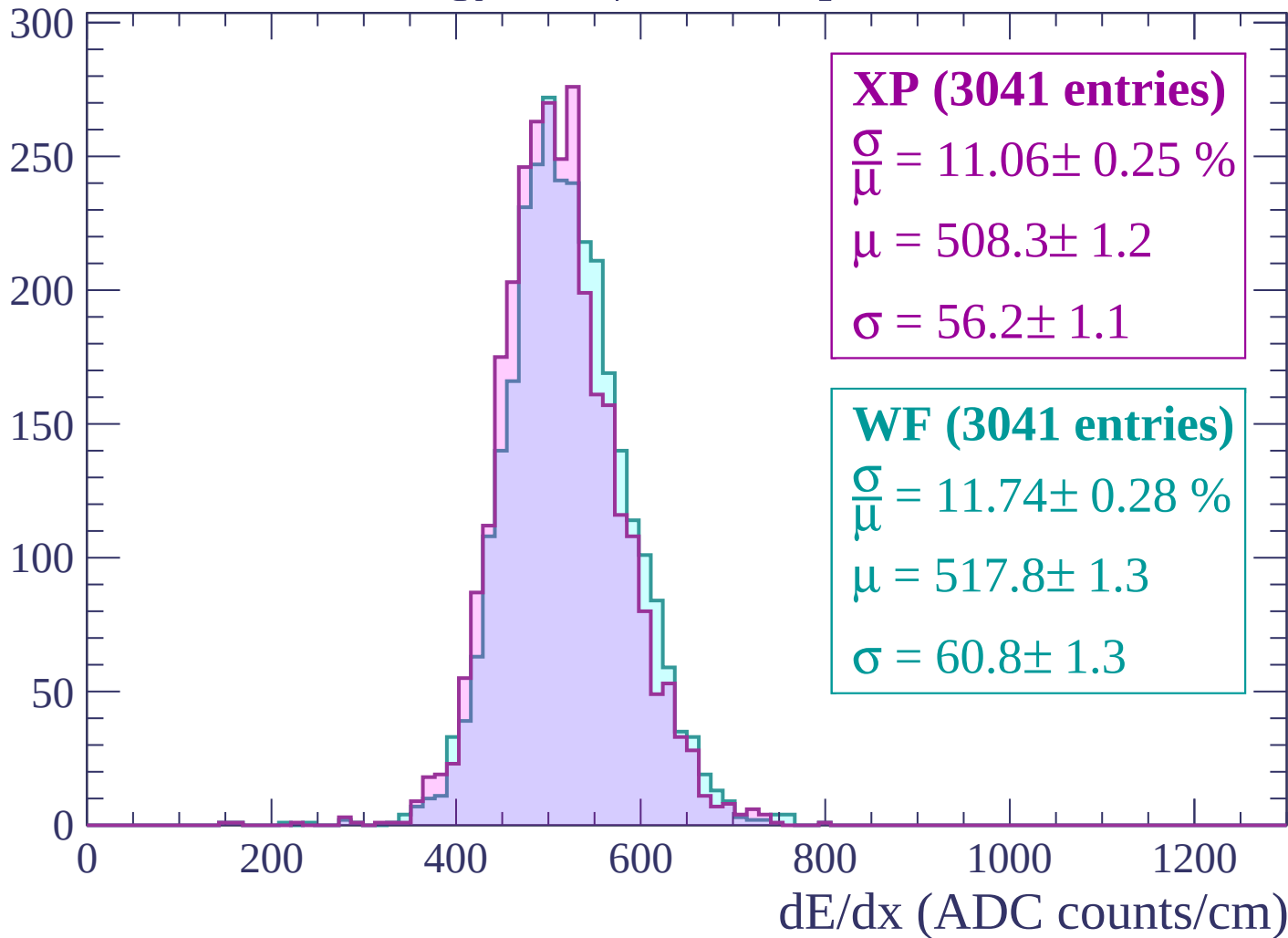
Energy loss | $-2640 < p < -2580$

Count



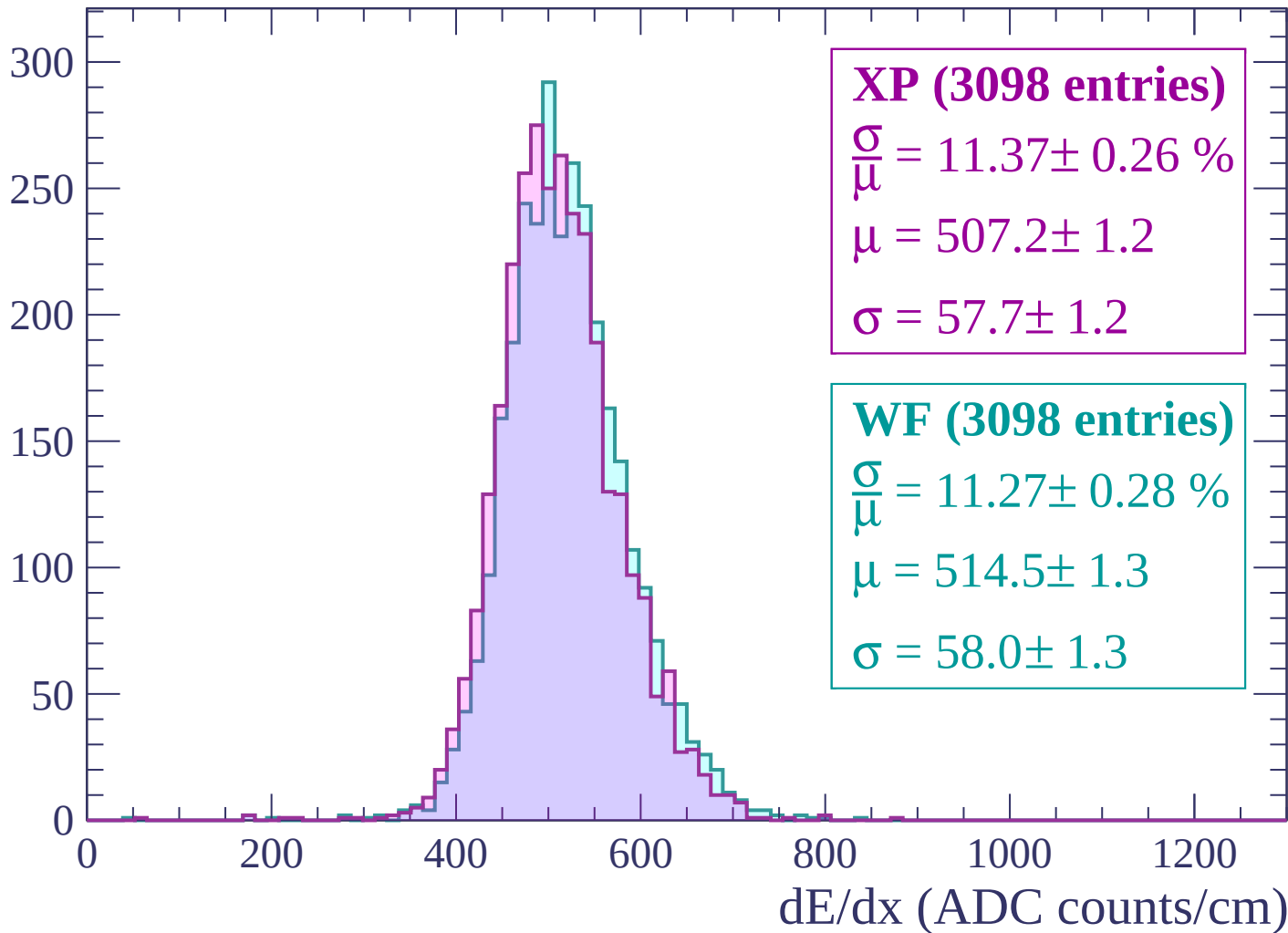
Energy loss | -2580 < p < -2520

Count



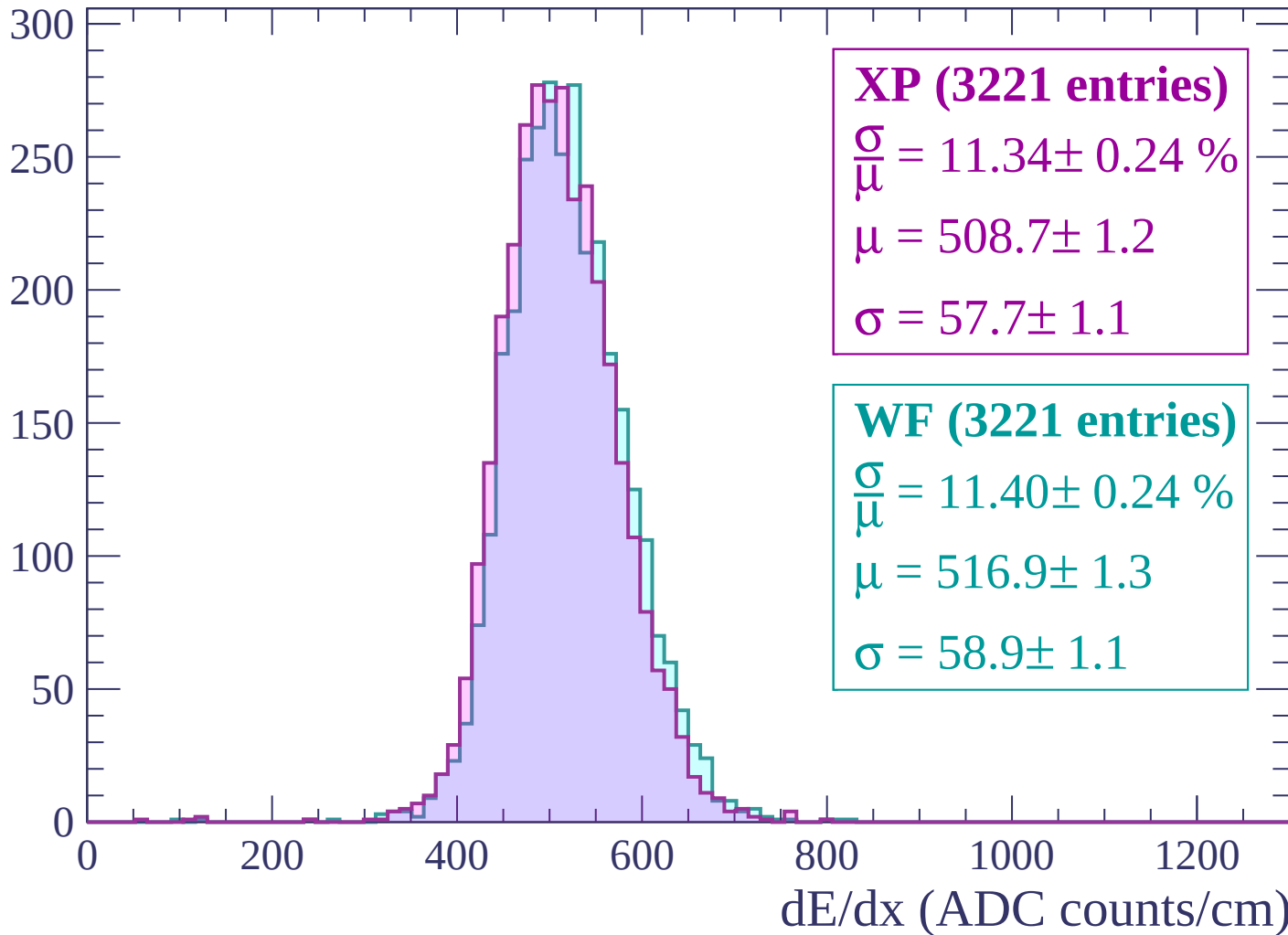
Energy loss | -2520 < p < -2460

Count



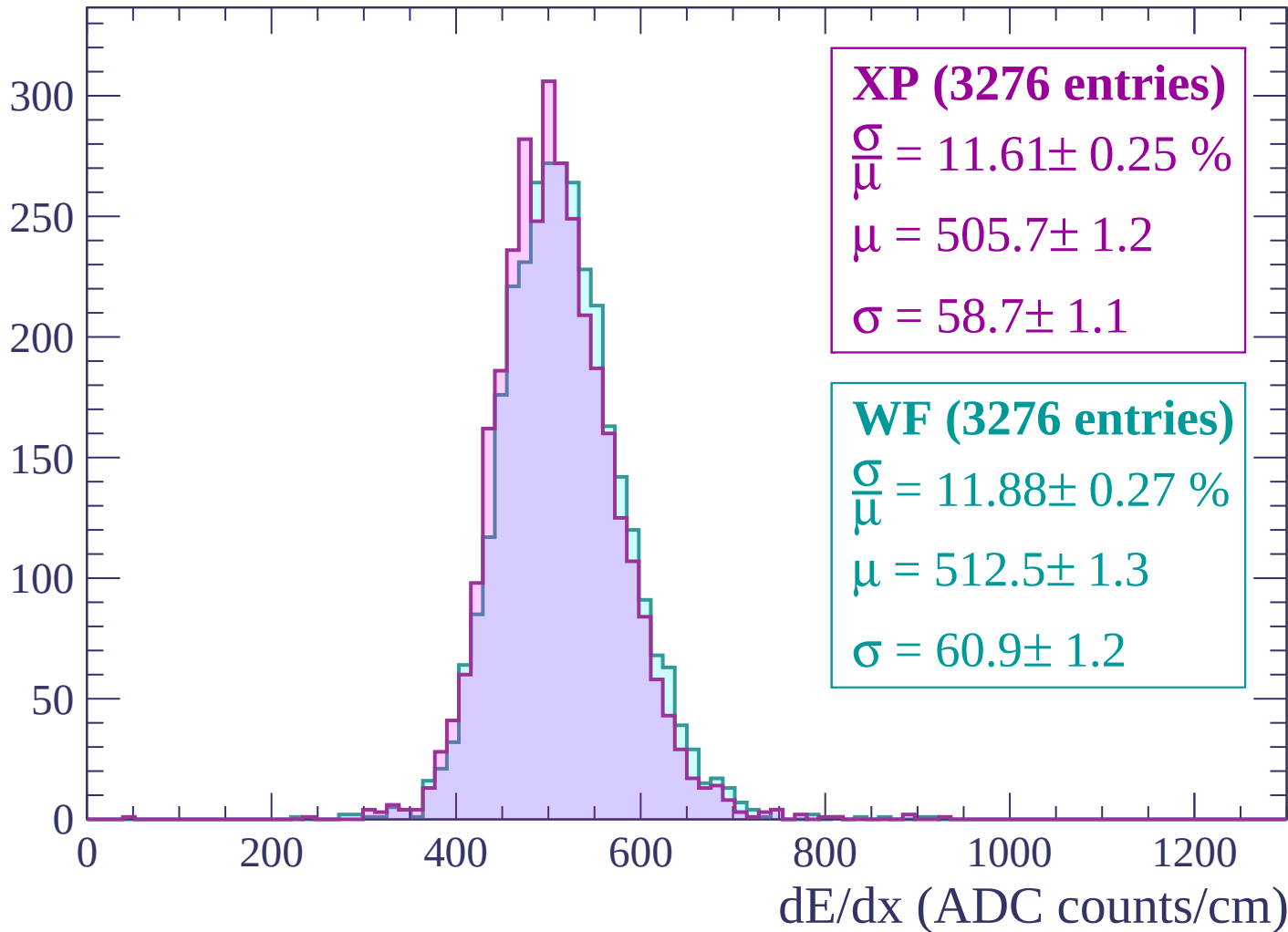
Energy loss | $-2460 < p < -2400$

Count



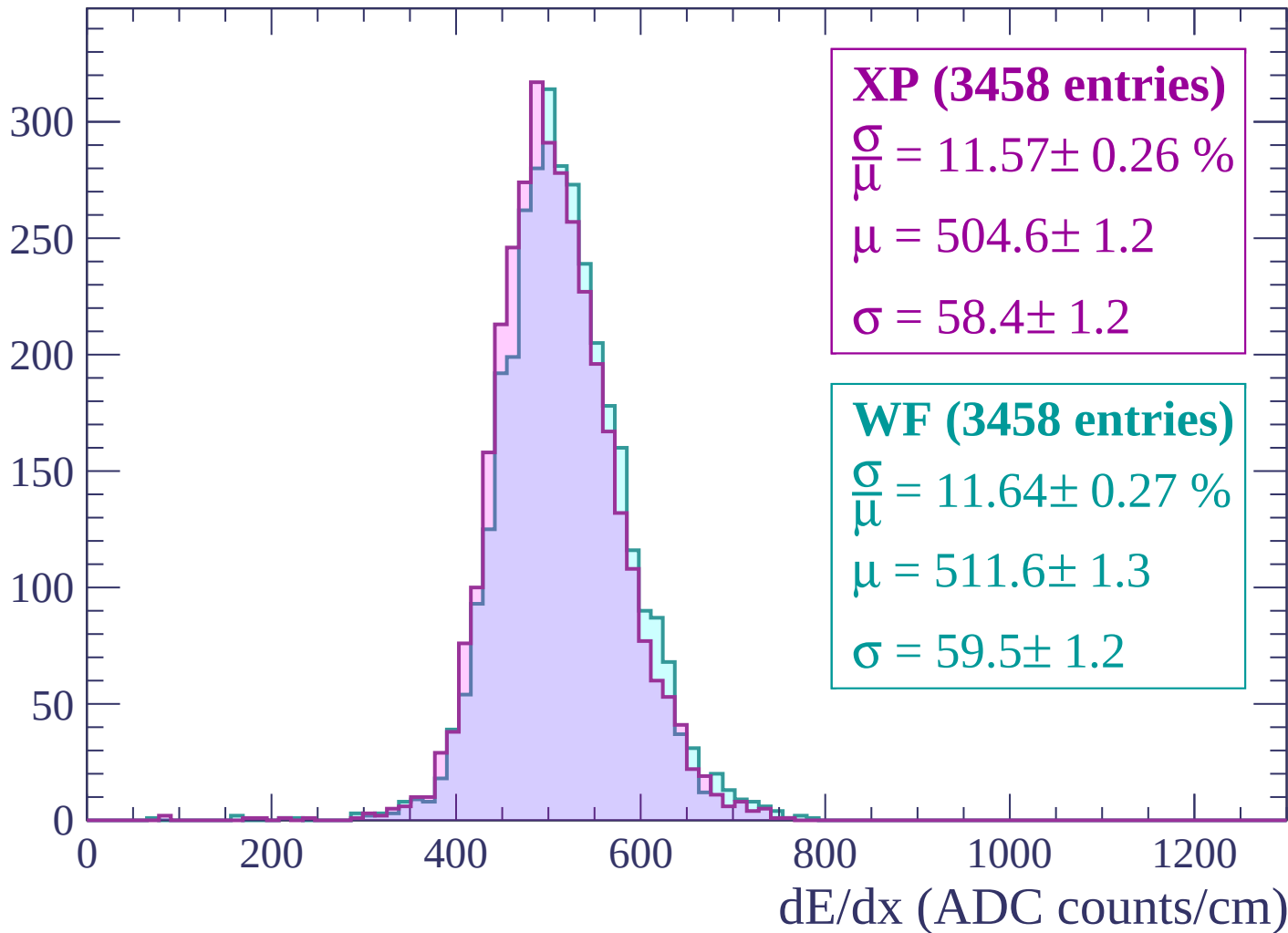
Energy loss | -2400 < p < -2340

Count



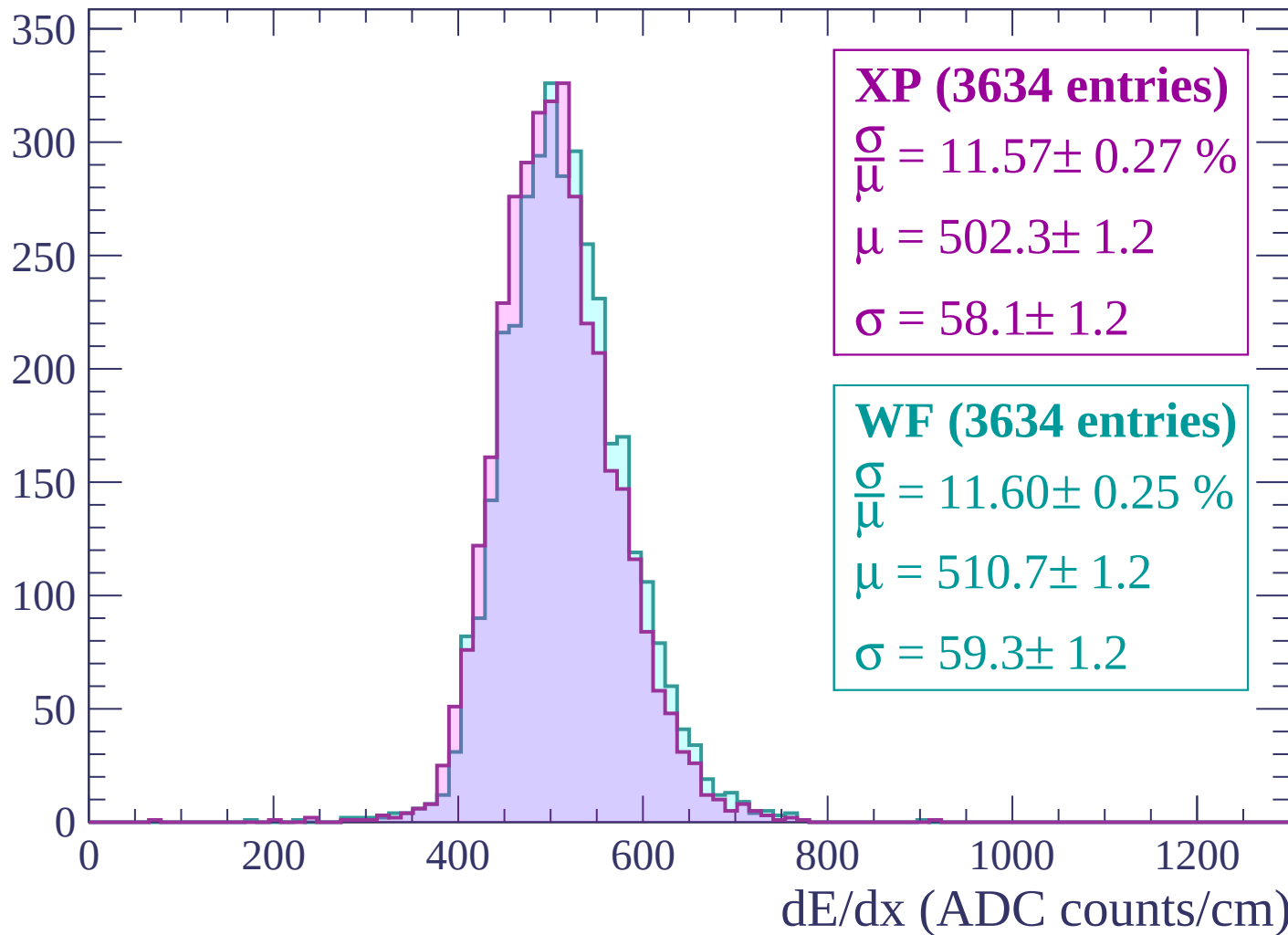
Energy loss | $-2340 < p < -2280$

Count



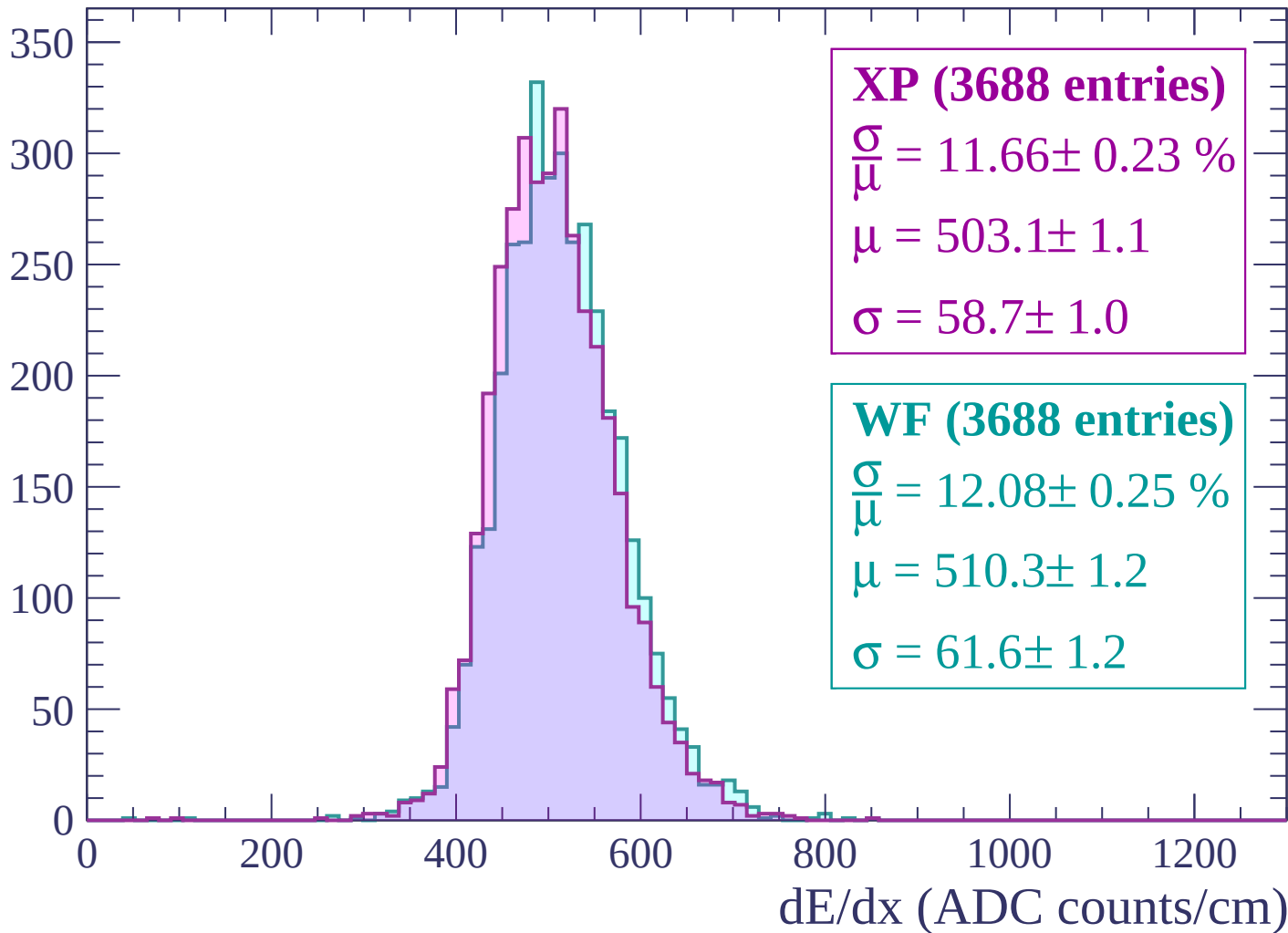
Energy loss | $-2280 < p < -2220$

Count



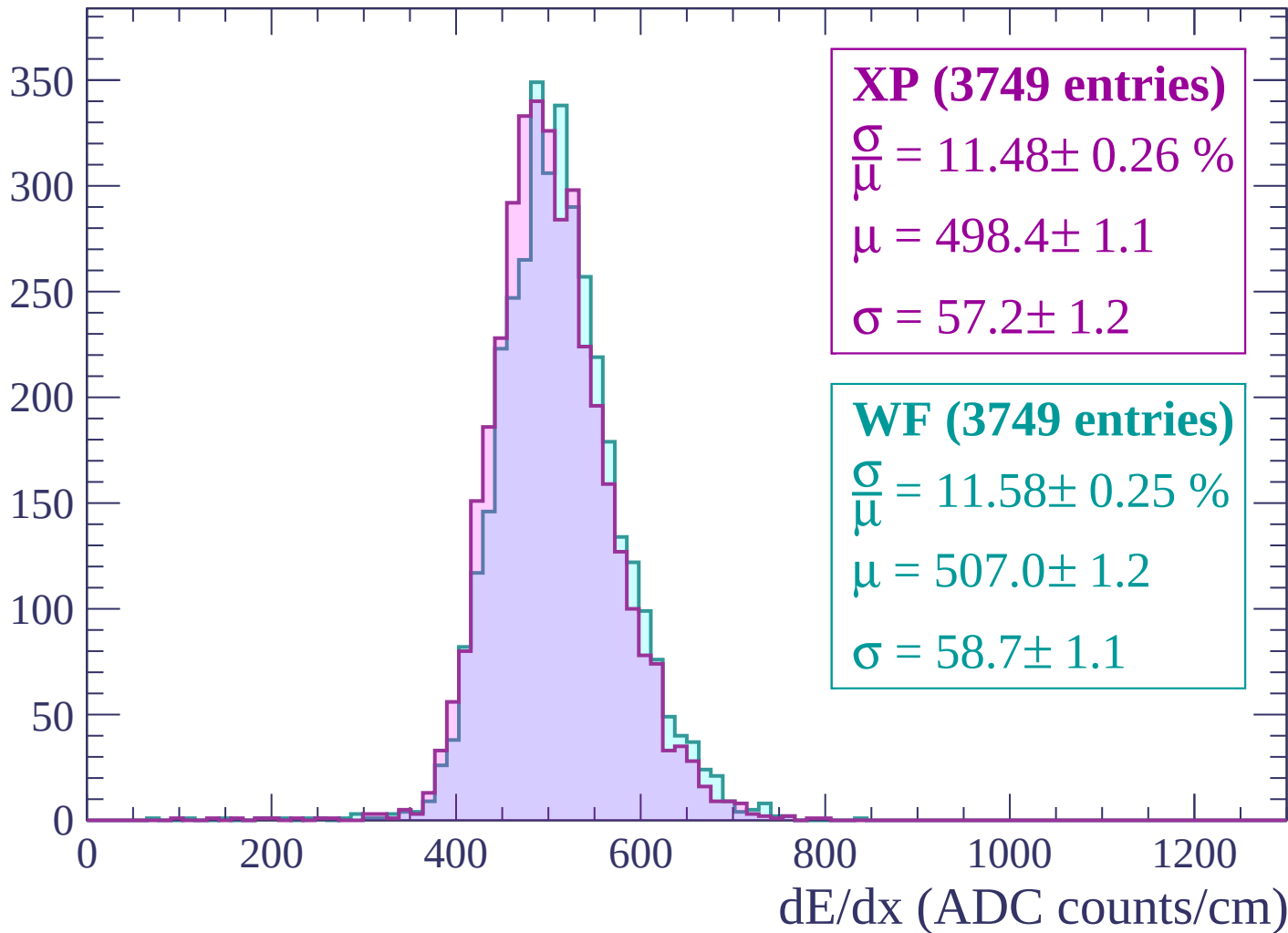
Energy loss | -2220 < p < -2160

Count



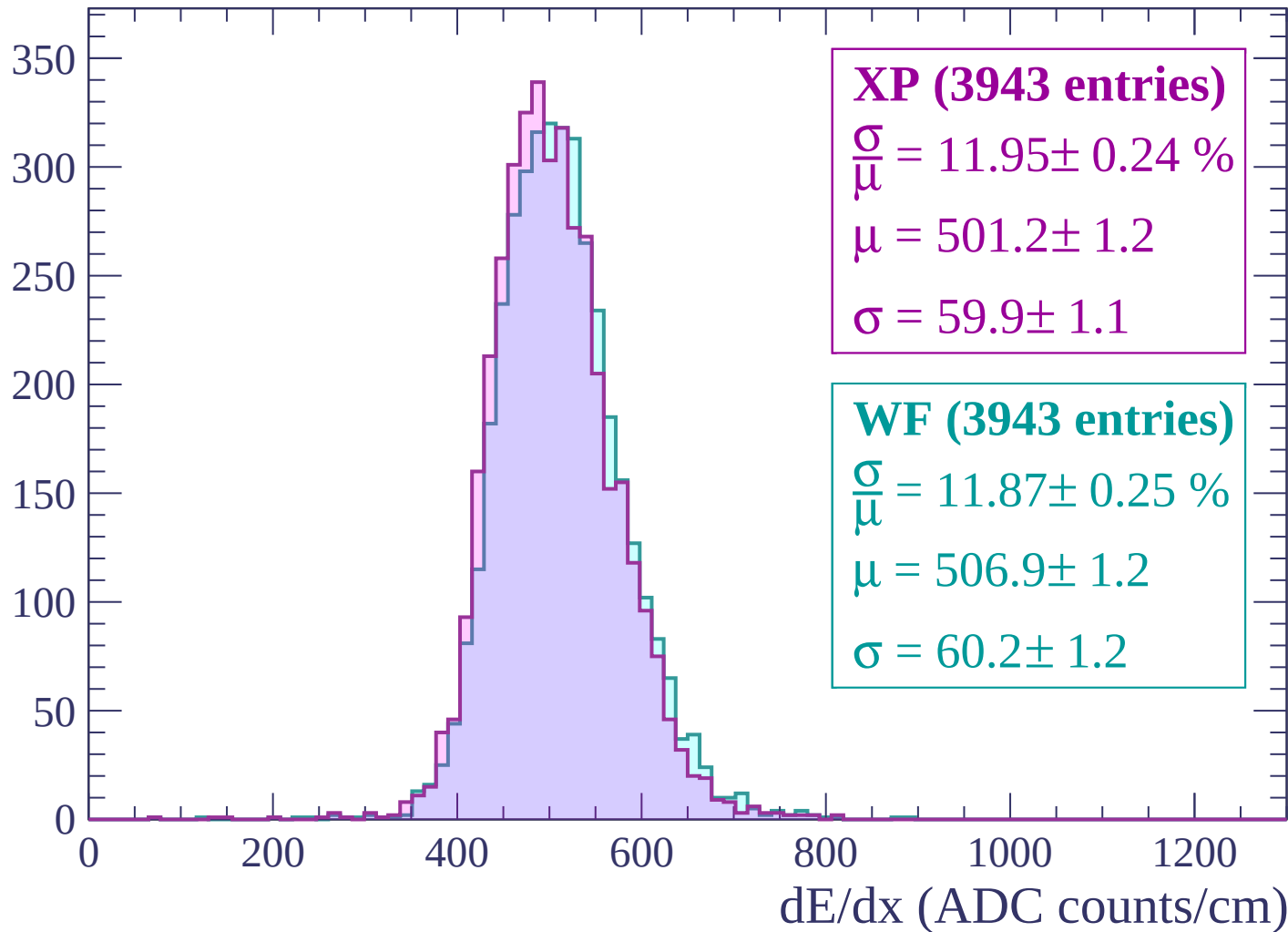
Energy loss | -2160 < p < -2100

Count



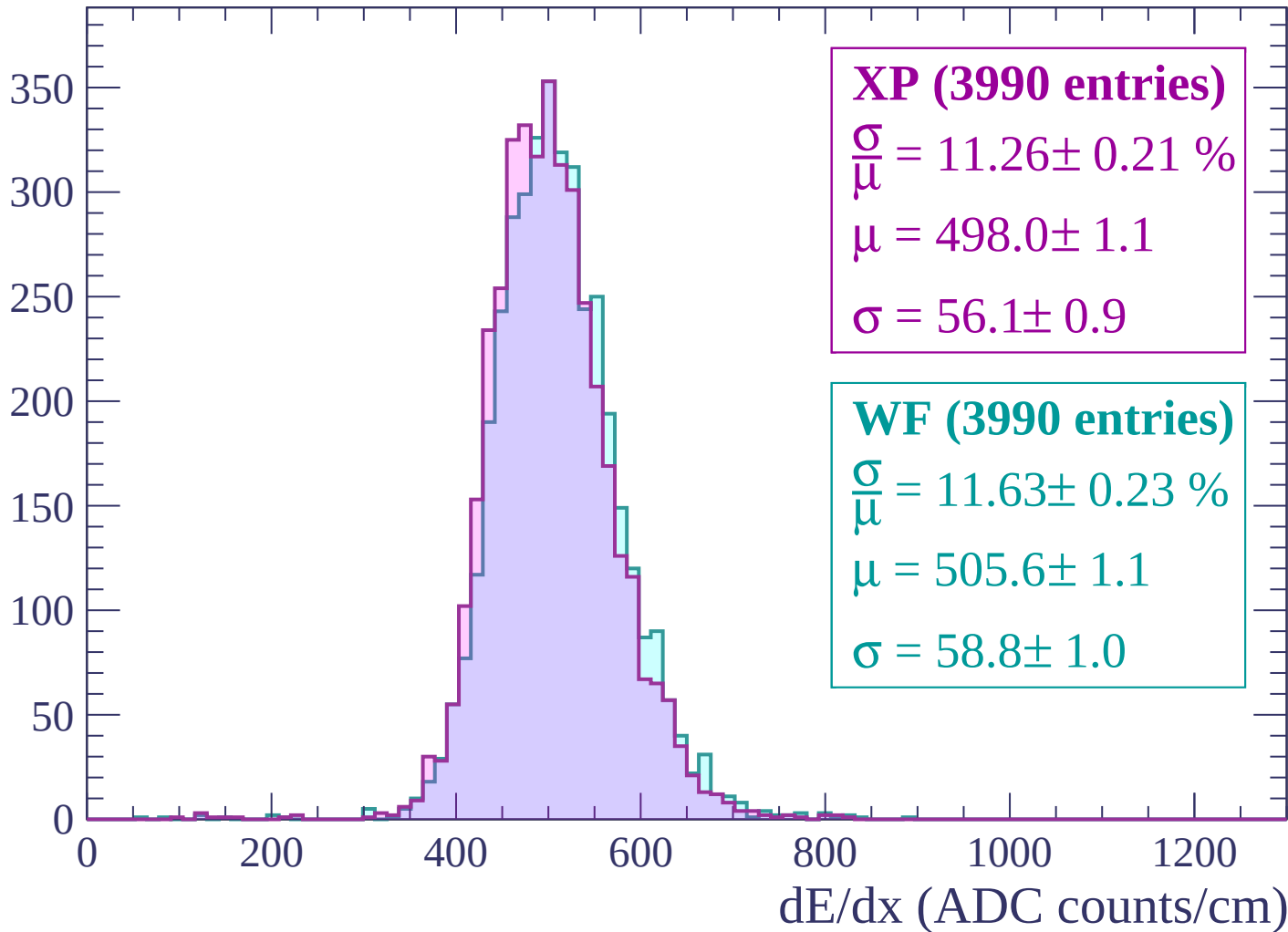
Energy loss | -2100 < p < -2040

Count



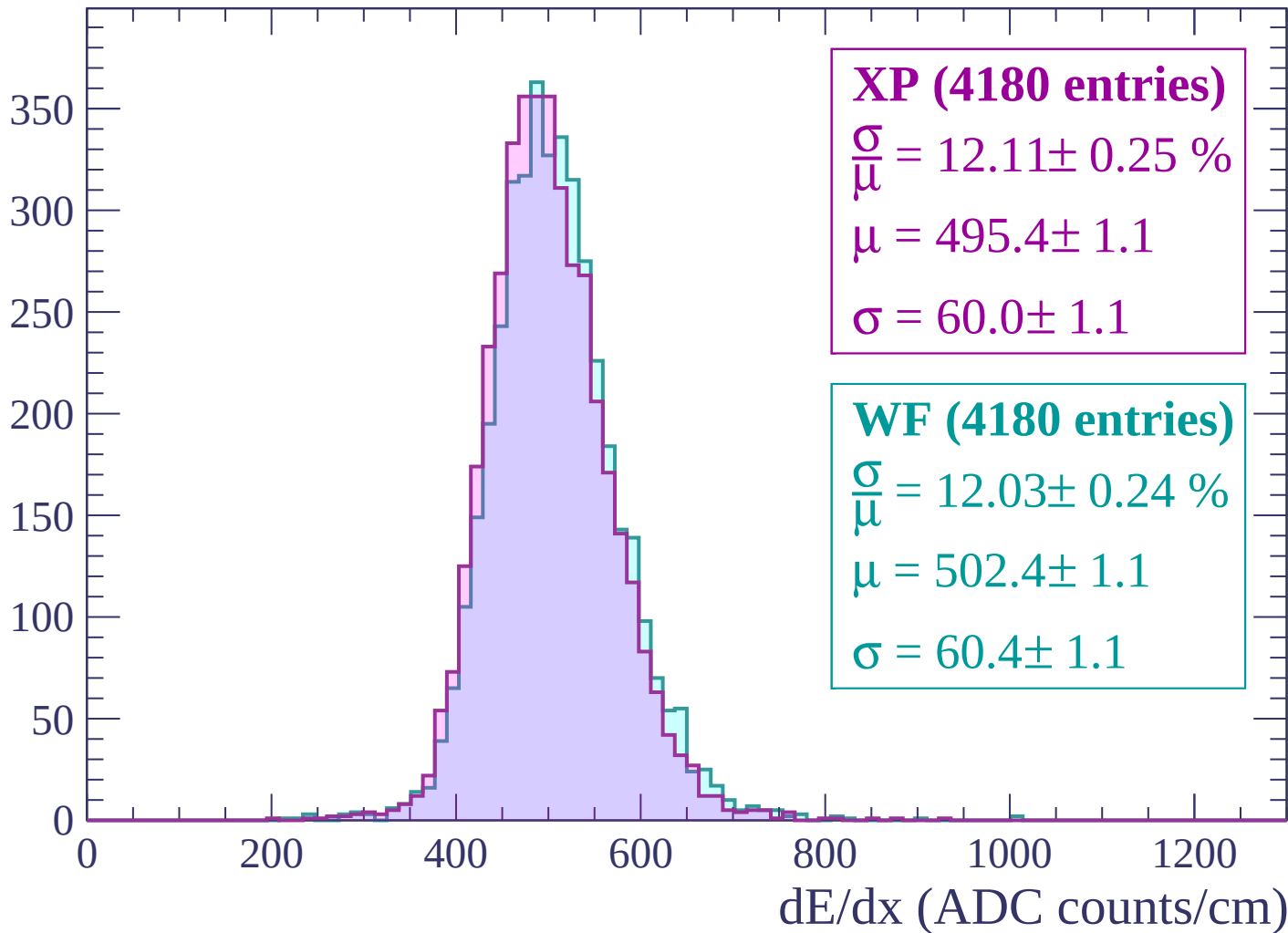
Energy loss | -2040 < p < -1980

Count



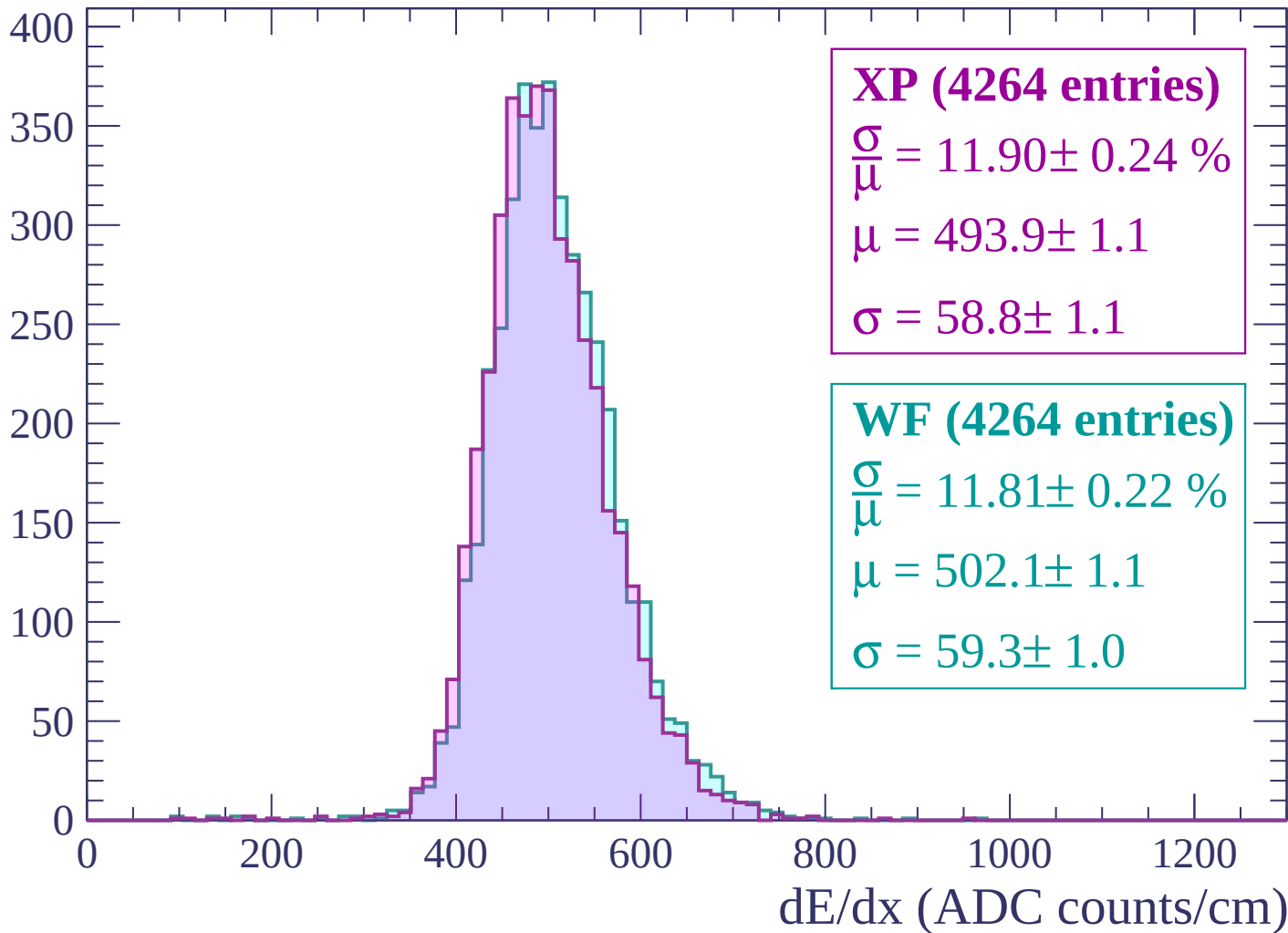
Energy loss | -1980 < p < -1920

Count



Energy loss | -1920 < p < -1860

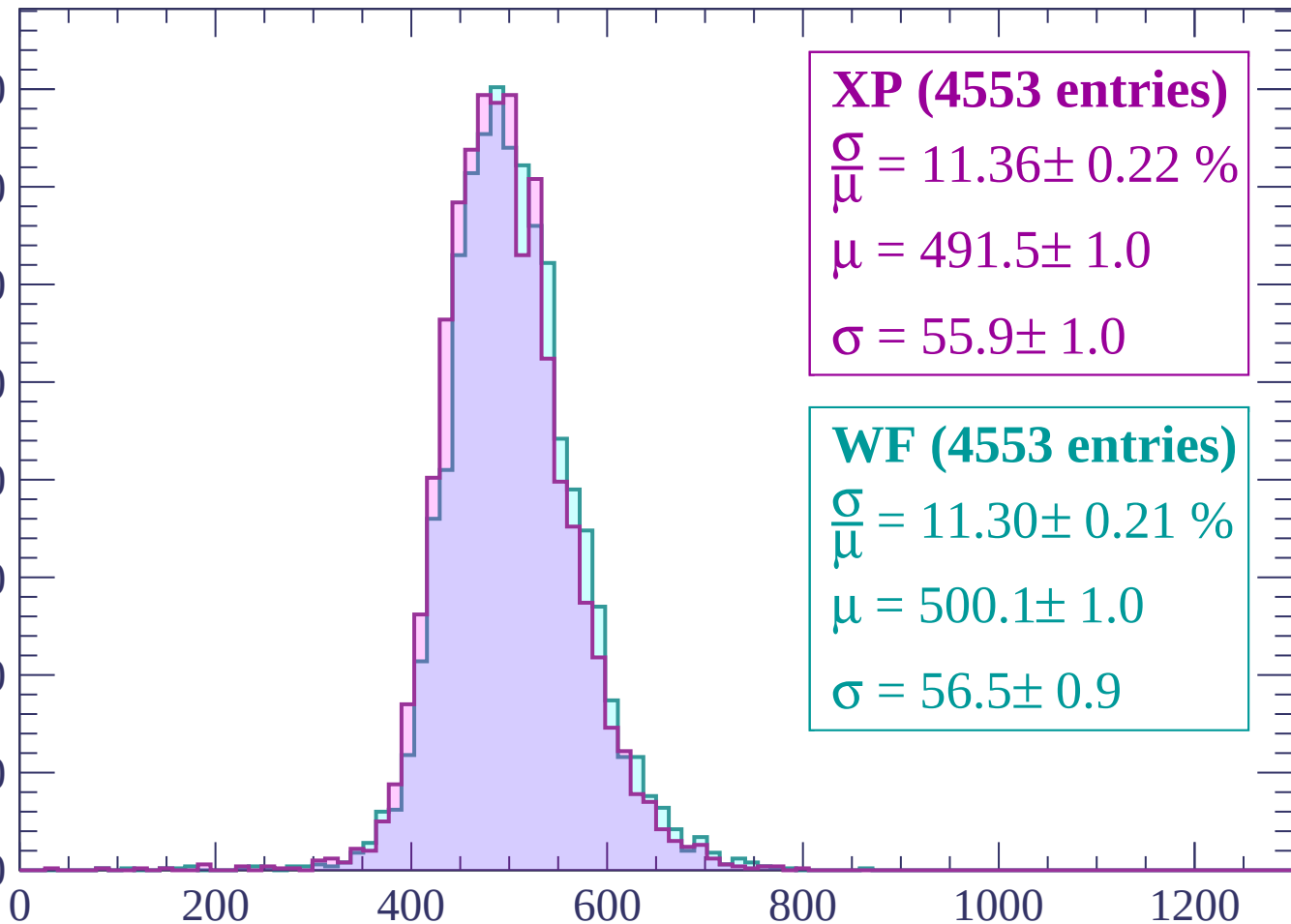
Count



Energy loss | -1860 < p < -1800

Count

400
350
300
250
200
150
100
50
0



XP (4553 entries)

$\frac{\sigma}{\mu} = 11.36 \pm 0.22 \%$

$\mu = 491.5 \pm 1.0$

$\sigma = 55.9 \pm 1.0$

WF (4553 entries)

$\frac{\sigma}{\mu} = 11.30 \pm 0.21 \%$

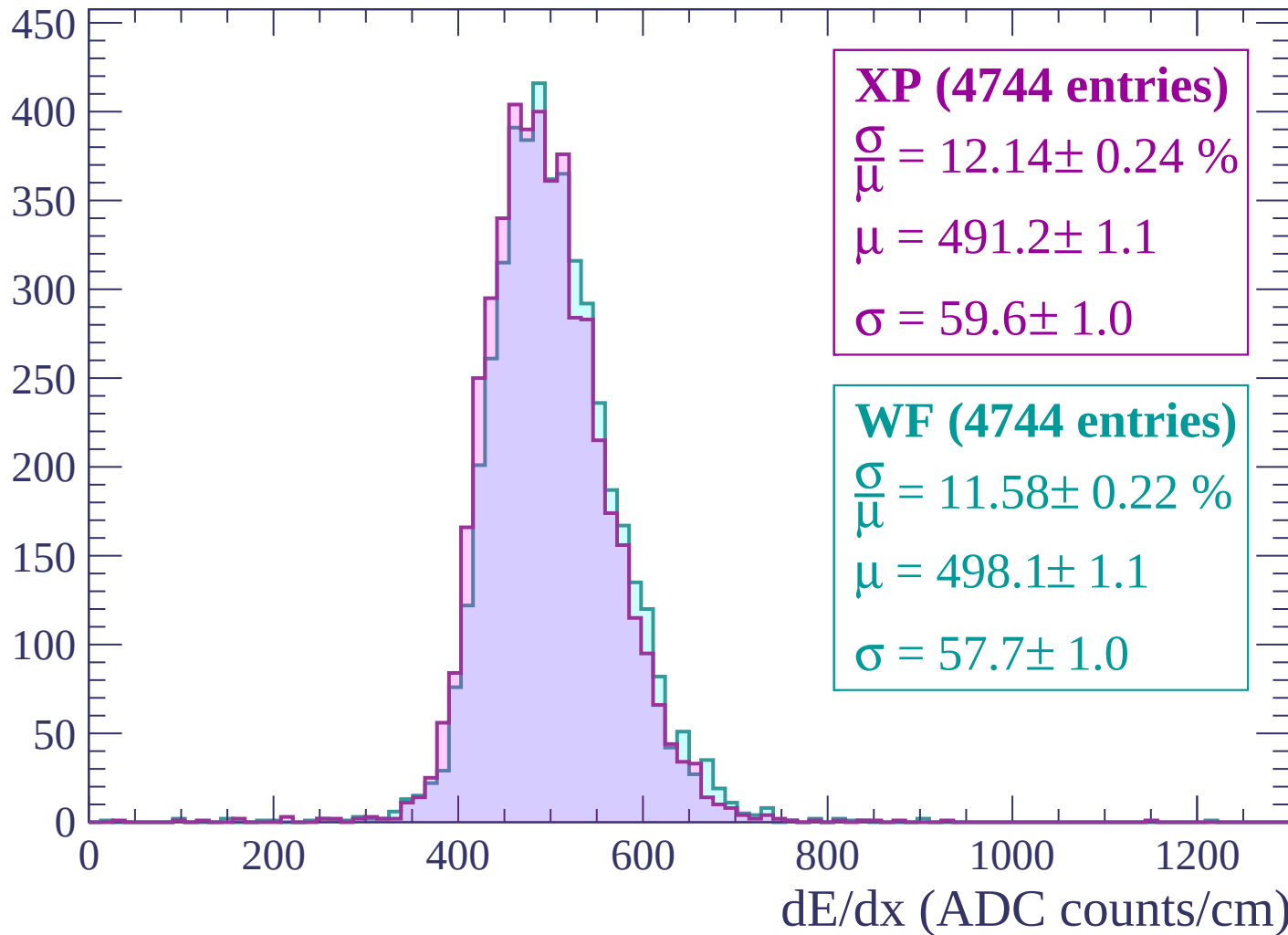
$\mu = 500.1 \pm 1.0$

$\sigma = 56.5 \pm 0.9$

dE/dx (ADC counts/cm)

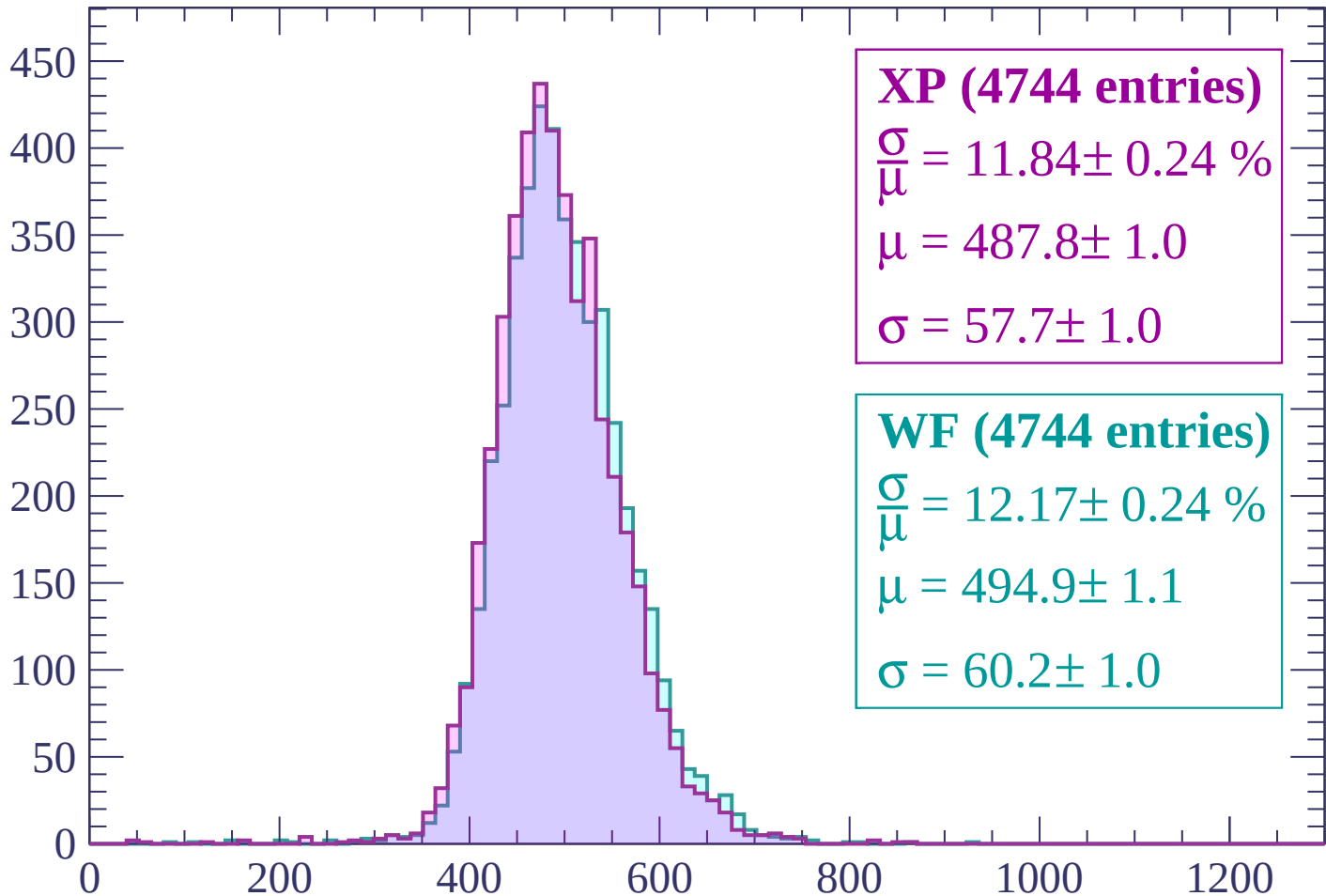
Energy loss | -1800 < p < -1740

Count



Energy loss | $-1740 < p < -1680$

Count



XP (4744 entries)

$$\frac{\sigma}{\mu} = 11.84 \pm 0.24 \%$$

$$\mu = 487.8 \pm 1.0$$

$$\sigma = 57.7 \pm 1.0$$

WF (4744 entries)

$$\frac{\sigma}{\mu} = 12.17 \pm 0.24 \%$$

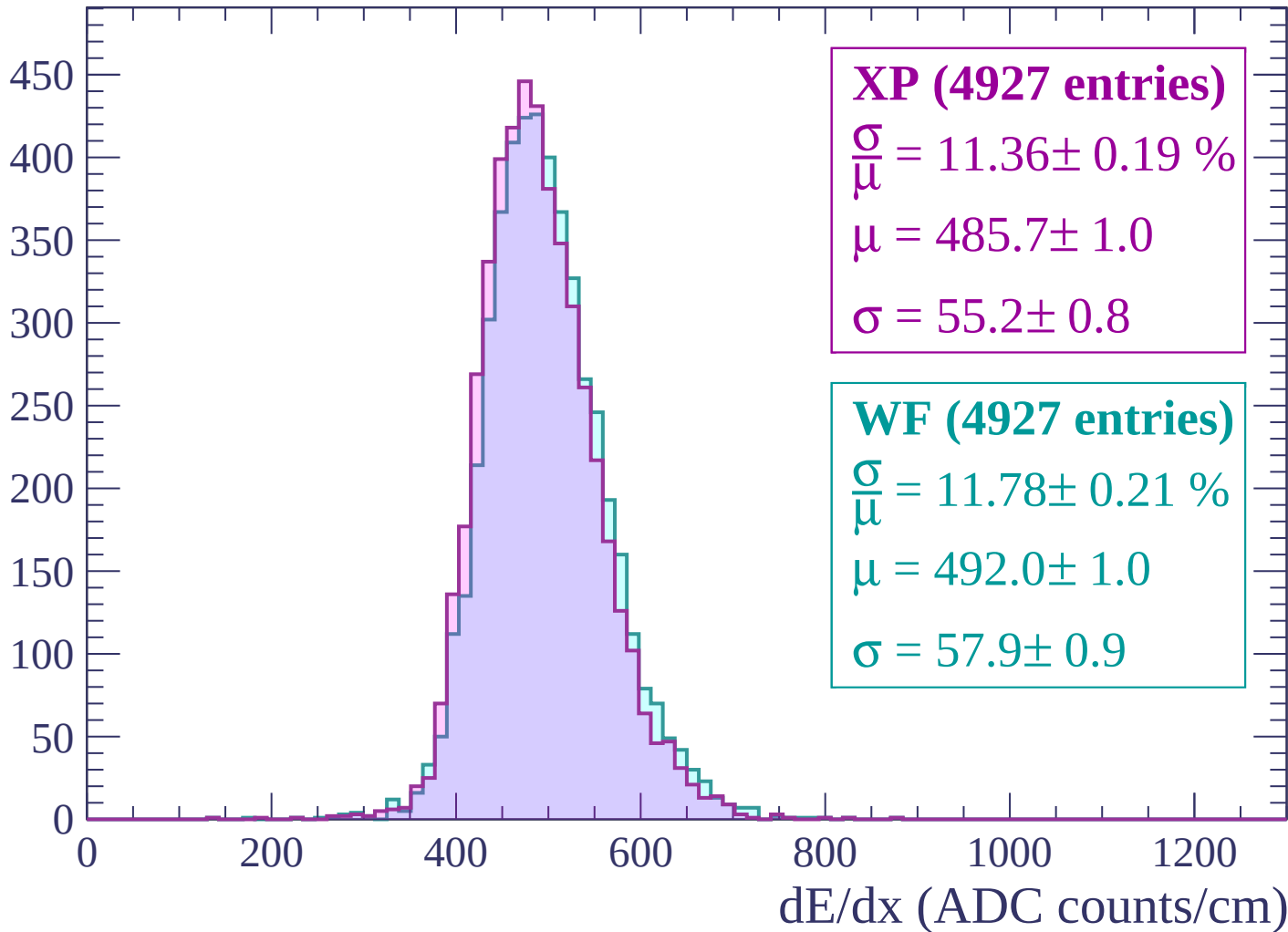
$$\mu = 494.9 \pm 1.1$$

$$\sigma = 60.2 \pm 1.0$$

dE/dx (ADC counts/cm)

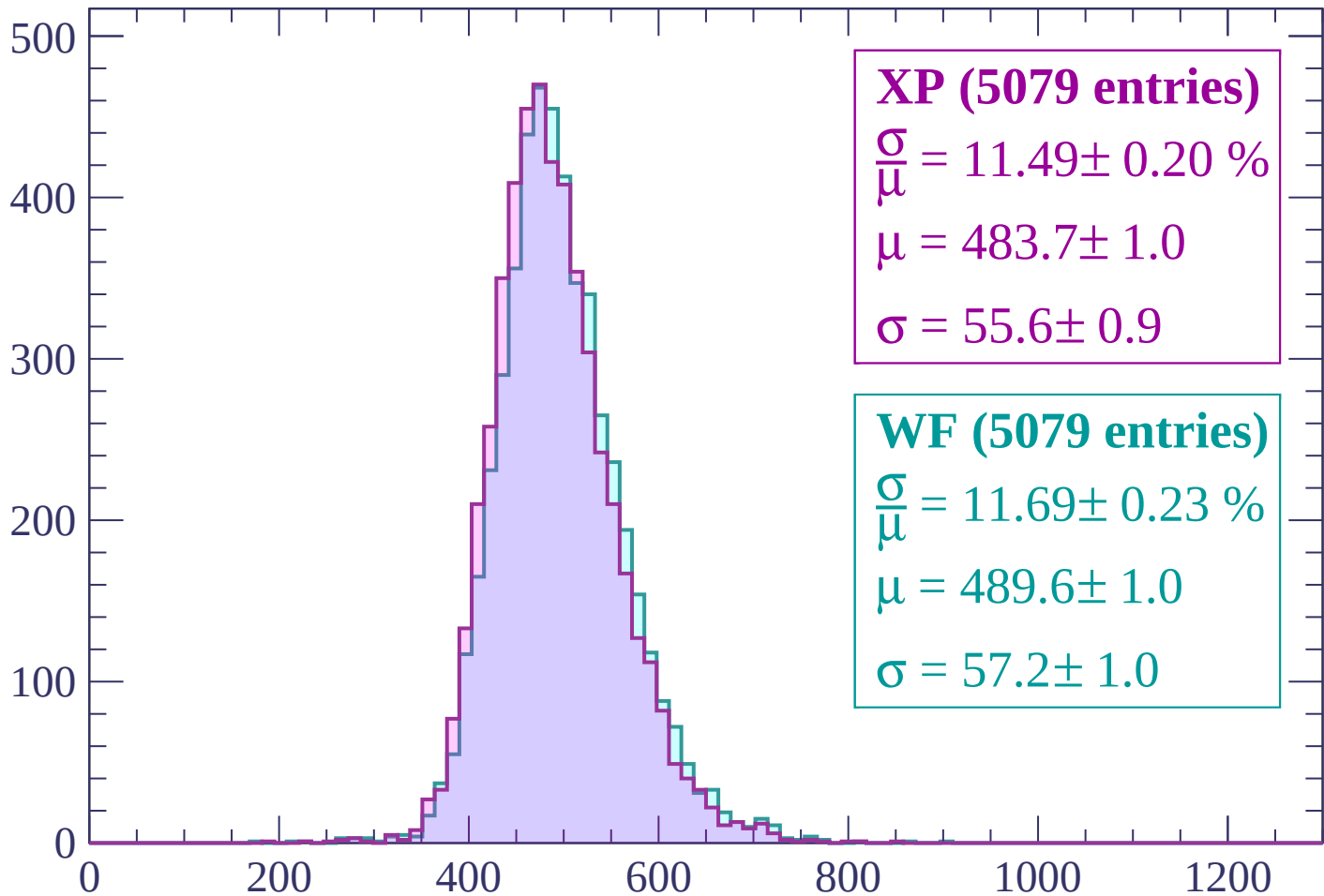
Energy loss | $-1680 < p < -1620$

Count



Energy loss | $-1620 < p < -1560$

Count



XP (5079 entries)

$$\frac{\sigma}{\mu} = 11.49 \pm 0.20 \%$$

$$\mu = 483.7 \pm 1.0$$

$$\sigma = 55.6 \pm 0.9$$

WF (5079 entries)

$$\frac{\sigma}{\mu} = 11.69 \pm 0.23 \%$$

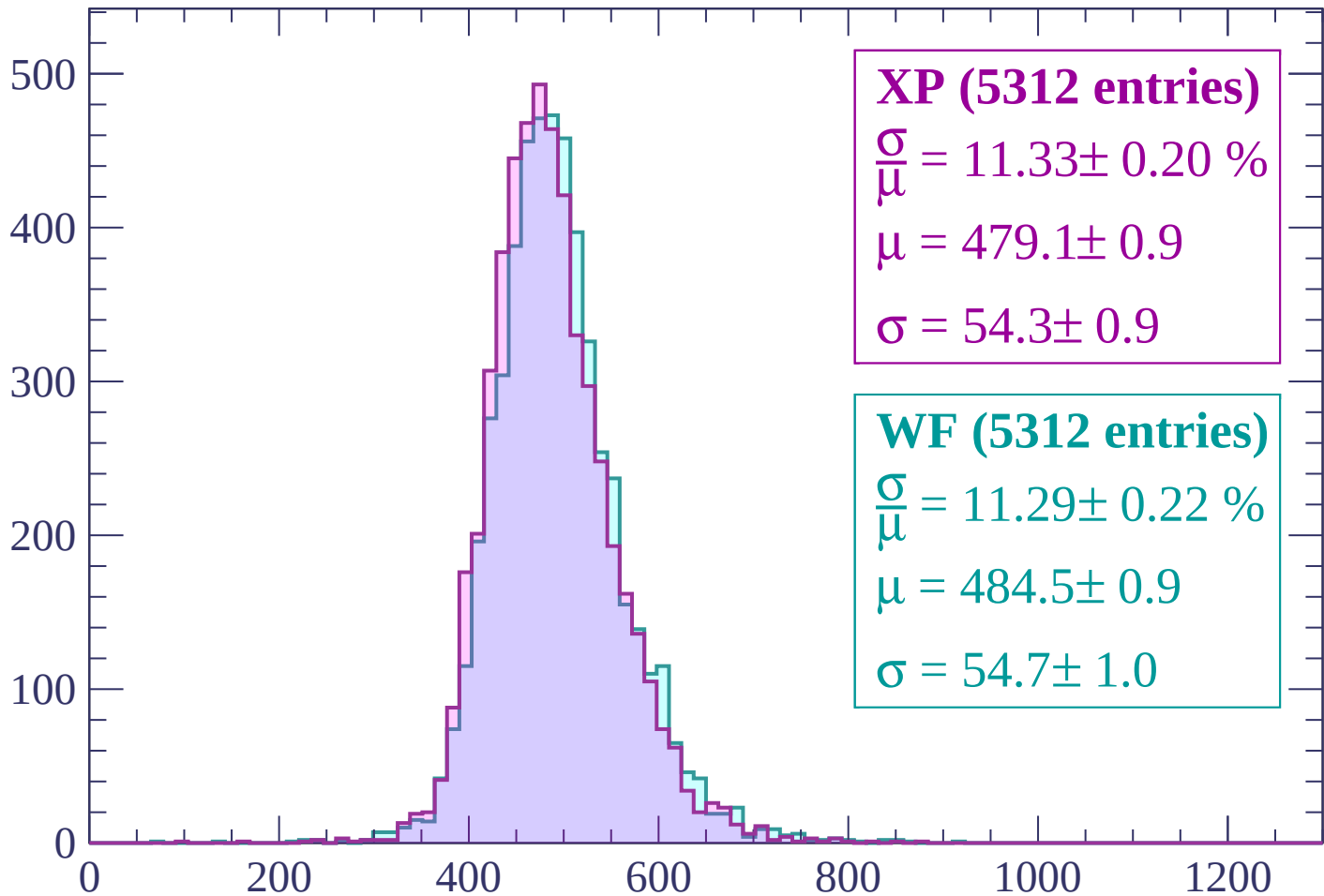
$$\mu = 489.6 \pm 1.0$$

$$\sigma = 57.2 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | -1560 < p < -1500

Count



XP (5312 entries)

$\frac{\sigma}{\mu} = 11.33 \pm 0.20 \%$

$\mu = 479.1 \pm 0.9$

$\sigma = 54.3 \pm 0.9$

WF (5312 entries)

$\frac{\sigma}{\mu} = 11.29 \pm 0.22 \%$

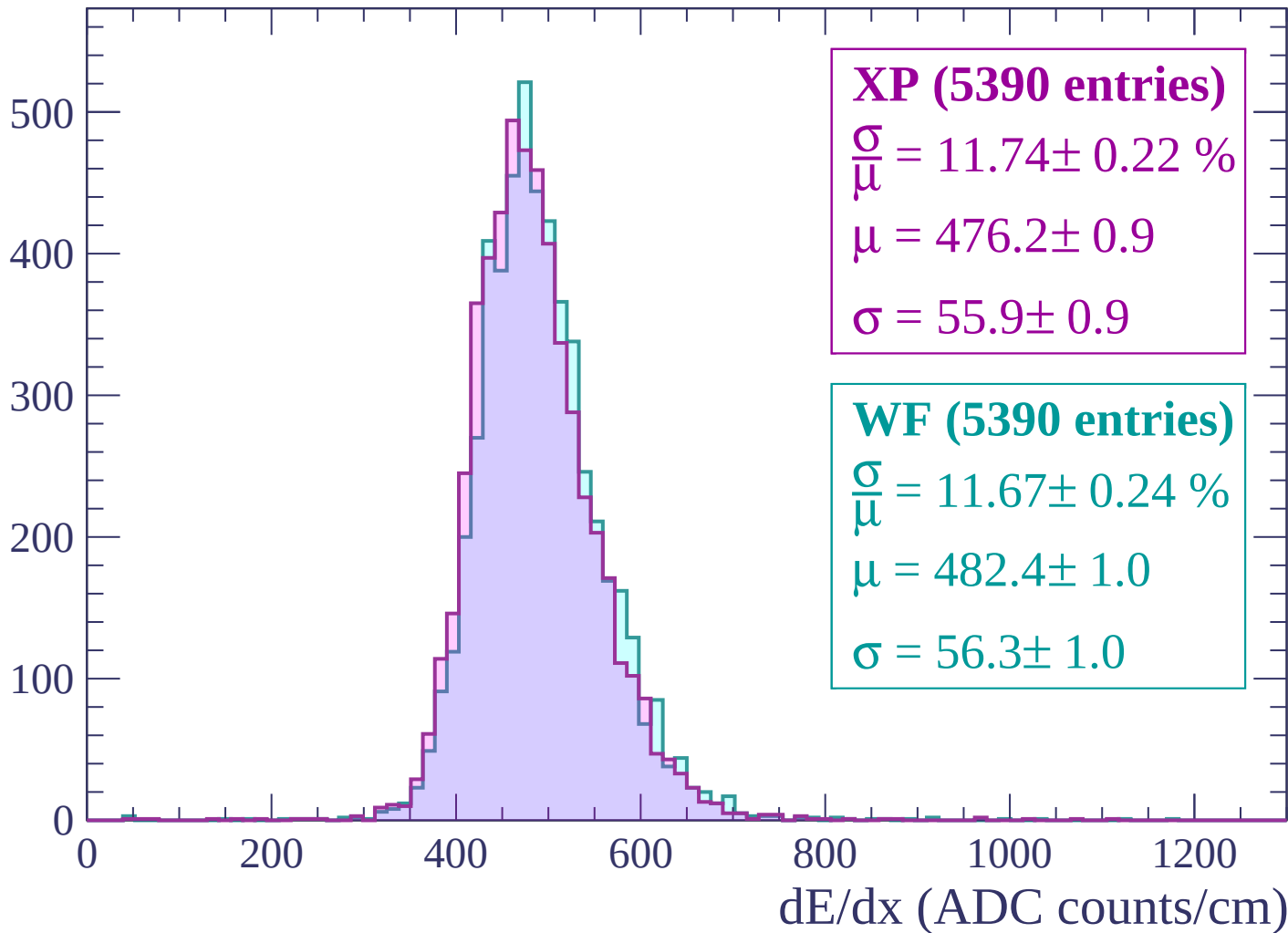
$\mu = 484.5 \pm 0.9$

$\sigma = 54.7 \pm 1.0$

dE/dx (ADC counts/cm)

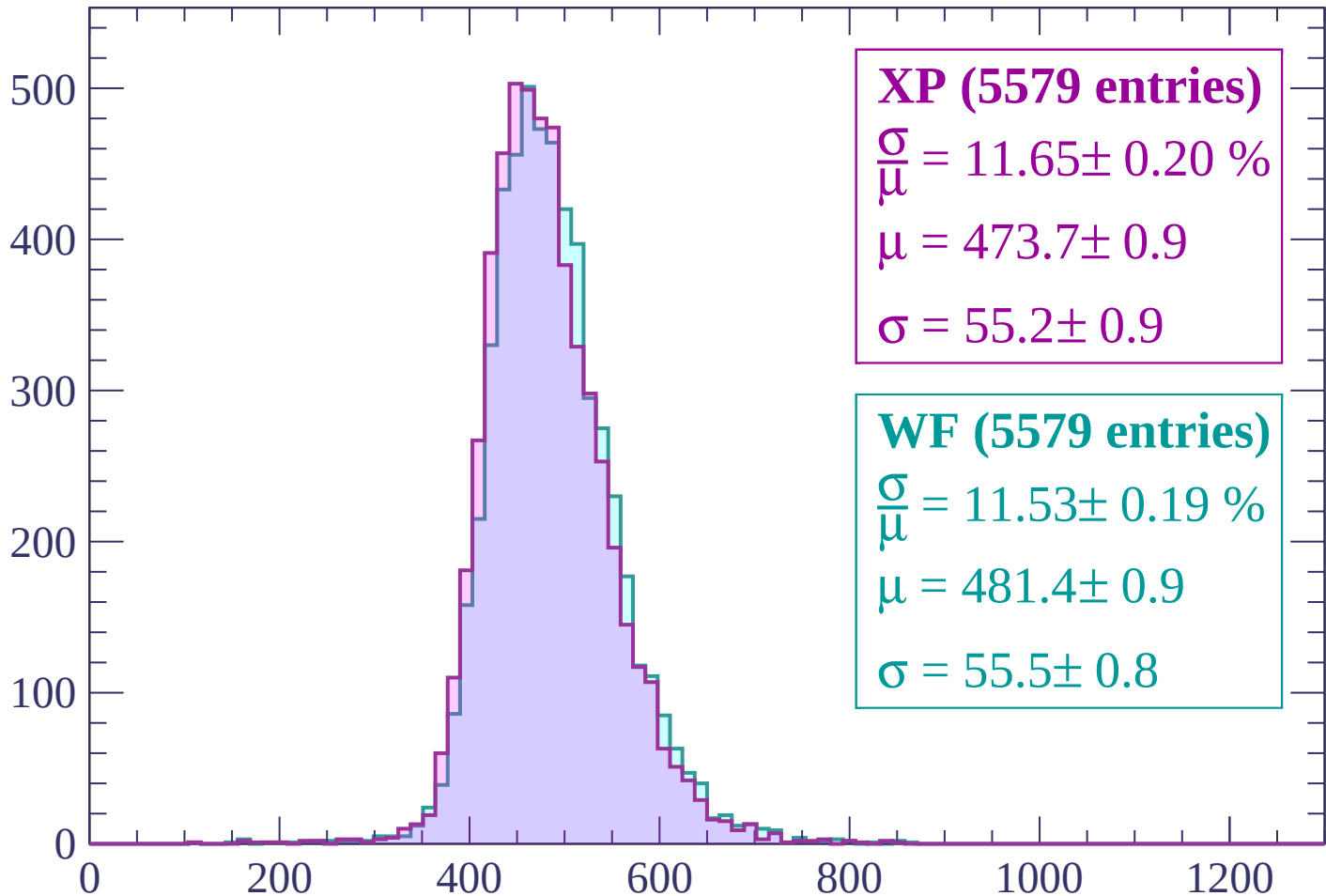
Energy loss | $-1500 < p < -1440$

Count



Energy loss | $-1440 < p < -1380$

Count



XP (5579 entries)

$\frac{\sigma}{\mu} = 11.65 \pm 0.20 \%$

$\mu = 473.7 \pm 0.9$

$\sigma = 55.2 \pm 0.9$

WF (5579 entries)

$\frac{\sigma}{\mu} = 11.53 \pm 0.19 \%$

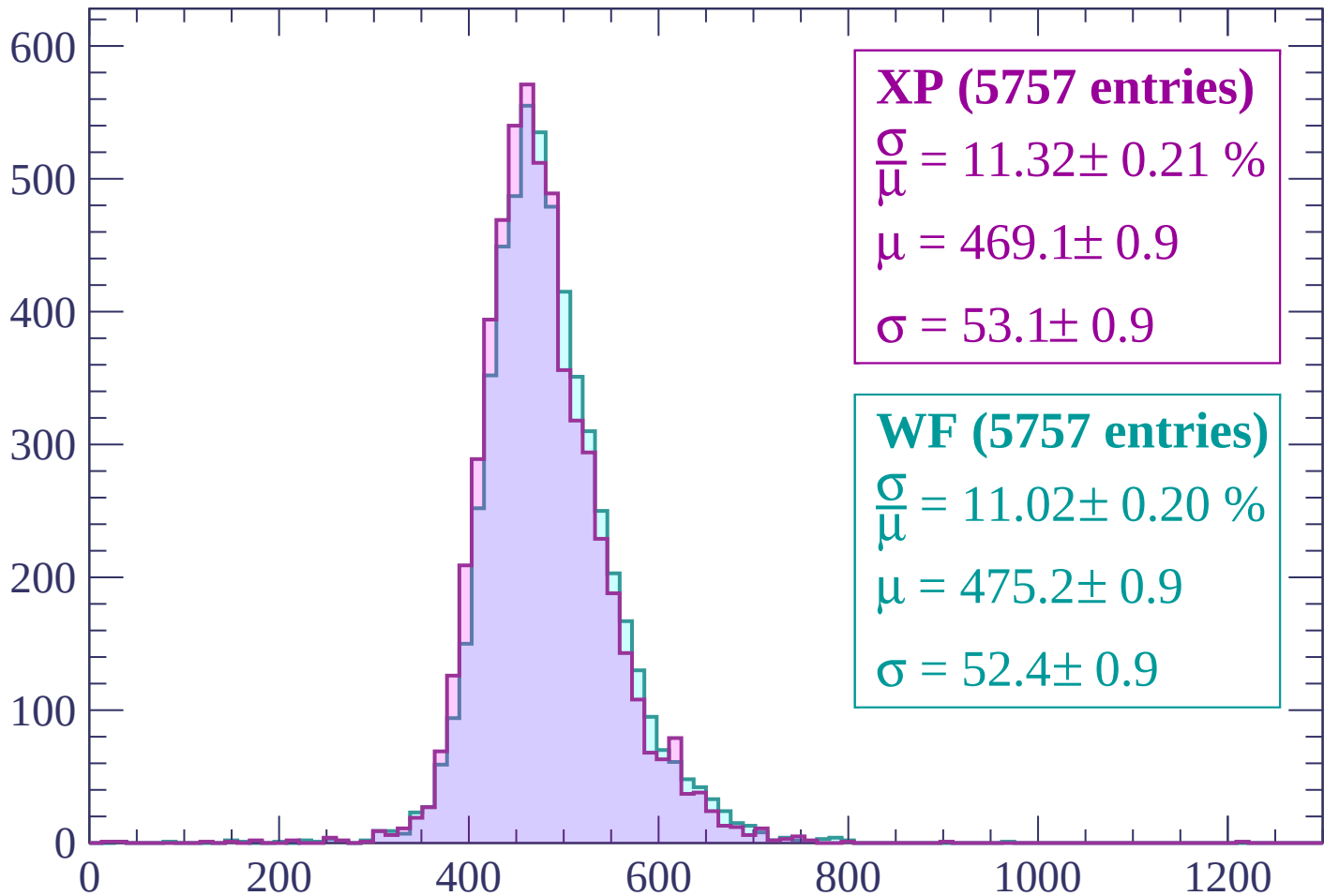
$\mu = 481.4 \pm 0.9$

$\sigma = 55.5 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | -1380 < p < -1320

Count



XP (5757 entries)

$$\frac{\sigma}{\mu} = 11.32 \pm 0.21 \%$$

$$\mu = 469.1 \pm 0.9$$

$$\sigma = 53.1 \pm 0.9$$

WF (5757 entries)

$$\frac{\sigma}{\mu} = 11.02 \pm 0.20 \%$$

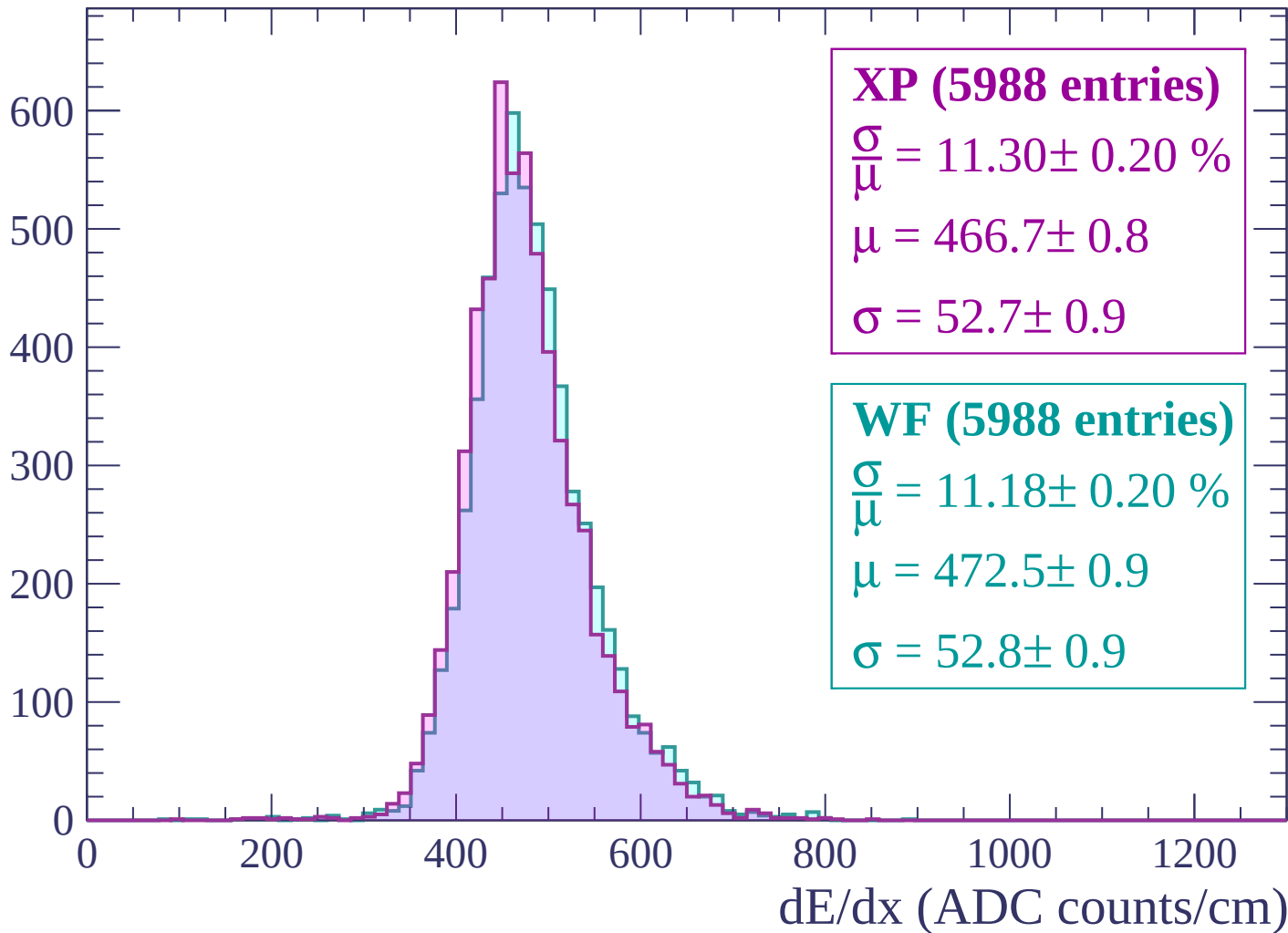
$$\mu = 475.2 \pm 0.9$$

$$\sigma = 52.4 \pm 0.9$$

dE/dx (ADC counts/cm)

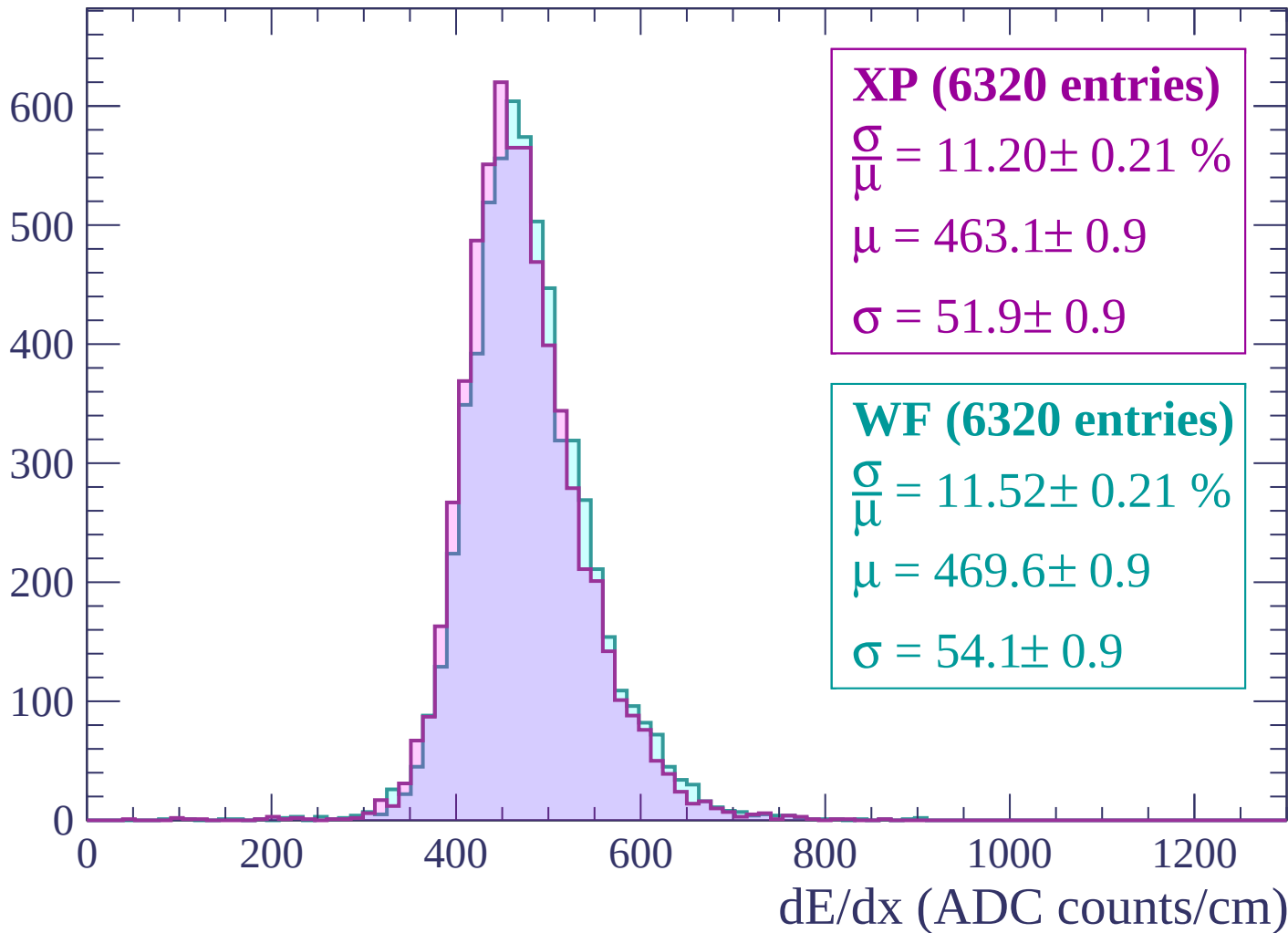
Energy loss | -1320 < p < -1260

Count



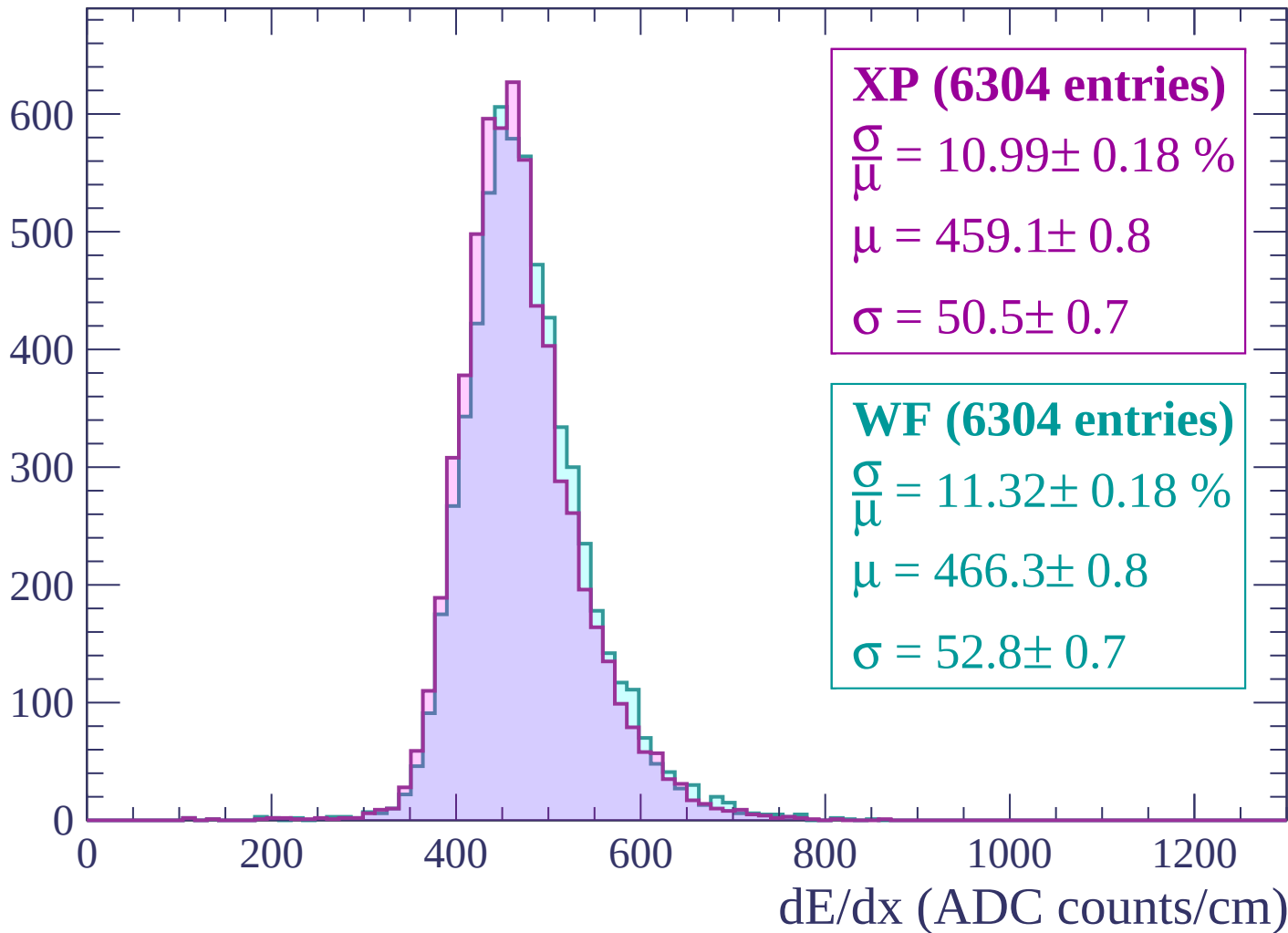
Energy loss | $-1260 < p < -1200$

Count



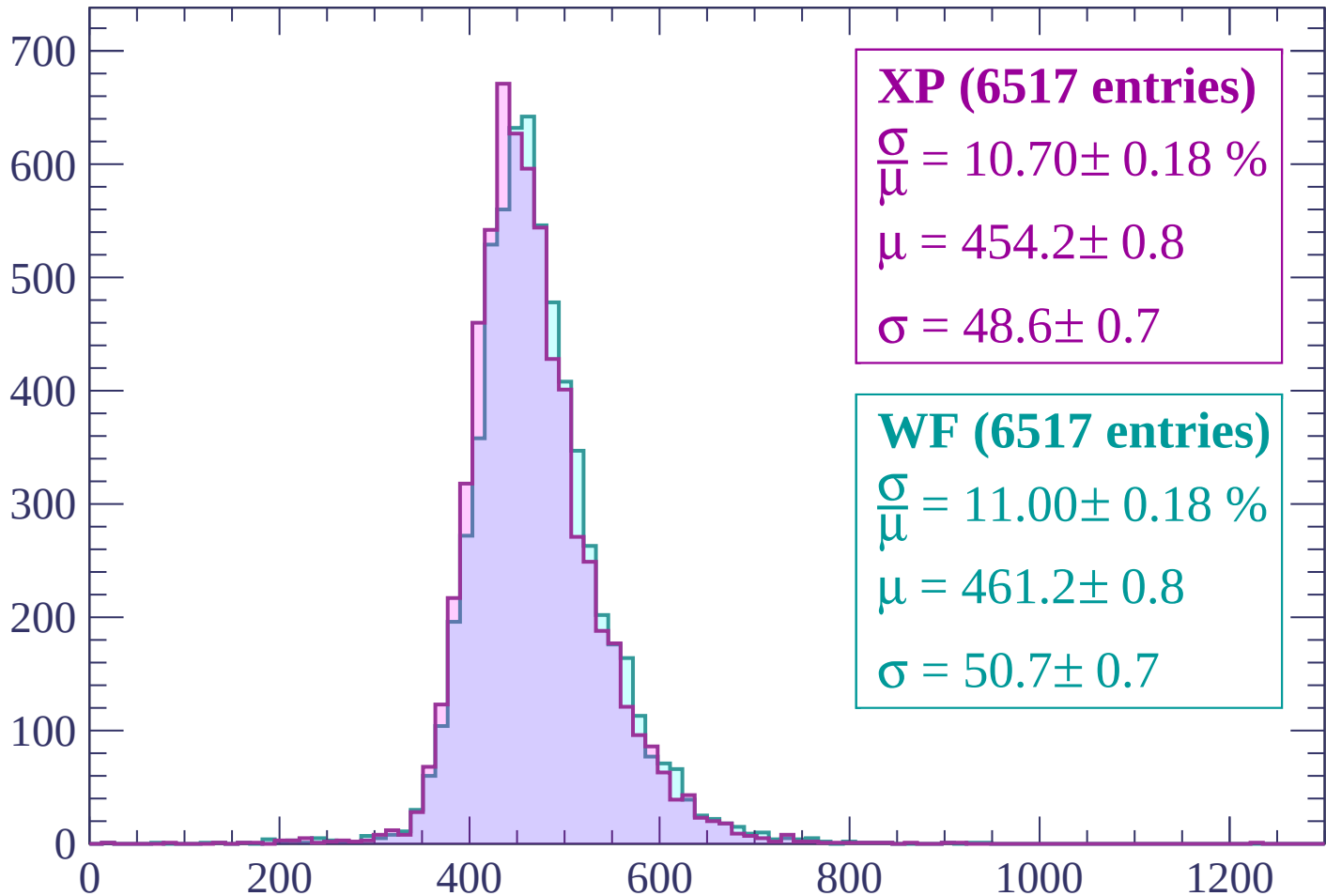
Energy loss | $-1200 < p < -1140$

Count



Energy loss | -1140 < p < -1080

Count



XP (6517 entries)

$\frac{\sigma}{\mu} = 10.70 \pm 0.18 \%$

$\mu = 454.2 \pm 0.8$

$\sigma = 48.6 \pm 0.7$

WF (6517 entries)

$\frac{\sigma}{\mu} = 11.00 \pm 0.18 \%$

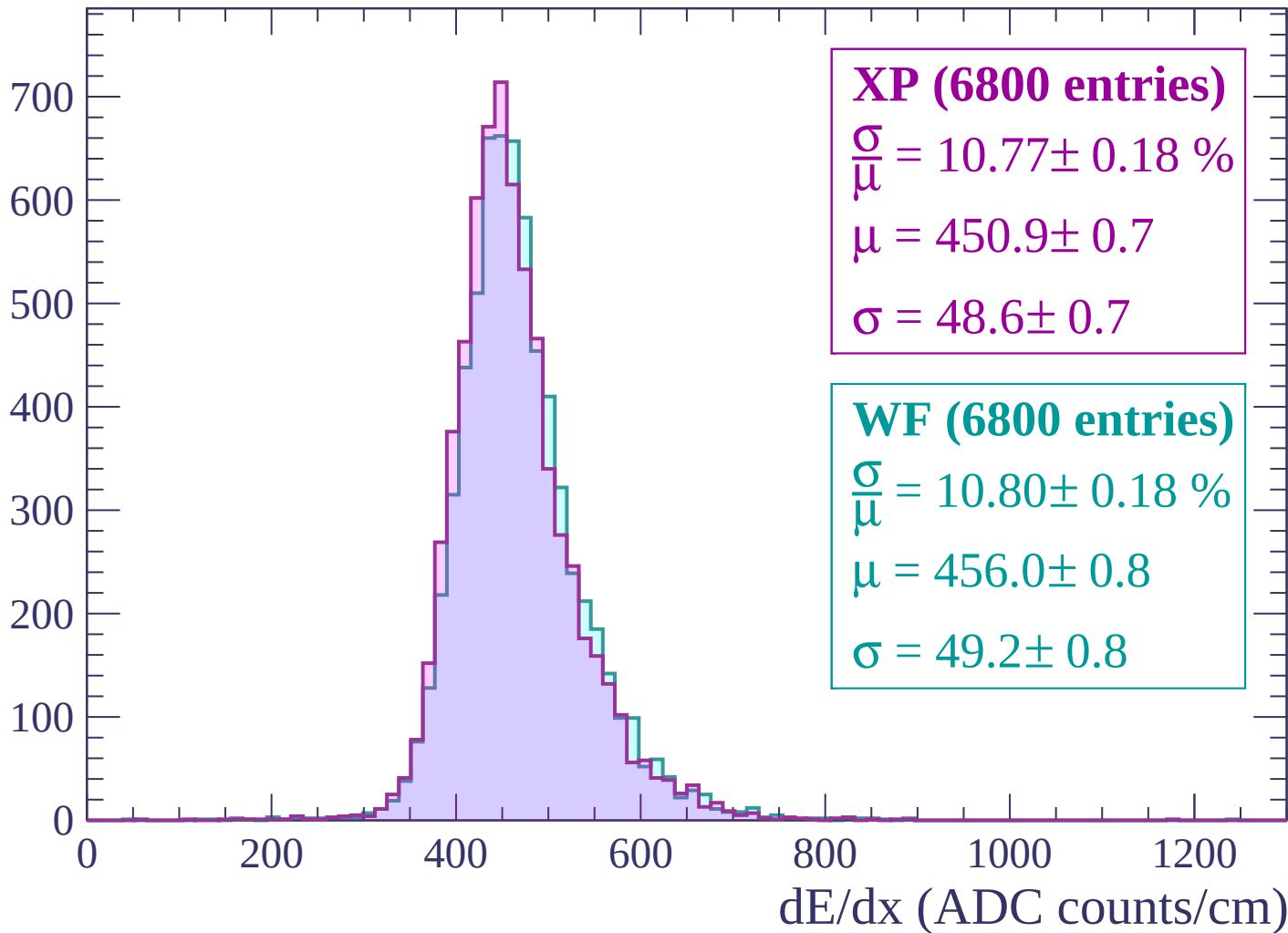
$\mu = 461.2 \pm 0.8$

$\sigma = 50.7 \pm 0.7$

dE/dx (ADC counts/cm)

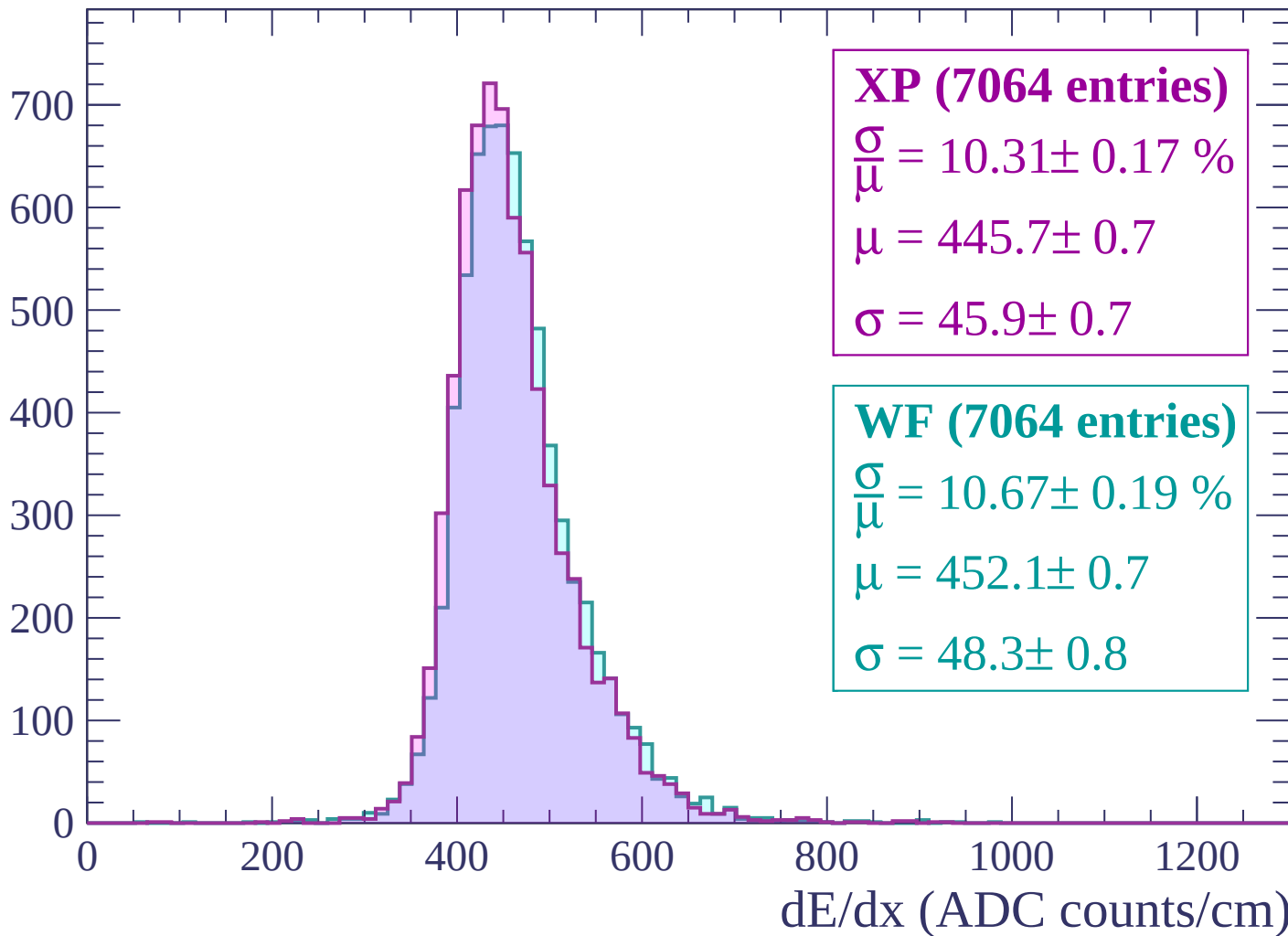
Energy loss | $-1080 < p < -1020$

Count



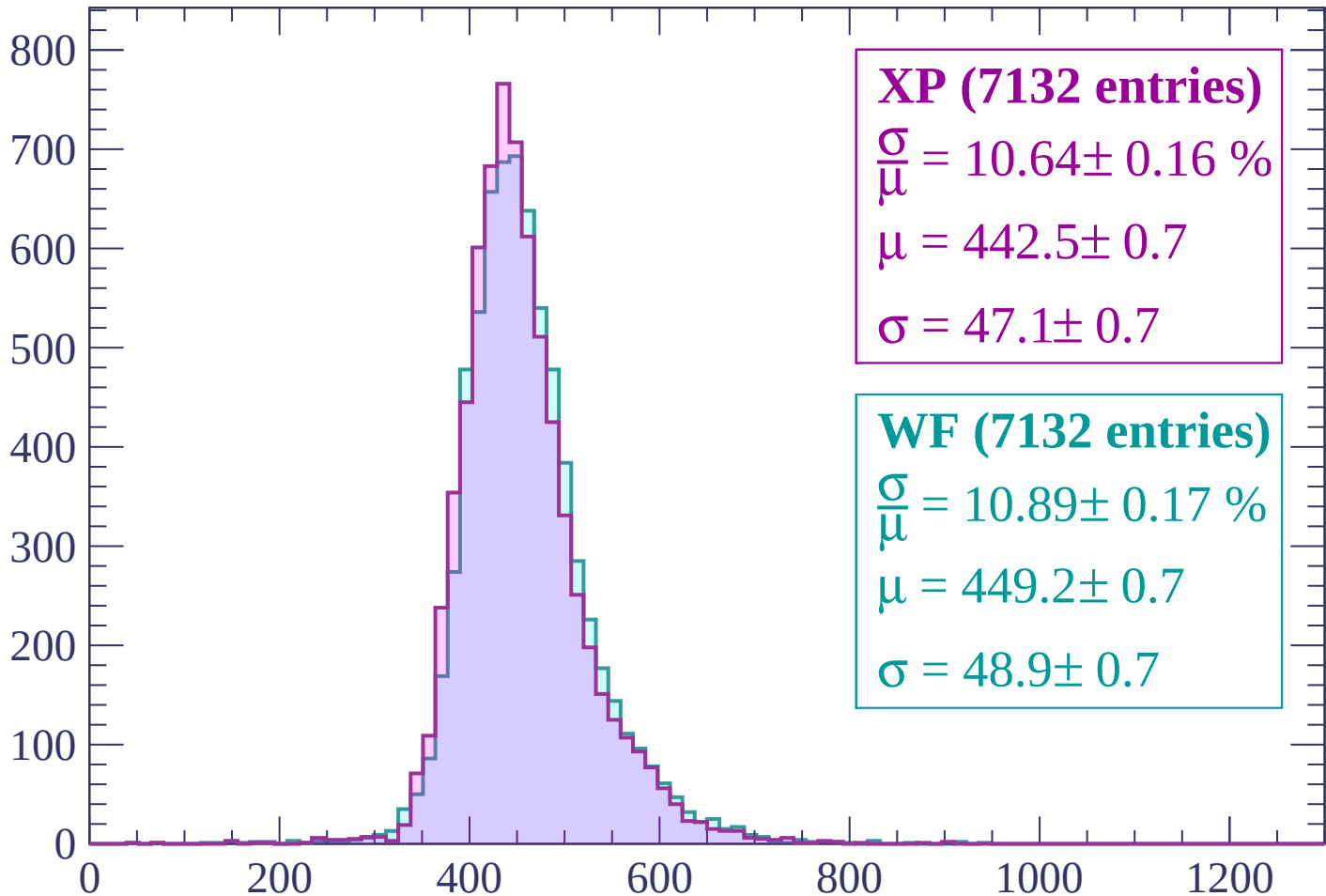
Energy loss | $-1020 < p < -960$

Count



Energy loss | $-960 < p < -900$

Count



XP (7132 entries)

$$\frac{\sigma}{\mu} = 10.64 \pm 0.16 \%$$

$$\mu = 442.5 \pm 0.7$$

$$\sigma = 47.1 \pm 0.7$$

WF (7132 entries)

$$\frac{\sigma}{\mu} = 10.89 \pm 0.17 \%$$

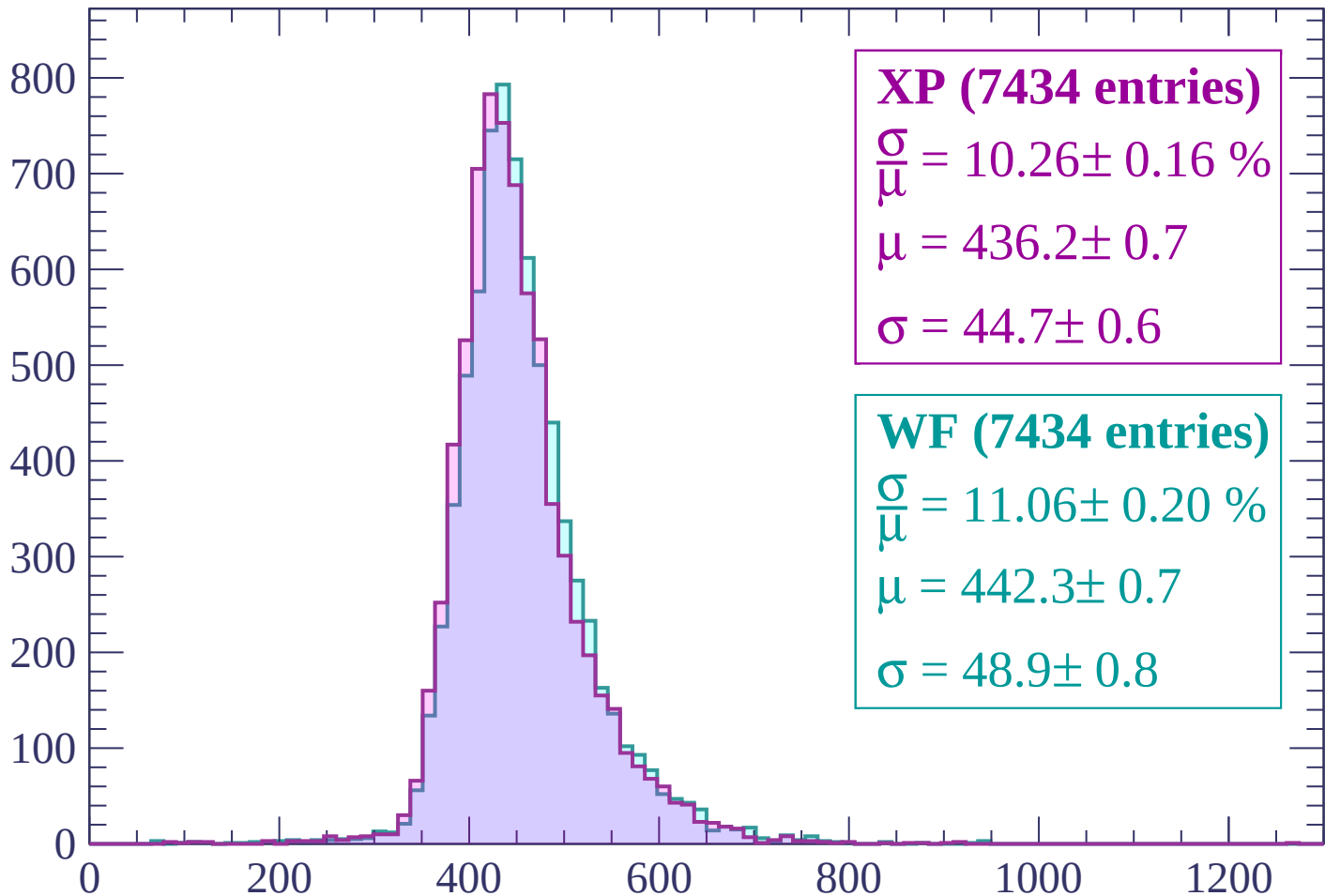
$$\mu = 449.2 \pm 0.7$$

$$\sigma = 48.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP (7434 entries)

$\frac{\sigma}{\mu} = 10.26 \pm 0.16 \%$

$\mu = 436.2 \pm 0.7$

$\sigma = 44.7 \pm 0.6$

WF (7434 entries)

$\frac{\sigma}{\mu} = 11.06 \pm 0.20 \%$

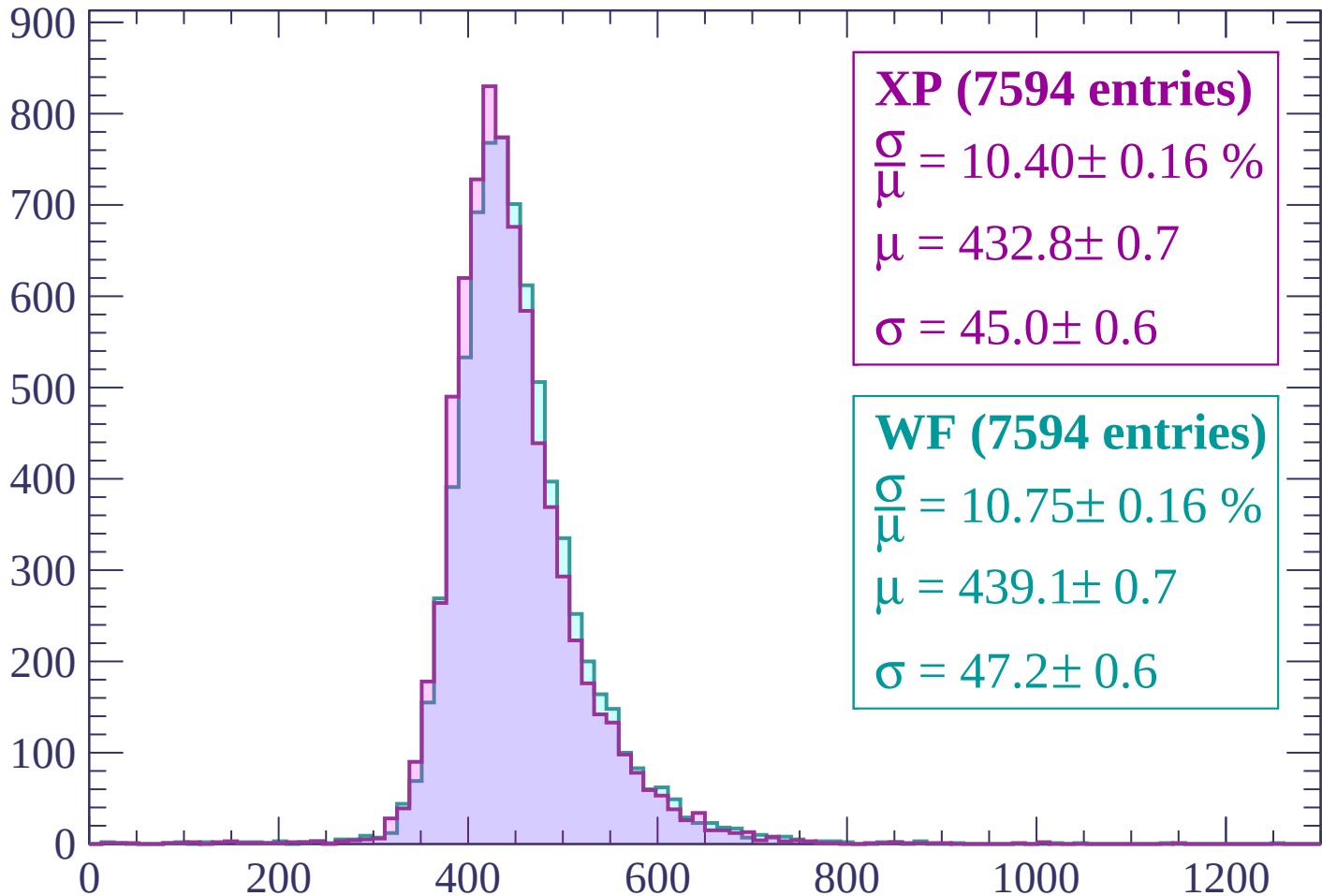
$\mu = 442.3 \pm 0.7$

$\sigma = 48.9 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count



XP (7594 entries)

$\frac{\sigma}{\mu} = 10.40 \pm 0.16 \%$

$\mu = 432.8 \pm 0.7$

$\sigma = 45.0 \pm 0.6$

WF (7594 entries)

$\frac{\sigma}{\mu} = 10.75 \pm 0.16 \%$

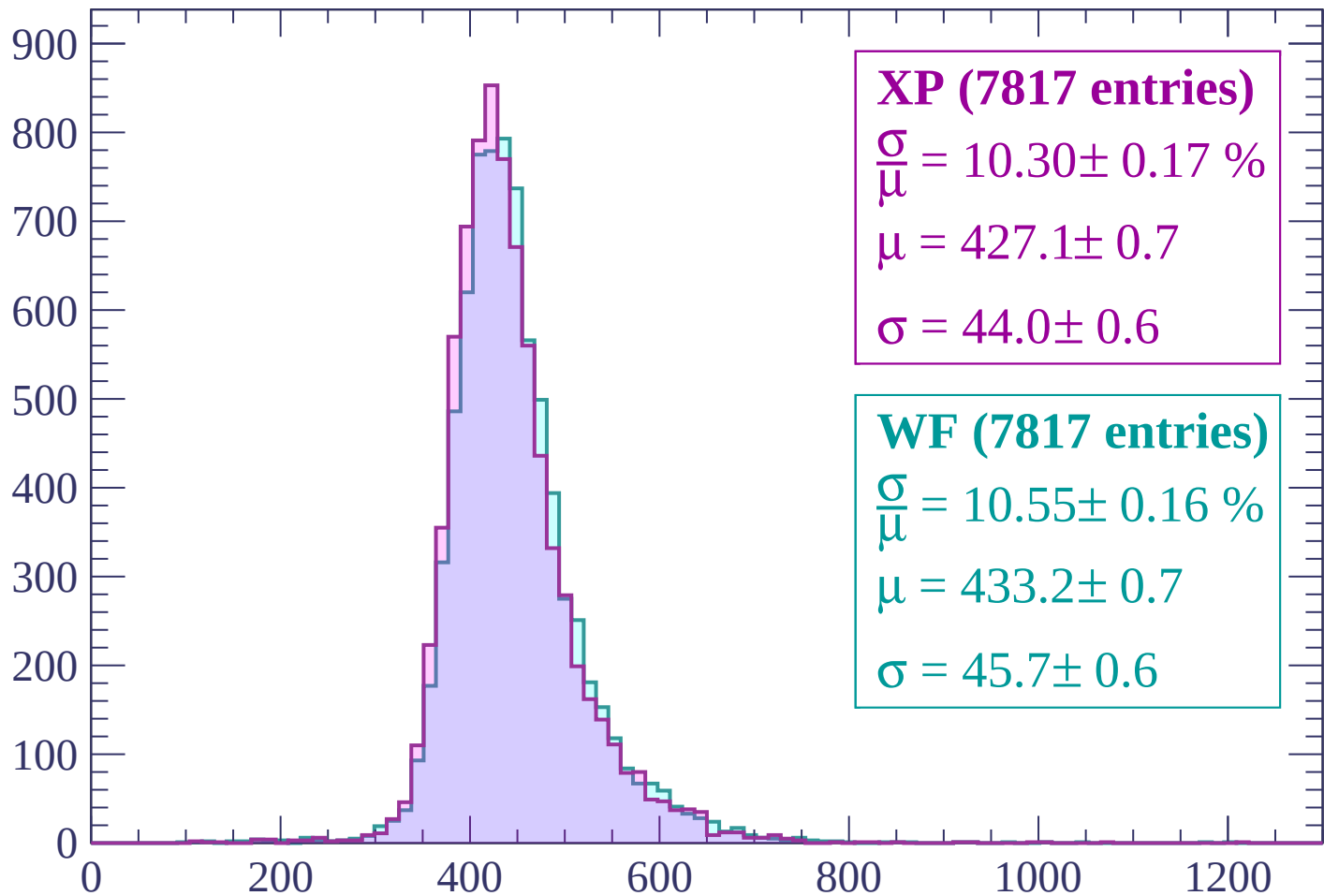
$\mu = 439.1 \pm 0.7$

$\sigma = 47.2 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$

Count



XP (7817 entries)

$$\frac{\sigma}{\mu} = 10.30 \pm 0.17 \%$$

$$\mu = 427.1 \pm 0.7$$

$$\sigma = 44.0 \pm 0.6$$

WF (7817 entries)

$$\frac{\sigma}{\mu} = 10.55 \pm 0.16 \%$$

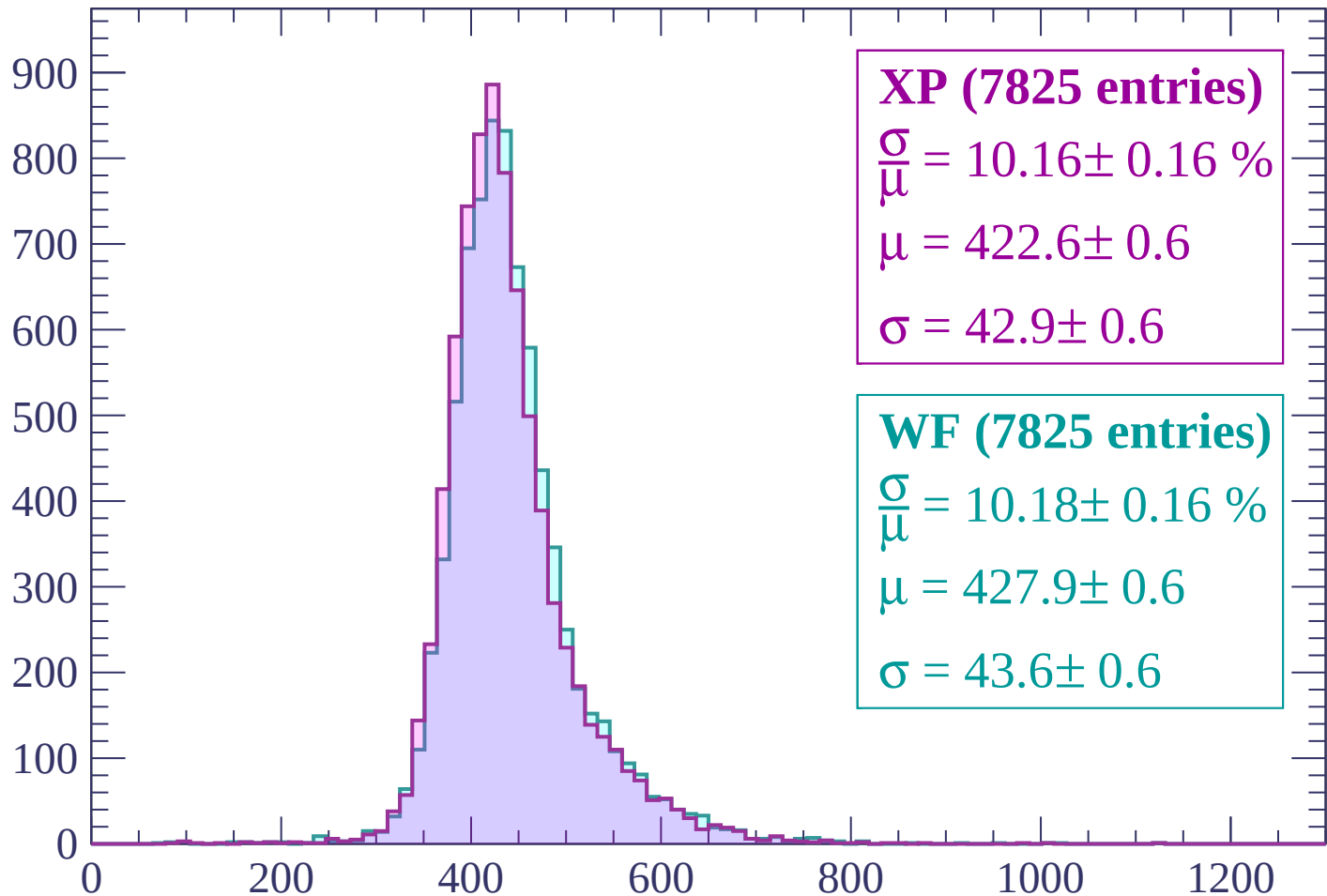
$$\mu = 433.2 \pm 0.7$$

$$\sigma = 45.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -660$

Count



XP (7825 entries)

$$\frac{\sigma}{\mu} = 10.16 \pm 0.16 \%$$

$$\mu = 422.6 \pm 0.6$$

$$\sigma = 42.9 \pm 0.6$$

WF (7825 entries)

$$\frac{\sigma}{\mu} = 10.18 \pm 0.16 \%$$

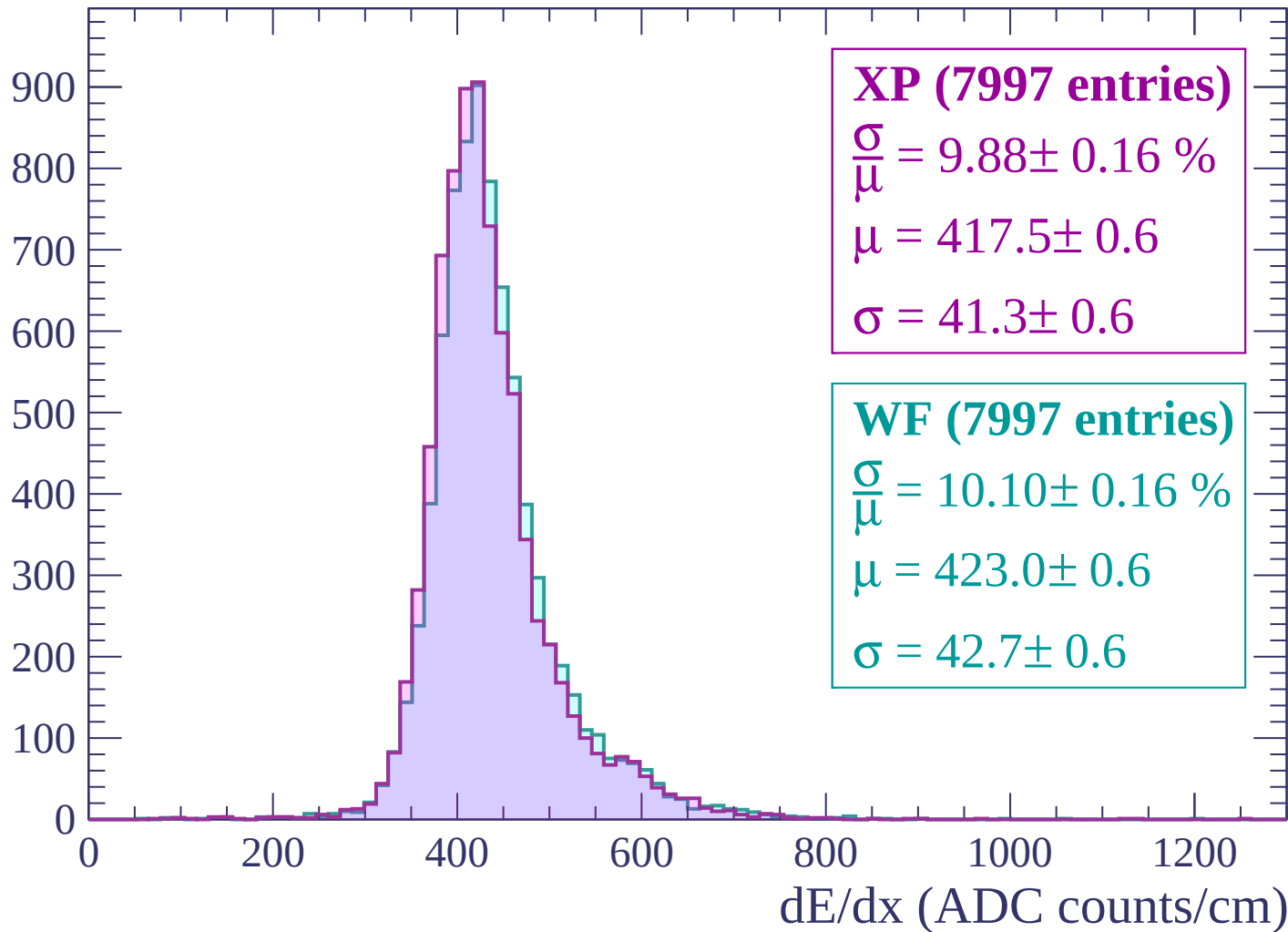
$$\mu = 427.9 \pm 0.6$$

$$\sigma = 43.6 \pm 0.6$$

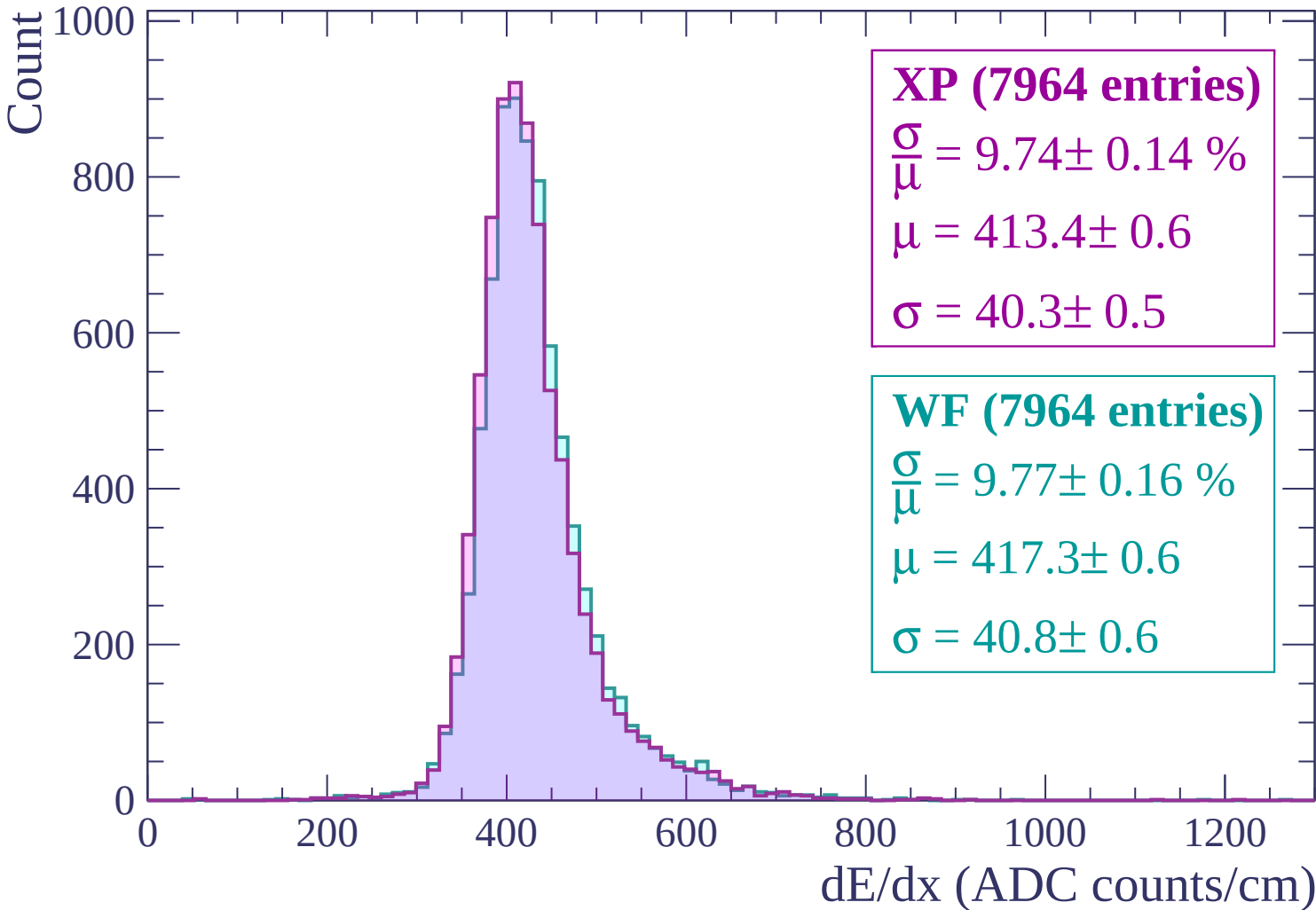
dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$

Count



Energy loss | $-600 < p < -540$



Energy loss | $-540 < p < -480$

Count

1000

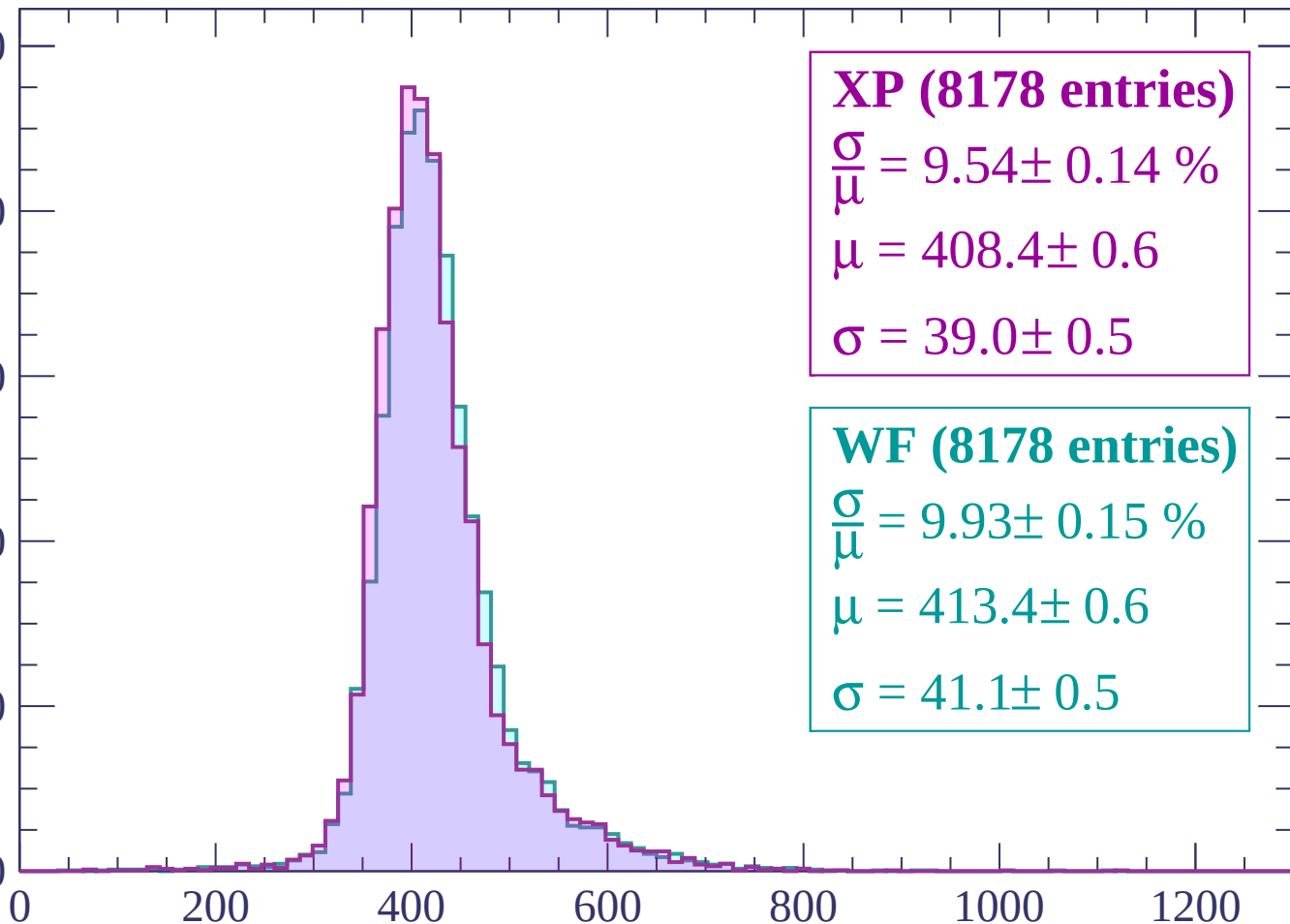
800

600

400

200

0



XP (8178 entries)

$\frac{\sigma}{\mu} = 9.54 \pm 0.14 \%$

$\mu = 408.4 \pm 0.6$

$\sigma = 39.0 \pm 0.5$

WF (8178 entries)

$\frac{\sigma}{\mu} = 9.93 \pm 0.15 \%$

$\mu = 413.4 \pm 0.6$

$\sigma = 41.1 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -420$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (8133 entries)

$$\frac{\sigma}{\mu} = 9.67 \pm 0.15 \%$$

$$\mu = 404.5 \pm 0.5$$

$$\sigma = 39.1 \pm 0.5$$

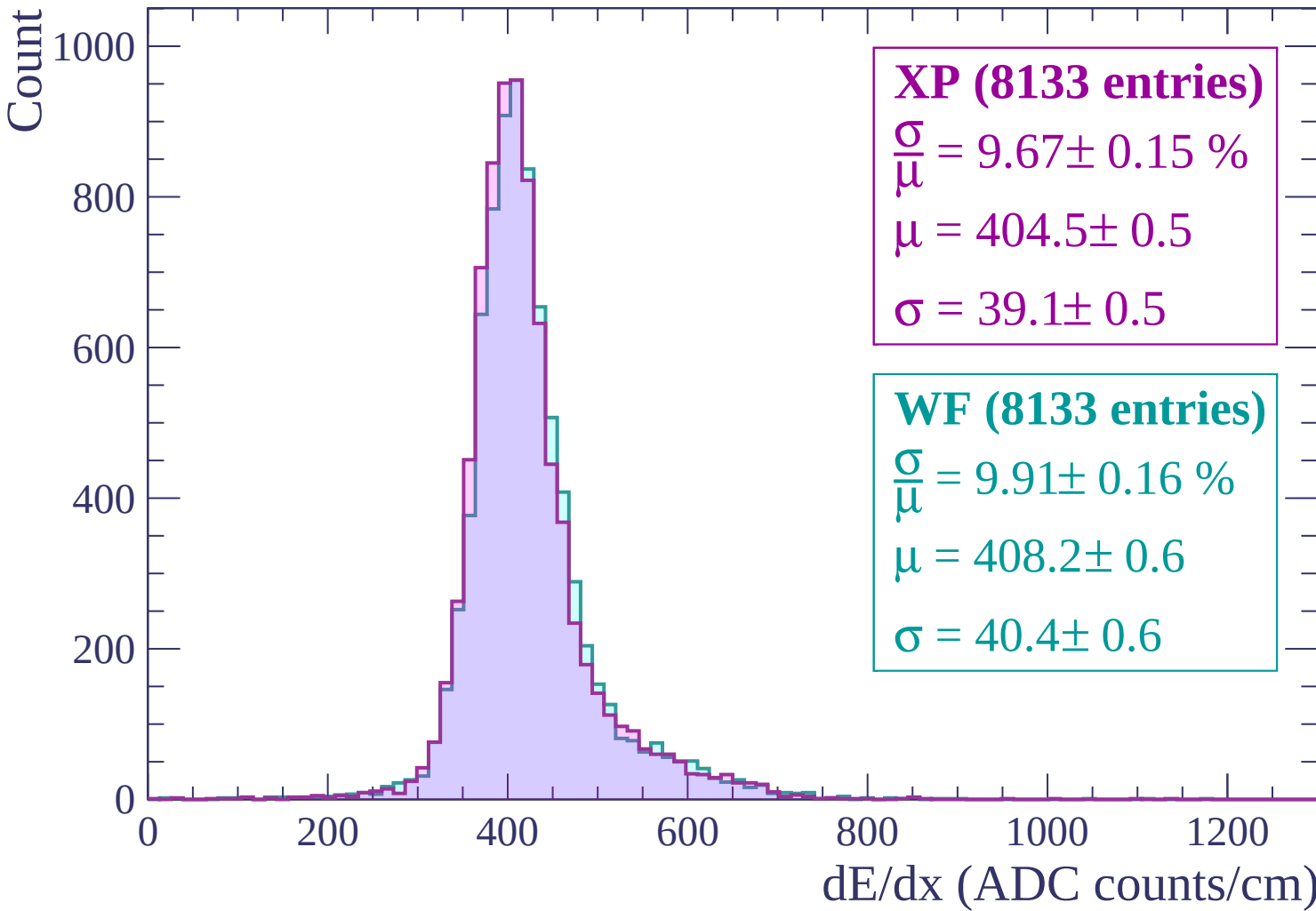
WF (8133 entries)

$$\frac{\sigma}{\mu} = 9.91 \pm 0.16 \%$$

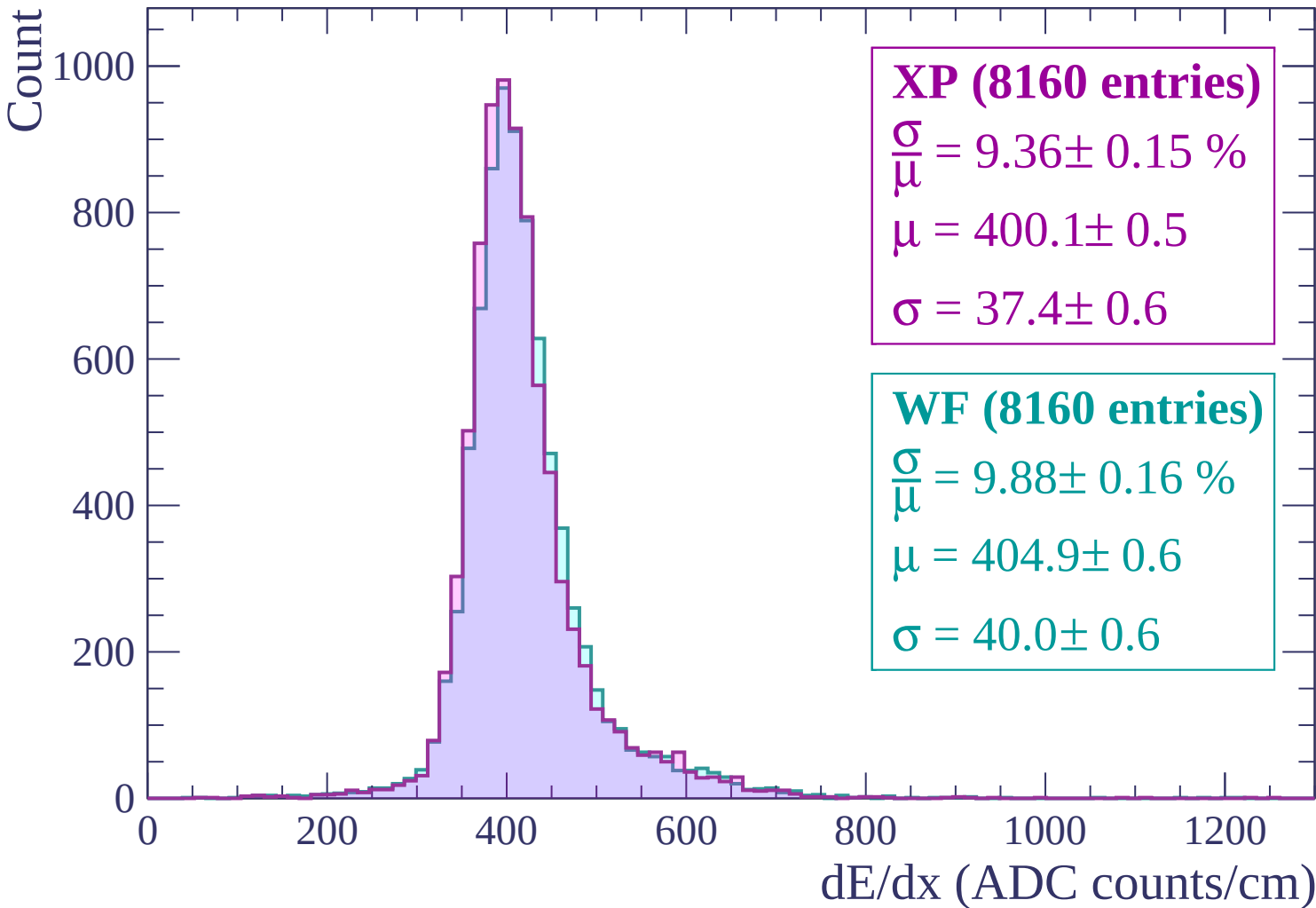
$$\mu = 408.2 \pm 0.6$$

$$\sigma = 40.4 \pm 0.6$$

dE/dx (ADC counts/cm)



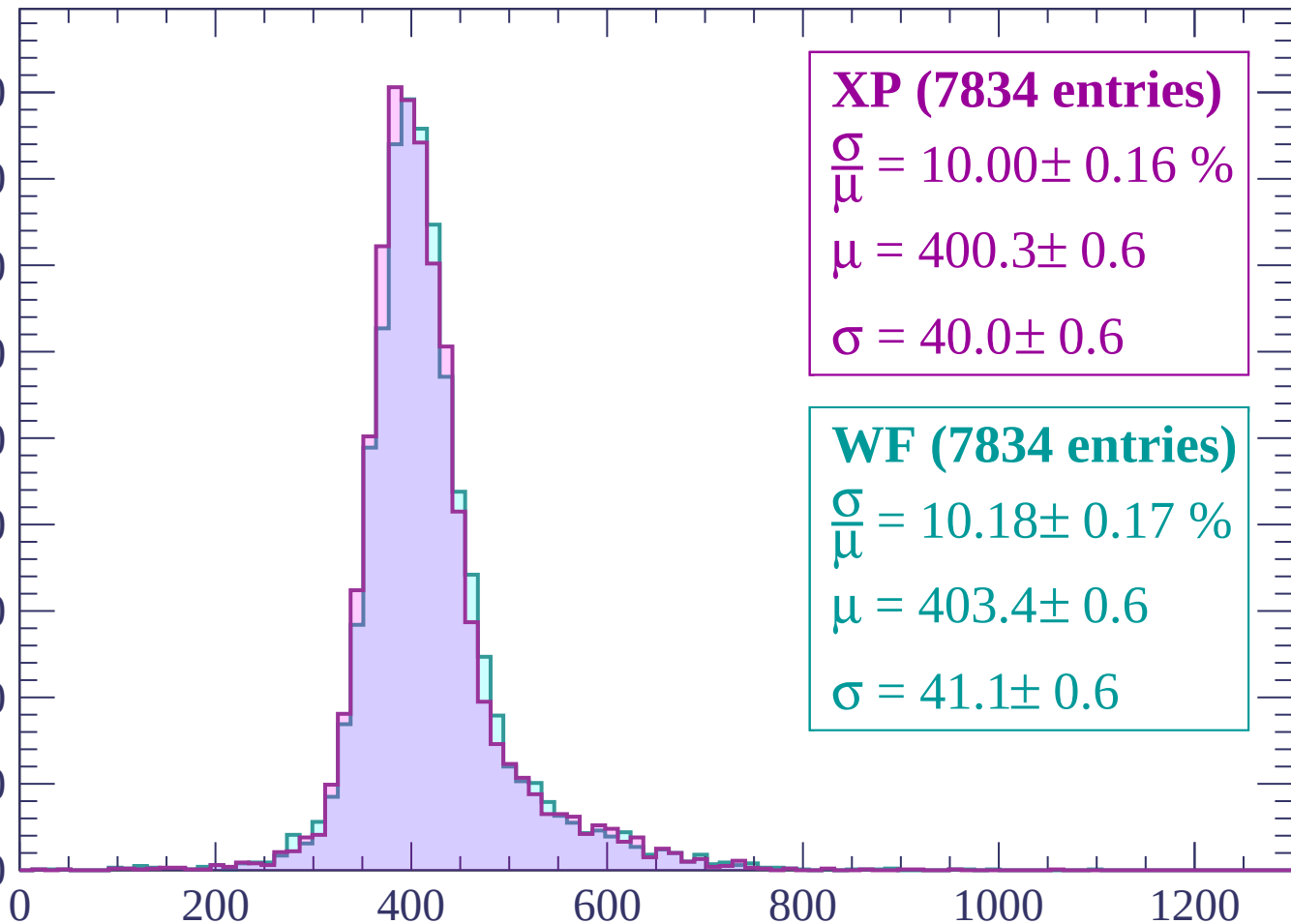
Energy loss | $-420 < p < -360$



Energy loss | $-360 < p < -300$

Count

900
800
700
600
500
400
300
200
100
0



XP (7834 entries)

$$\frac{\sigma}{\mu} = 10.00 \pm 0.16 \%$$

$$\mu = 400.3 \pm 0.6$$

$$\sigma = 40.0 \pm 0.6$$

WF (7834 entries)

$$\frac{\sigma}{\mu} = 10.18 \pm 0.17 \%$$

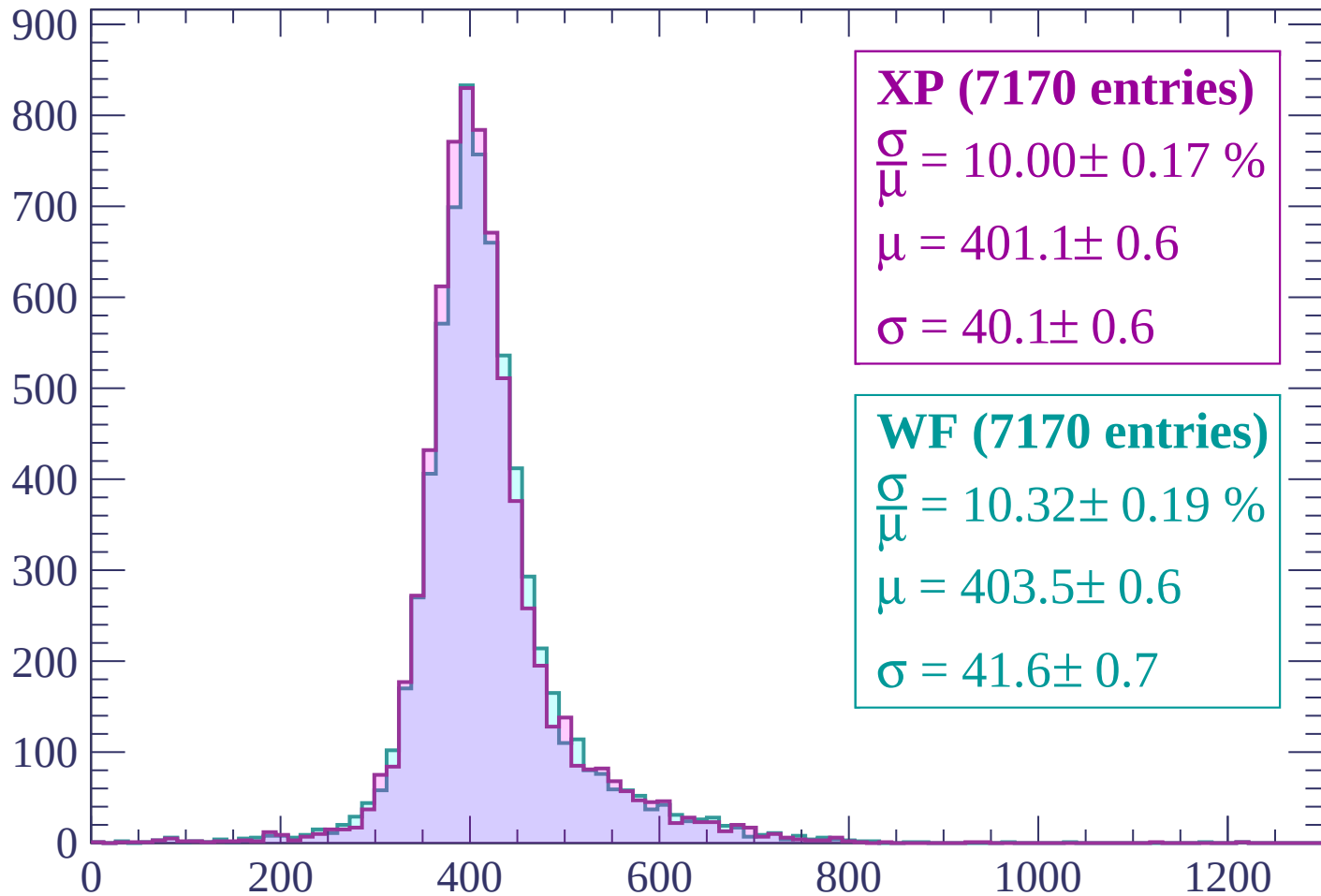
$$\mu = 403.4 \pm 0.6$$

$$\sigma = 41.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-300 < p < -240$

Count



XP (7170 entries)

$$\frac{\sigma}{\mu} = 10.00 \pm 0.17 \%$$

$$\mu = 401.1 \pm 0.6$$

$$\sigma = 40.1 \pm 0.6$$

WF (7170 entries)

$$\frac{\sigma}{\mu} = 10.32 \pm 0.19 \%$$

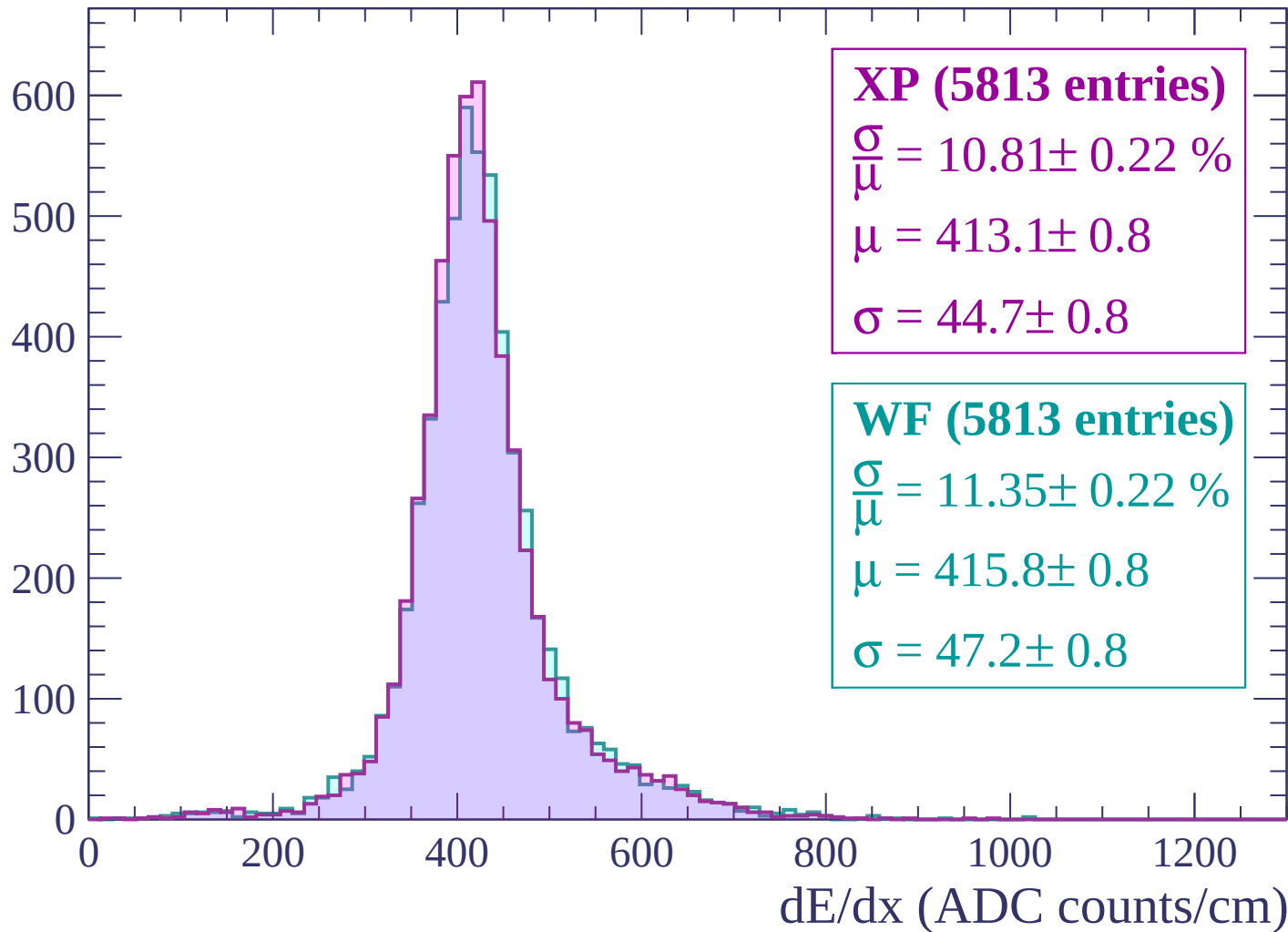
$$\mu = 403.5 \pm 0.6$$

$$\sigma = 41.6 \pm 0.7$$

dE/dx (ADC counts/cm)

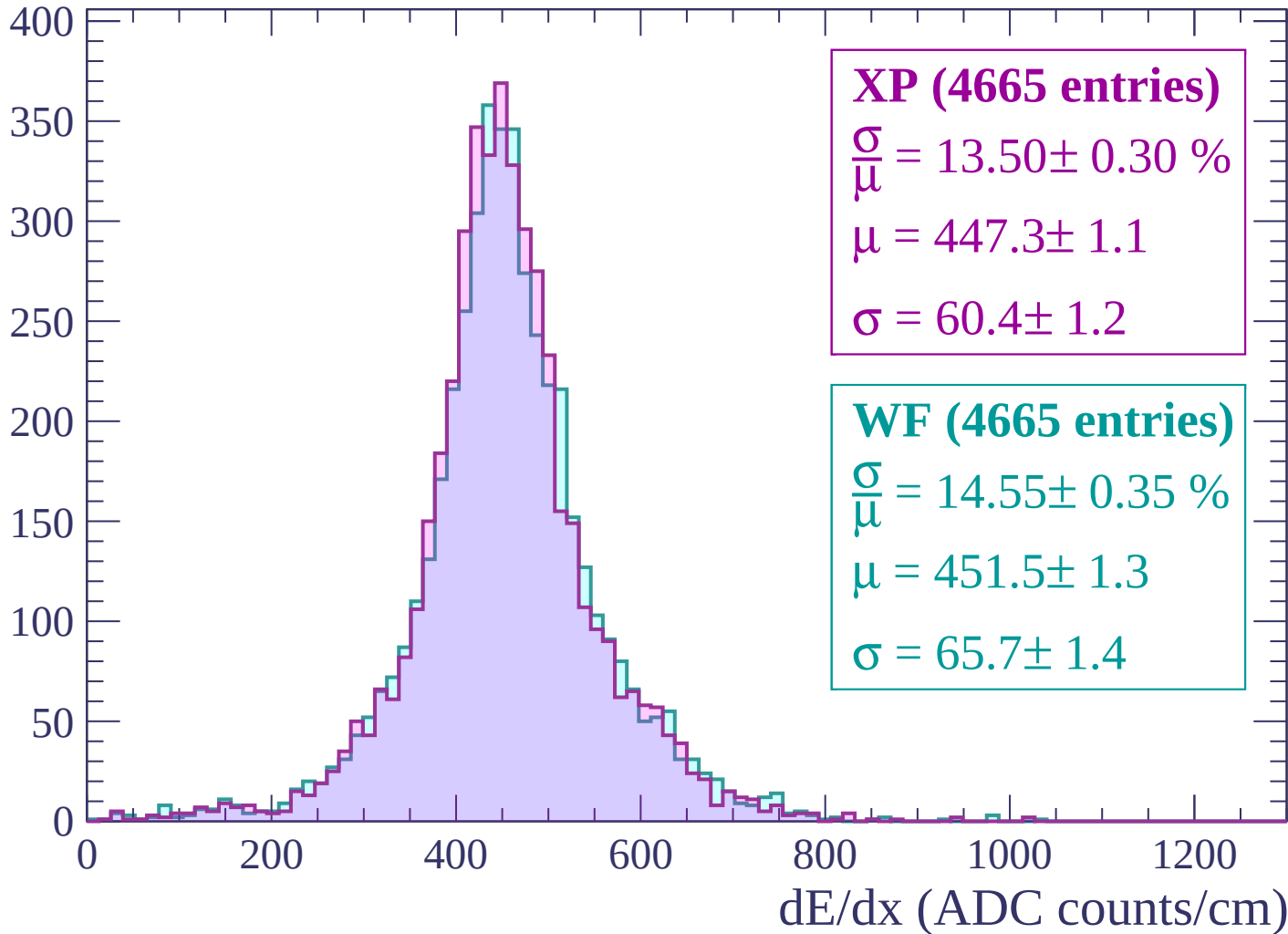
Energy loss | $-240 < p < -180$

Count



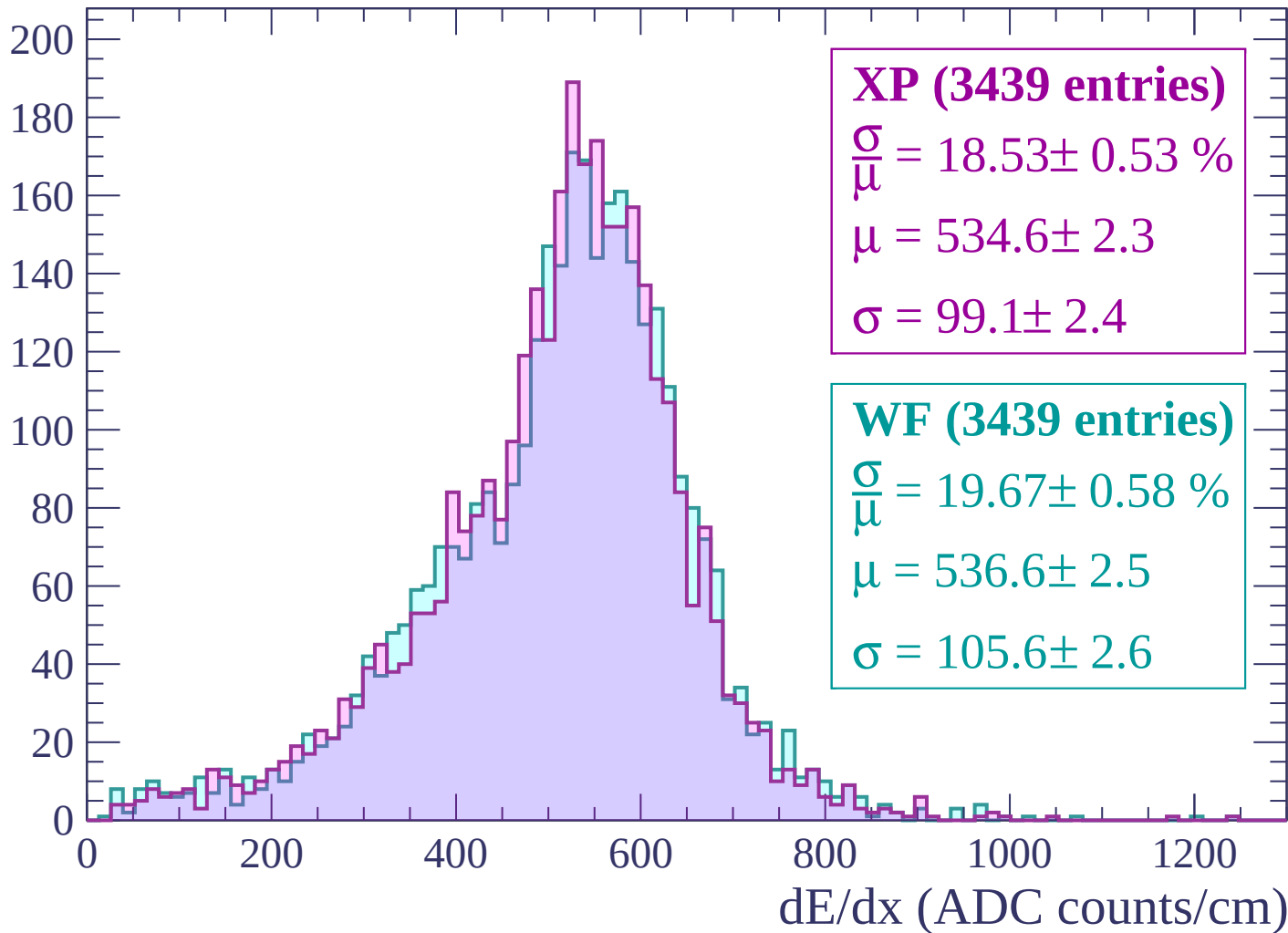
Energy loss | -180 < p < -120

Count



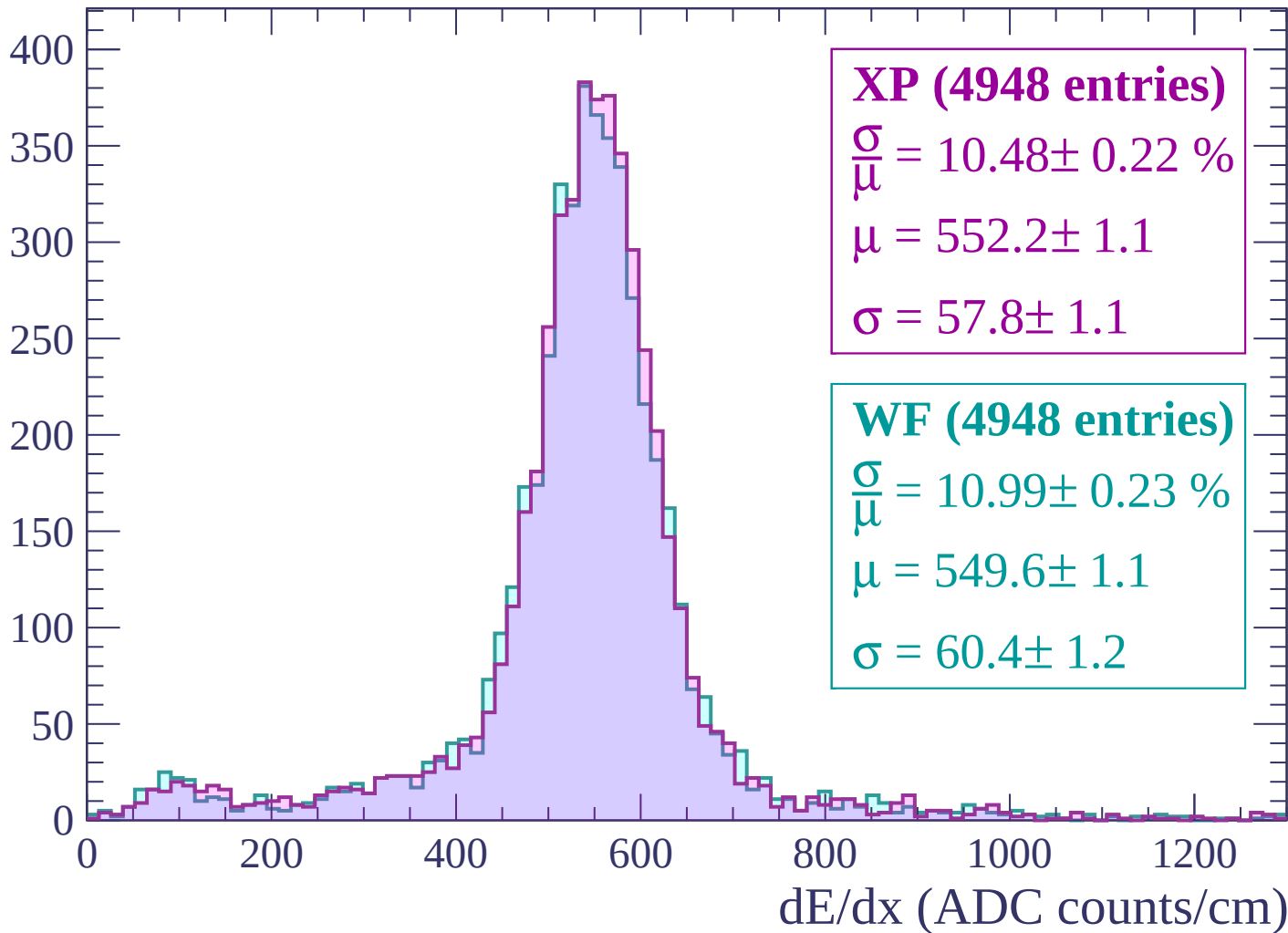
Energy loss | $-120 < p < -60$

Count



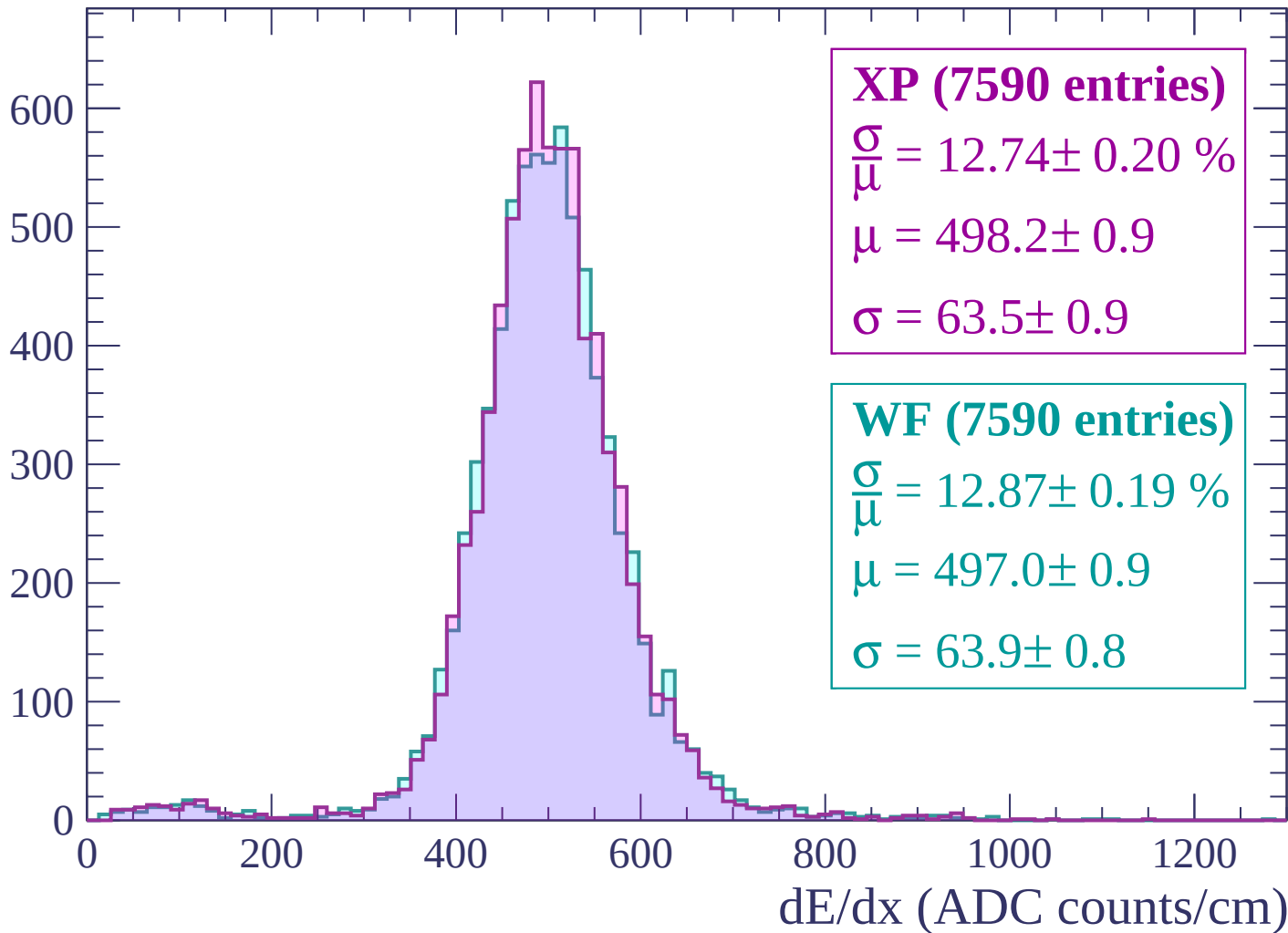
Energy loss | $-60 < p < 0$

Count



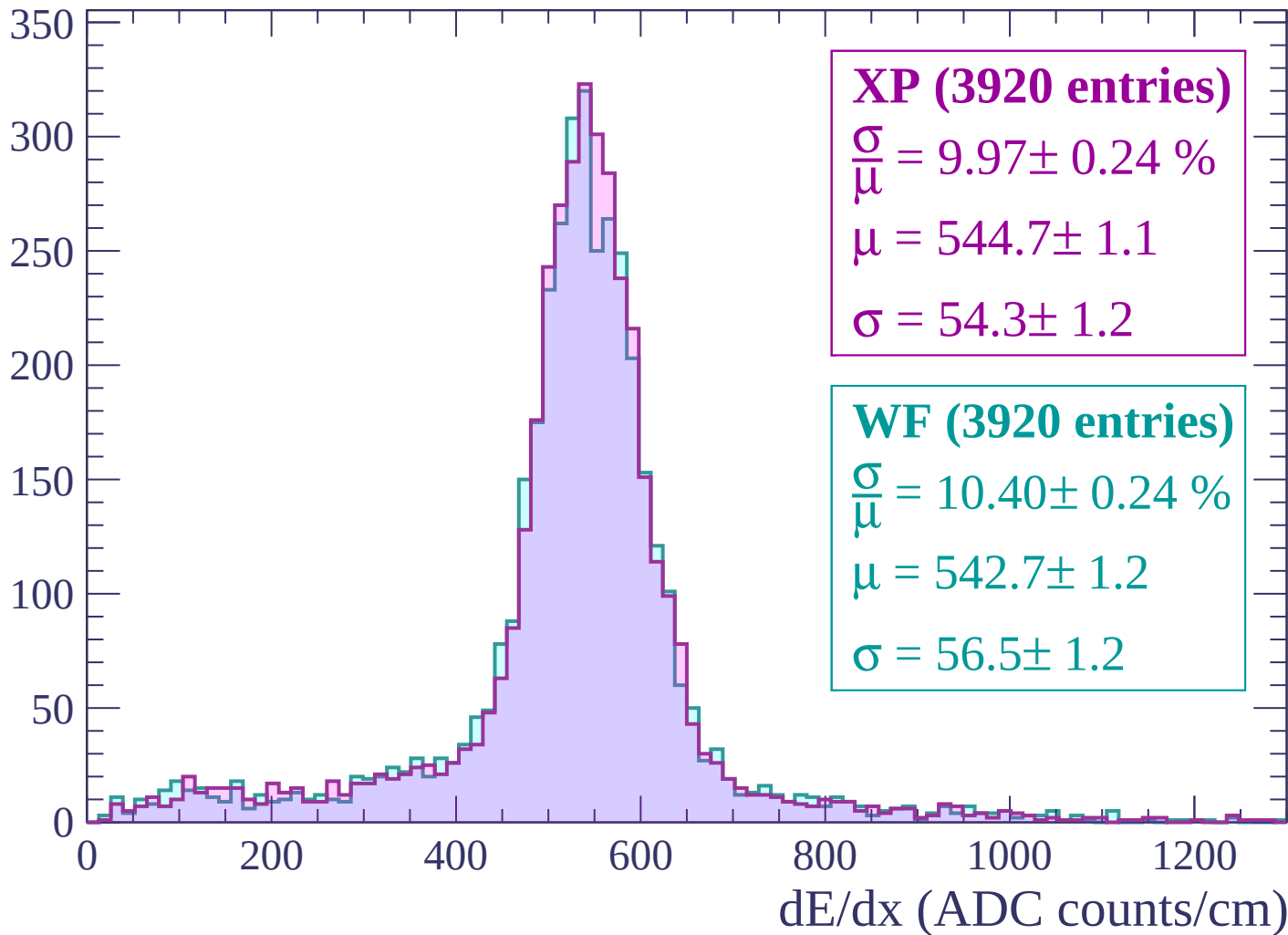
Energy loss | $0 < p < 60$

Count



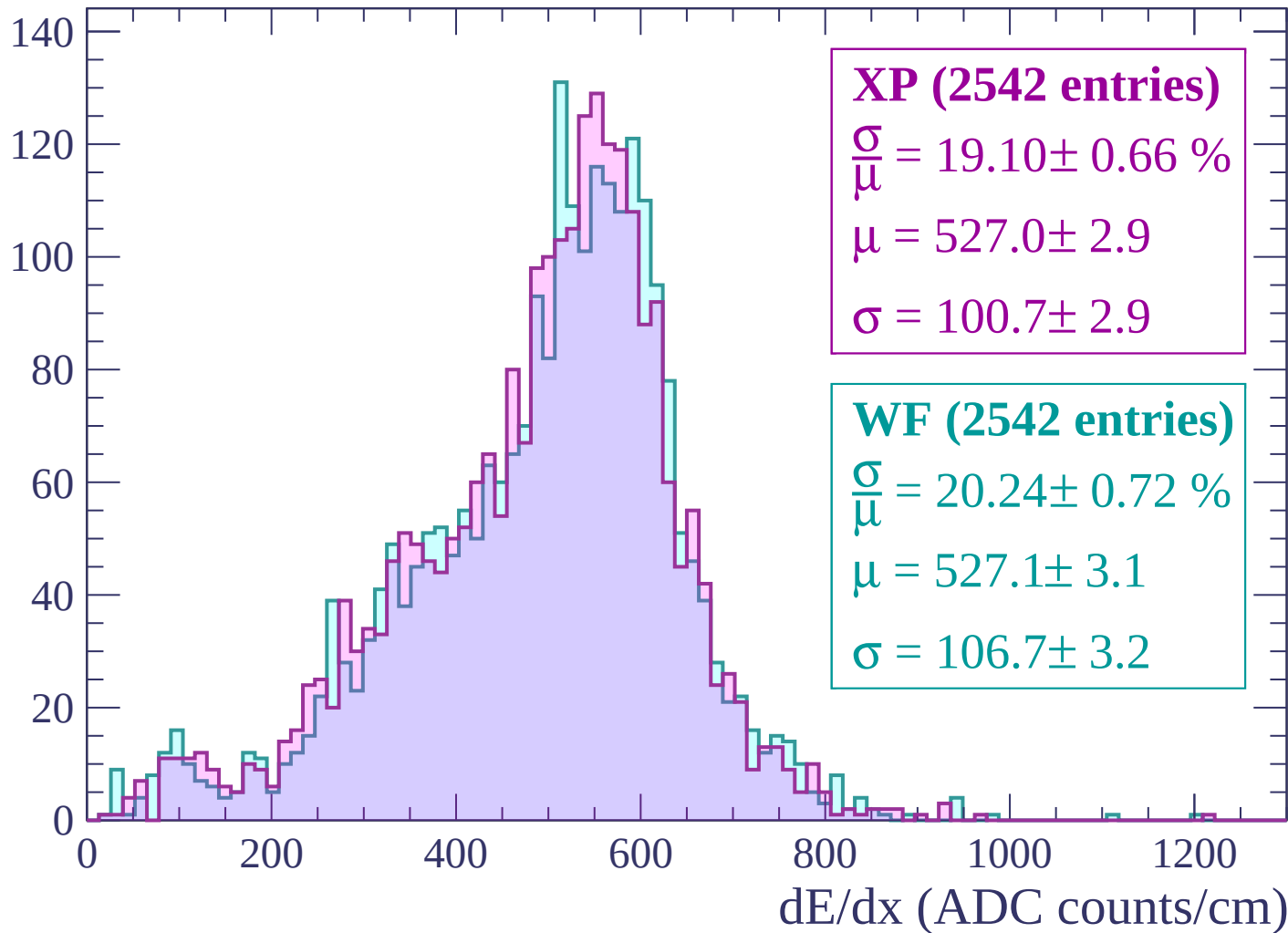
Energy loss | $60 < p < 120$

Count



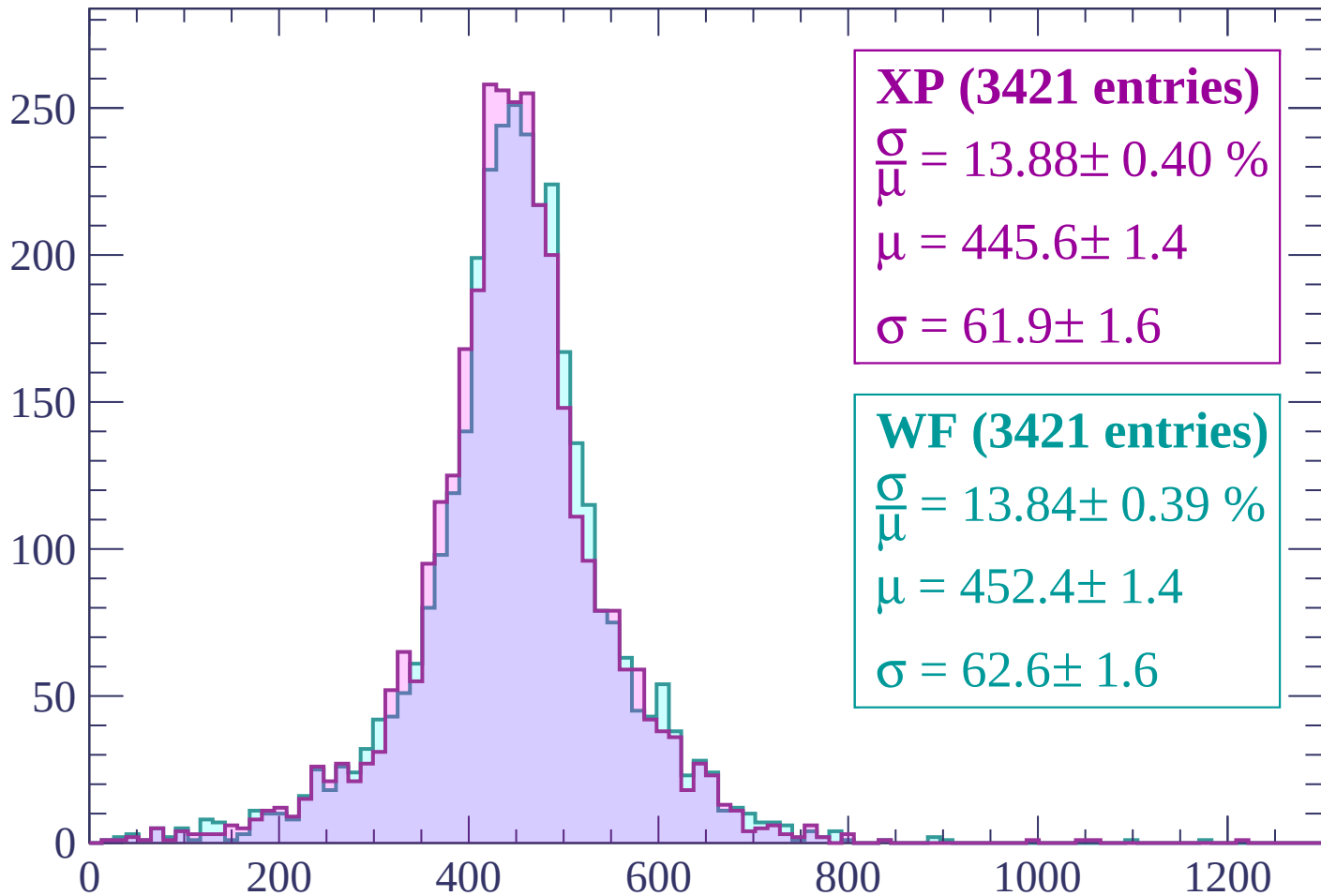
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP (3421 entries)

$$\frac{\sigma}{\mu} = 13.88 \pm 0.40 \%$$

$$\mu = 445.6 \pm 1.4$$

$$\sigma = 61.9 \pm 1.6$$

WF (3421 entries)

$$\frac{\sigma}{\mu} = 13.84 \pm 0.39 \%$$

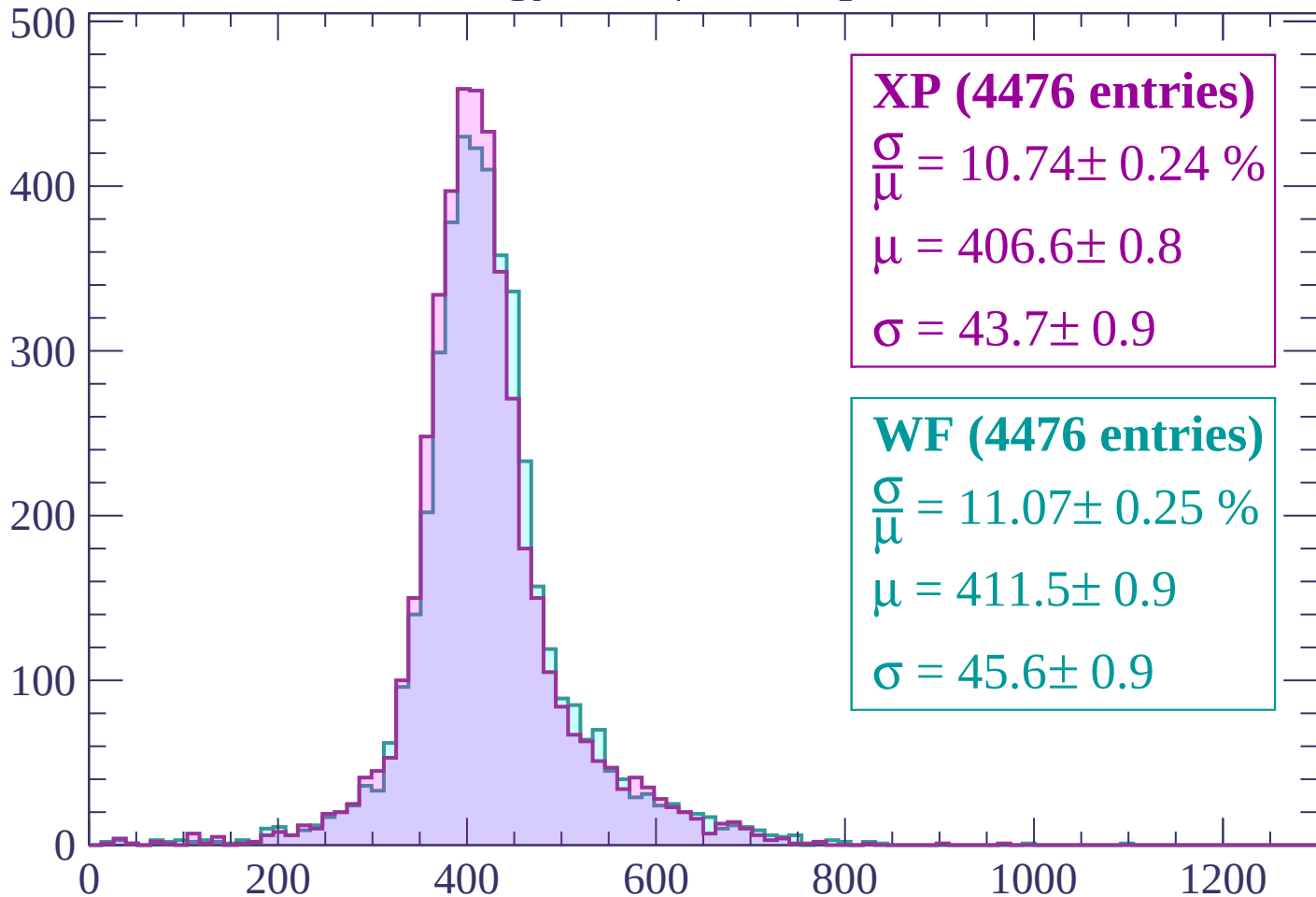
$$\mu = 452.4 \pm 1.4$$

$$\sigma = 62.6 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count



XP (4476 entries)

$$\frac{\sigma}{\mu} = 10.74 \pm 0.24 \%$$

$$\mu = 406.6 \pm 0.8$$

$$\sigma = 43.7 \pm 0.9$$

WF (4476 entries)

$$\frac{\sigma}{\mu} = 11.07 \pm 0.25 \%$$

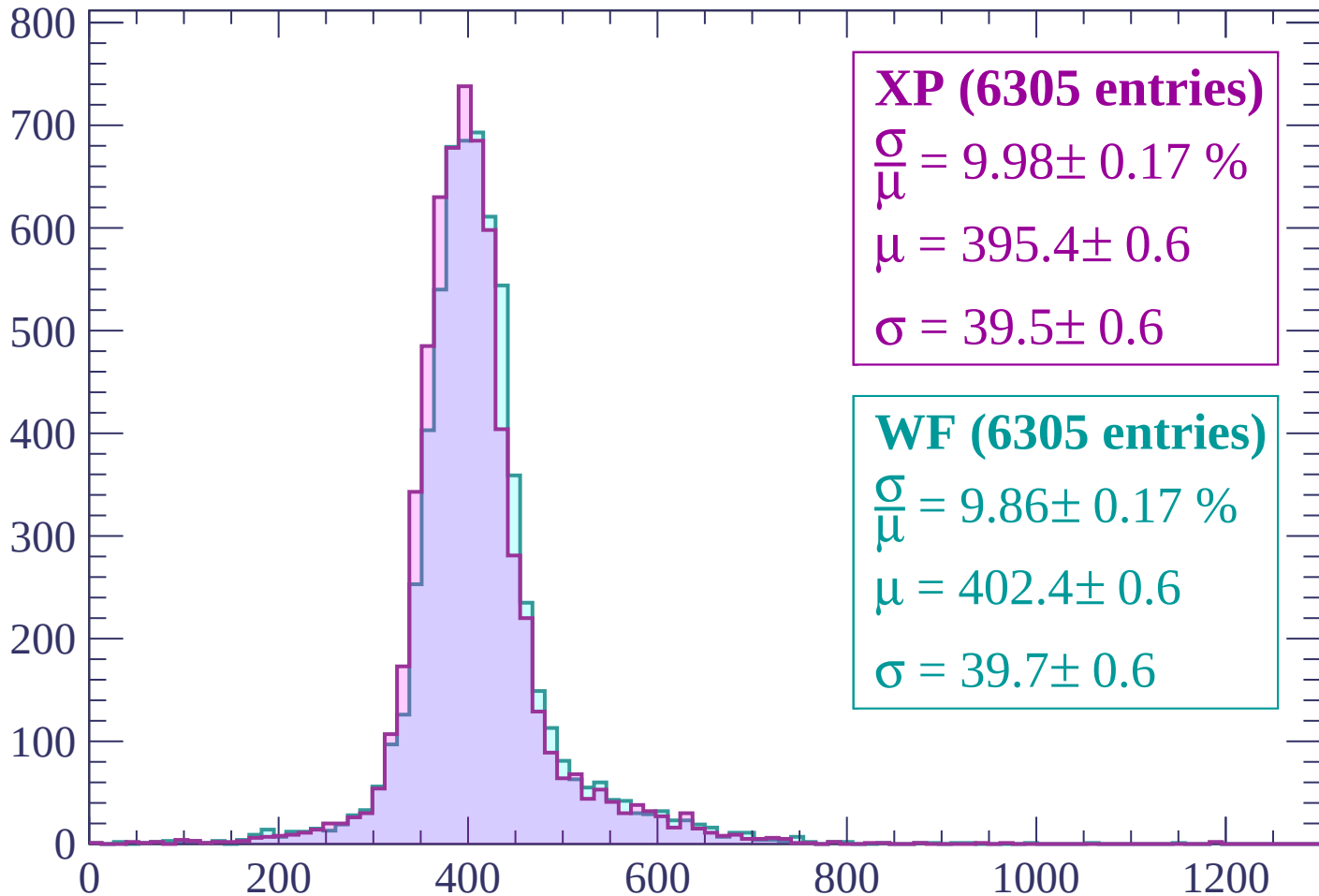
$$\mu = 411.5 \pm 0.9$$

$$\sigma = 45.6 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count



XP (6305 entries)

$$\frac{\sigma}{\mu} = 9.98 \pm 0.17 \%$$

$$\mu = 395.4 \pm 0.6$$

$$\sigma = 39.5 \pm 0.6$$

WF (6305 entries)

$$\frac{\sigma}{\mu} = 9.86 \pm 0.17 \%$$

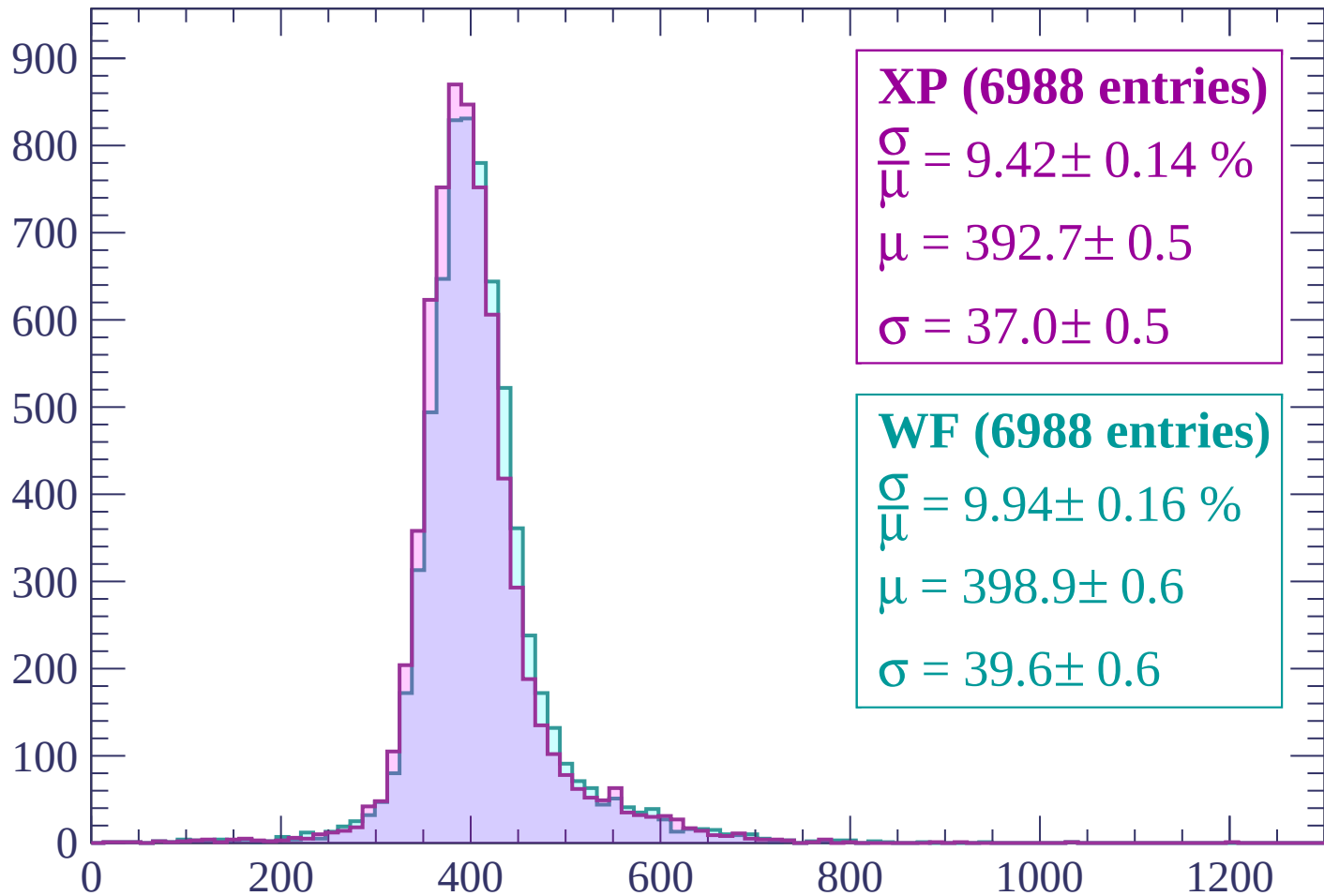
$$\mu = 402.4 \pm 0.6$$

$$\sigma = 39.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

Count



XP (6988 entries)

$$\frac{\sigma}{\mu} = 9.42 \pm 0.14 \%$$

$$\mu = 392.7 \pm 0.5$$

$$\sigma = 37.0 \pm 0.5$$

WF (6988 entries)

$$\frac{\sigma}{\mu} = 9.94 \pm 0.16 \%$$

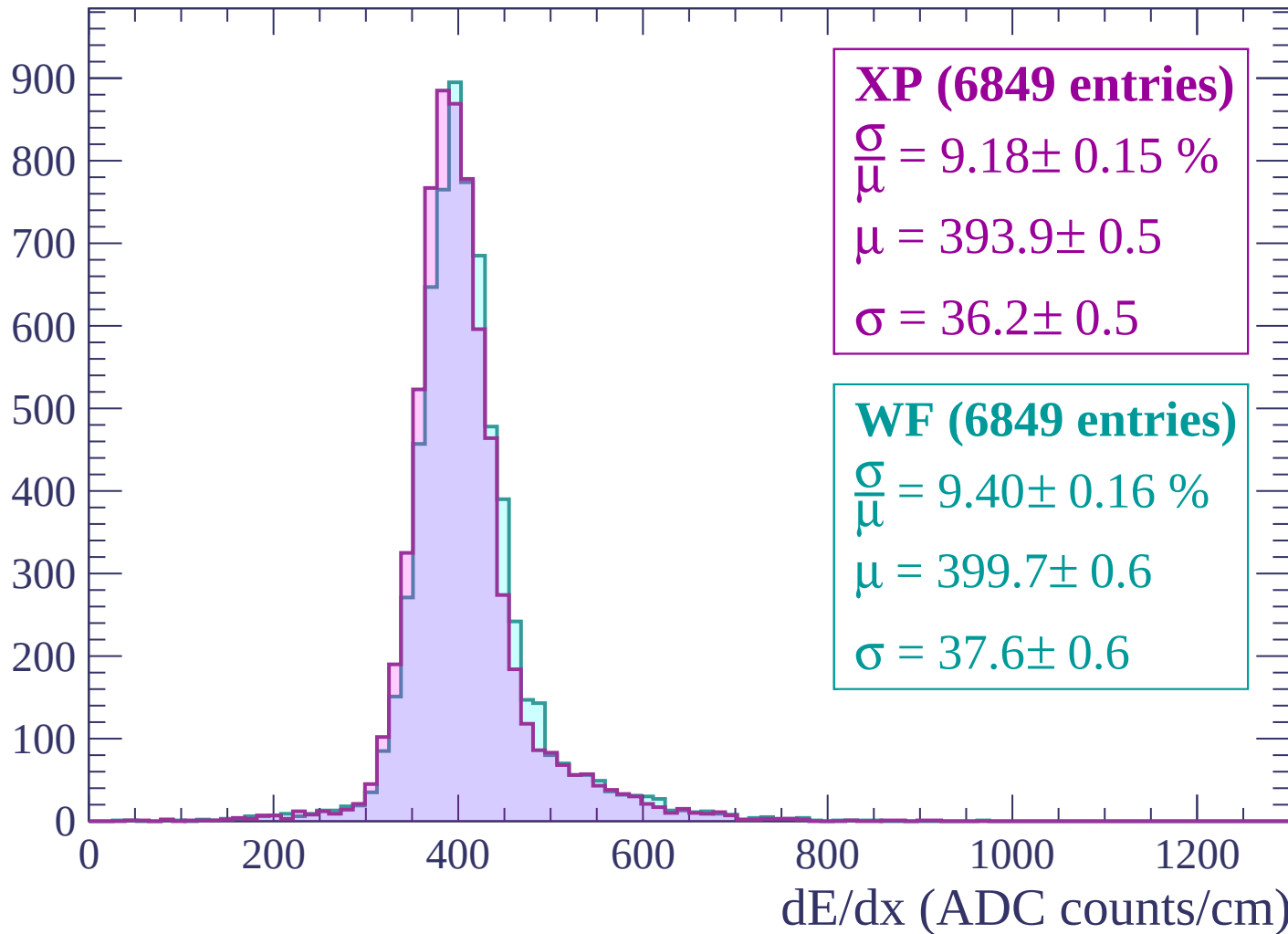
$$\mu = 398.9 \pm 0.6$$

$$\sigma = 39.6 \pm 0.6$$

dE/dx (ADC counts/cm)

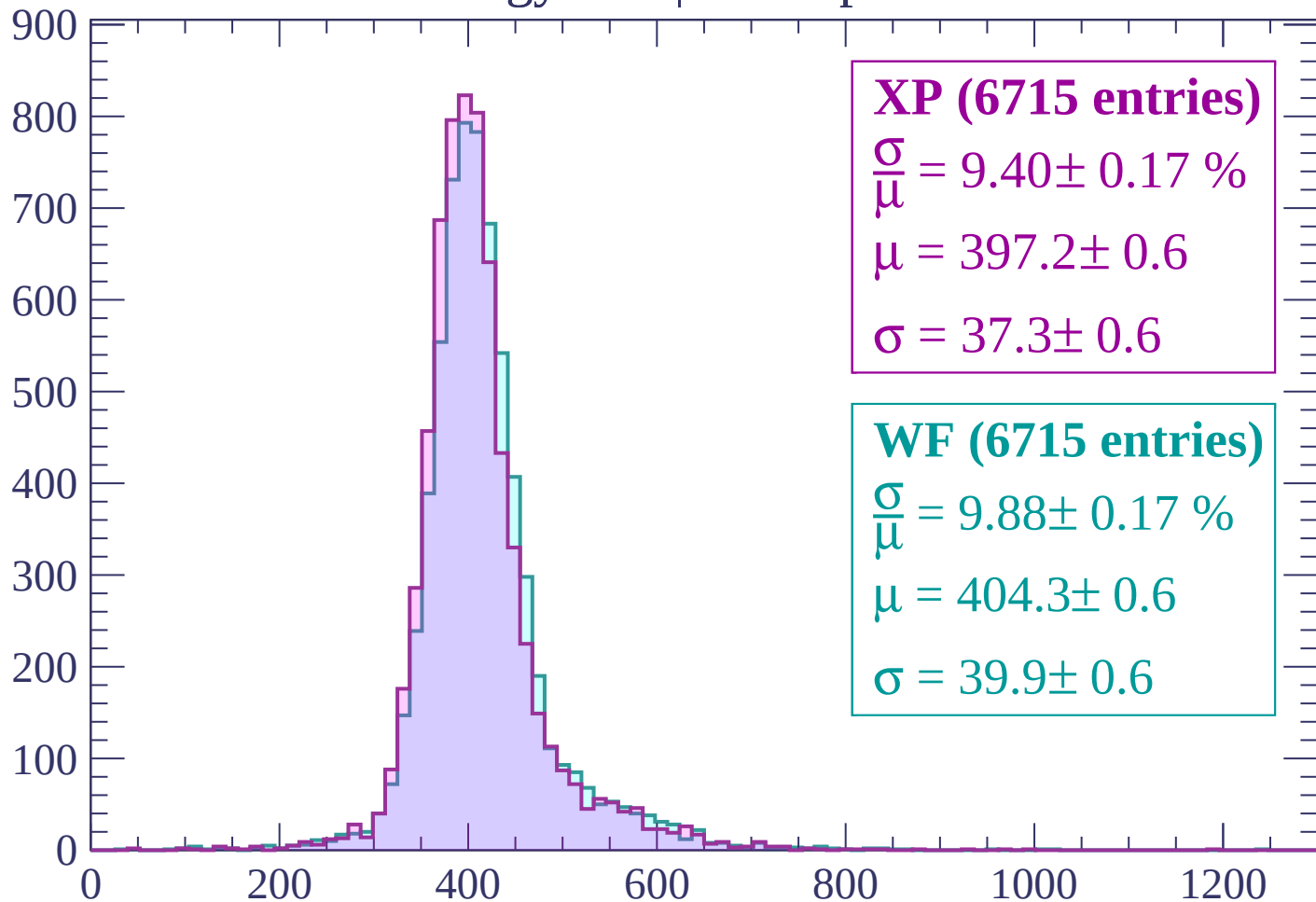
Energy loss | $420 < p < 480$

Count



Energy loss | $480 < p < 540$

Count



XP (6715 entries)

$$\frac{\sigma}{\mu} = 9.40 \pm 0.17 \%$$

$$\mu = 397.2 \pm 0.6$$

$$\sigma = 37.3 \pm 0.6$$

WF (6715 entries)

$$\frac{\sigma}{\mu} = 9.88 \pm 0.17 \%$$

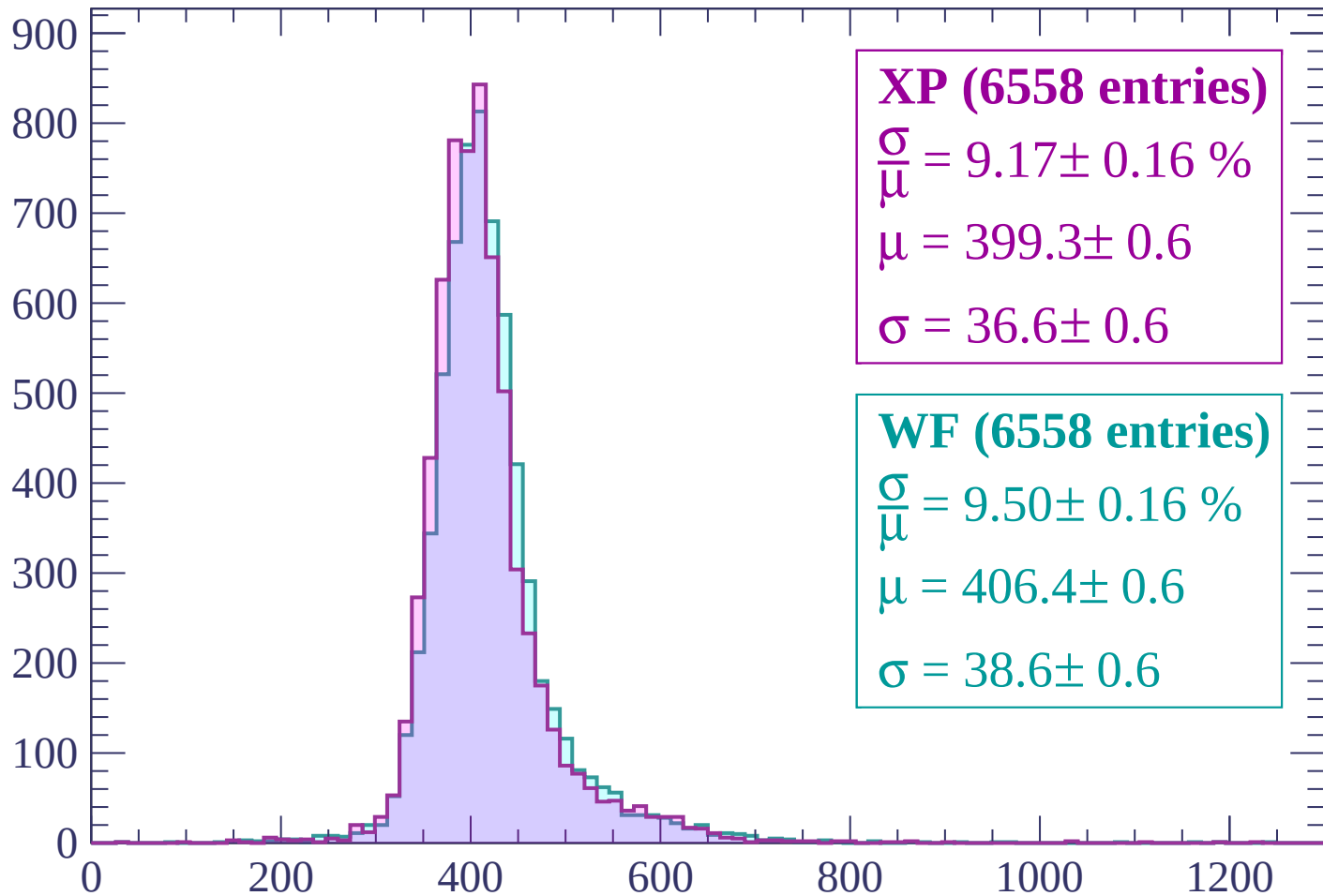
$$\mu = 404.3 \pm 0.6$$

$$\sigma = 39.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$

Count



XP (6558 entries)

$$\frac{\sigma}{\mu} = 9.17 \pm 0.16 \%$$

$$\mu = 399.3 \pm 0.6$$

$$\sigma = 36.6 \pm 0.6$$

WF (6558 entries)

$$\frac{\sigma}{\mu} = 9.50 \pm 0.16 \%$$

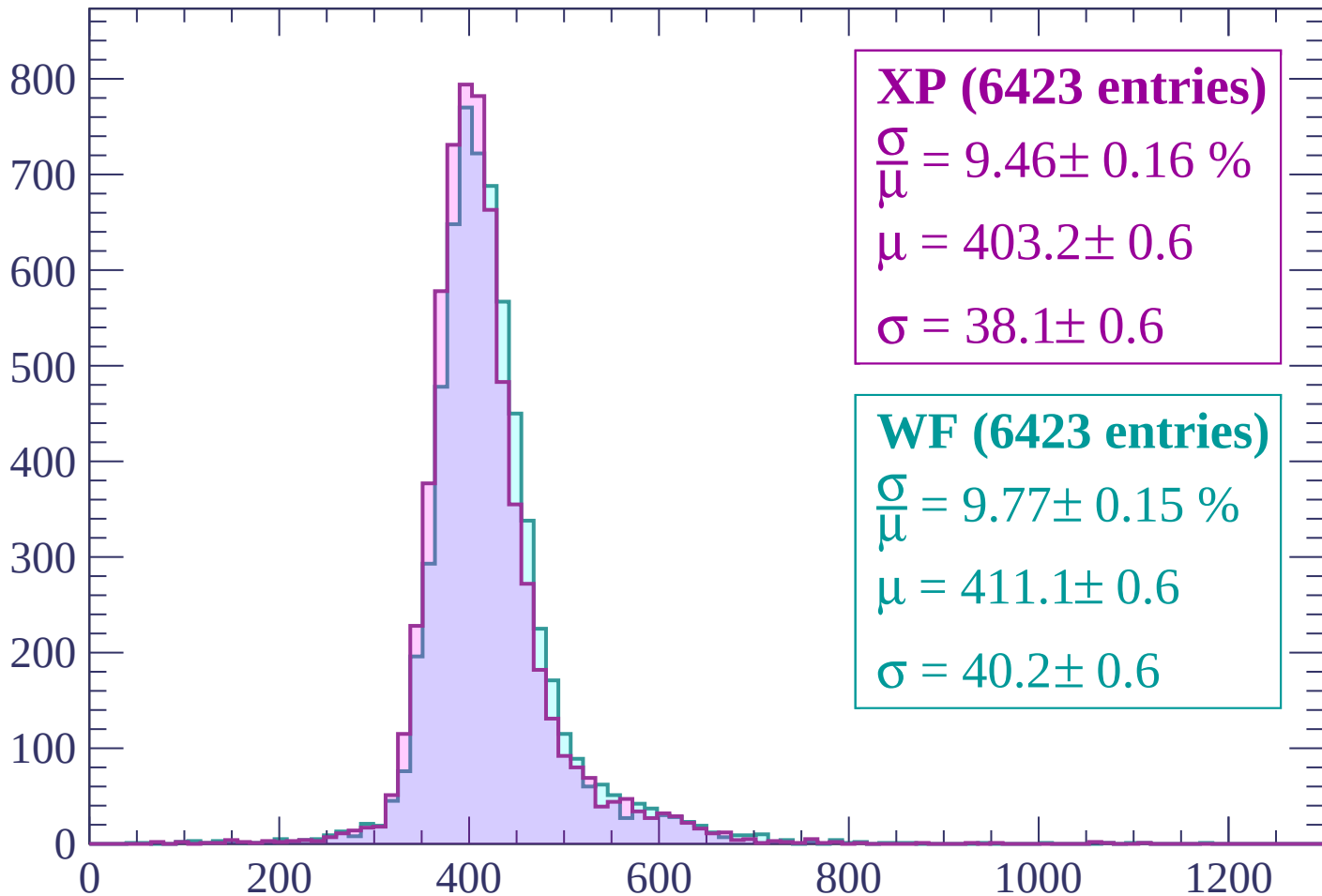
$$\mu = 406.4 \pm 0.6$$

$$\sigma = 38.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$

Count



XP (6423 entries)

$$\frac{\sigma}{\mu} = 9.46 \pm 0.16 \%$$

$$\mu = 403.2 \pm 0.6$$

$$\sigma = 38.1 \pm 0.6$$

WF (6423 entries)

$$\frac{\sigma}{\mu} = 9.77 \pm 0.15 \%$$

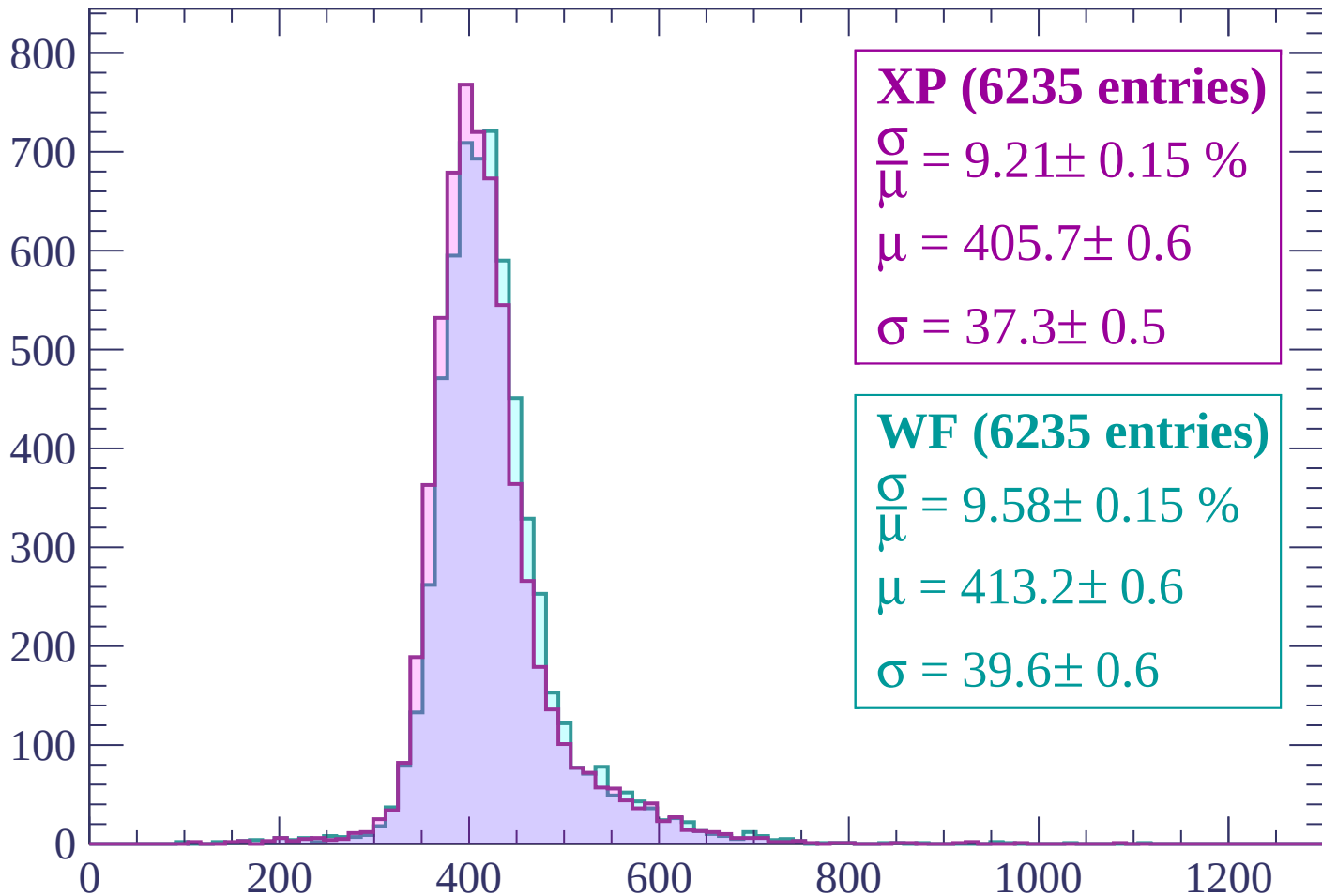
$$\mu = 411.1 \pm 0.6$$

$$\sigma = 40.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$

Count



XP (6235 entries)

$$\frac{\sigma}{\mu} = 9.21 \pm 0.15 \%$$

$$\mu = 405.7 \pm 0.6$$

$$\sigma = 37.3 \pm 0.5$$

WF (6235 entries)

$$\frac{\sigma}{\mu} = 9.58 \pm 0.15 \%$$

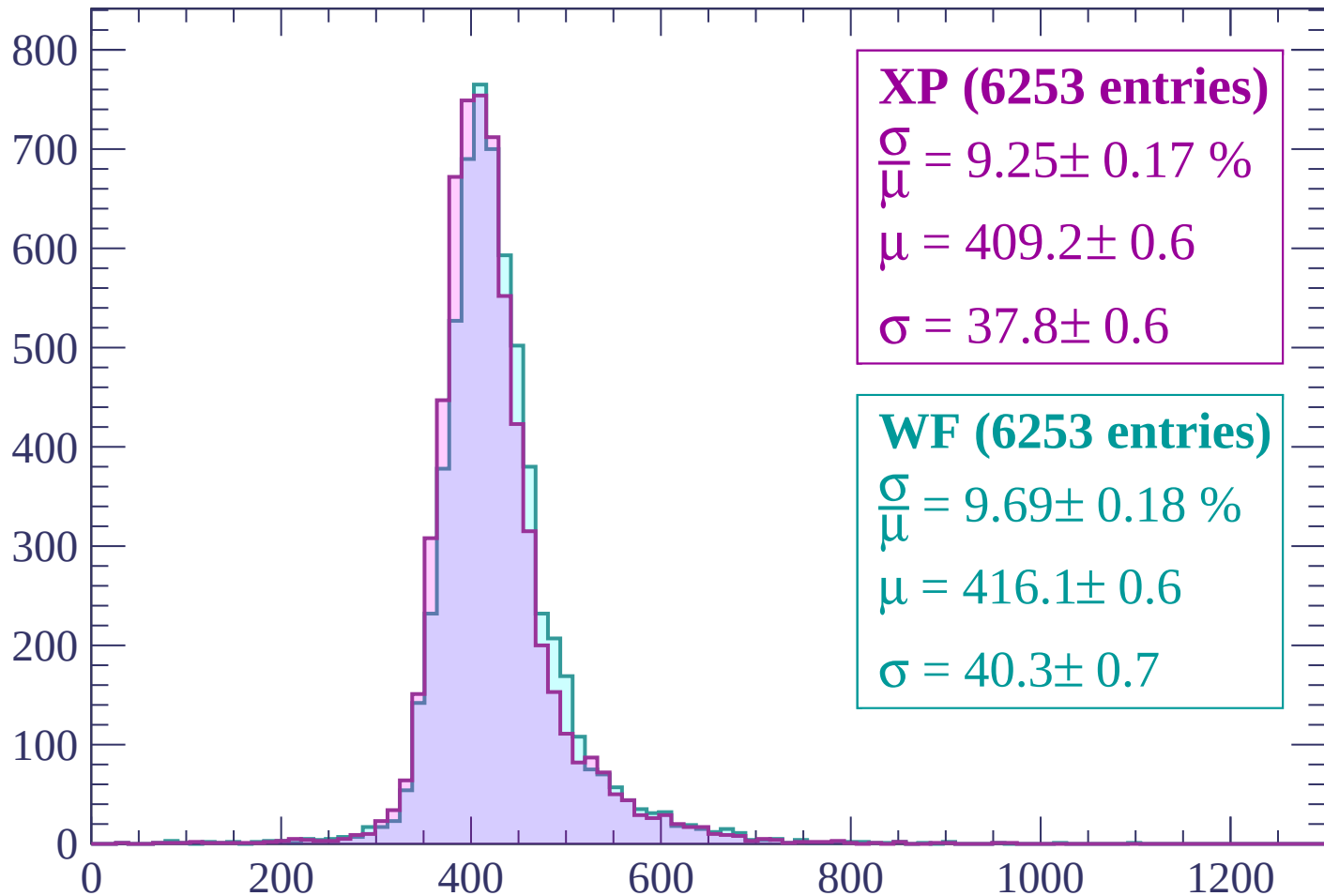
$$\mu = 413.2 \pm 0.6$$

$$\sigma = 39.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count



XP (6253 entries)

$\frac{\sigma}{\mu} = 9.25 \pm 0.17 \%$

$\mu = 409.2 \pm 0.6$

$\sigma = 37.8 \pm 0.6$

WF (6253 entries)

$\frac{\sigma}{\mu} = 9.69 \pm 0.18 \%$

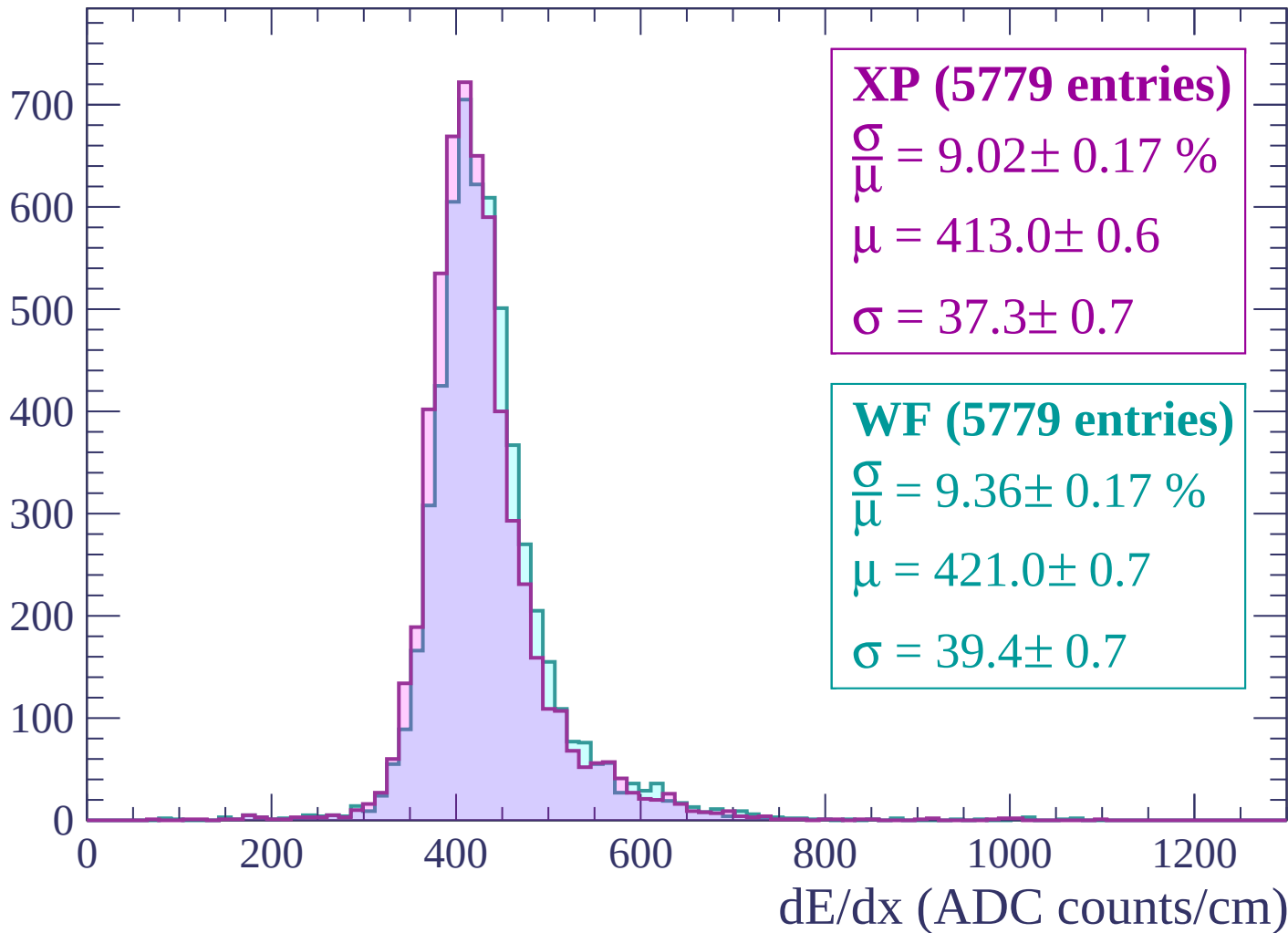
$\mu = 416.1 \pm 0.6$

$\sigma = 40.3 \pm 0.7$

dE/dx (ADC counts/cm)

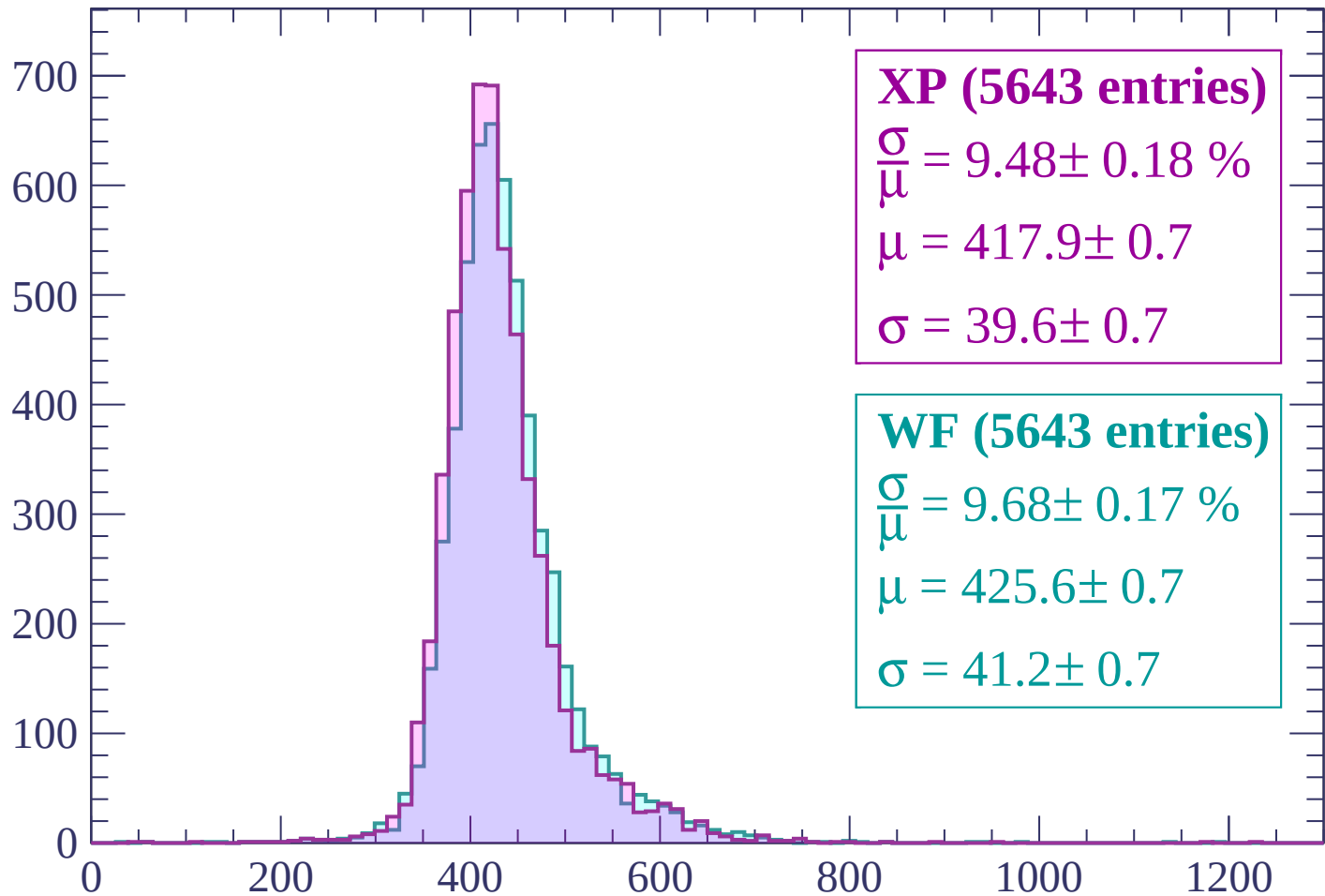
Energy loss | $780 < p < 840$

Count



Energy loss | $840 < p < 900$

Count



XP (5643 entries)

$$\frac{\sigma}{\mu} = 9.48 \pm 0.18 \%$$

$$\mu = 417.9 \pm 0.7$$

$$\sigma = 39.6 \pm 0.7$$

WF (5643 entries)

$$\frac{\sigma}{\mu} = 9.68 \pm 0.17 \%$$

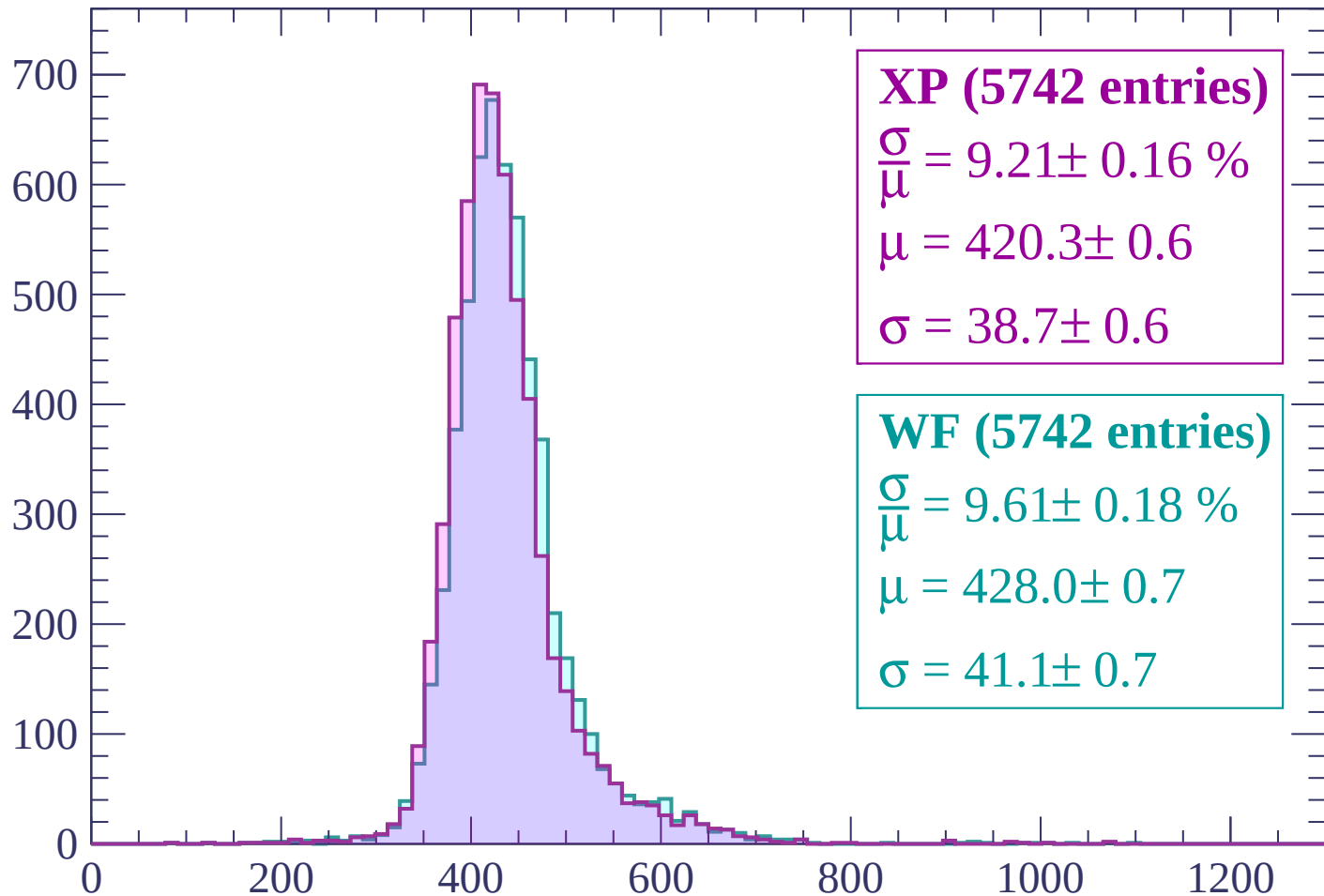
$$\mu = 425.6 \pm 0.7$$

$$\sigma = 41.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count



XP (5742 entries)

$\frac{\sigma}{\mu} = 9.21 \pm 0.16 \%$

$\mu = 420.3 \pm 0.6$

$\sigma = 38.7 \pm 0.6$

WF (5742 entries)

$\frac{\sigma}{\mu} = 9.61 \pm 0.18 \%$

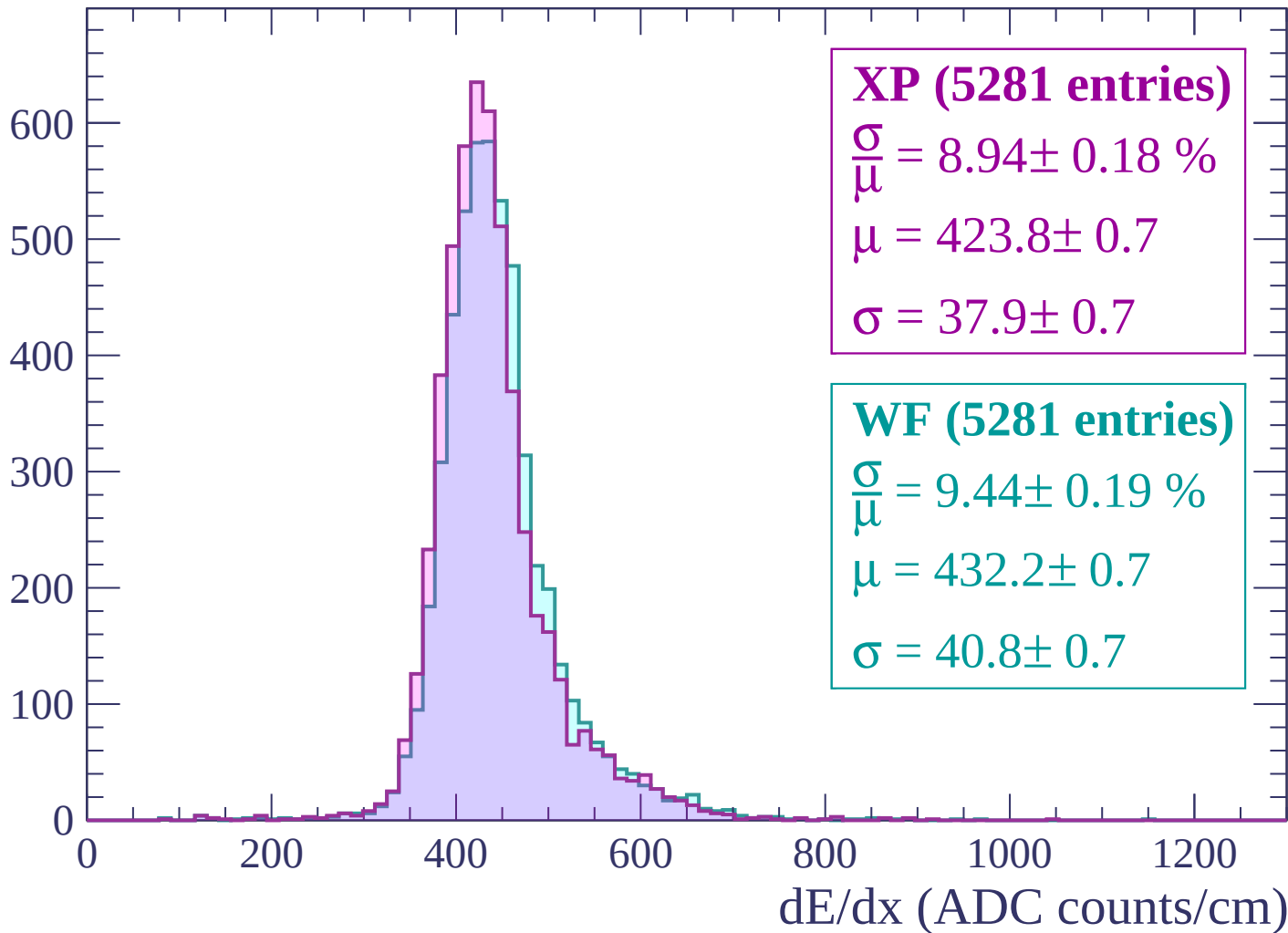
$\mu = 428.0 \pm 0.7$

$\sigma = 41.1 \pm 0.7$

dE/dx (ADC counts/cm)

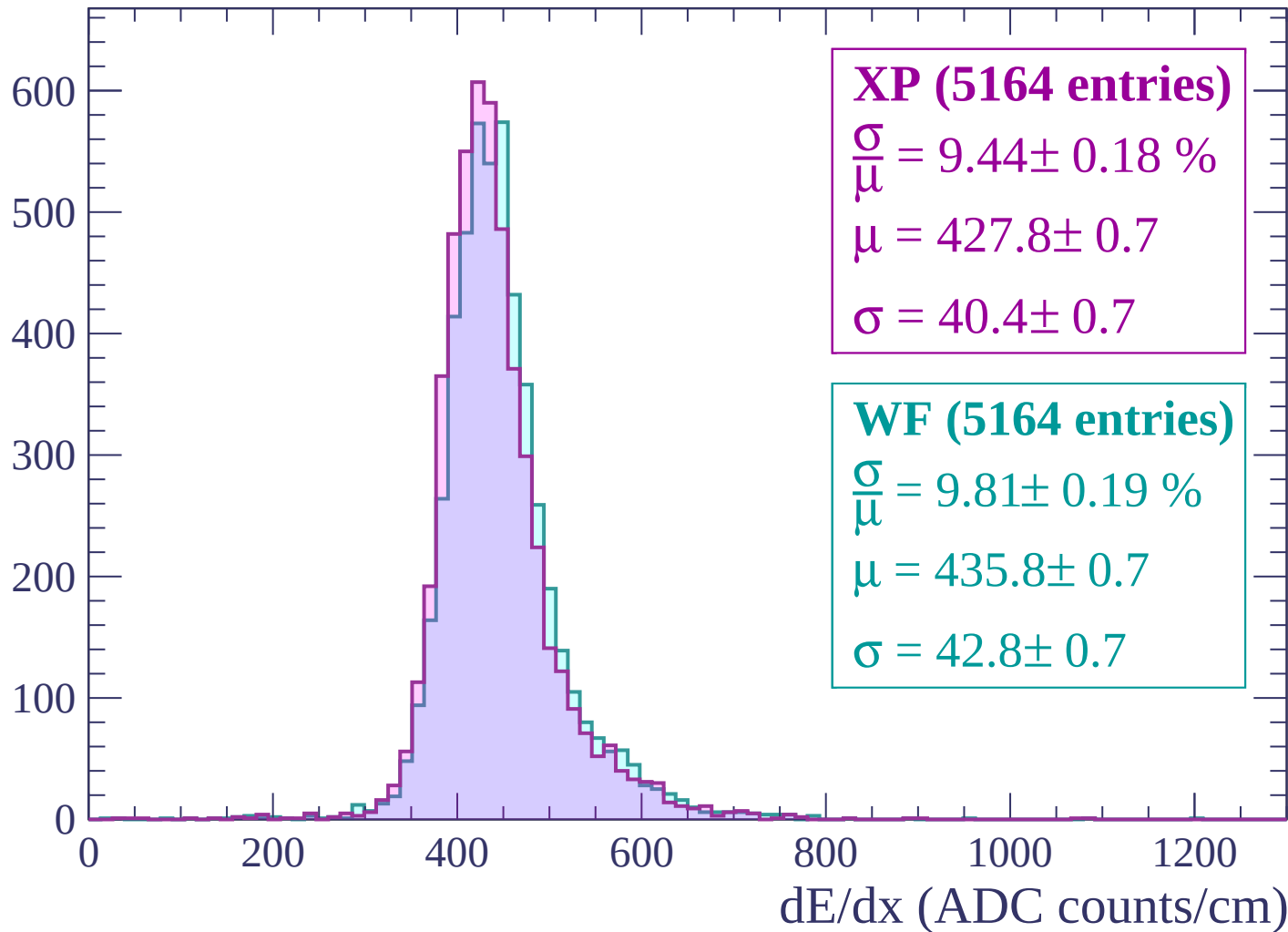
Energy loss | $960 < p < 1020$

Count



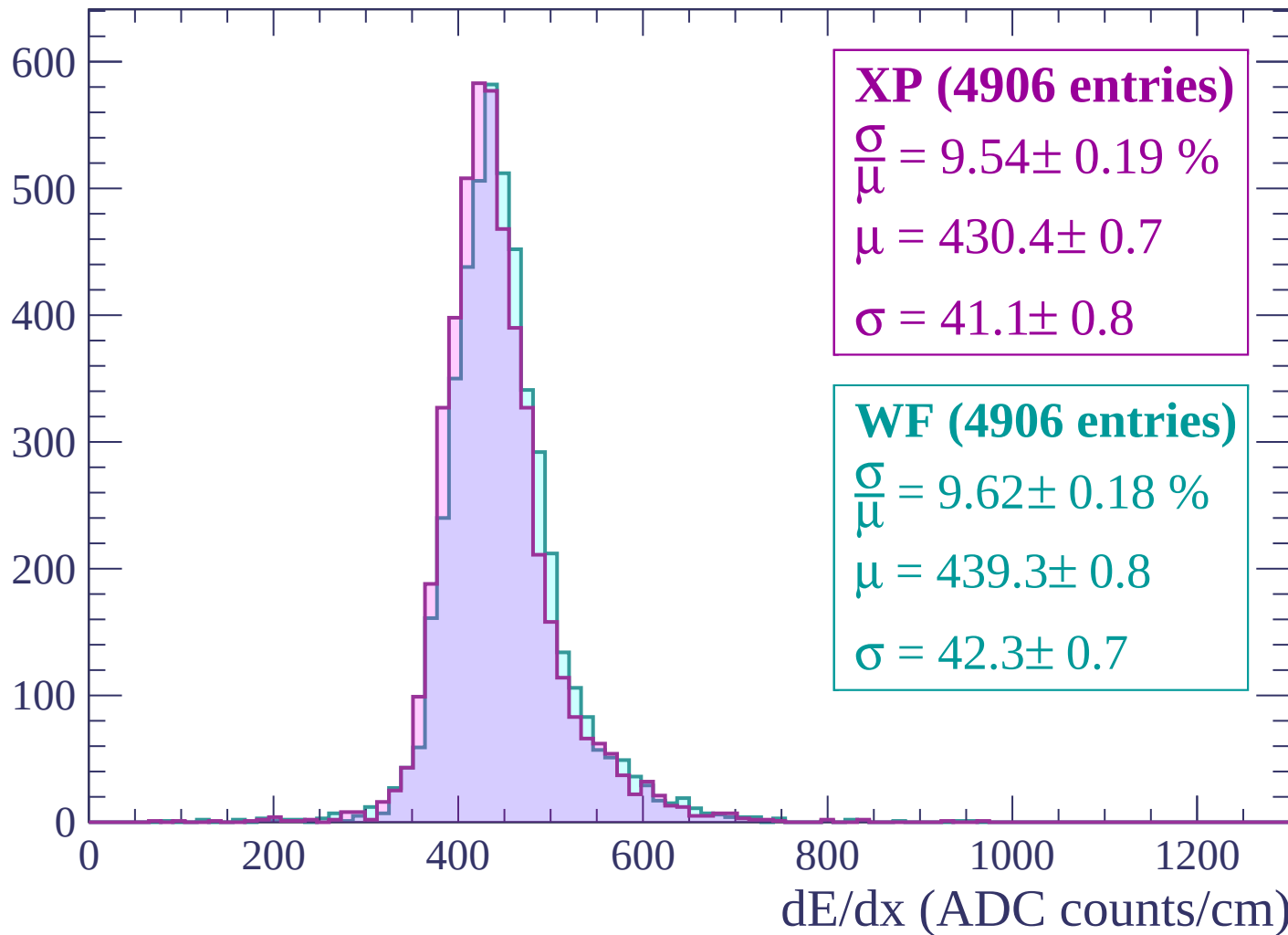
Energy loss | $1020 < p < 1080$

Count



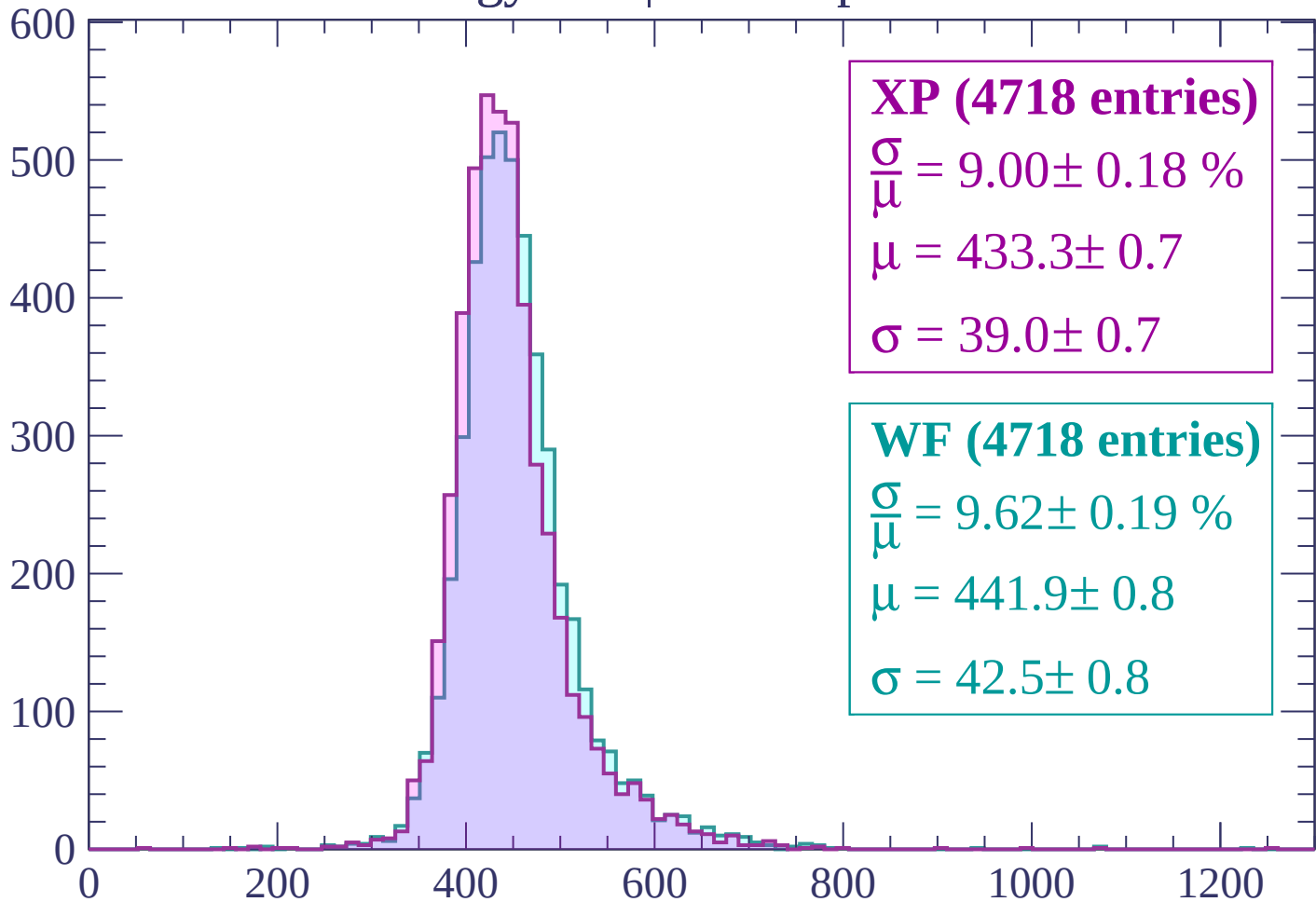
Energy loss | $1080 < p < 1140$

Count



Energy loss | $1140 < p < 1200$

Count



XP (4718 entries)

$$\frac{\sigma}{\mu} = 9.00 \pm 0.18 \%$$

$$\mu = 433.3 \pm 0.7$$

$$\sigma = 39.0 \pm 0.7$$

WF (4718 entries)

$$\frac{\sigma}{\mu} = 9.62 \pm 0.19 \%$$

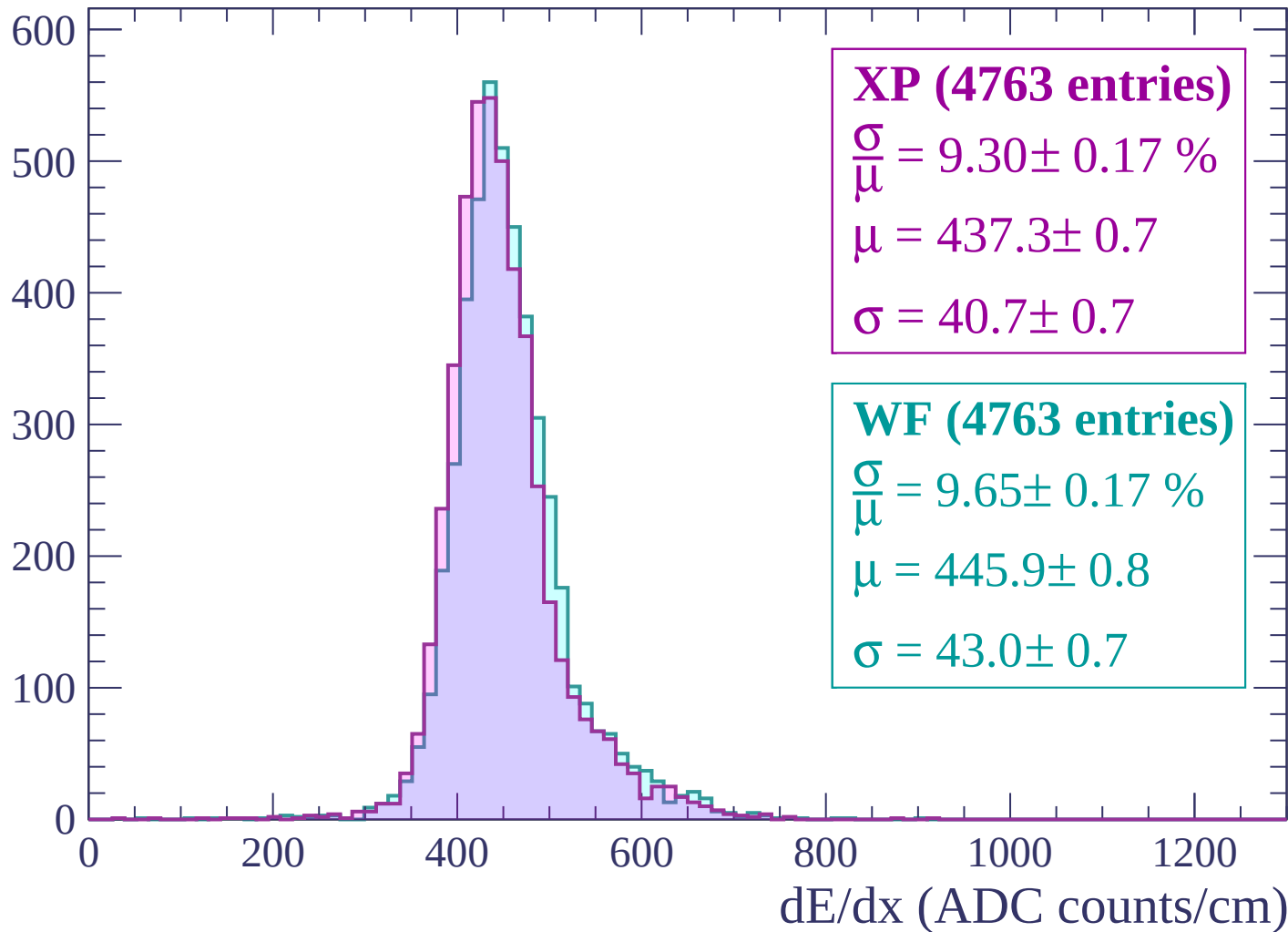
$$\mu = 441.9 \pm 0.8$$

$$\sigma = 42.5 \pm 0.8$$

dE/dx (ADC counts/cm)

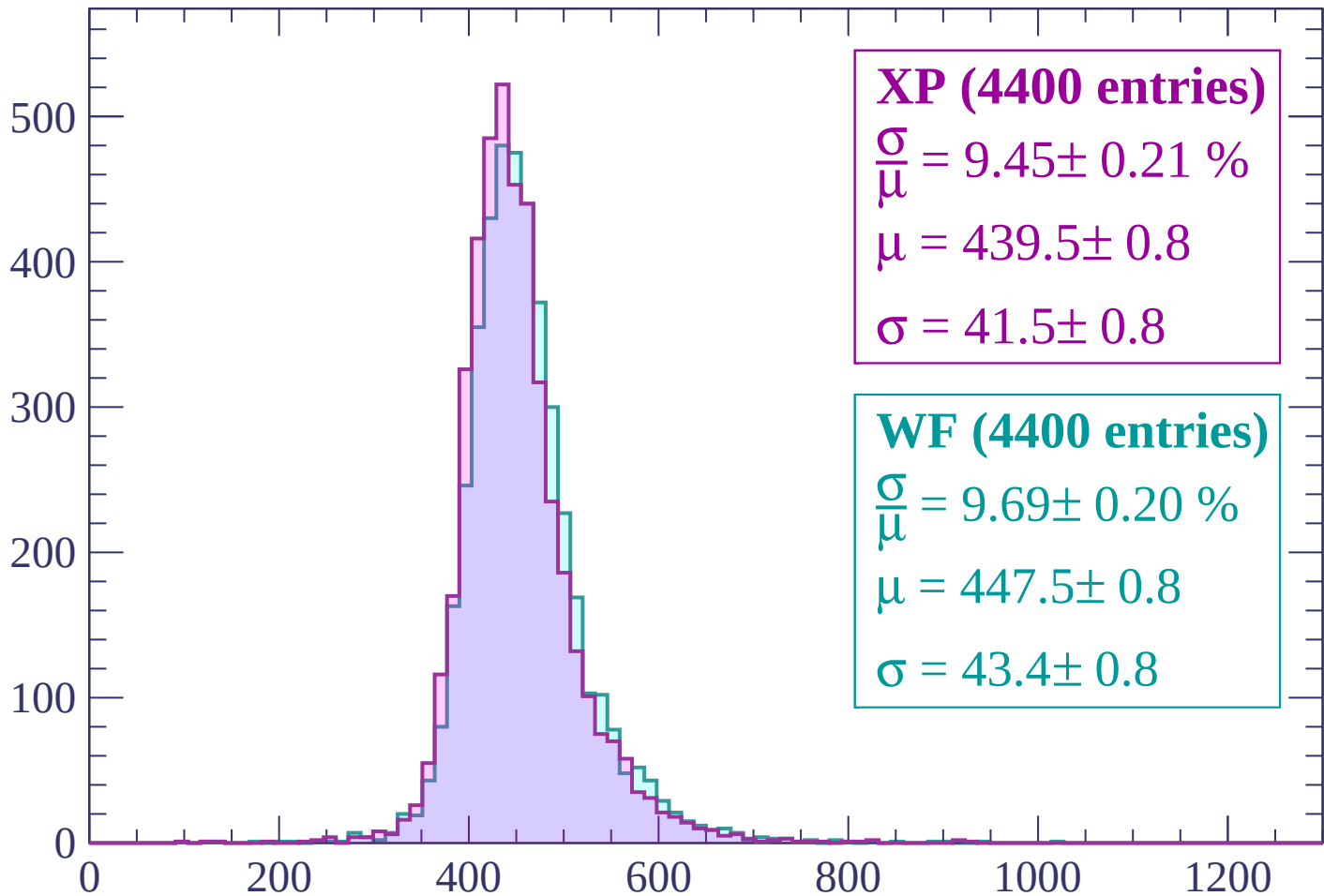
Energy loss | $1200 < p < 1260$

Count



Energy loss | $1260 < p < 1320$

Count



XP (4400 entries)

$$\frac{\sigma}{\mu} = 9.45 \pm 0.21 \%$$

$$\mu = 439.5 \pm 0.8$$

$$\sigma = 41.5 \pm 0.8$$

WF (4400 entries)

$$\frac{\sigma}{\mu} = 9.69 \pm 0.20 \%$$

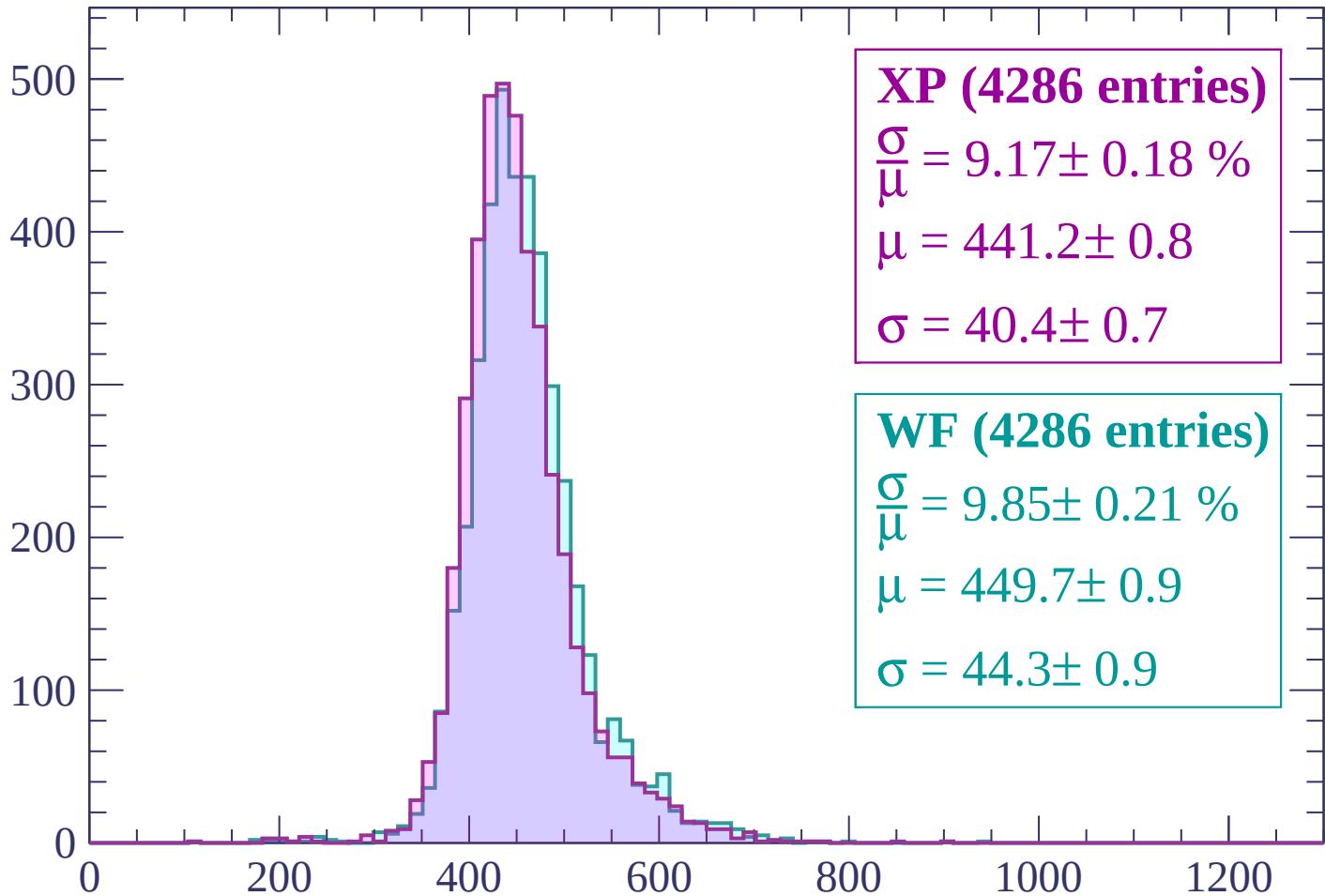
$$\mu = 447.5 \pm 0.8$$

$$\sigma = 43.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count



XP (4286 entries)

$$\frac{\sigma}{\mu} = 9.17 \pm 0.18 \%$$

$$\mu = 441.2 \pm 0.8$$

$$\sigma = 40.4 \pm 0.7$$

WF (4286 entries)

$$\frac{\sigma}{\mu} = 9.85 \pm 0.21 \%$$

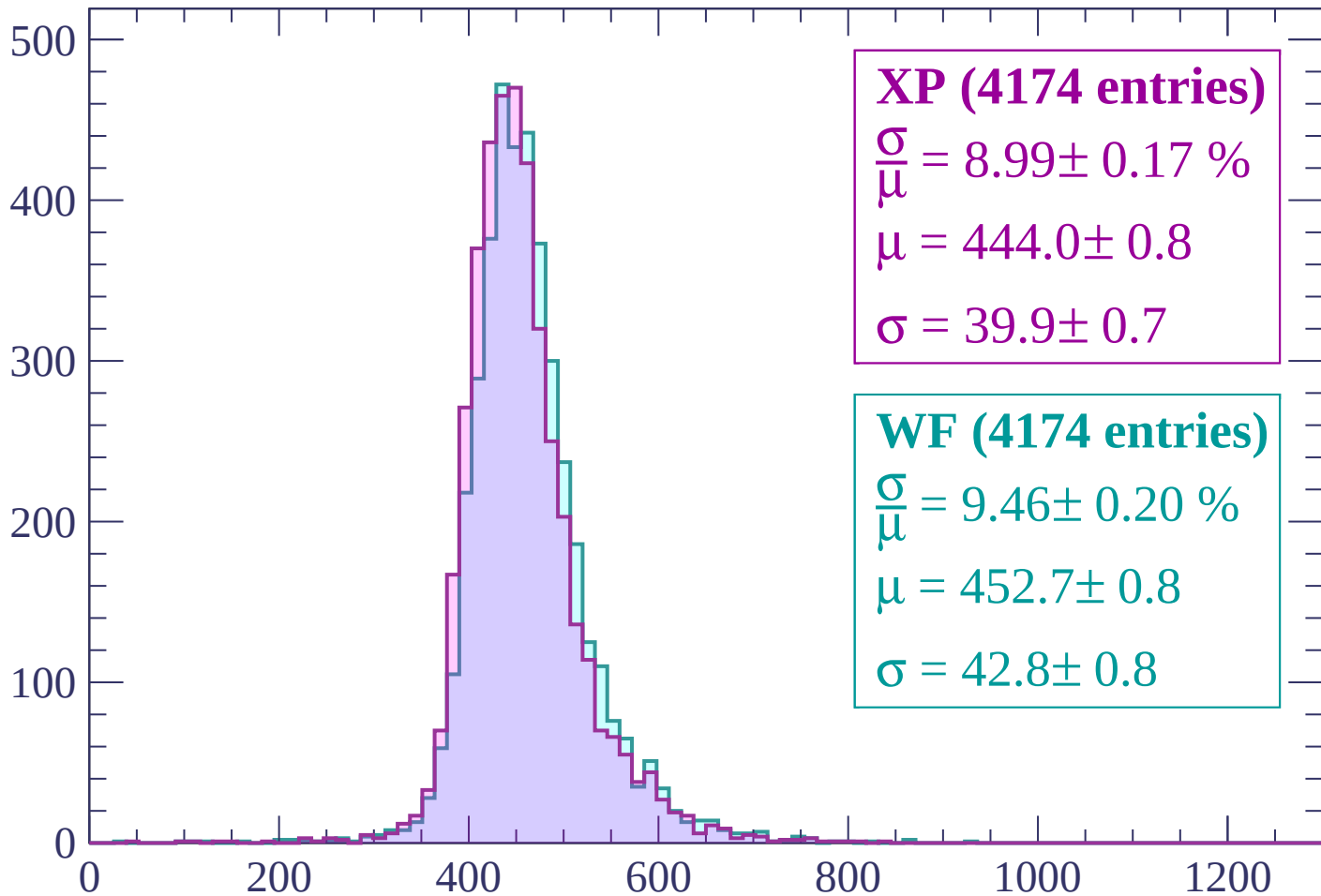
$$\mu = 449.7 \pm 0.9$$

$$\sigma = 44.3 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1380 < p < 1440$

Count



XP (4174 entries)

$$\frac{\sigma}{\mu} = 8.99 \pm 0.17 \%$$

$$\mu = 444.0 \pm 0.8$$

$$\sigma = 39.9 \pm 0.7$$

WF (4174 entries)

$$\frac{\sigma}{\mu} = 9.46 \pm 0.20 \%$$

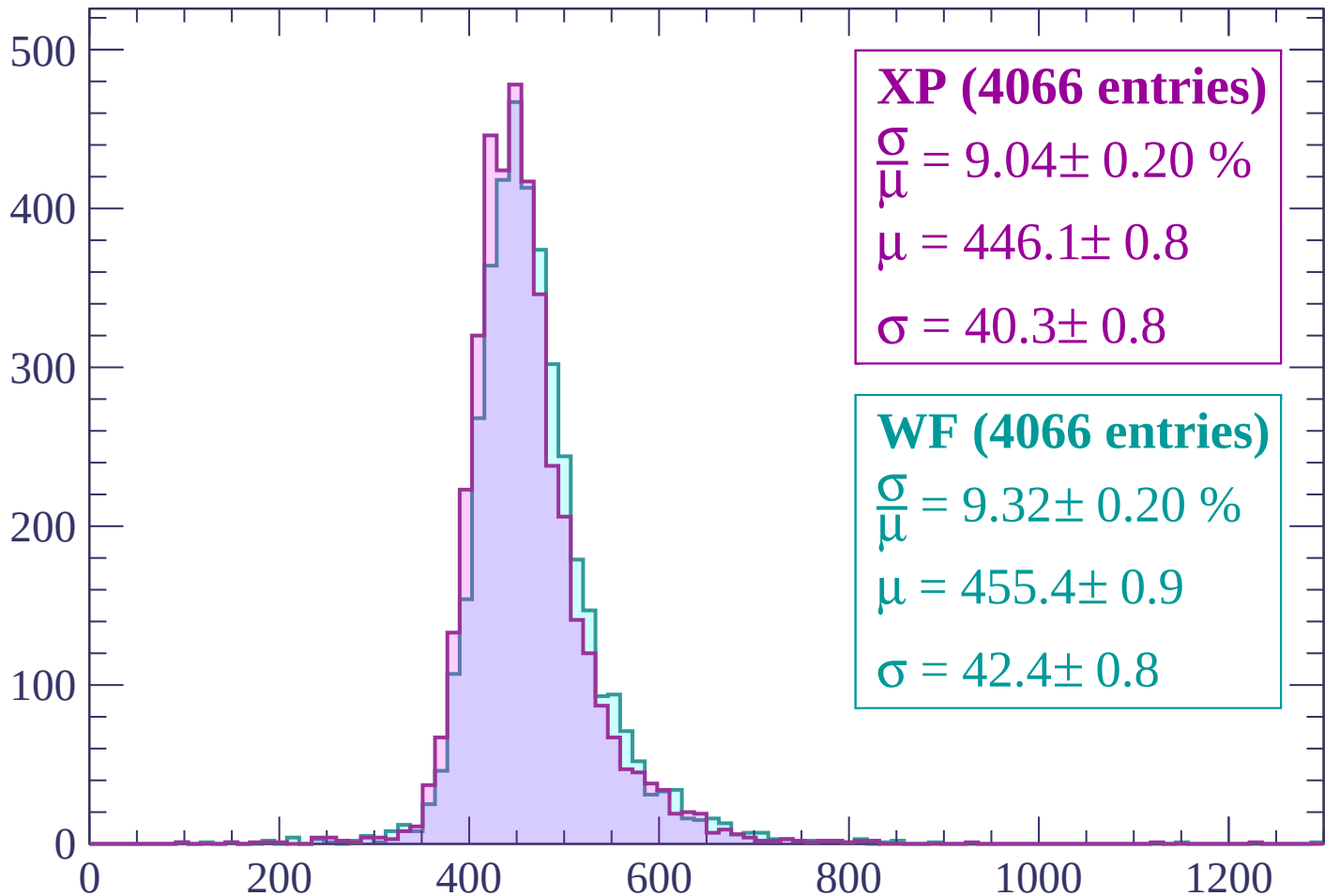
$$\mu = 452.7 \pm 0.8$$

$$\sigma = 42.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1500$

Count



XP (4066 entries)

$\frac{\sigma}{\mu} = 9.04 \pm 0.20 \%$

$\mu = 446.1 \pm 0.8$

$\sigma = 40.3 \pm 0.8$

WF (4066 entries)

$\frac{\sigma}{\mu} = 9.32 \pm 0.20 \%$

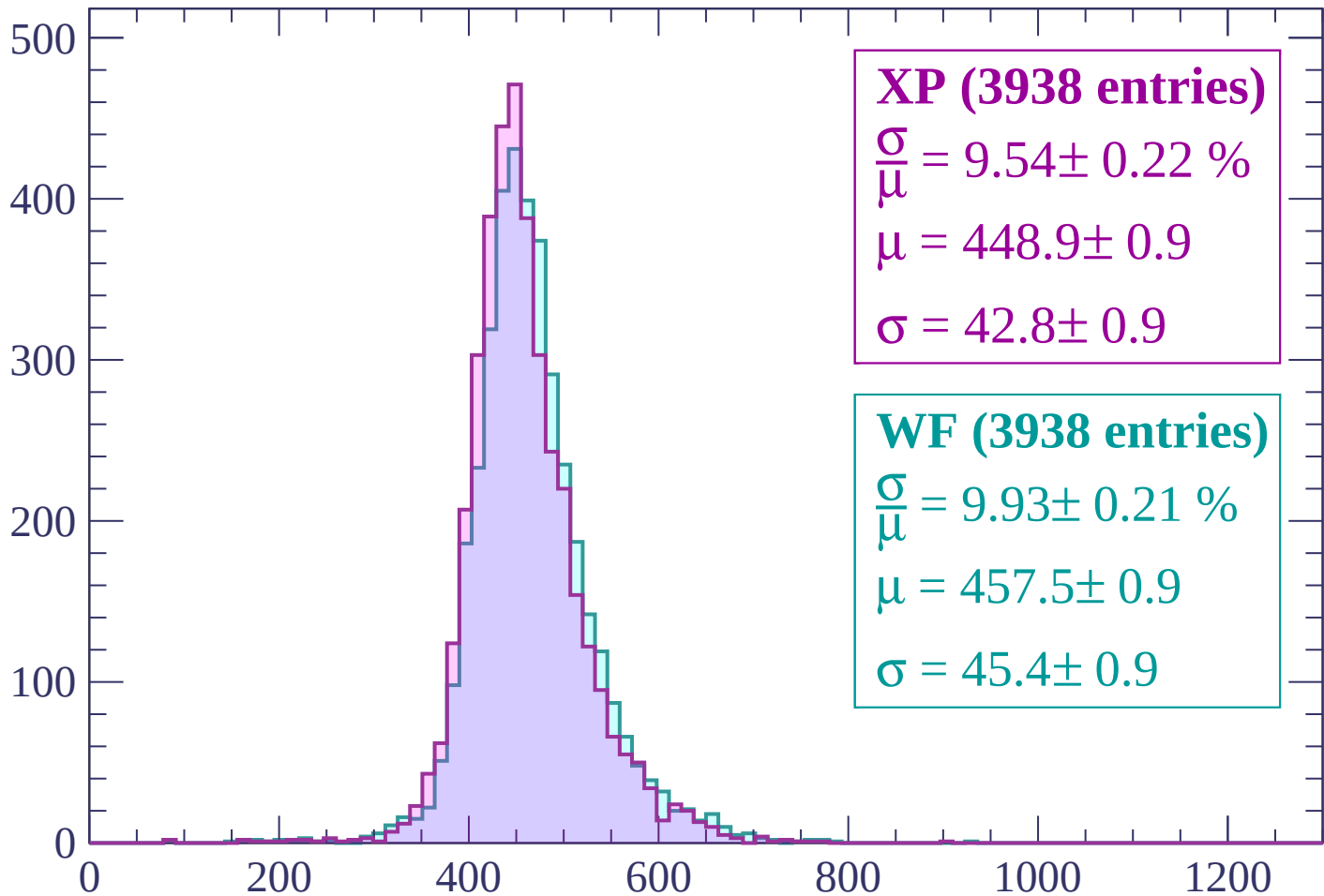
$\mu = 455.4 \pm 0.9$

$\sigma = 42.4 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $1500 < p < 1560$

Count



XP (3938 entries)

$\frac{\sigma}{\mu} = 9.54 \pm 0.22 \%$

$\mu = 448.9 \pm 0.9$

$\sigma = 42.8 \pm 0.9$

WF (3938 entries)

$\frac{\sigma}{\mu} = 9.93 \pm 0.21 \%$

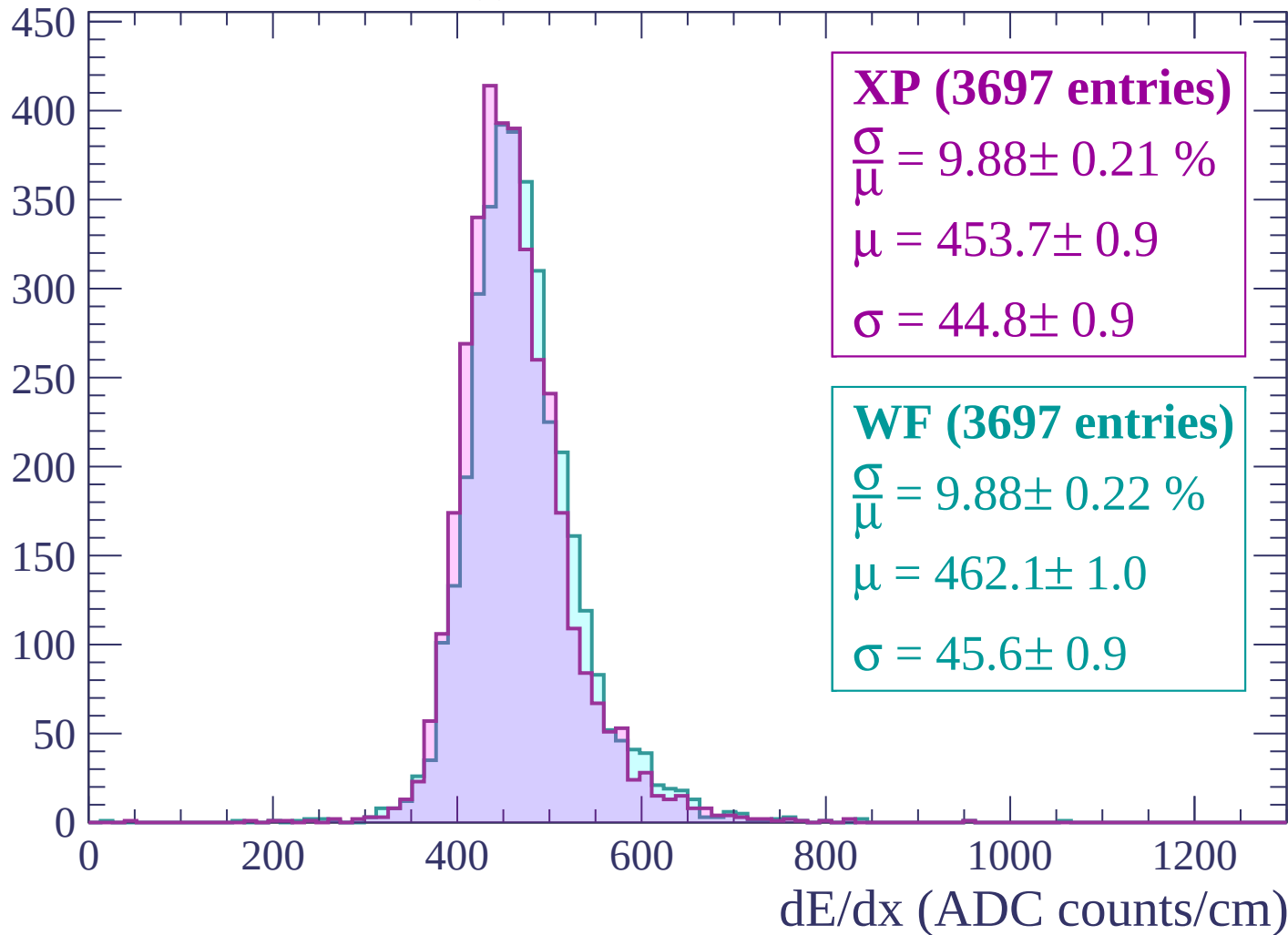
$\mu = 457.5 \pm 0.9$

$\sigma = 45.4 \pm 0.9$

dE/dx (ADC counts/cm)

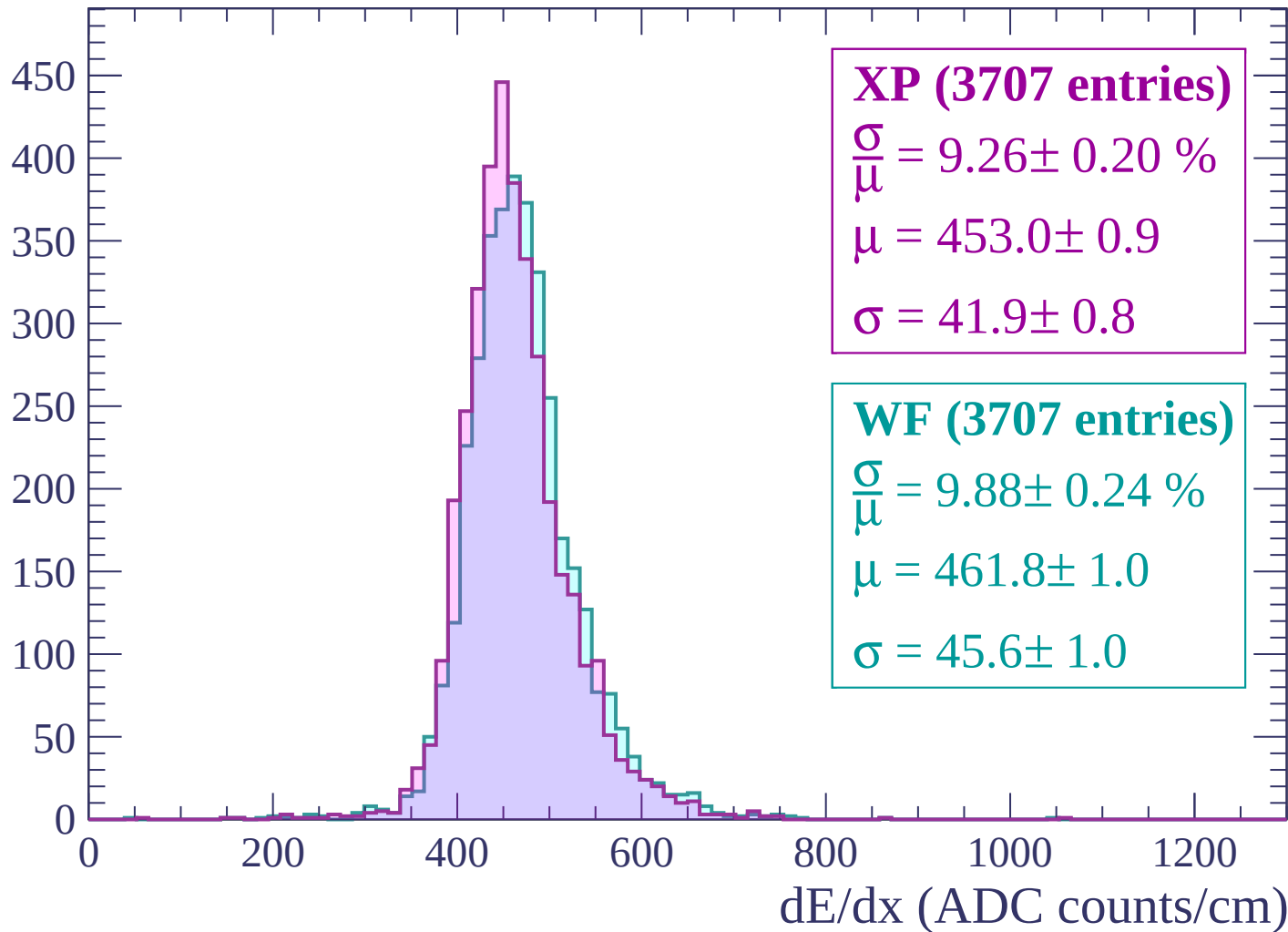
Energy loss | $1560 < p < 1620$

Count



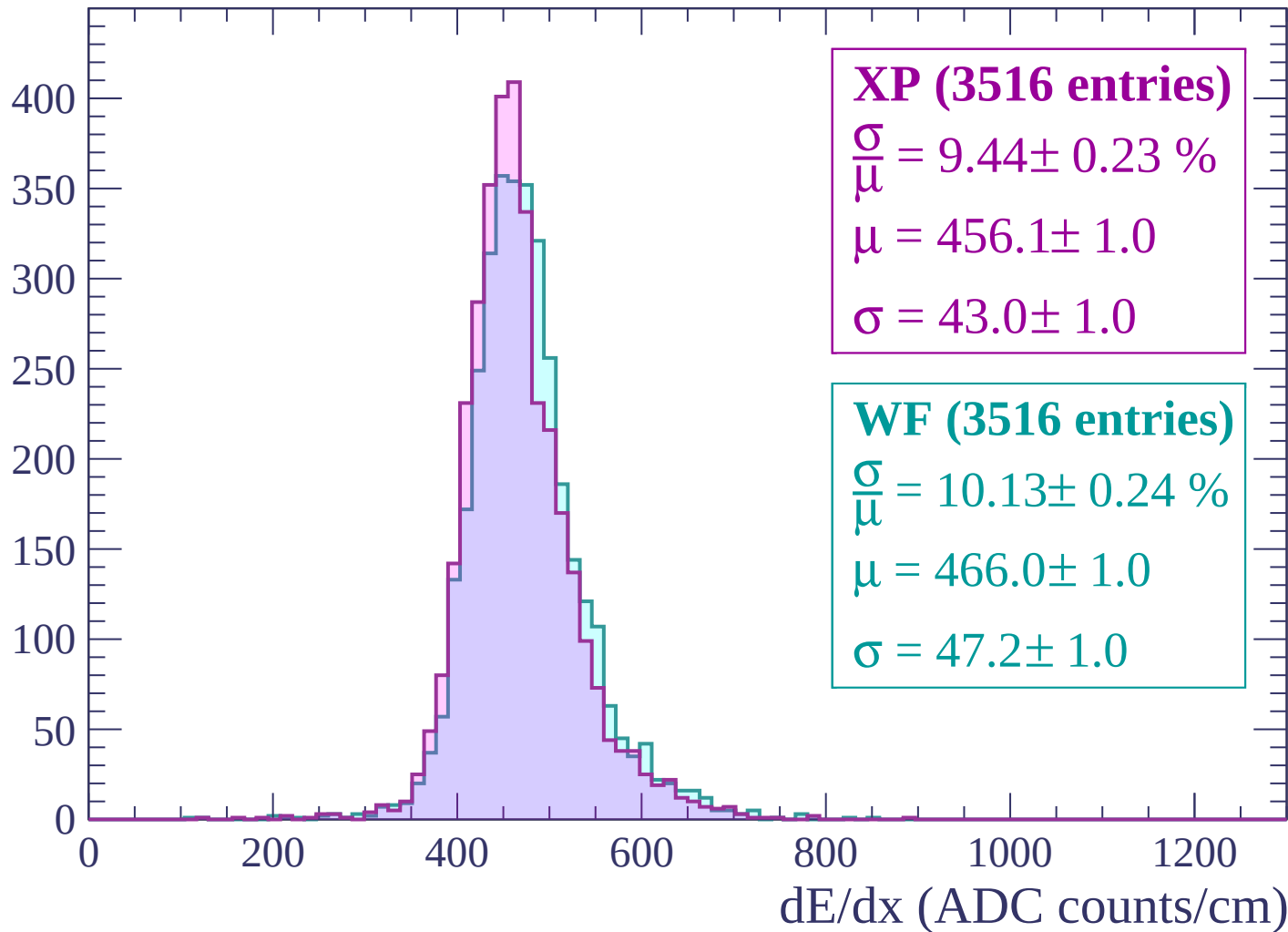
Energy loss | $1620 < p < 1680$

Count



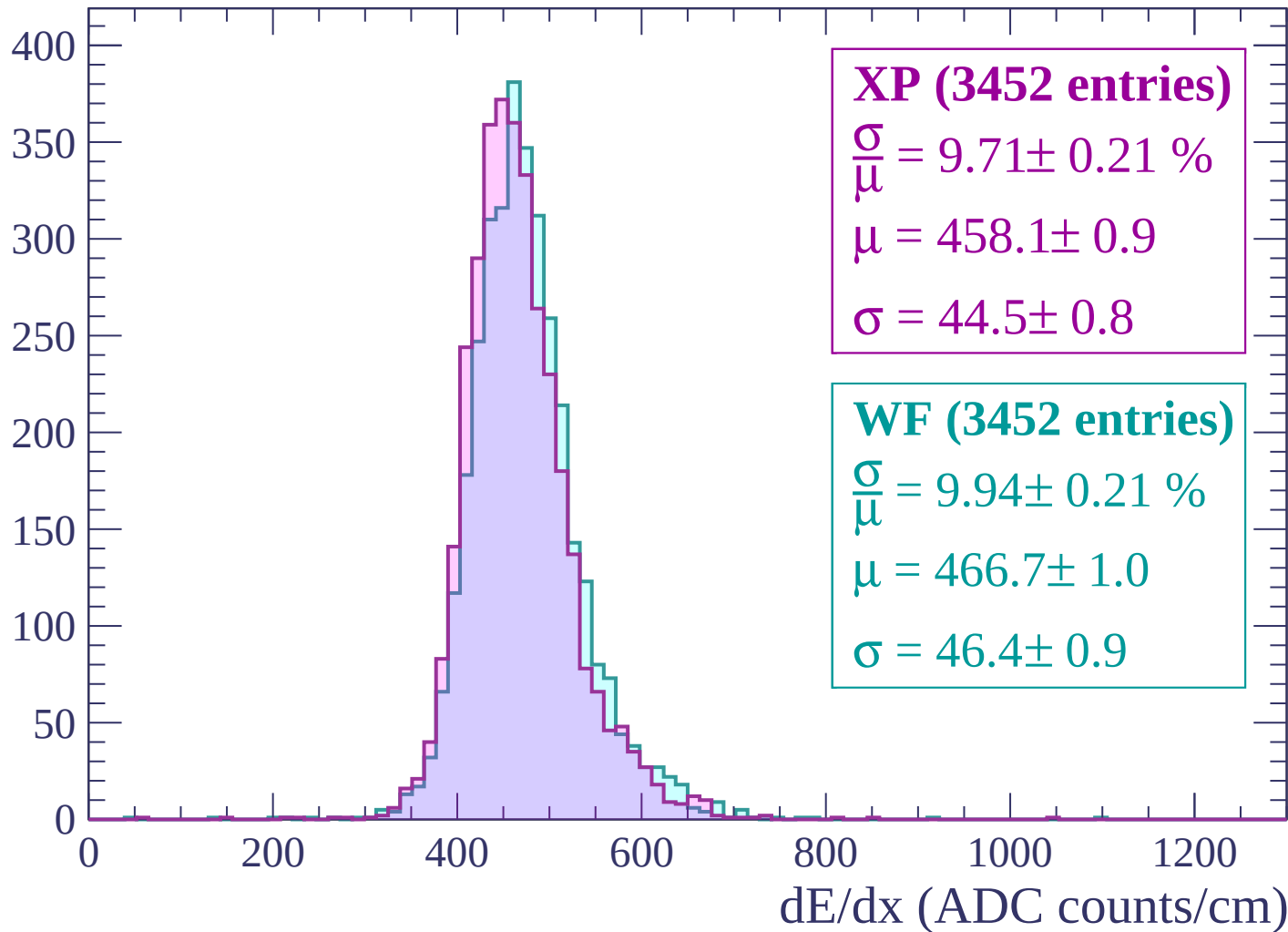
Energy loss | $1680 < p < 1740$

Count



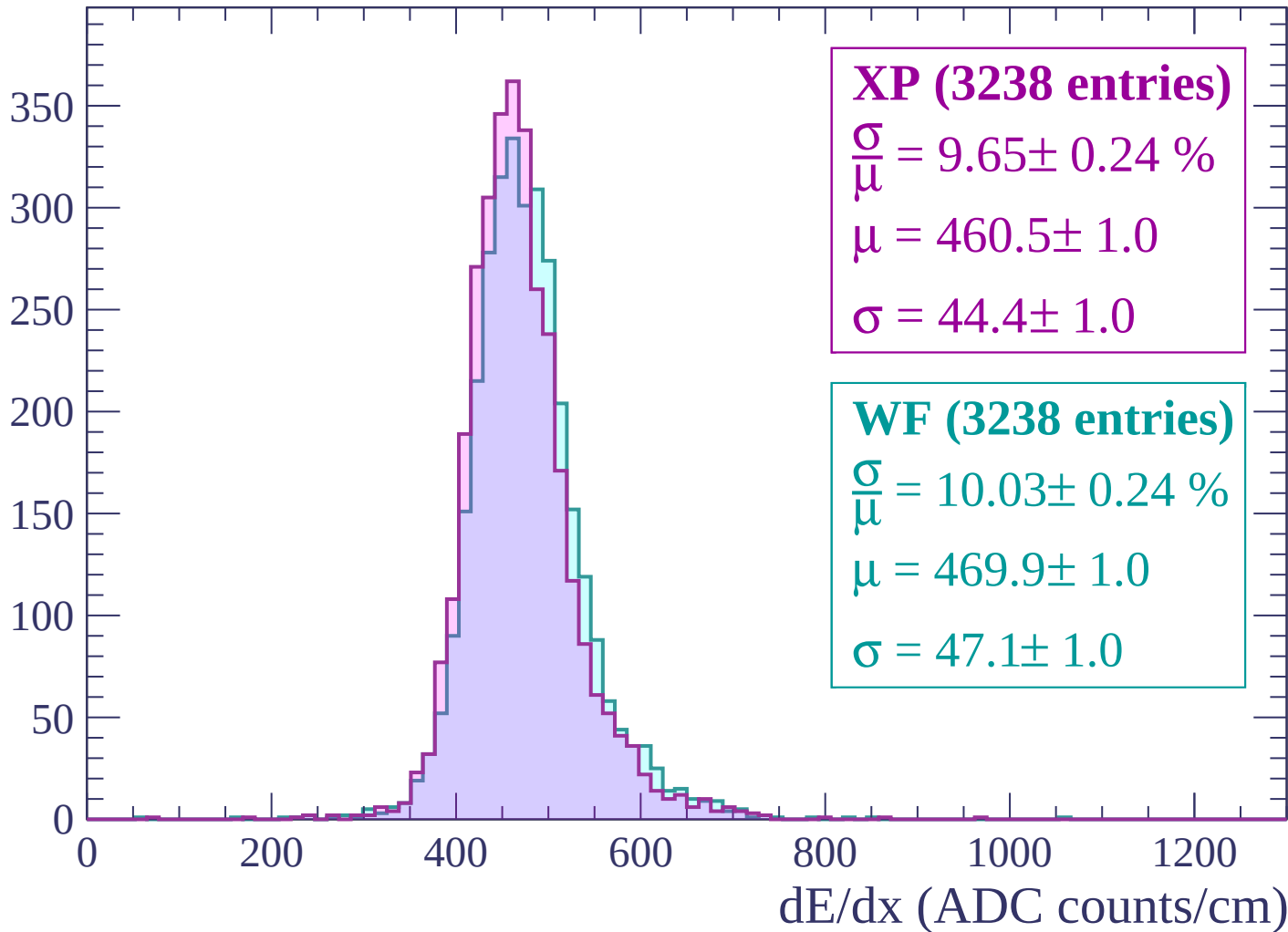
Energy loss | $1740 < p < 1800$

Count



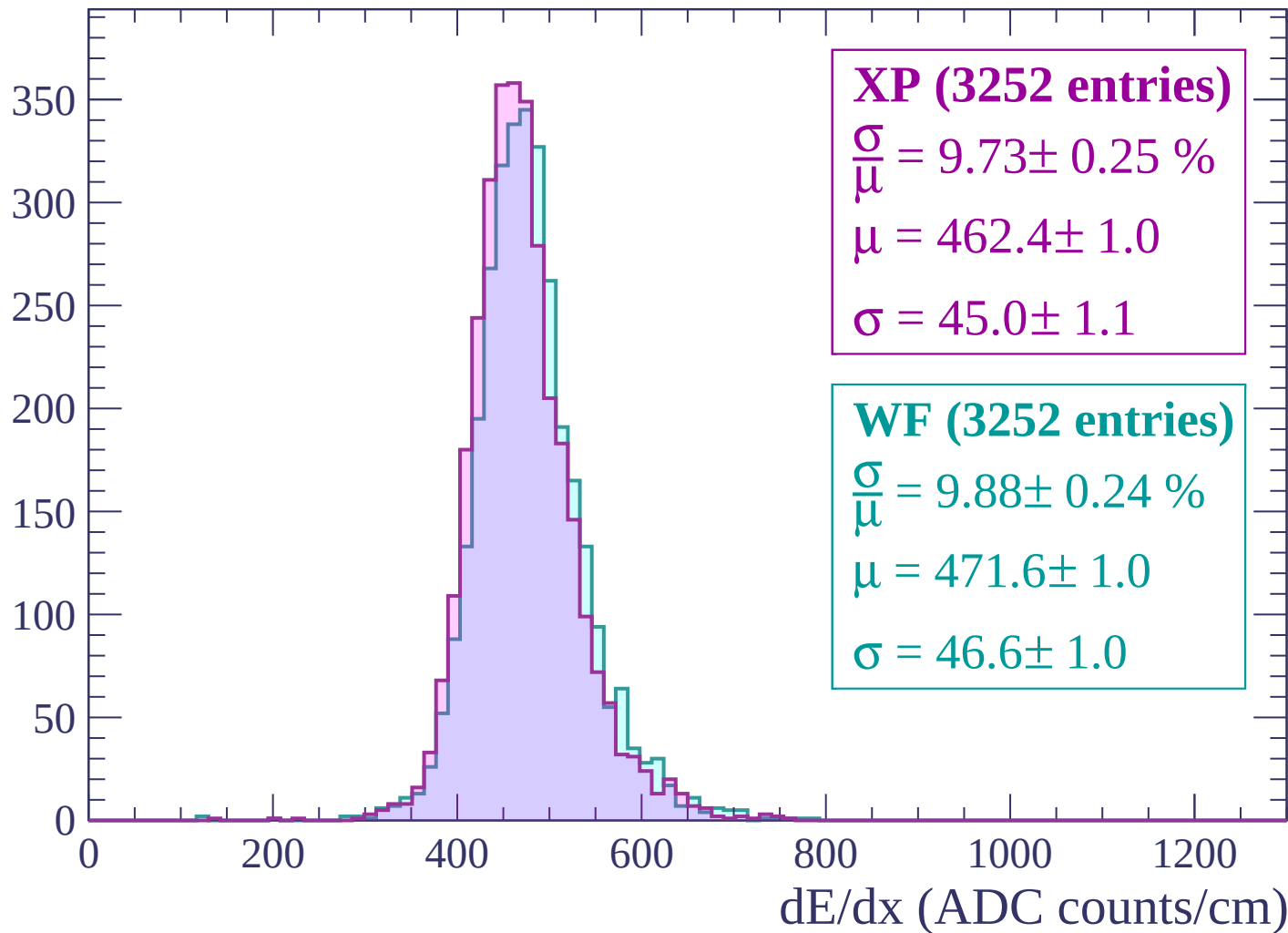
Energy loss | $1800 < p < 1860$

Count



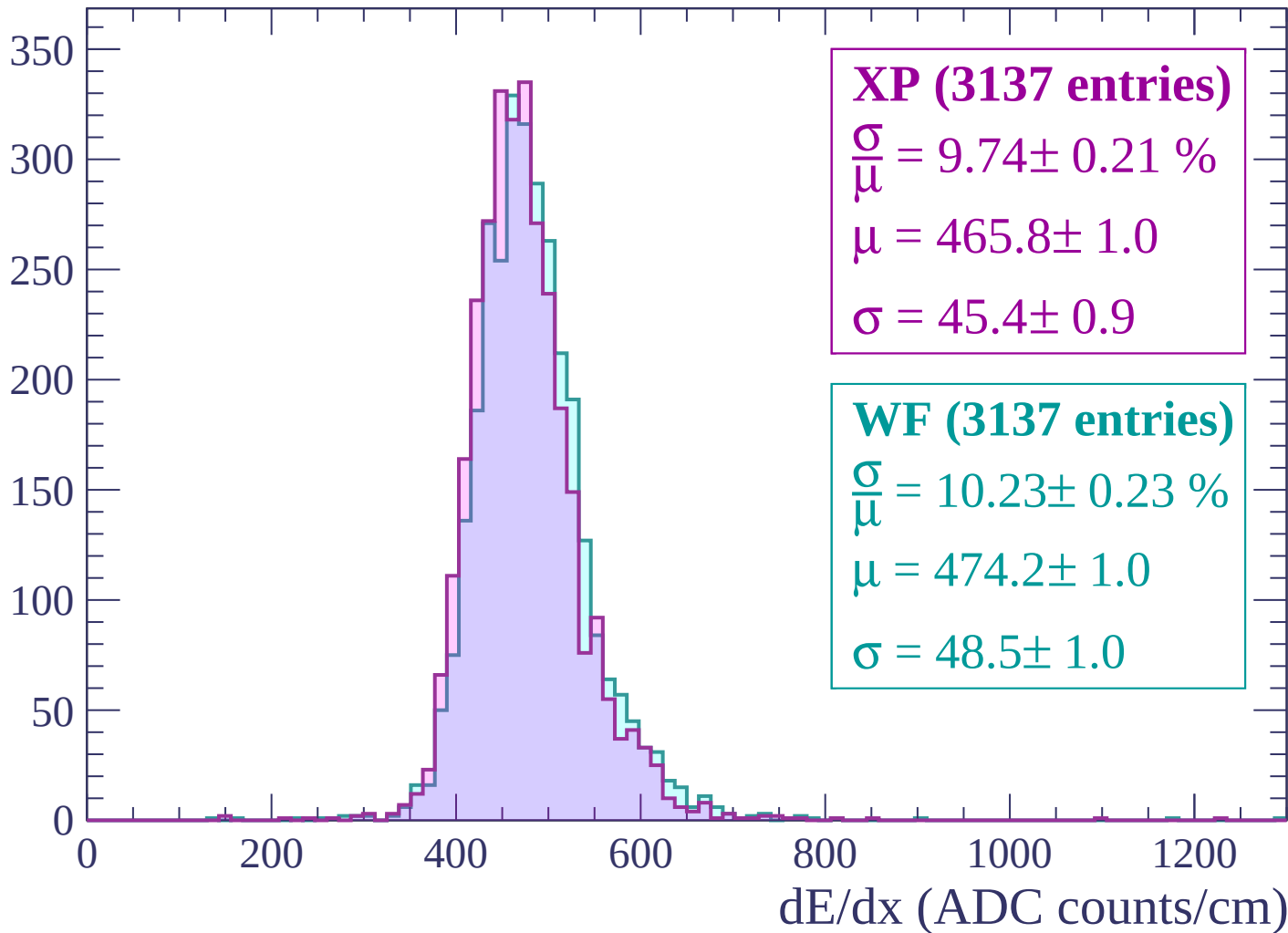
Energy loss | $1860 < p < 1920$

Count



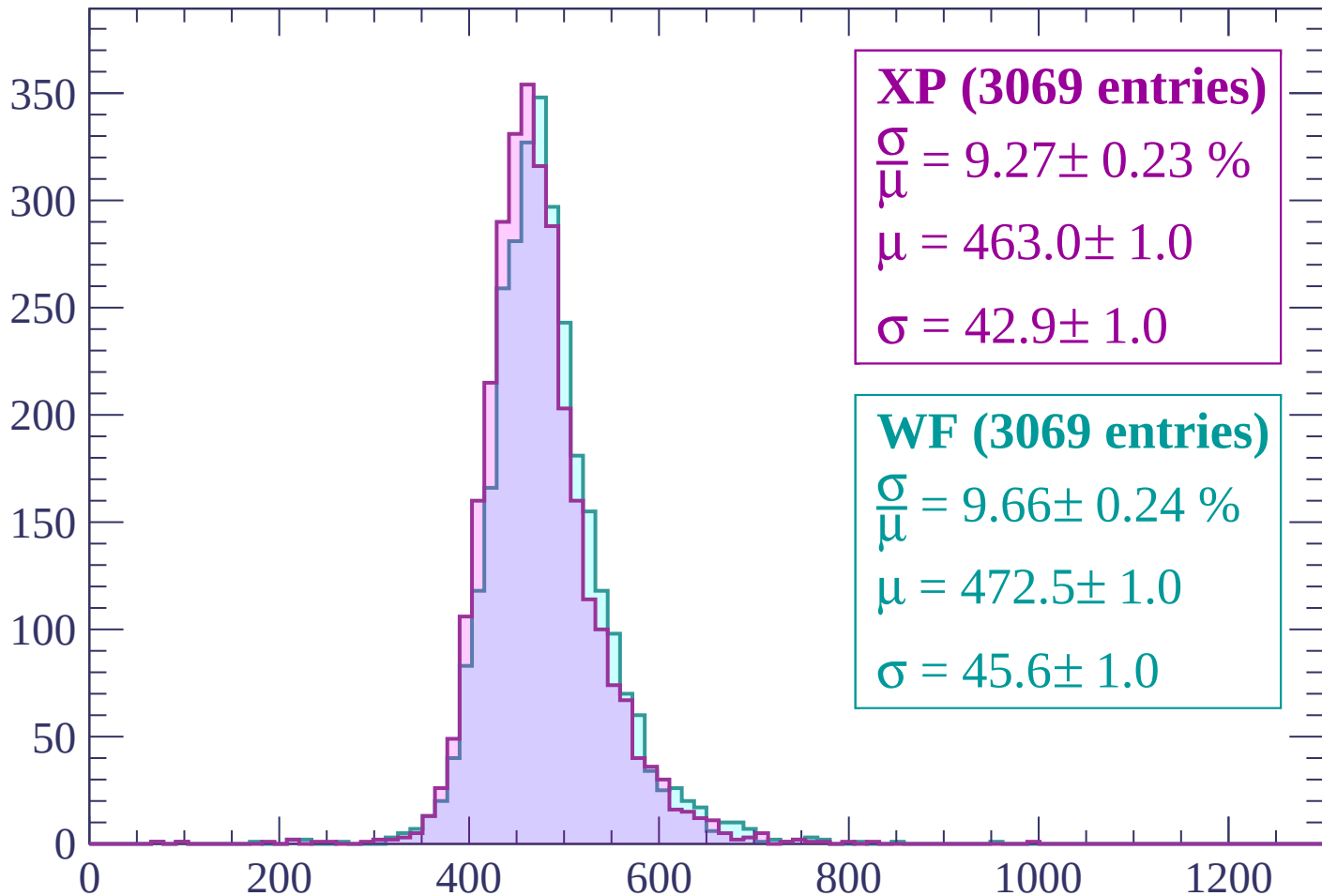
Energy loss | $1920 < p < 1980$

Count



Energy loss | $1980 < p < 2040$

Count



XP (3069 entries)

$$\frac{\sigma}{\mu} = 9.27 \pm 0.23 \%$$

$$\mu = 463.0 \pm 1.0$$

$$\sigma = 42.9 \pm 1.0$$

WF (3069 entries)

$$\frac{\sigma}{\mu} = 9.66 \pm 0.24 \%$$

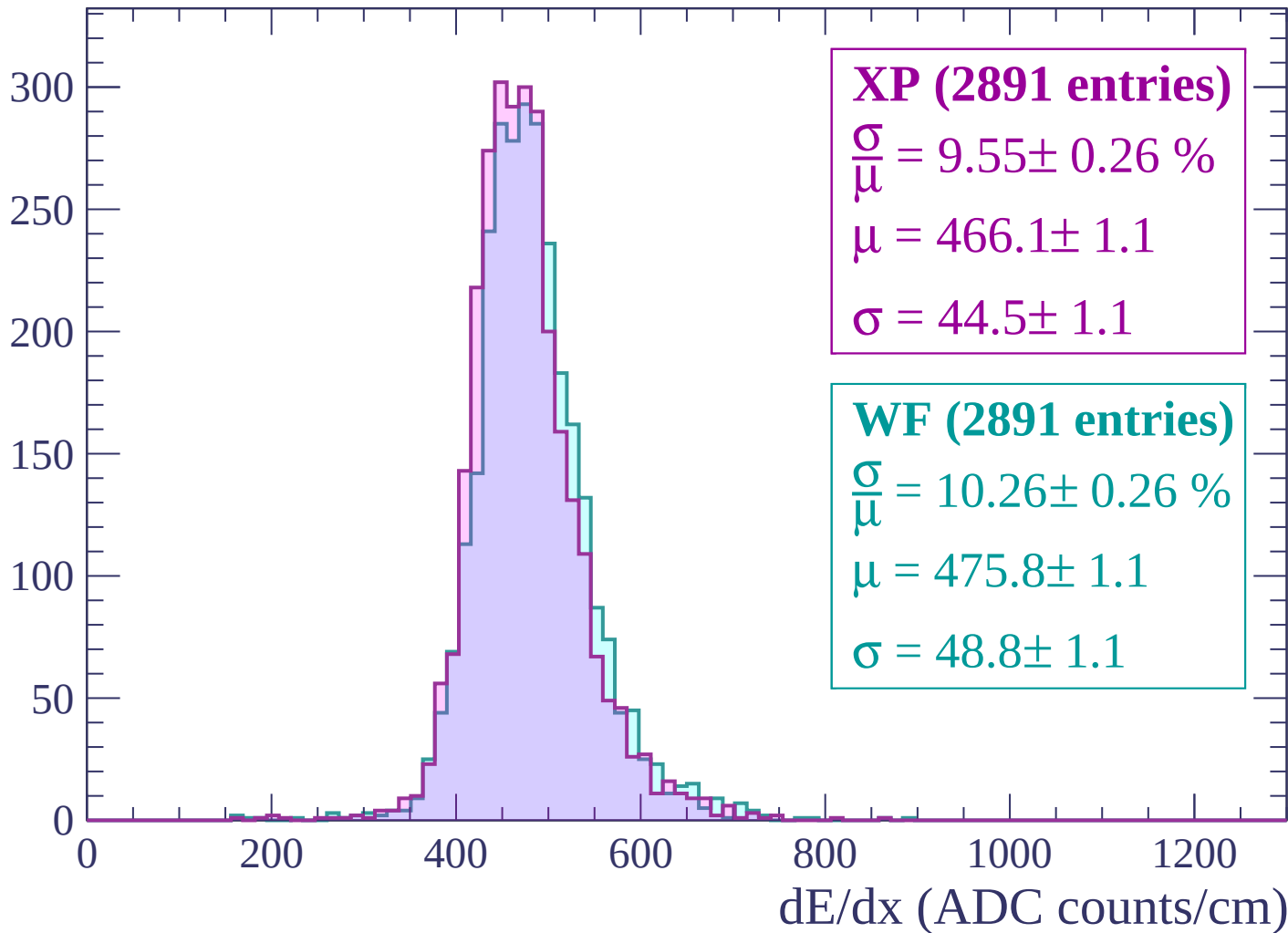
$$\mu = 472.5 \pm 1.0$$

$$\sigma = 45.6 \pm 1.0$$

dE/dx (ADC counts/cm)

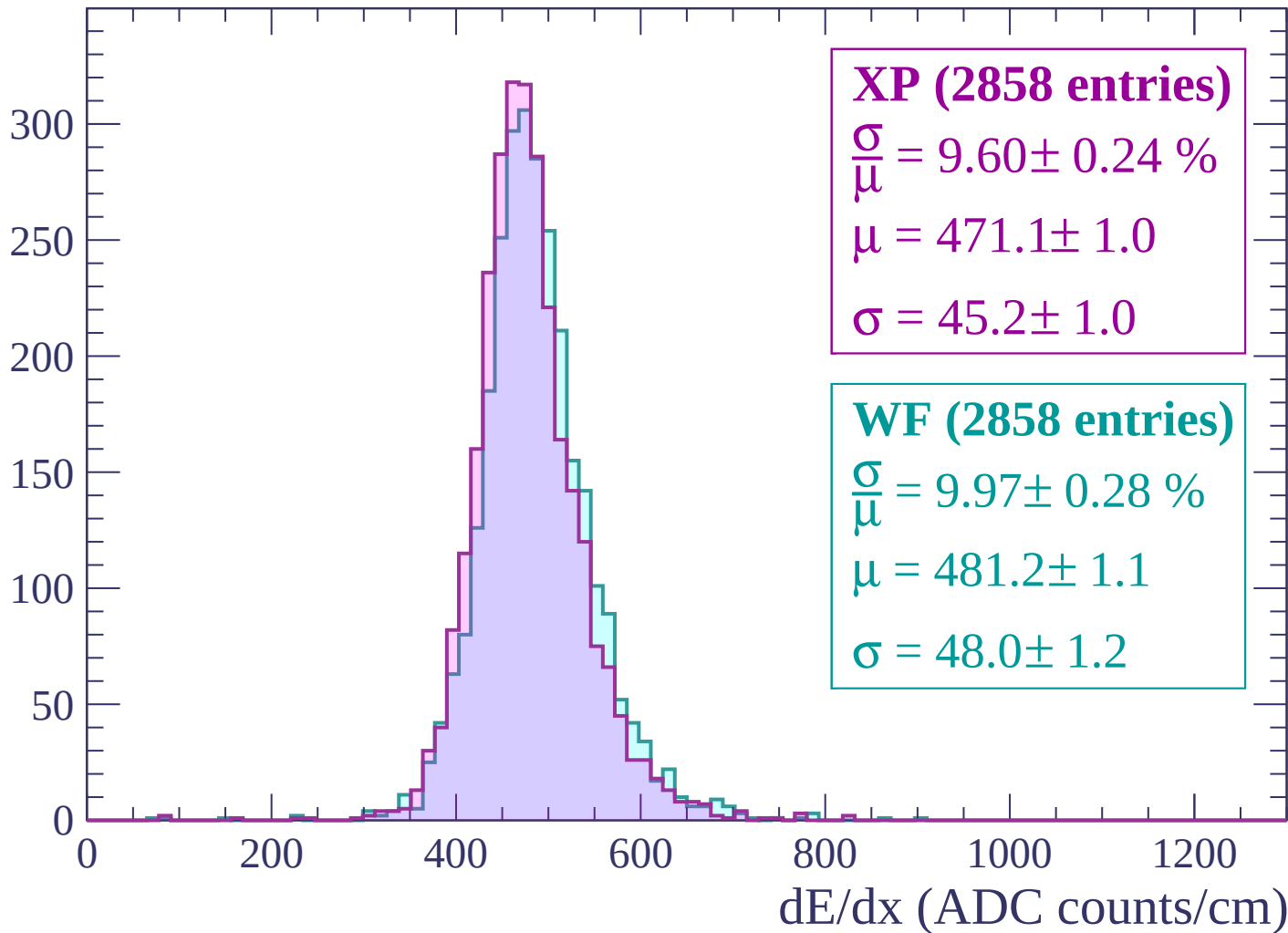
Energy loss | $2040 < p < 2100$

Count



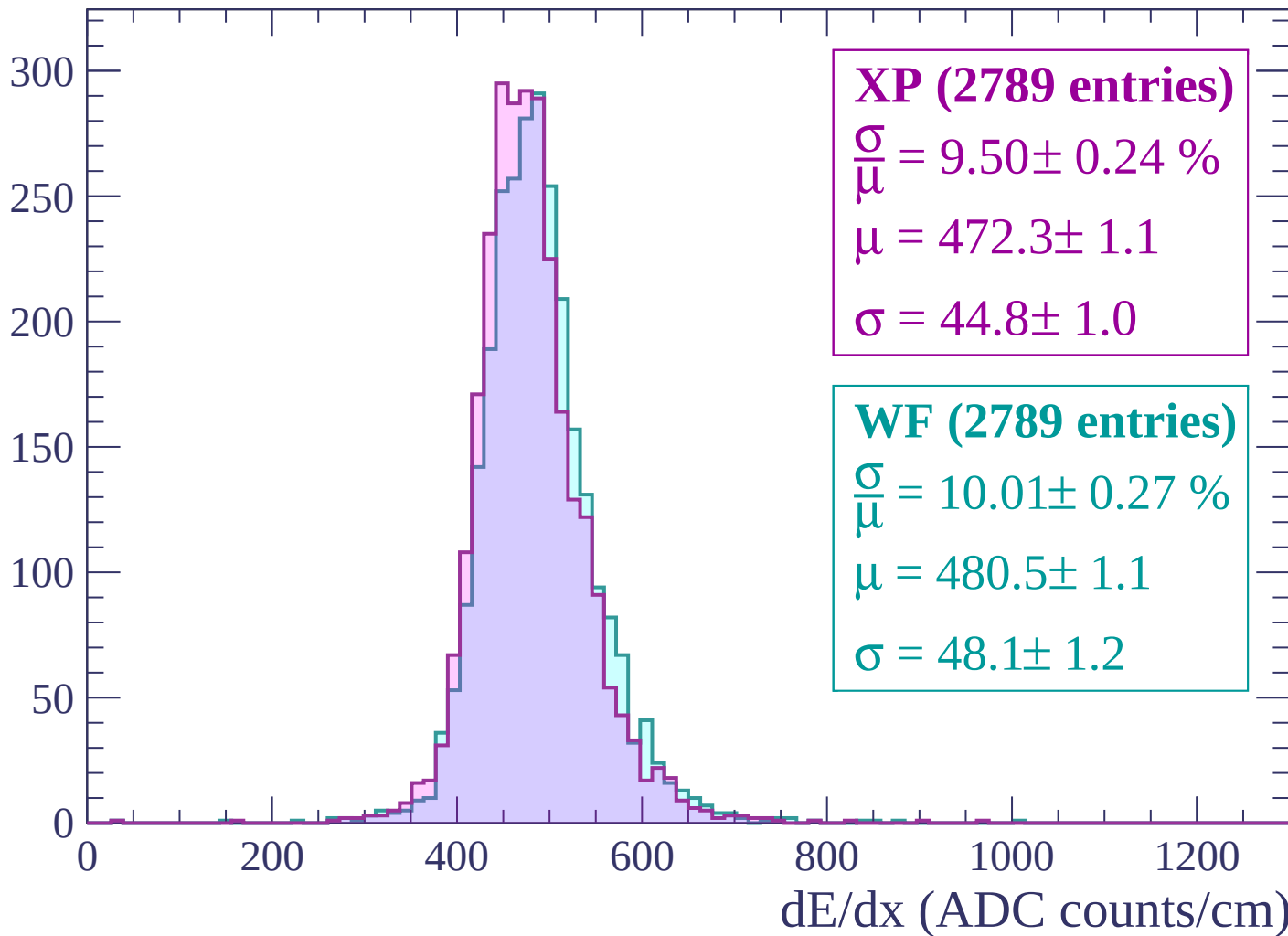
Energy loss | $2100 < p < 2160$

Count



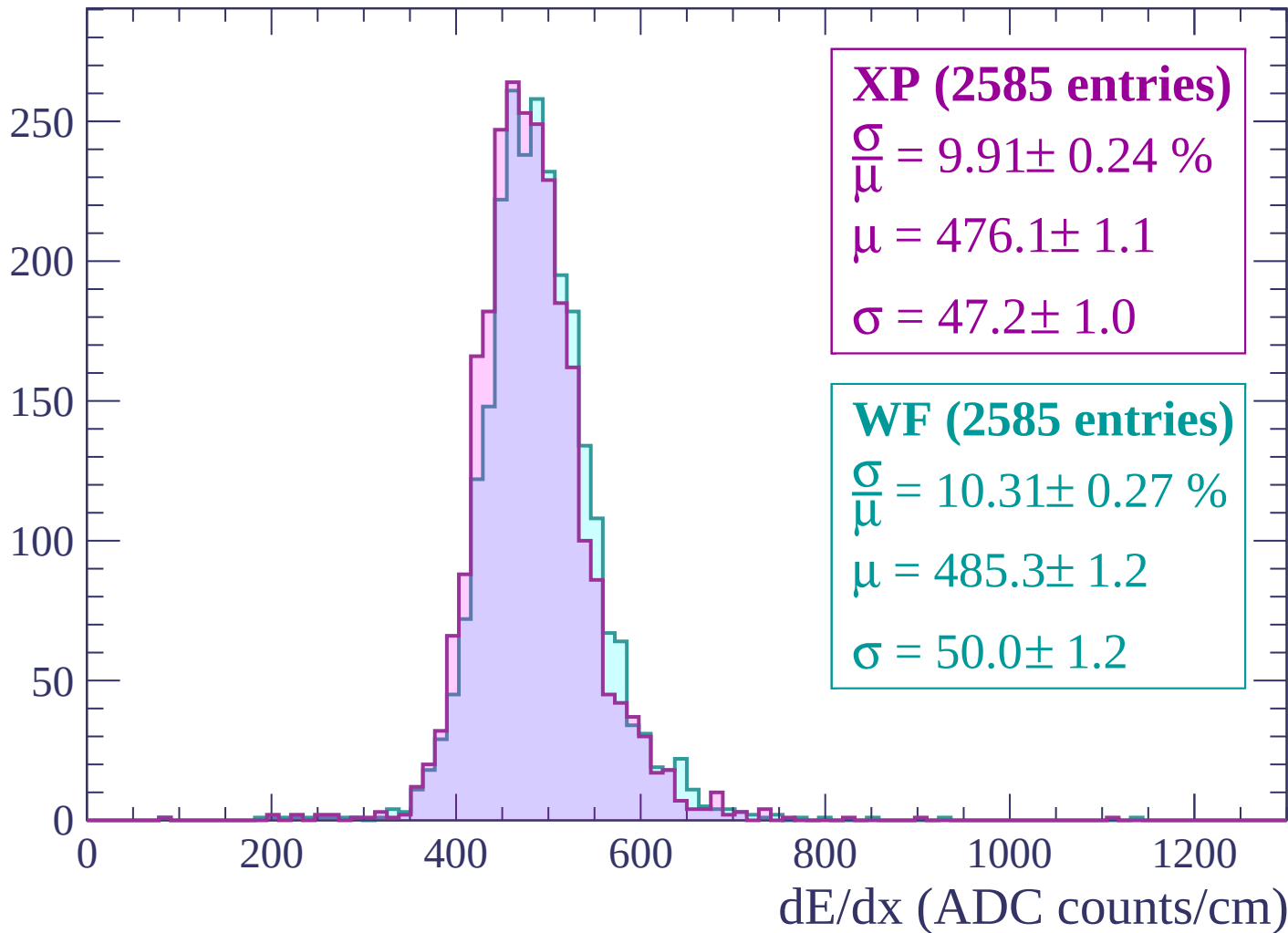
Energy loss | $2160 < p < 2220$

Count



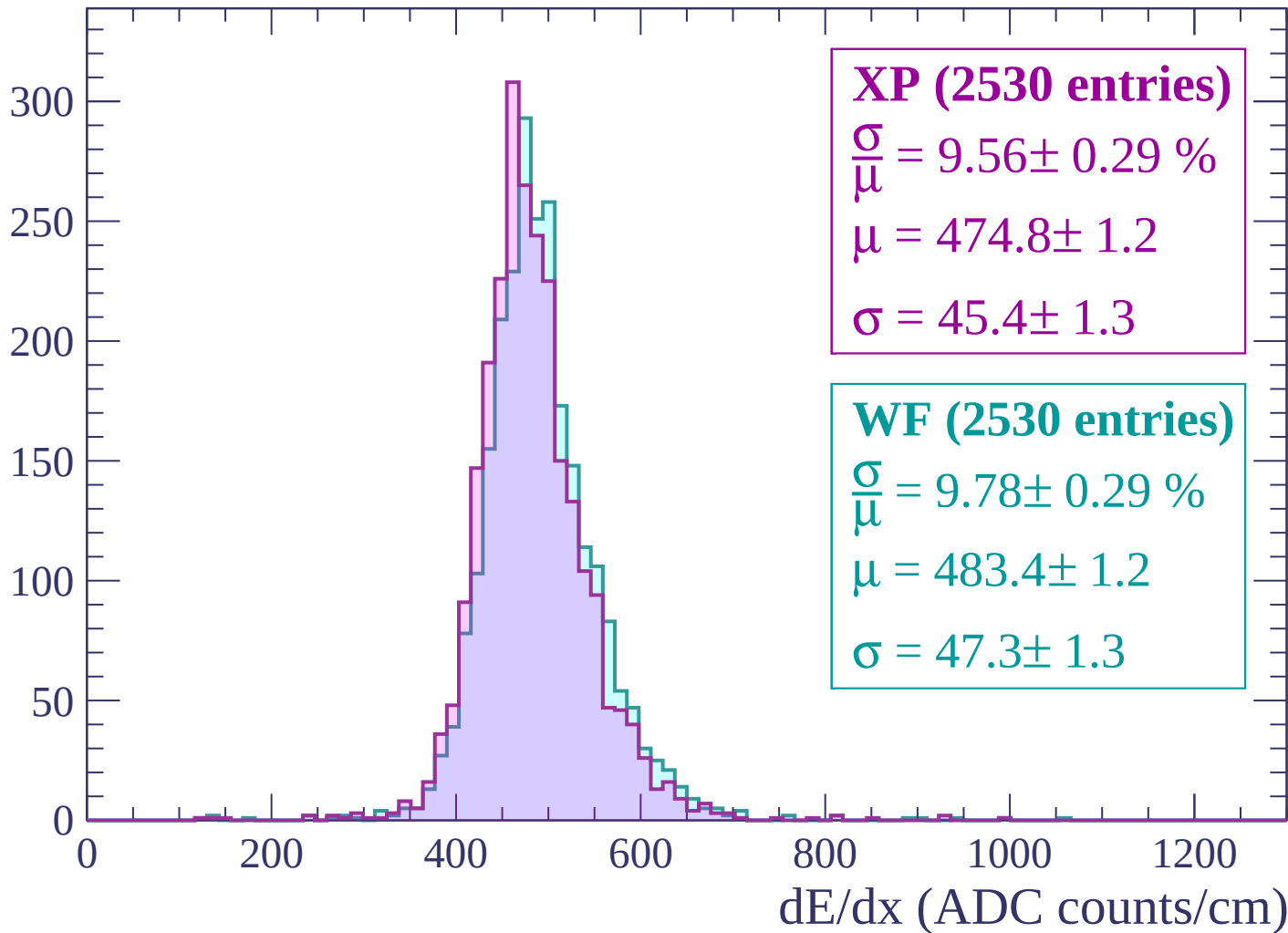
Energy loss | $2220 < p < 2280$

Count



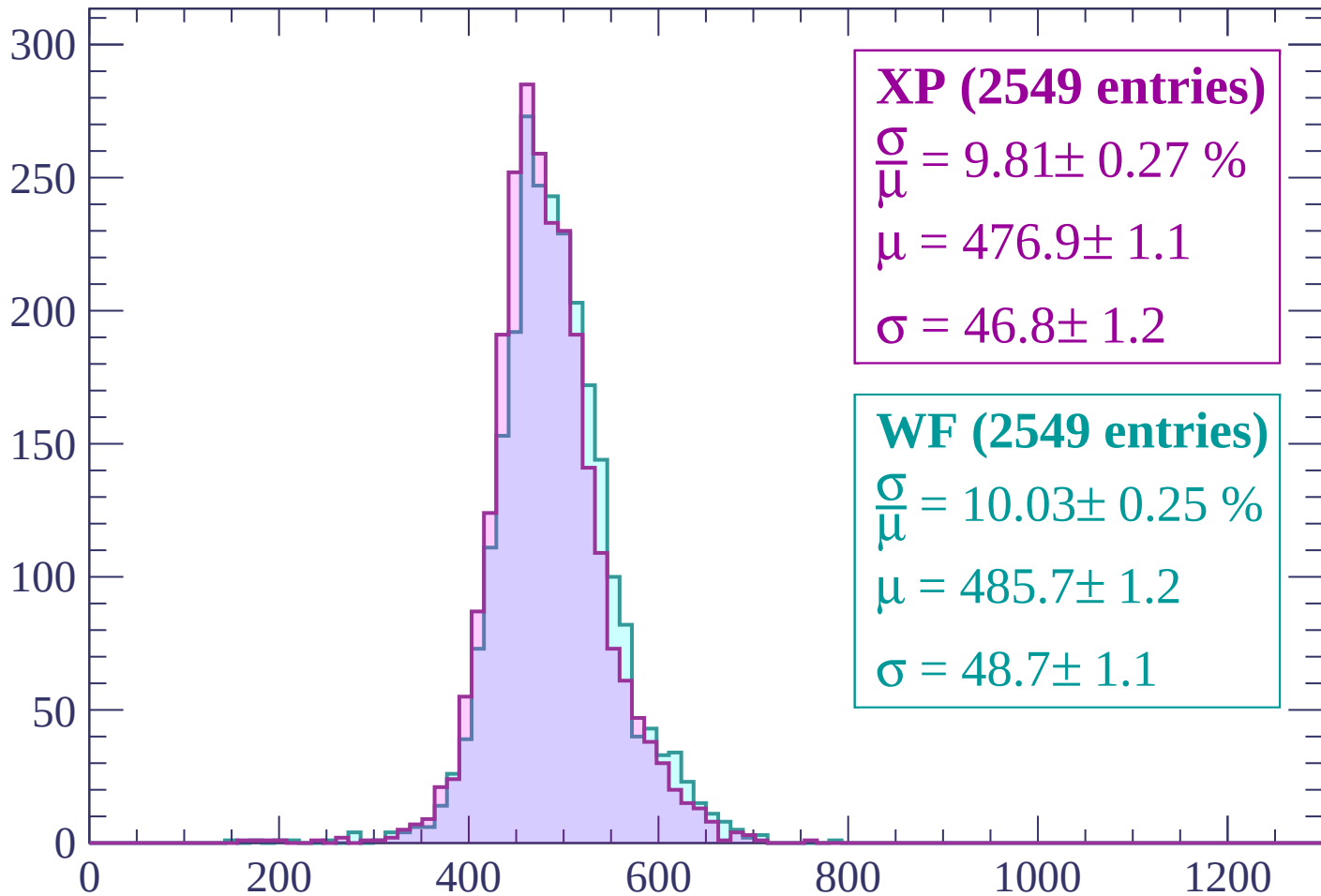
Energy loss | $2280 < p < 2340$

Count



Energy loss | $2340 < p < 2400$

Count



XP (2549 entries)

$$\frac{\sigma}{\mu} = 9.81 \pm 0.27 \%$$

$$\mu = 476.9 \pm 1.1$$

$$\sigma = 46.8 \pm 1.2$$

WF (2549 entries)

$$\frac{\sigma}{\mu} = 10.03 \pm 0.25 \%$$

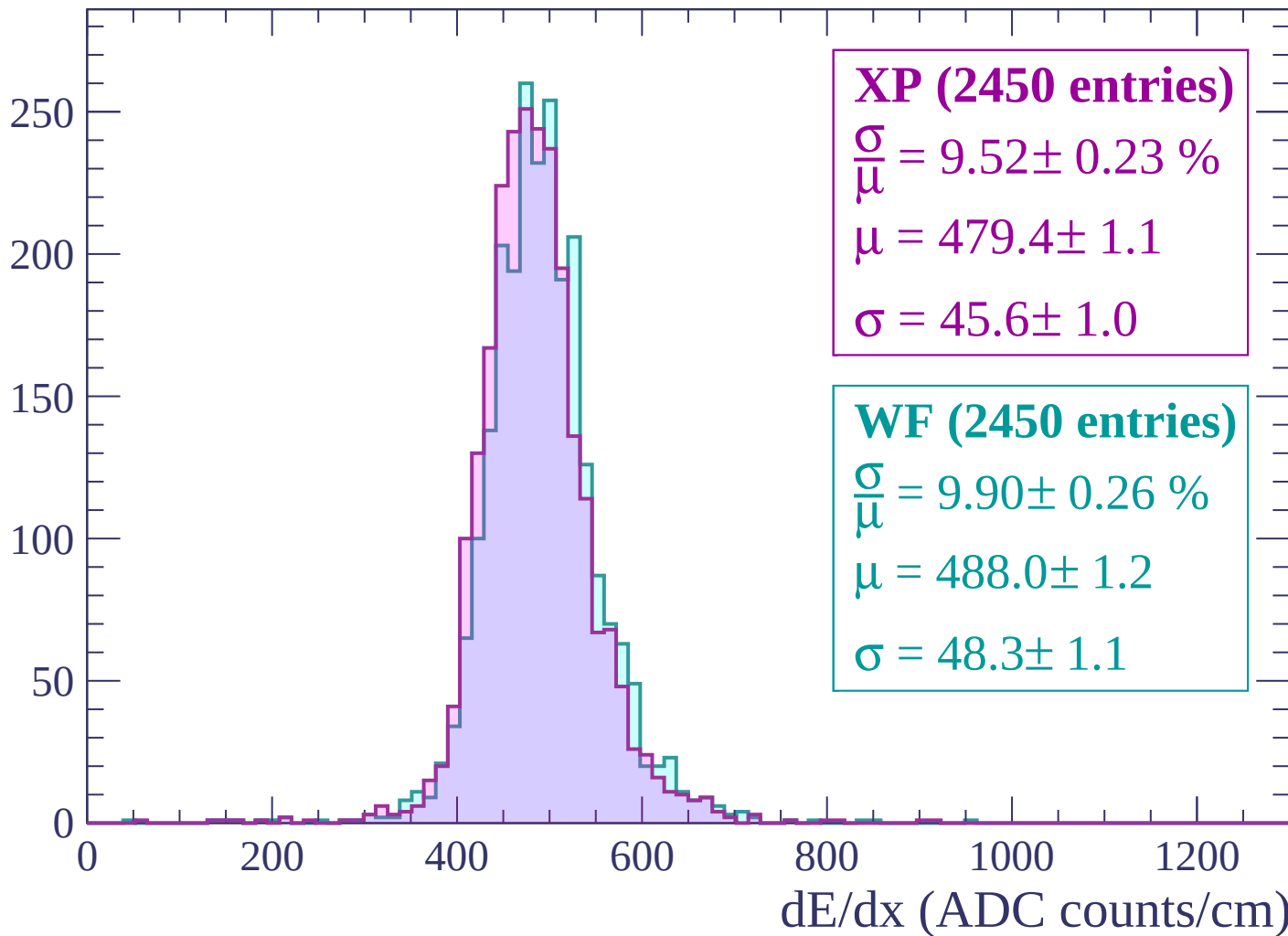
$$\mu = 485.7 \pm 1.2$$

$$\sigma = 48.7 \pm 1.1$$

dE/dx (ADC counts/cm)

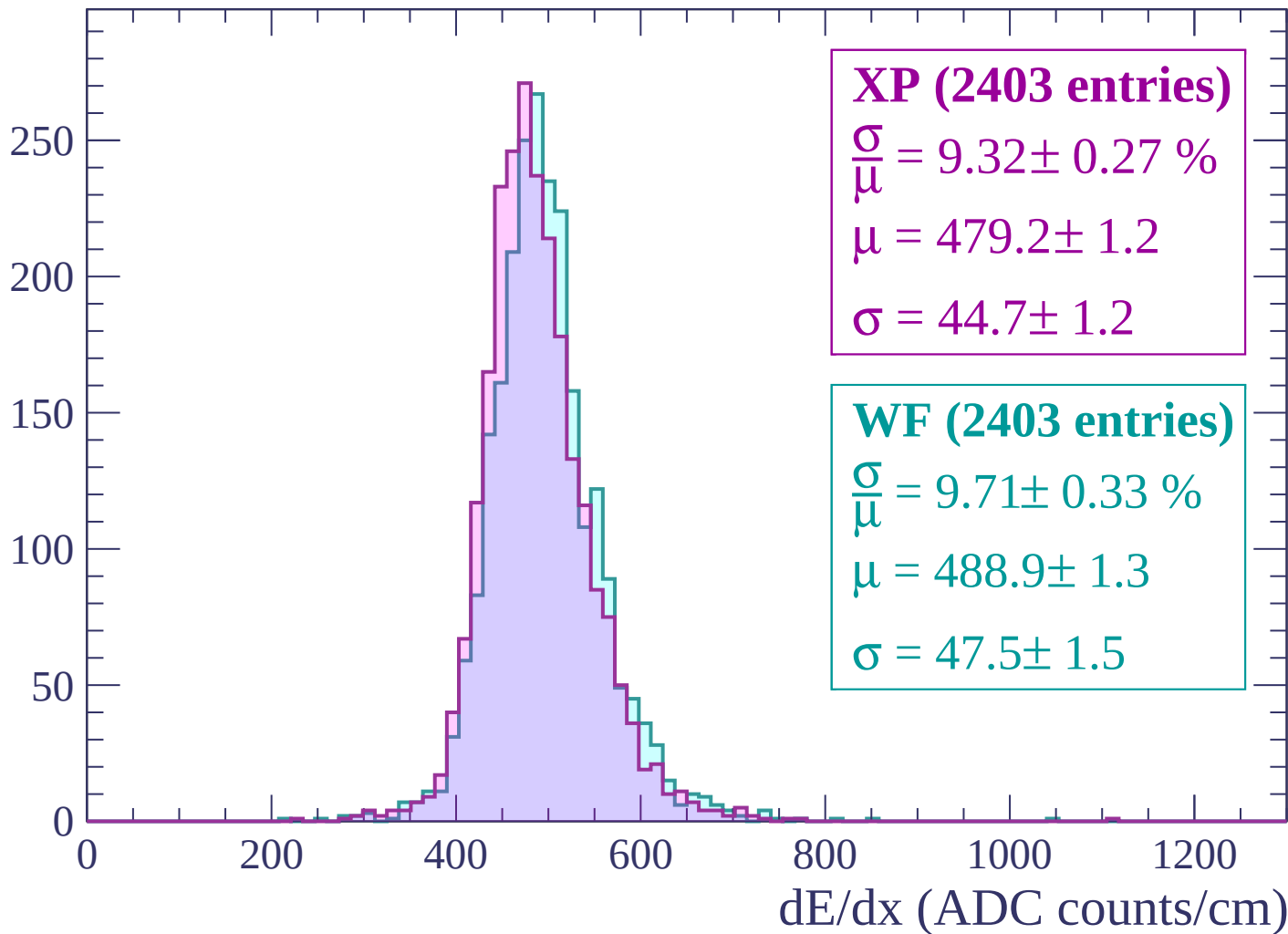
Energy loss | $2400 < p < 2460$

Count



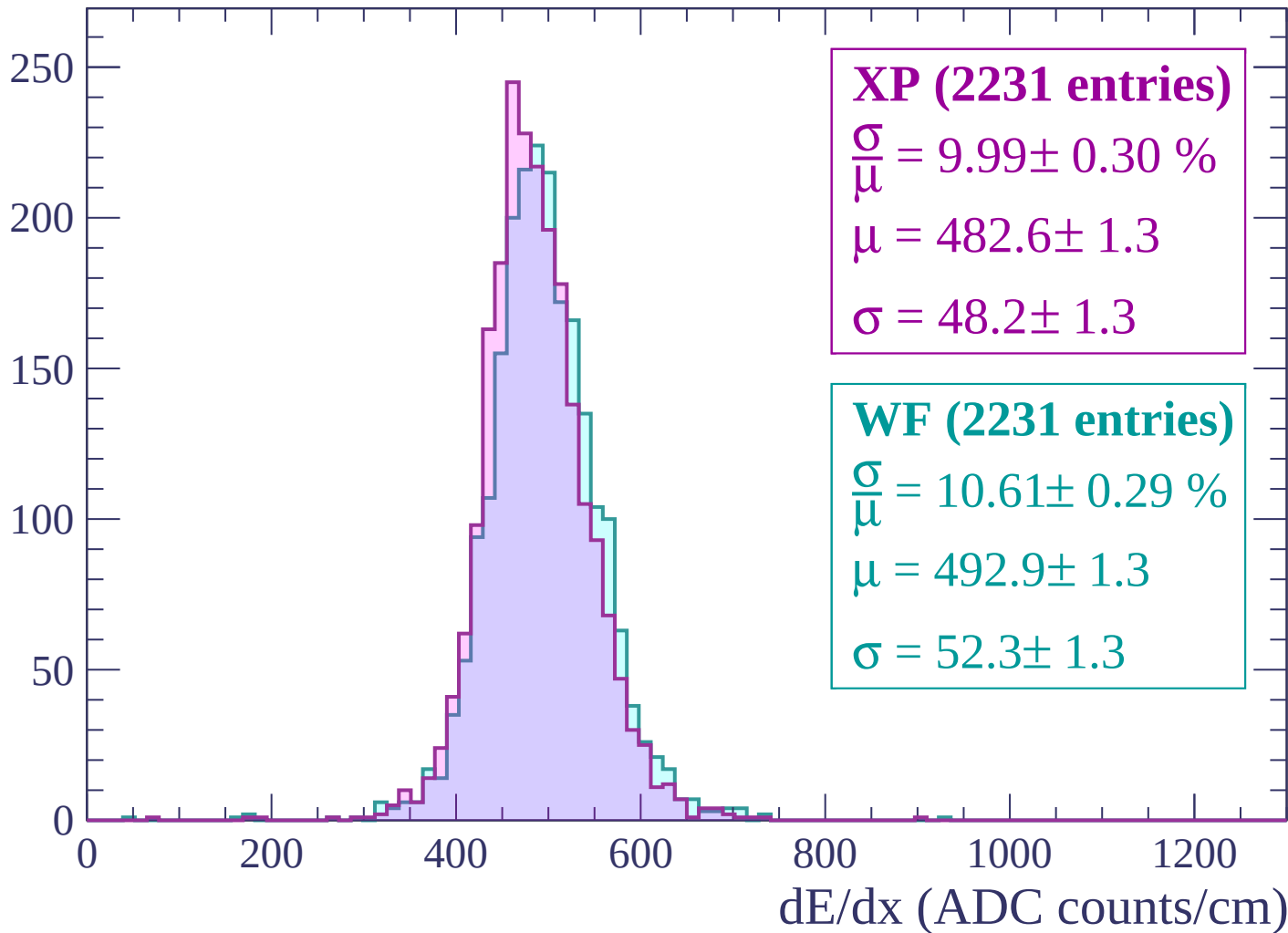
Energy loss | $2460 < p < 2520$

Count



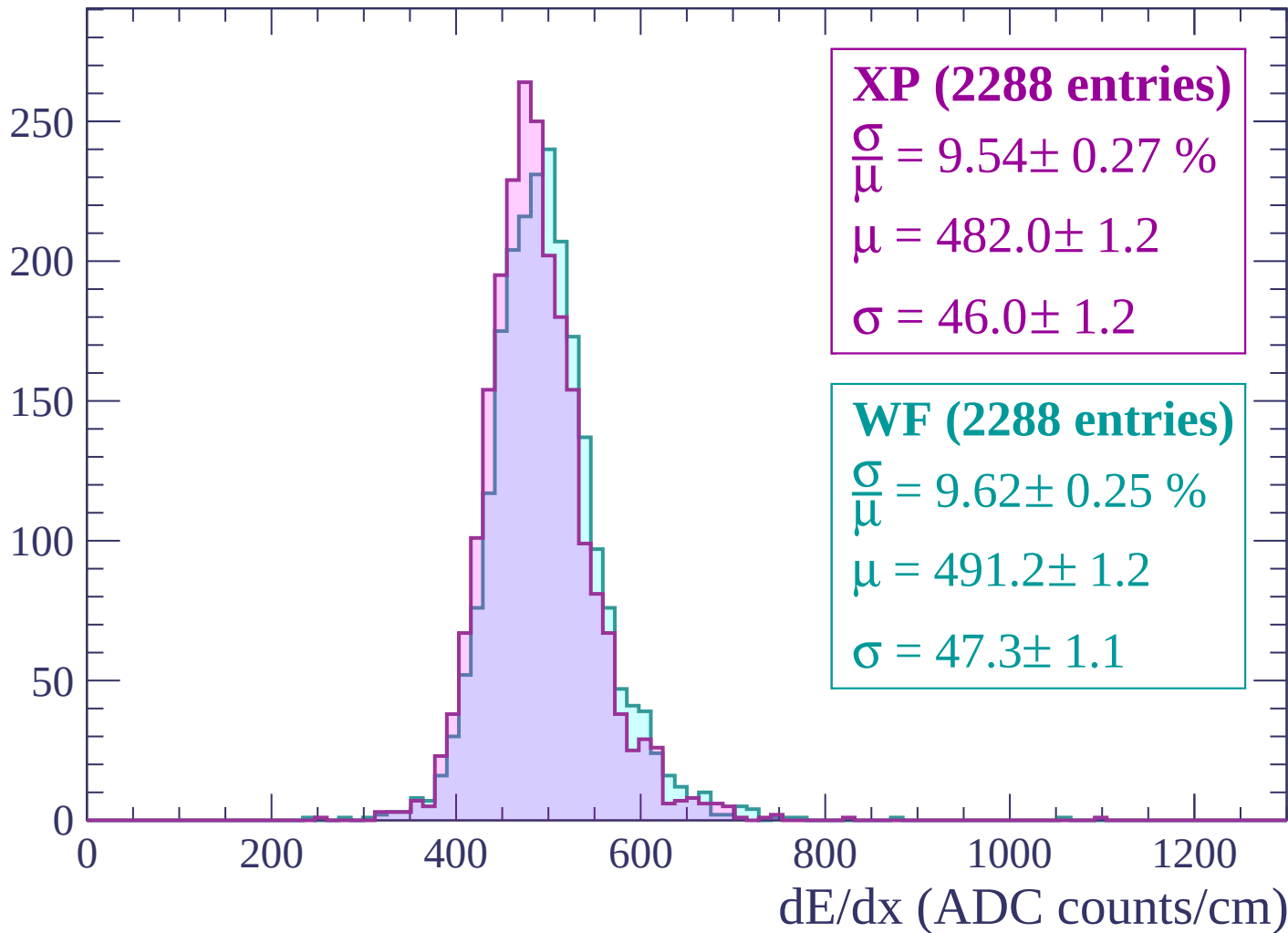
Energy loss | $2520 < p < 2580$

Count



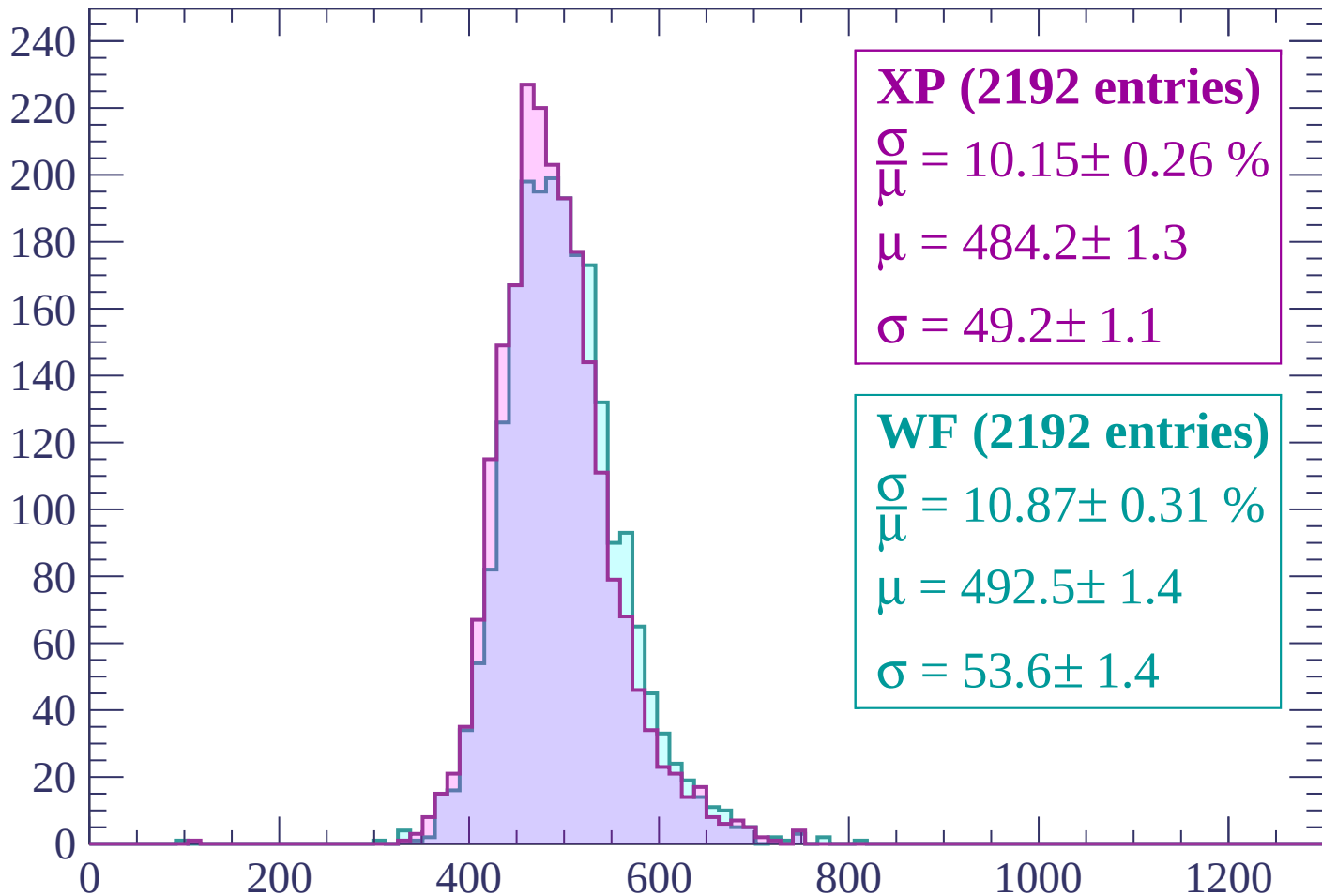
Energy loss | $2580 < p < 2640$

Count



Energy loss | $2640 < p < 2700$

Count



XP (2192 entries)

$\frac{\sigma}{\mu} = 10.15 \pm 0.26 \%$

$\mu = 484.2 \pm 1.3$

$\sigma = 49.2 \pm 1.1$

WF (2192 entries)

$\frac{\sigma}{\mu} = 10.87 \pm 0.31 \%$

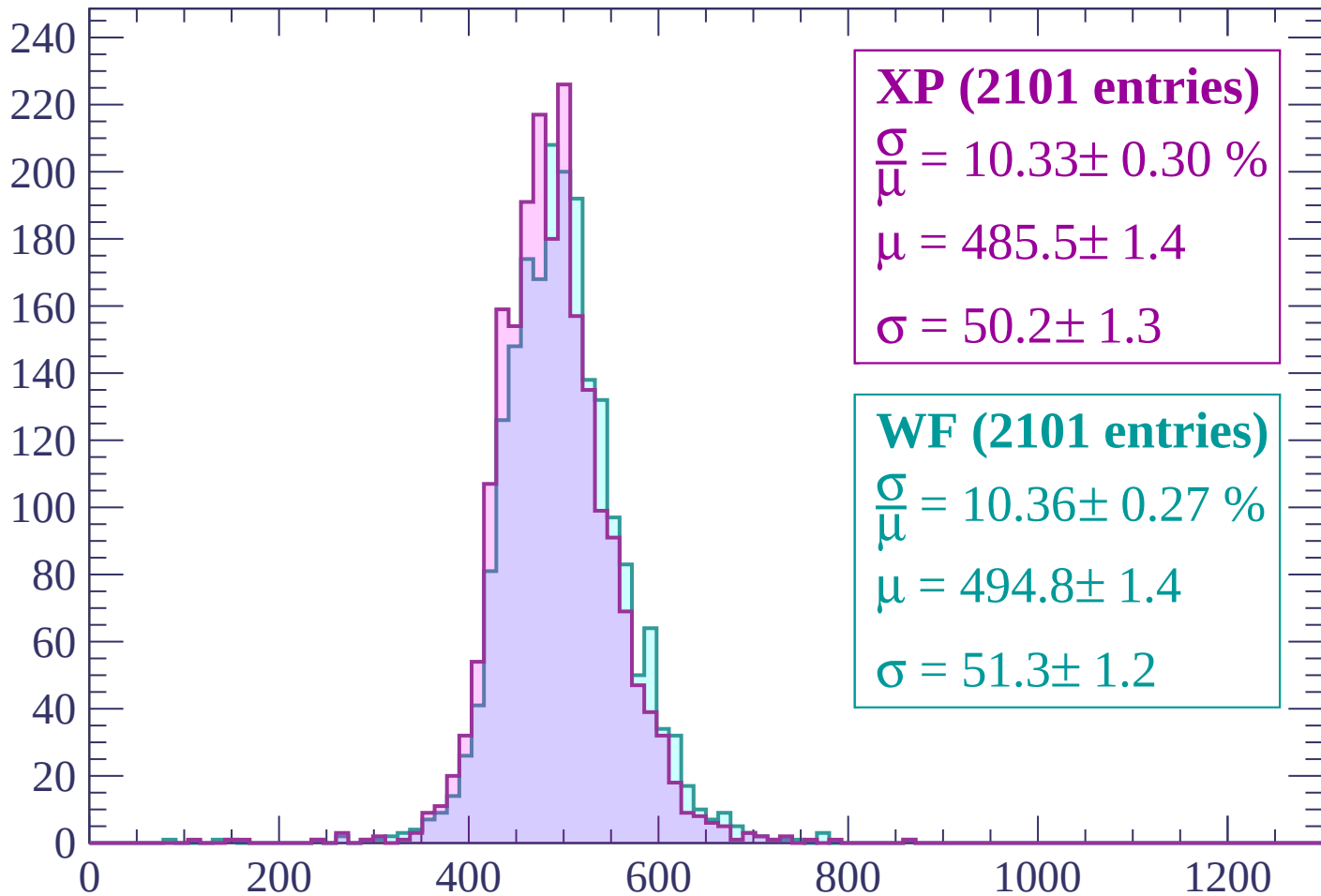
$\mu = 492.5 \pm 1.4$

$\sigma = 53.6 \pm 1.4$

dE/dx (ADC counts/cm)

Energy loss | $2700 < p < 2760$

Count



XP (2101 entries)

$$\frac{\sigma}{\mu} = 10.33 \pm 0.30 \%$$

$$\mu = 485.5 \pm 1.4$$

$$\sigma = 50.2 \pm 1.3$$

WF (2101 entries)

$$\frac{\sigma}{\mu} = 10.36 \pm 0.27 \%$$

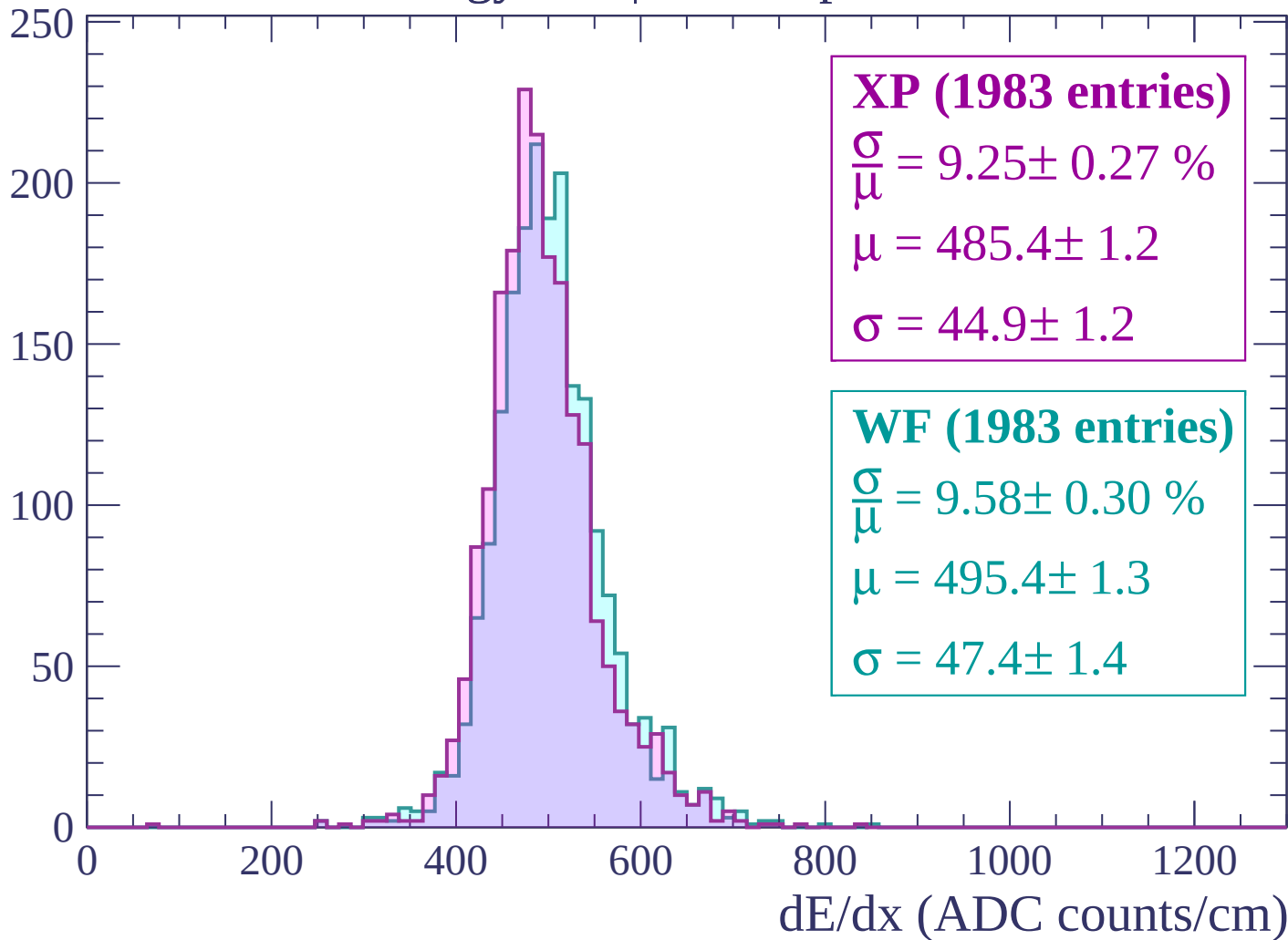
$$\mu = 494.8 \pm 1.4$$

$$\sigma = 51.3 \pm 1.2$$

dE/dx (ADC counts/cm)

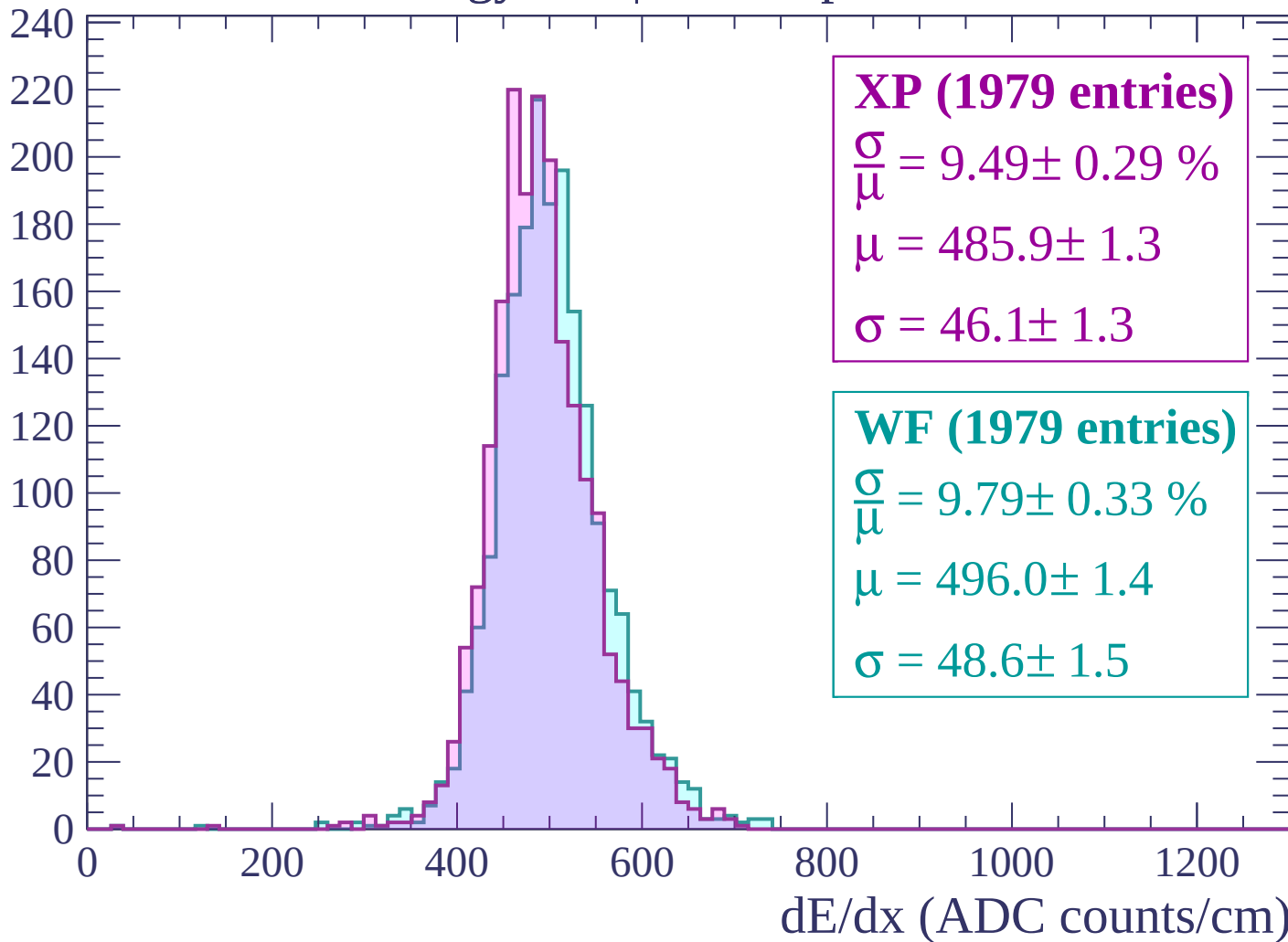
Energy loss | $2760 < p < 2820$

Count



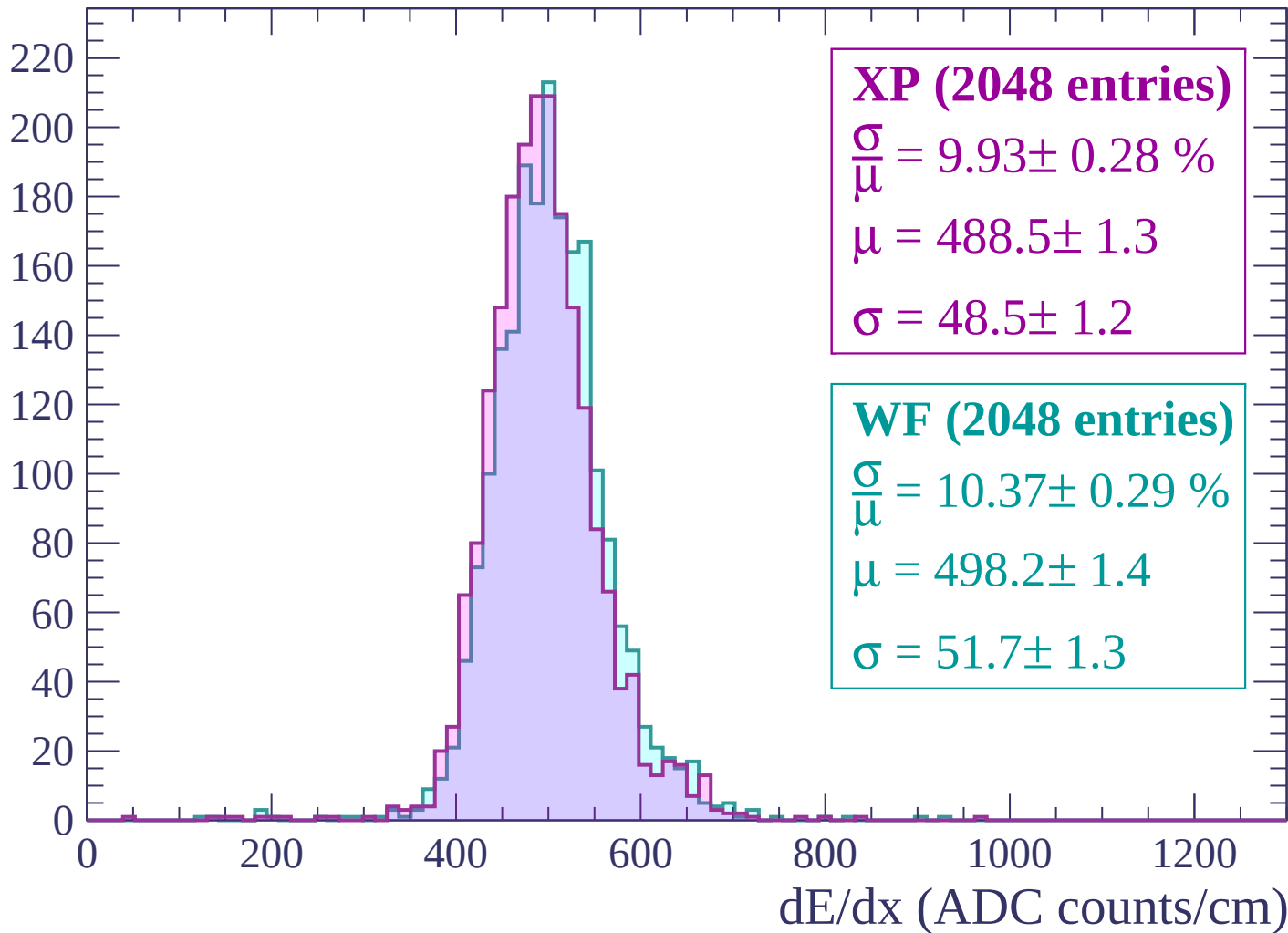
Energy loss | $2820 < p < 2880$

Count



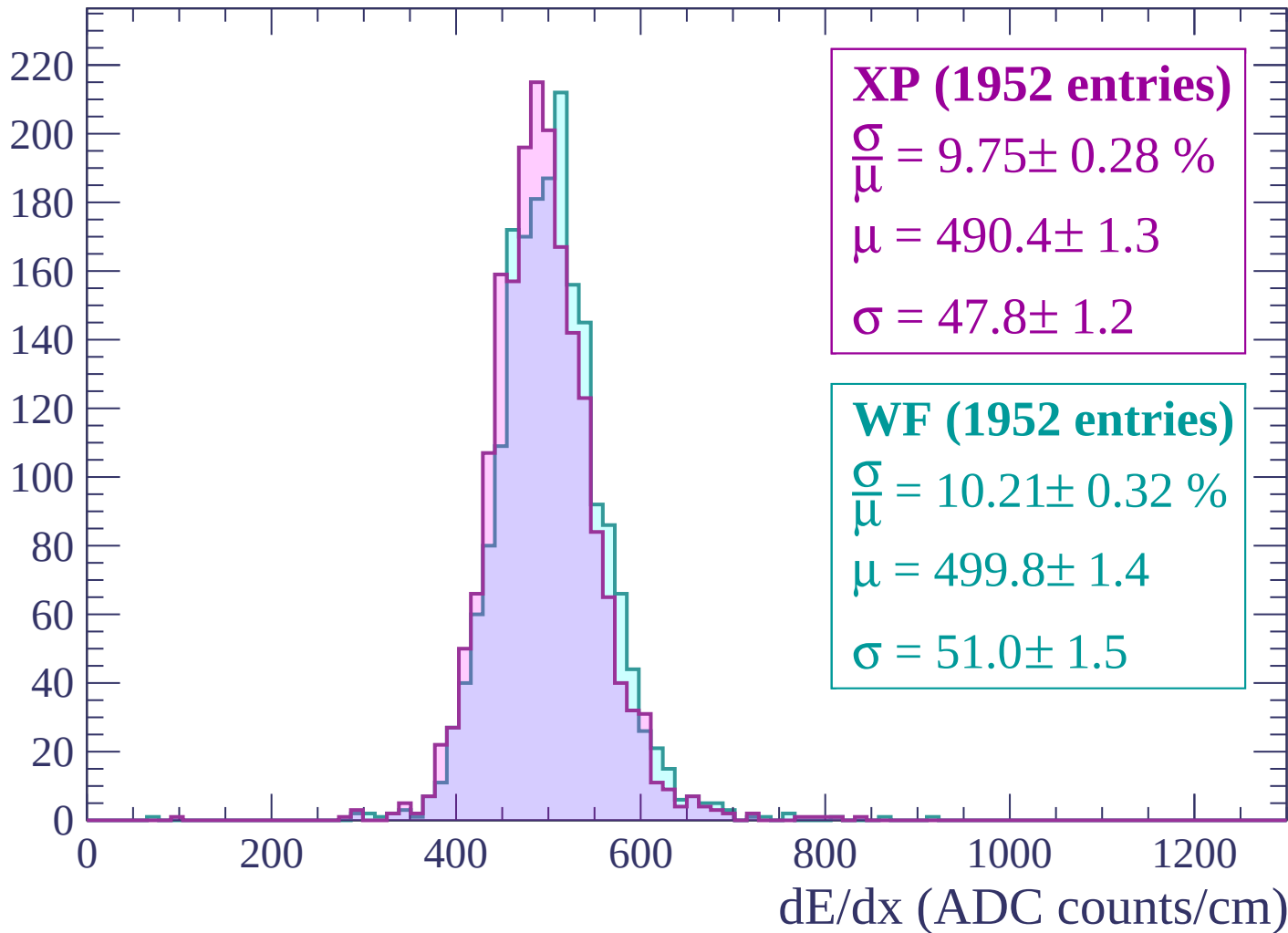
Energy loss | $2880 < p < 2940$

Count



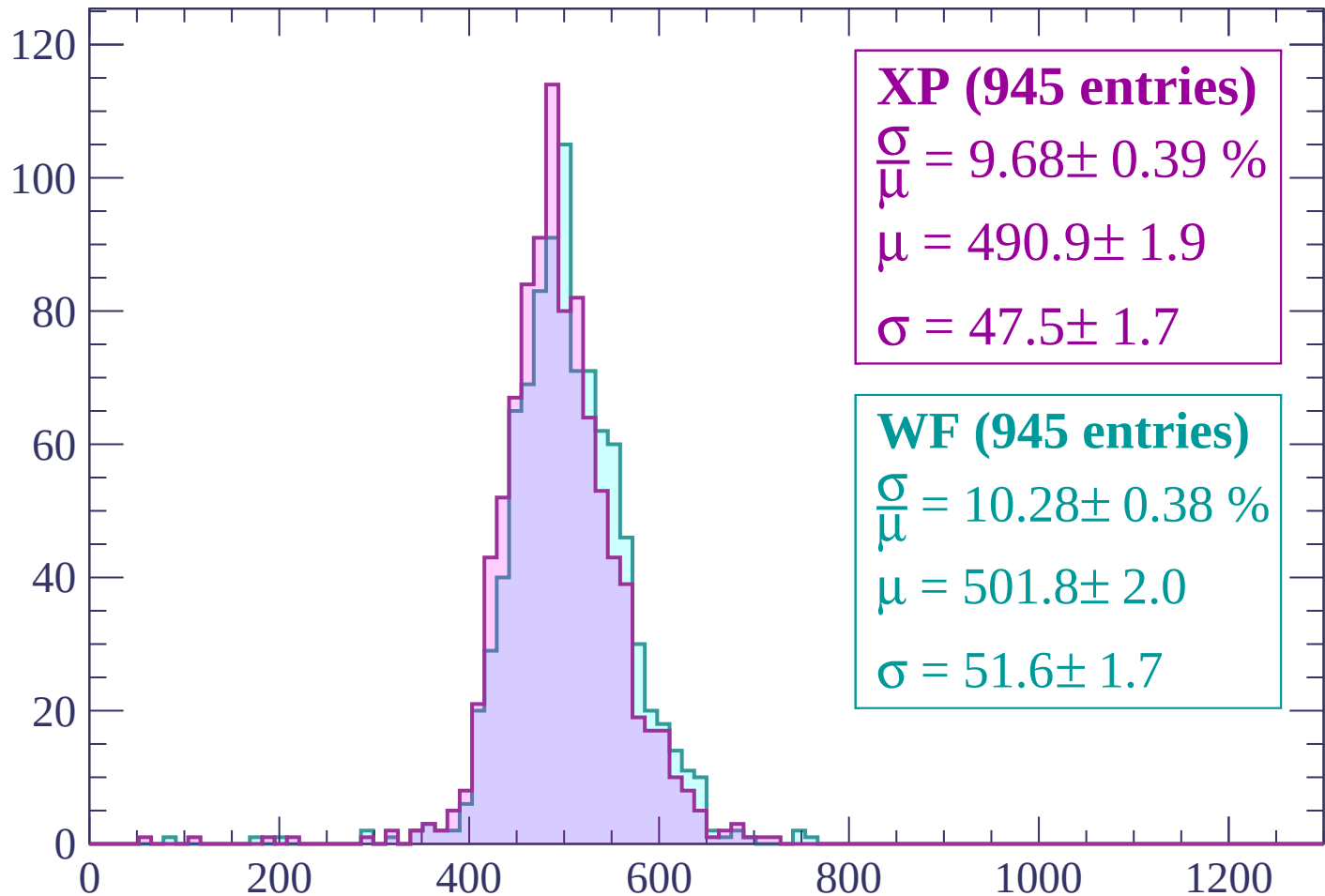
Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count



XP (945 entries)

$$\frac{\sigma}{\mu} = 9.68 \pm 0.39 \%$$

$$\mu = 490.9 \pm 1.9$$

$$\sigma = 47.5 \pm 1.7$$

WF (945 entries)

$$\frac{\sigma}{\mu} = 10.28 \pm 0.38 \%$$

$$\mu = 501.8 \pm 2.0$$

$$\sigma = 51.6 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $-90 < \theta < -88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-88 < \theta < -86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-86 < \theta < -84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-84 < \theta < -82$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-82 < \theta < -80$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

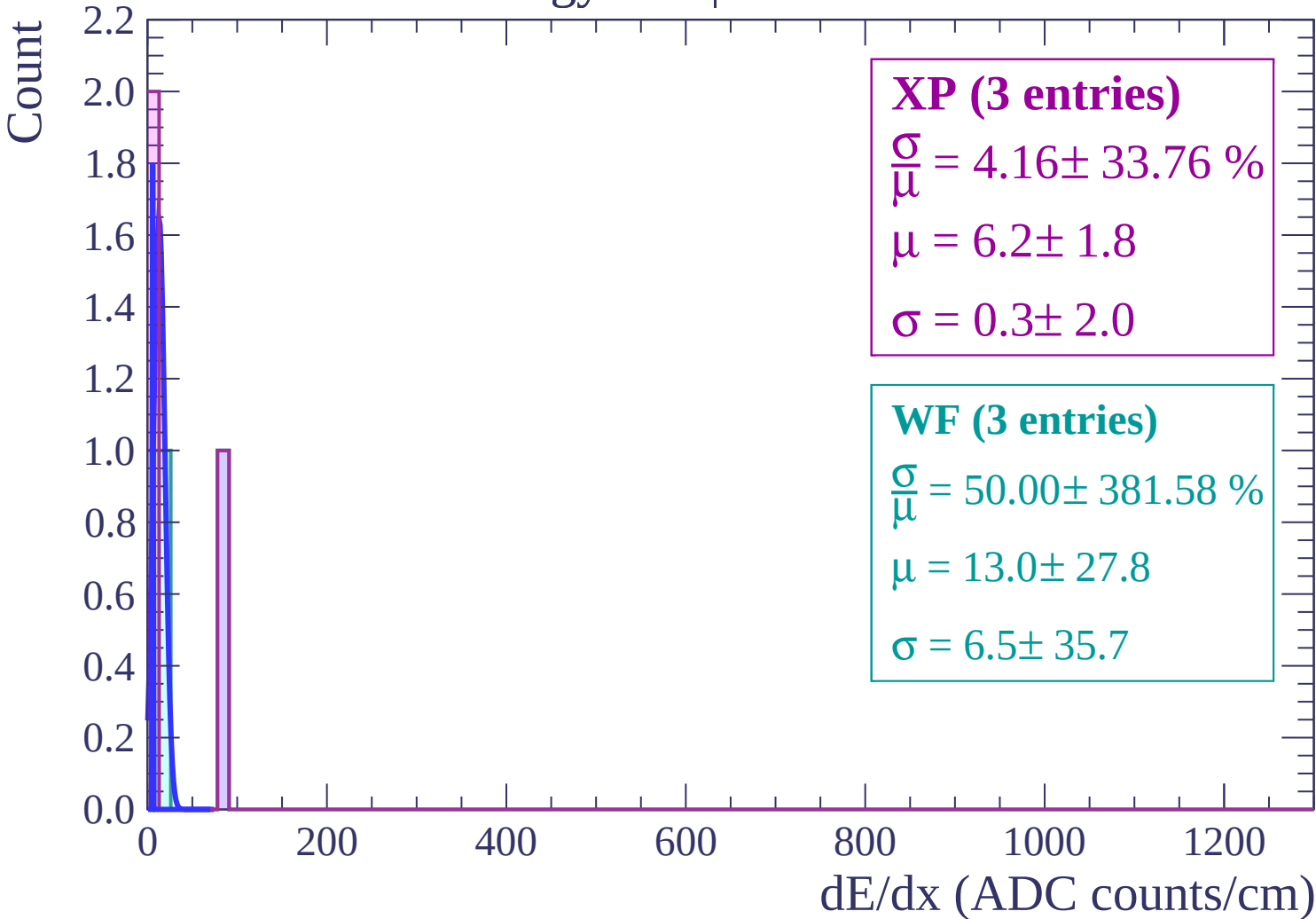
WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

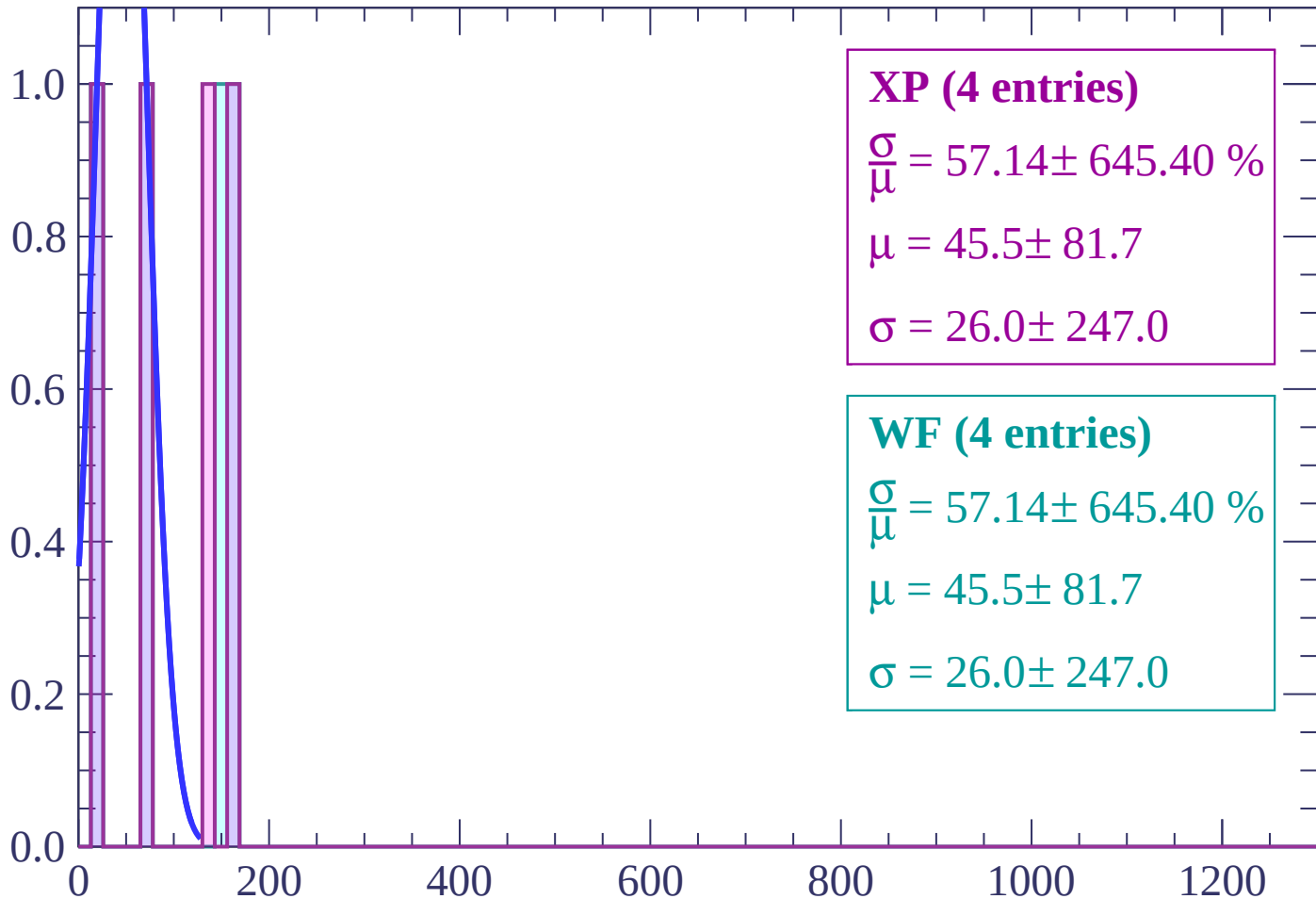
$\sigma = 0.0 \pm 0.0$

Energy loss | $-80 < \theta < -78$



Energy loss | $-78 < \theta < -76$

Count



XP (4 entries)

$$\frac{\sigma}{\mu} = 57.14 \pm 645.40 \%$$

$$\mu = 45.5 \pm 81.7$$

$$\sigma = 26.0 \pm 247.0$$

WF (4 entries)

$$\frac{\sigma}{\mu} = 57.14 \pm 645.40 \%$$

$$\mu = 45.5 \pm 81.7$$

$$\sigma = 26.0 \pm 247.0$$

dE/dx (ADC counts/cm)

Energy loss | $-76 < \theta < -74$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (3 entries)

$\frac{\sigma}{\mu} = 87.50 \pm 1041.46 \%$

$\mu = 52.0 \pm 138.3$

$\sigma = 45.5 \pm 420.6$

WF (3 entries)

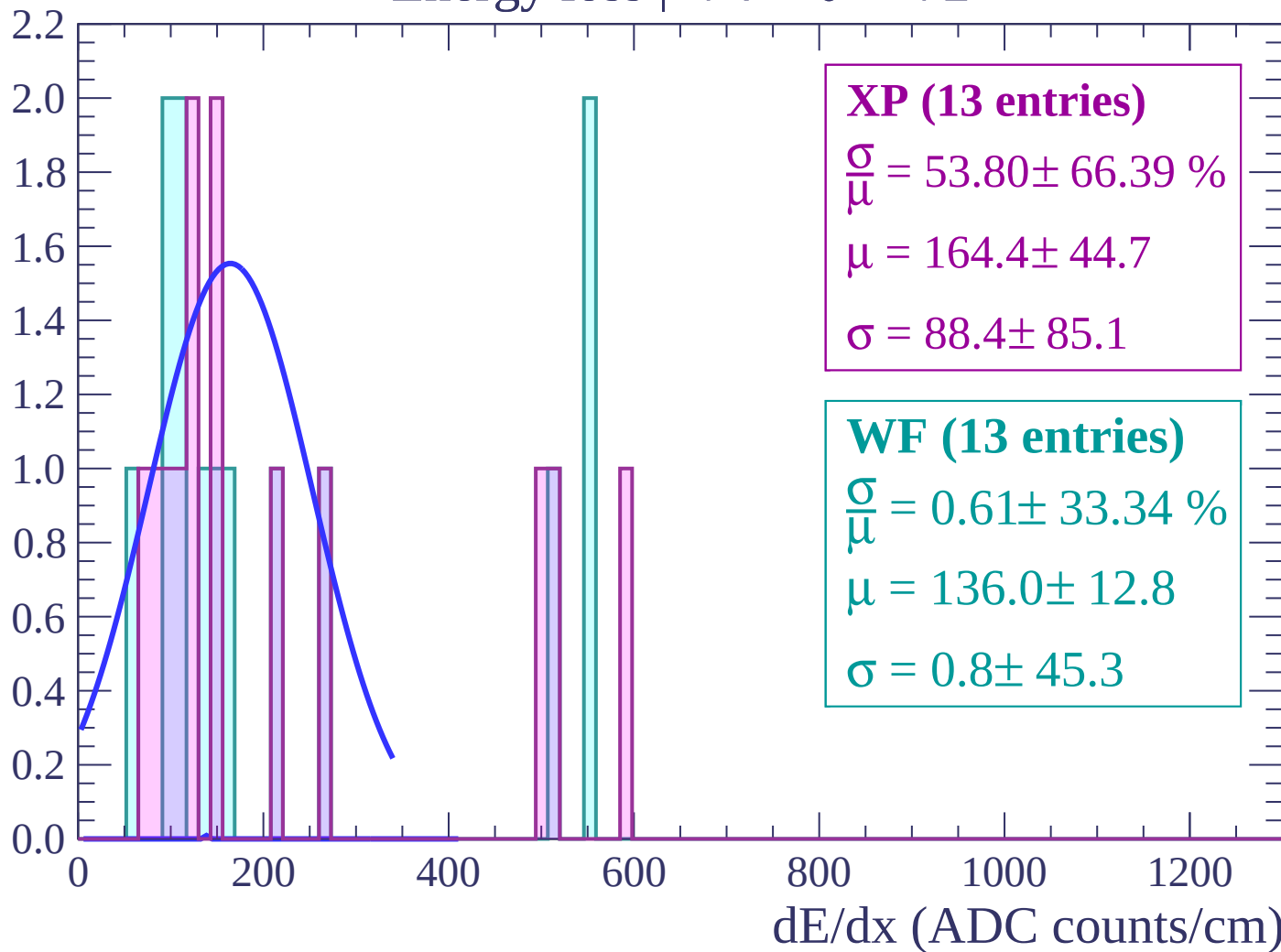
$\frac{\sigma}{\mu} = 87.50 \pm 1041.46 \%$

$\mu = 52.0 \pm 138.3$

$\sigma = 45.5 \pm 420.6$

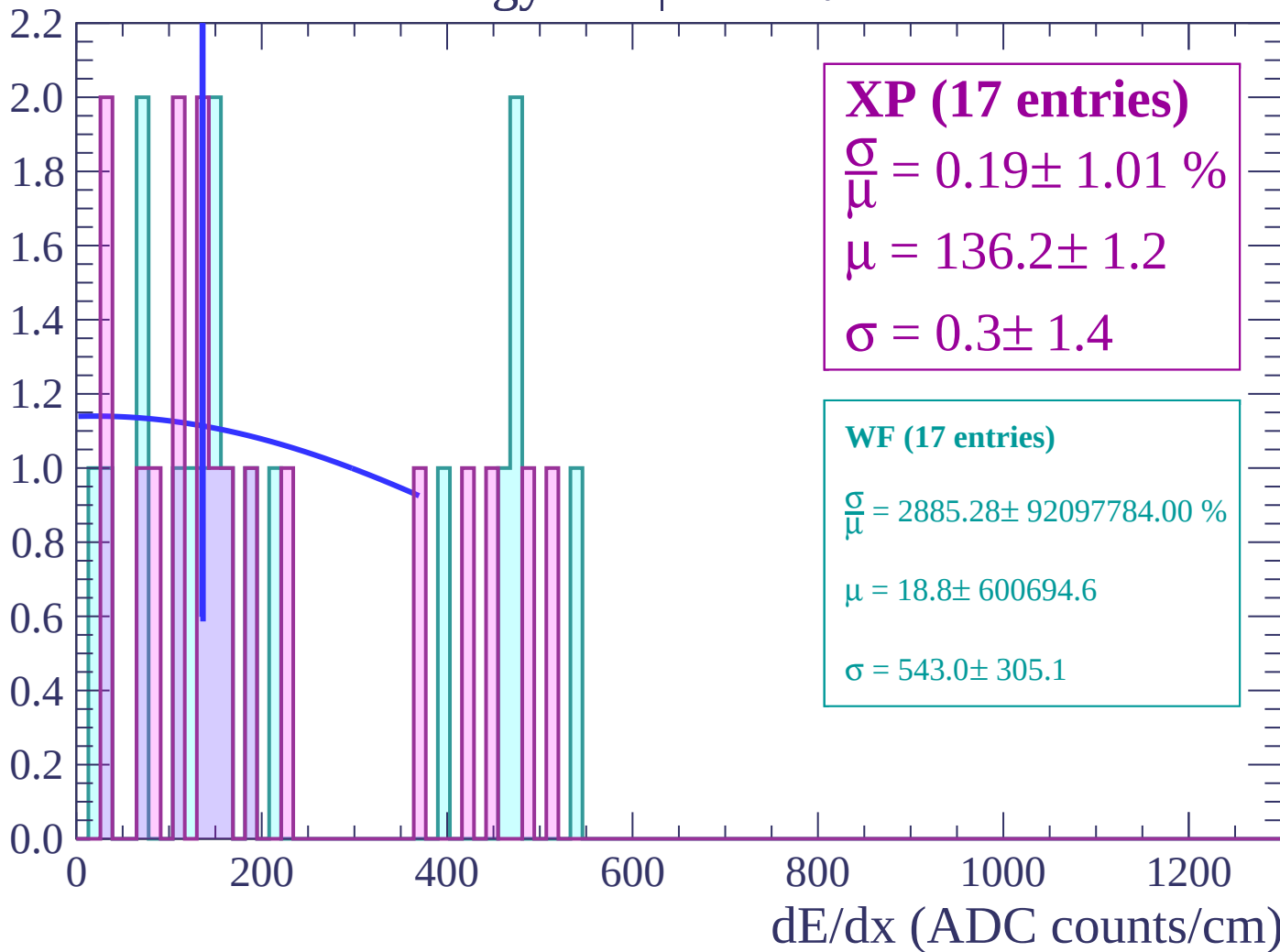
Energy loss | $-74 < \theta < -72$

Count



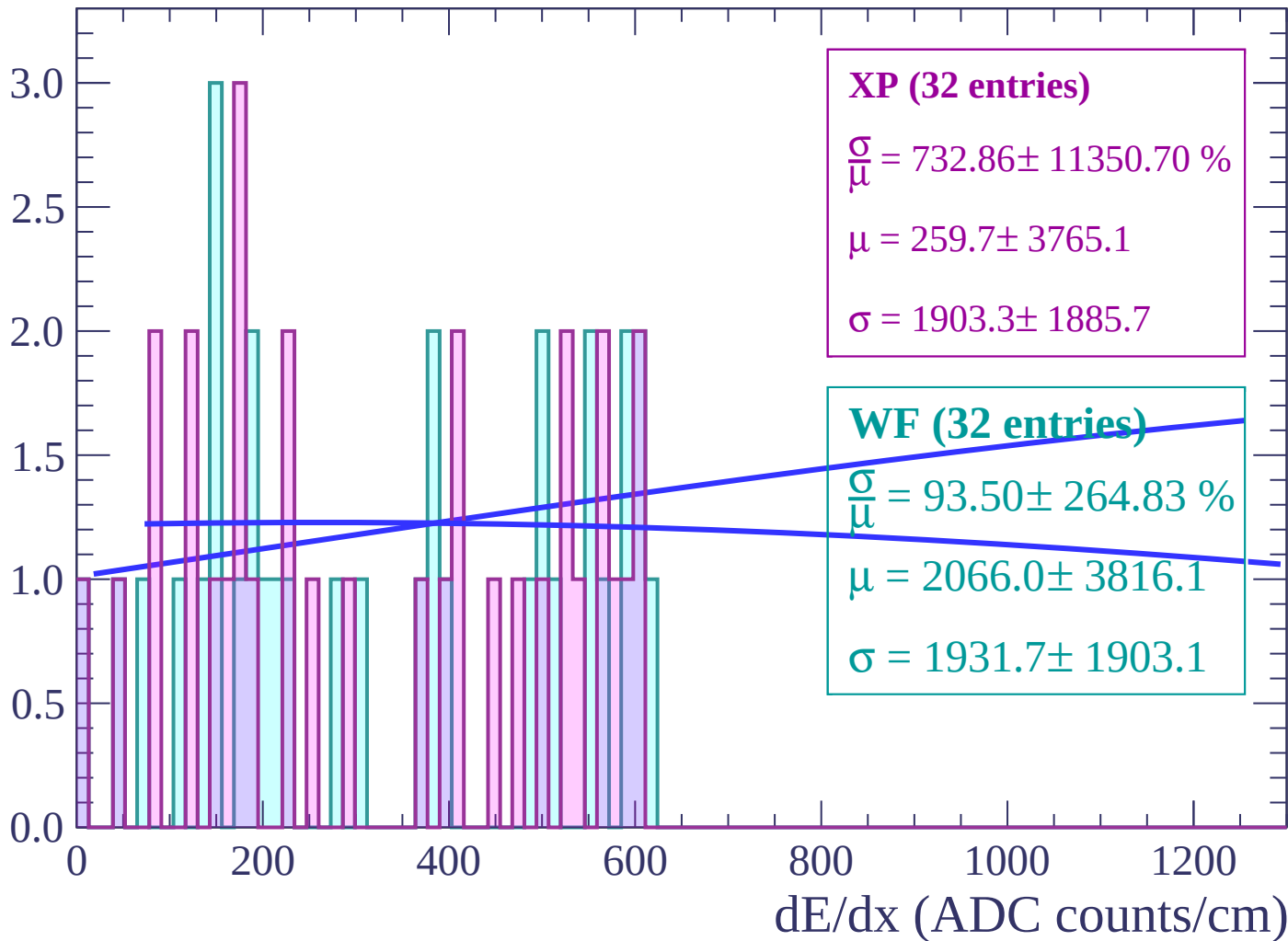
Energy loss | $-72 < \theta < -70$

Count



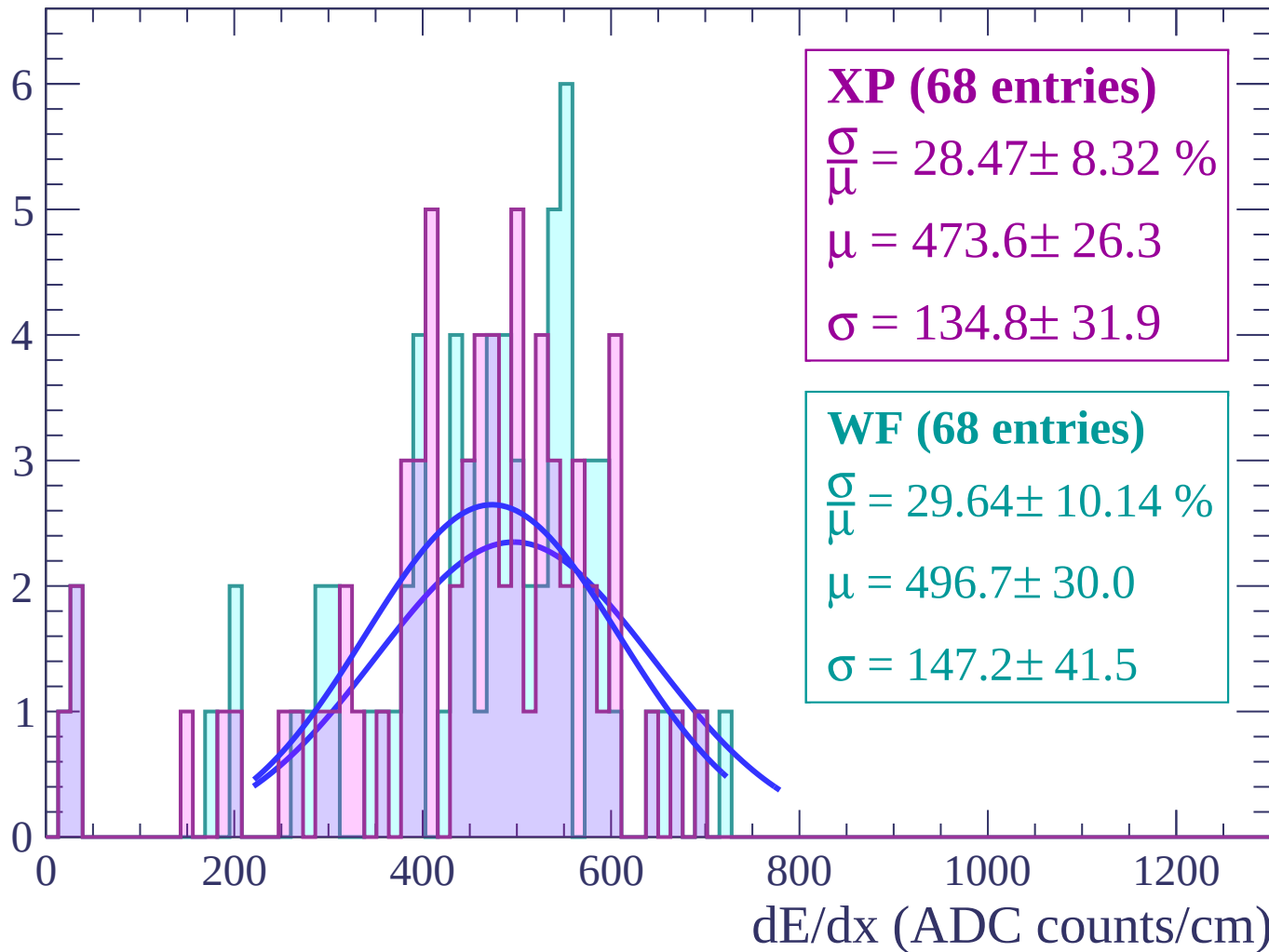
Energy loss | $-70 < \theta < -68$

Count



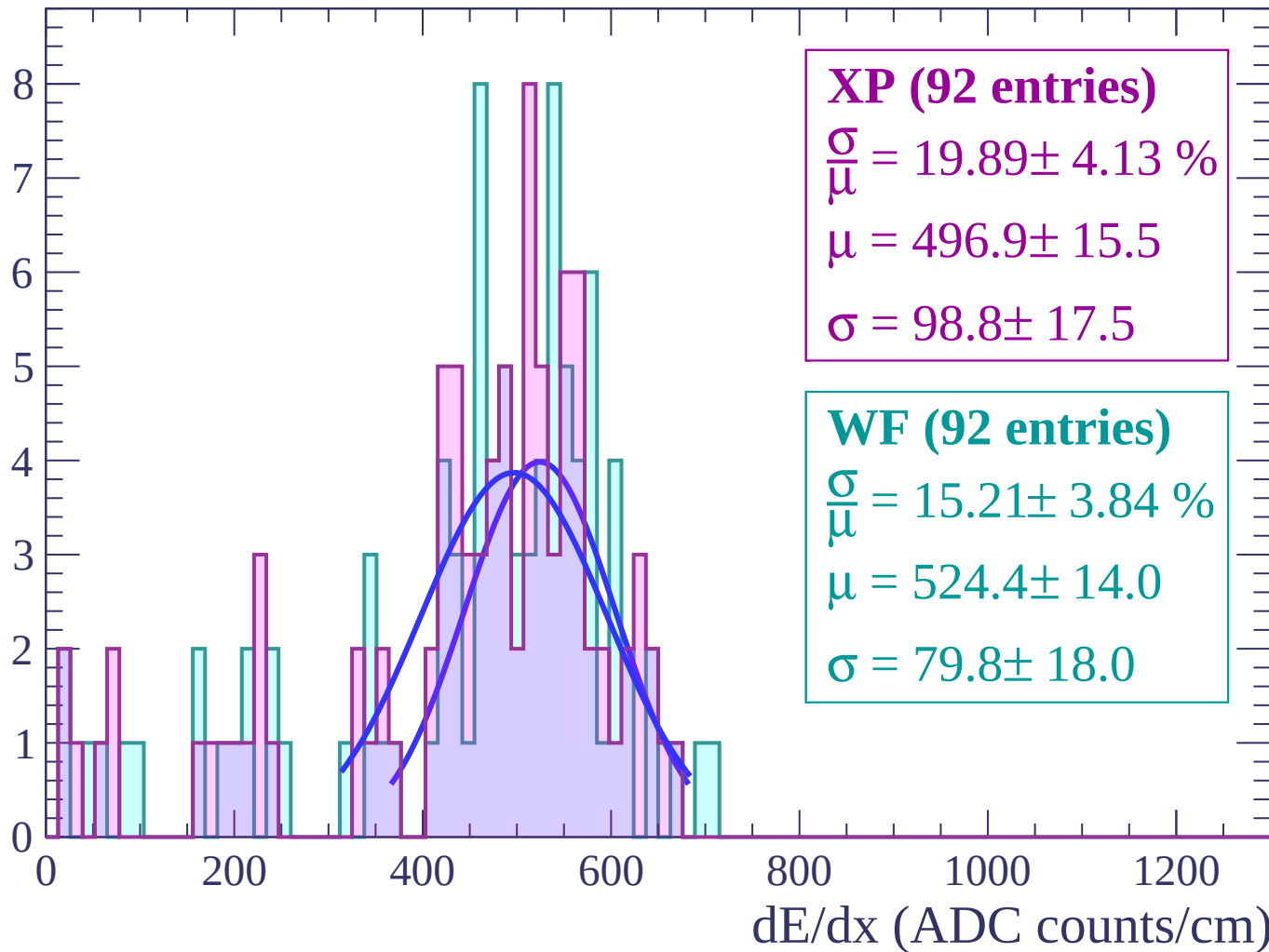
Energy loss | $-68 < \theta < -66$

Count



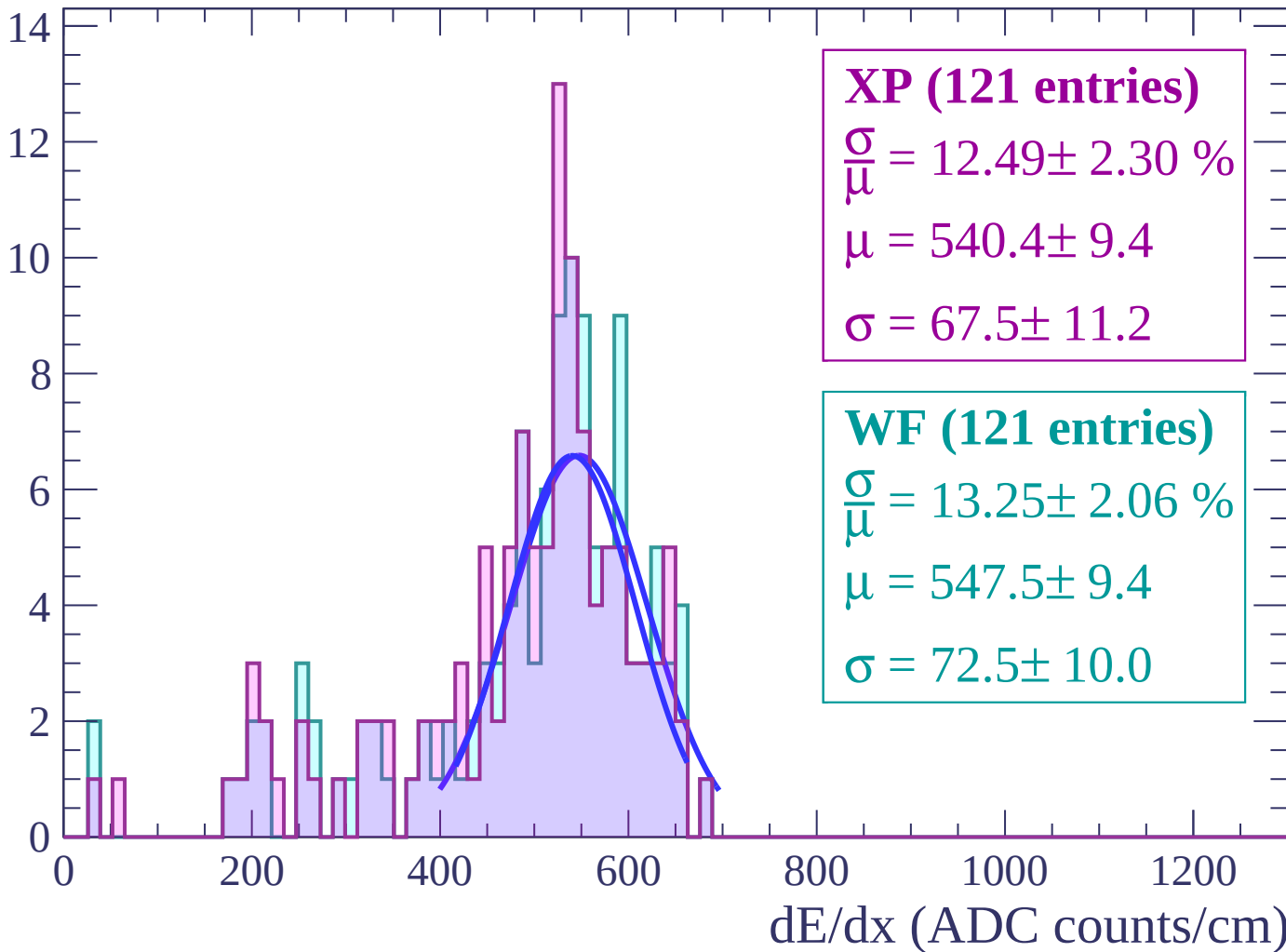
Energy loss | $-66 < \theta < -64$

Count



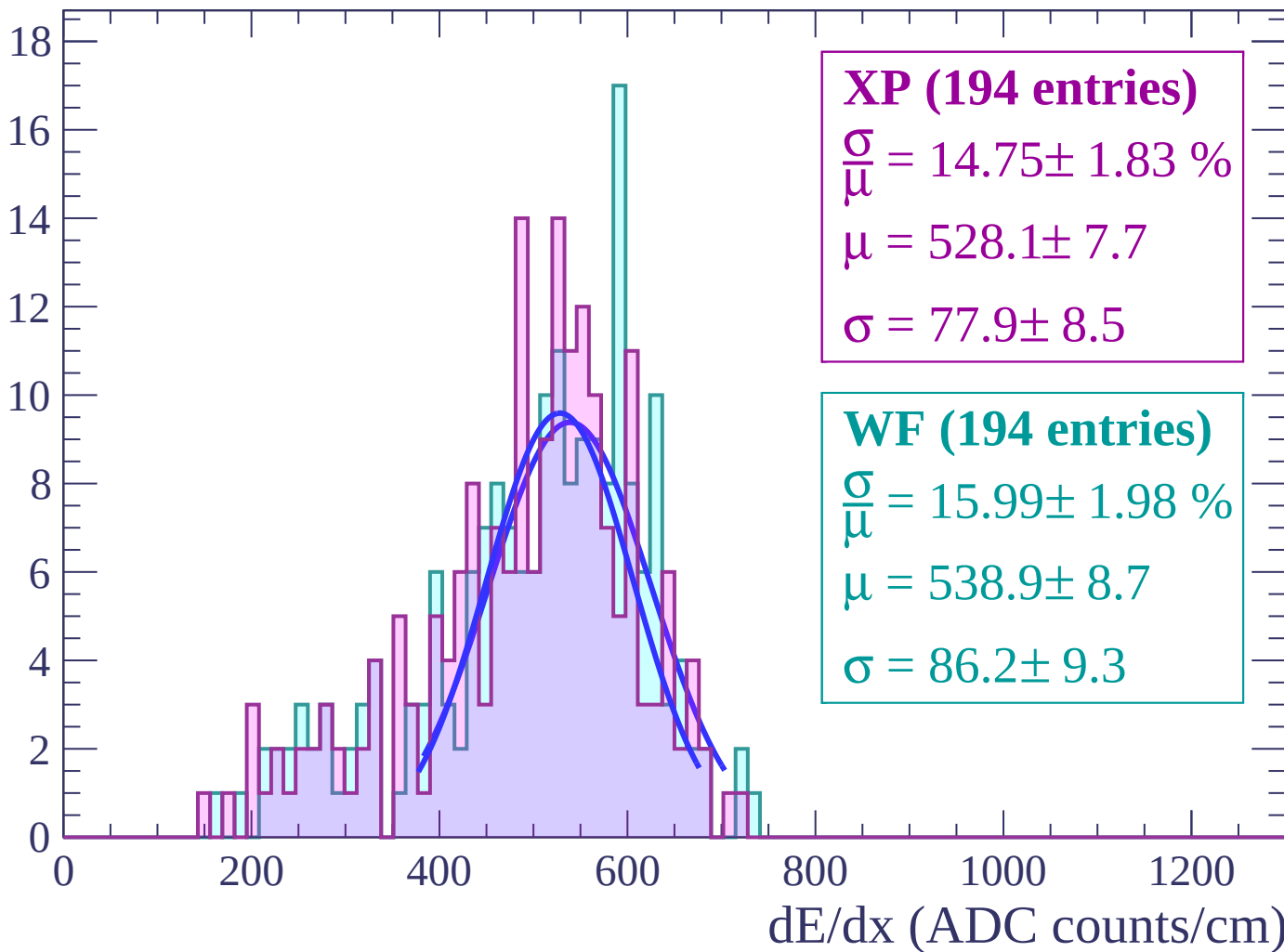
Energy loss | $-64 < \theta < -62$

Count



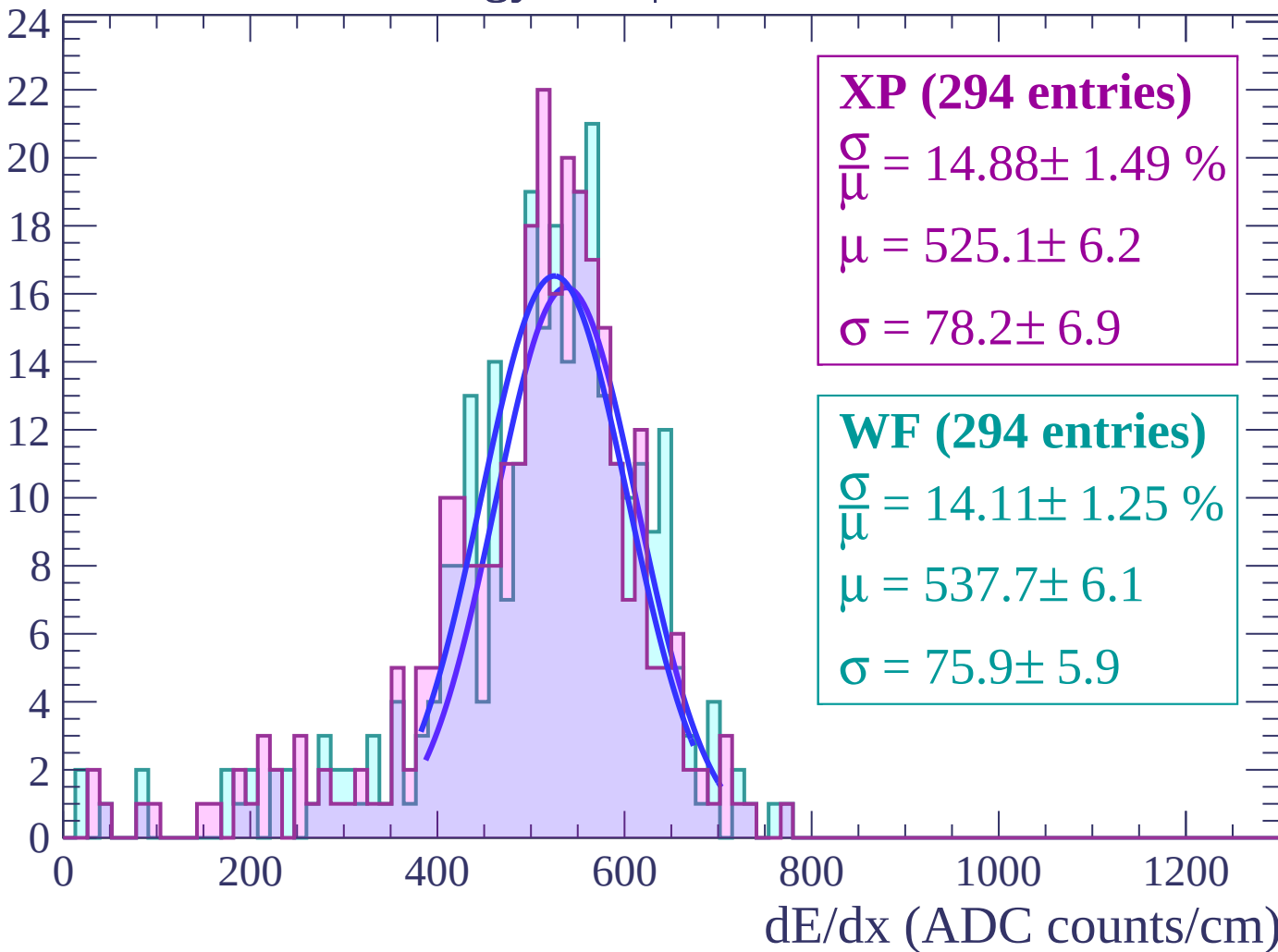
Energy loss | $-62 < \theta < -60$

Count



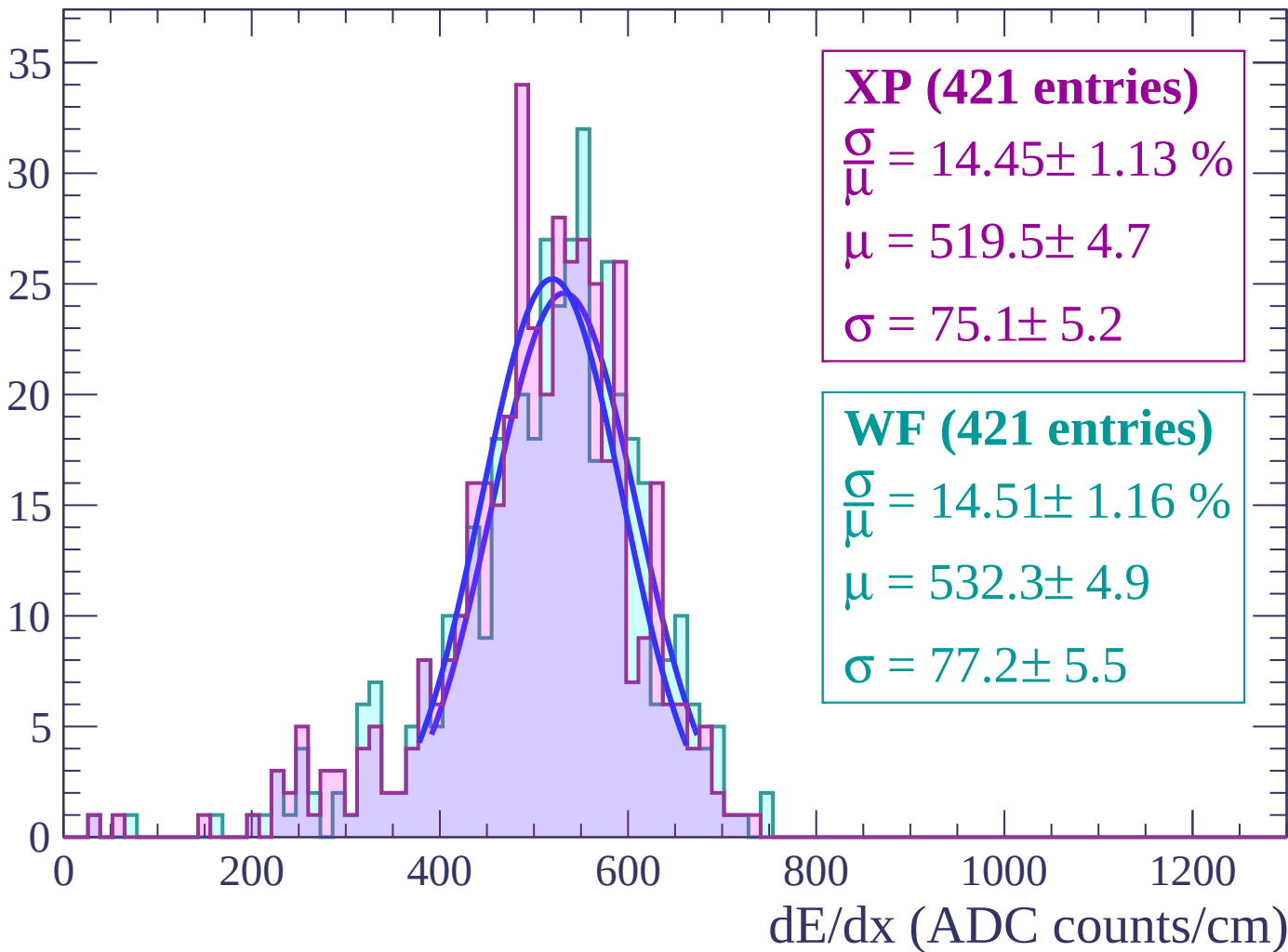
Energy loss | $-60 < \theta < -58$

Count



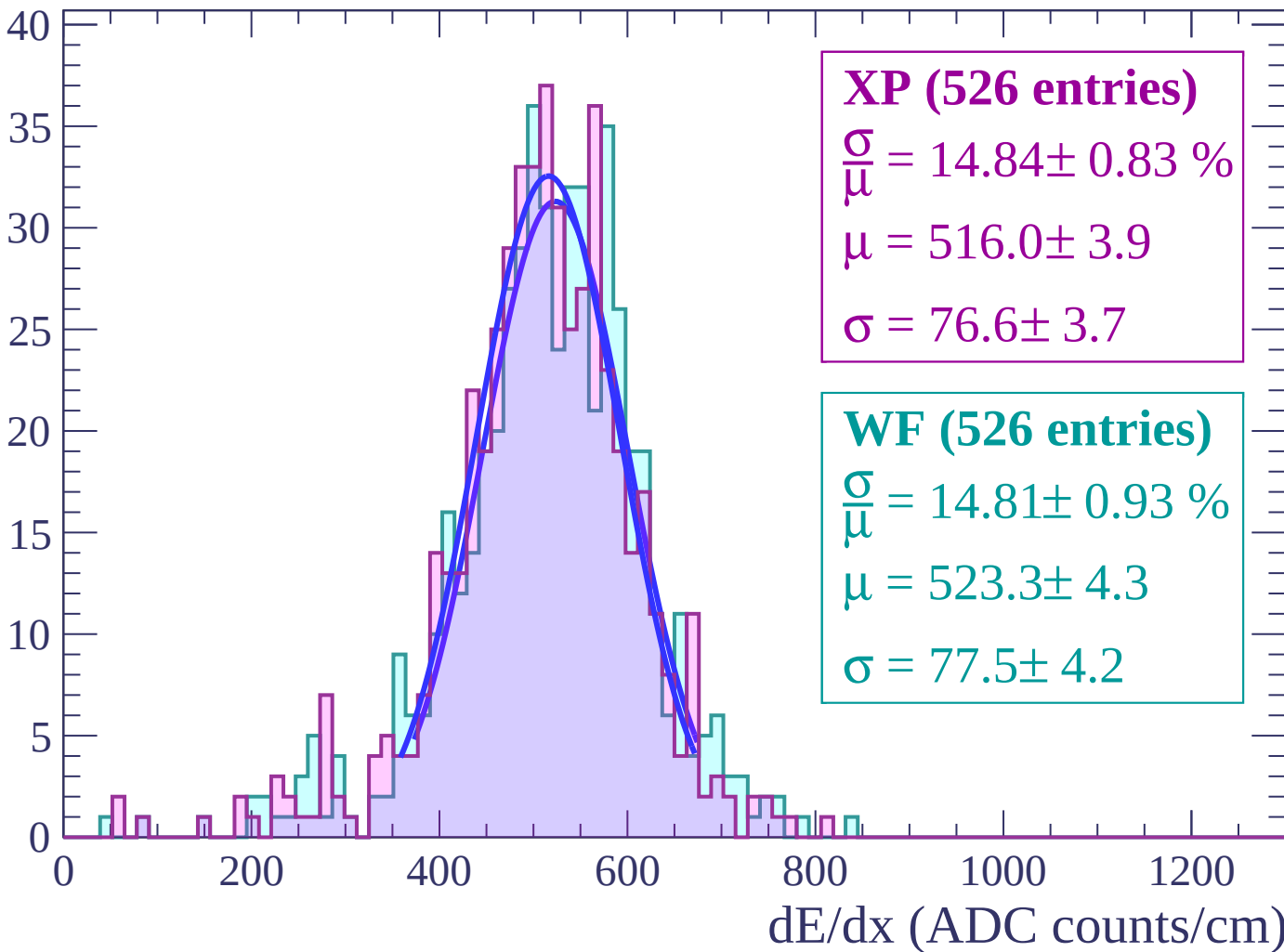
Energy loss | $-58 < \theta < -56$

Count



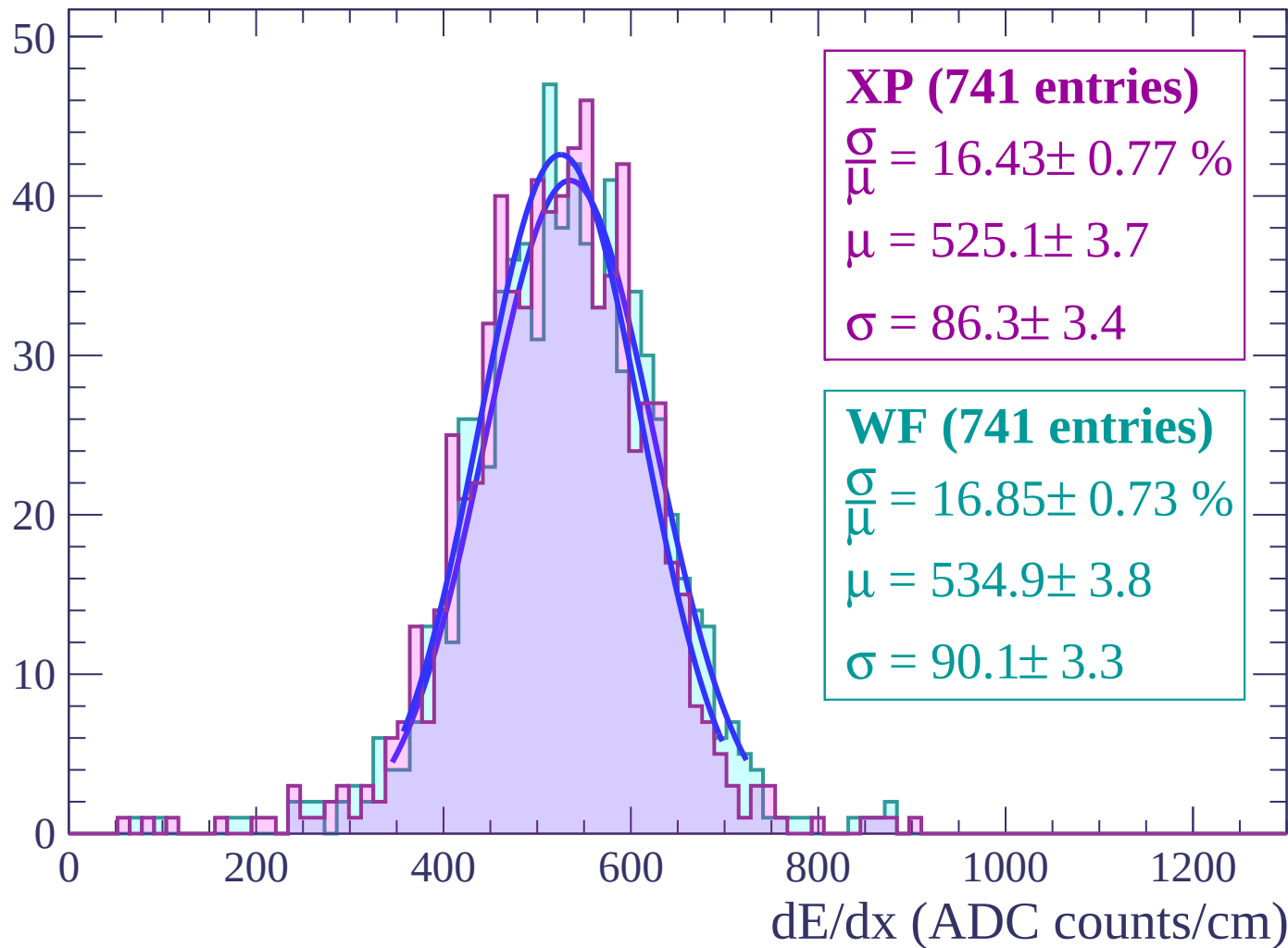
Energy loss | $-56 < \theta < -54$

Count



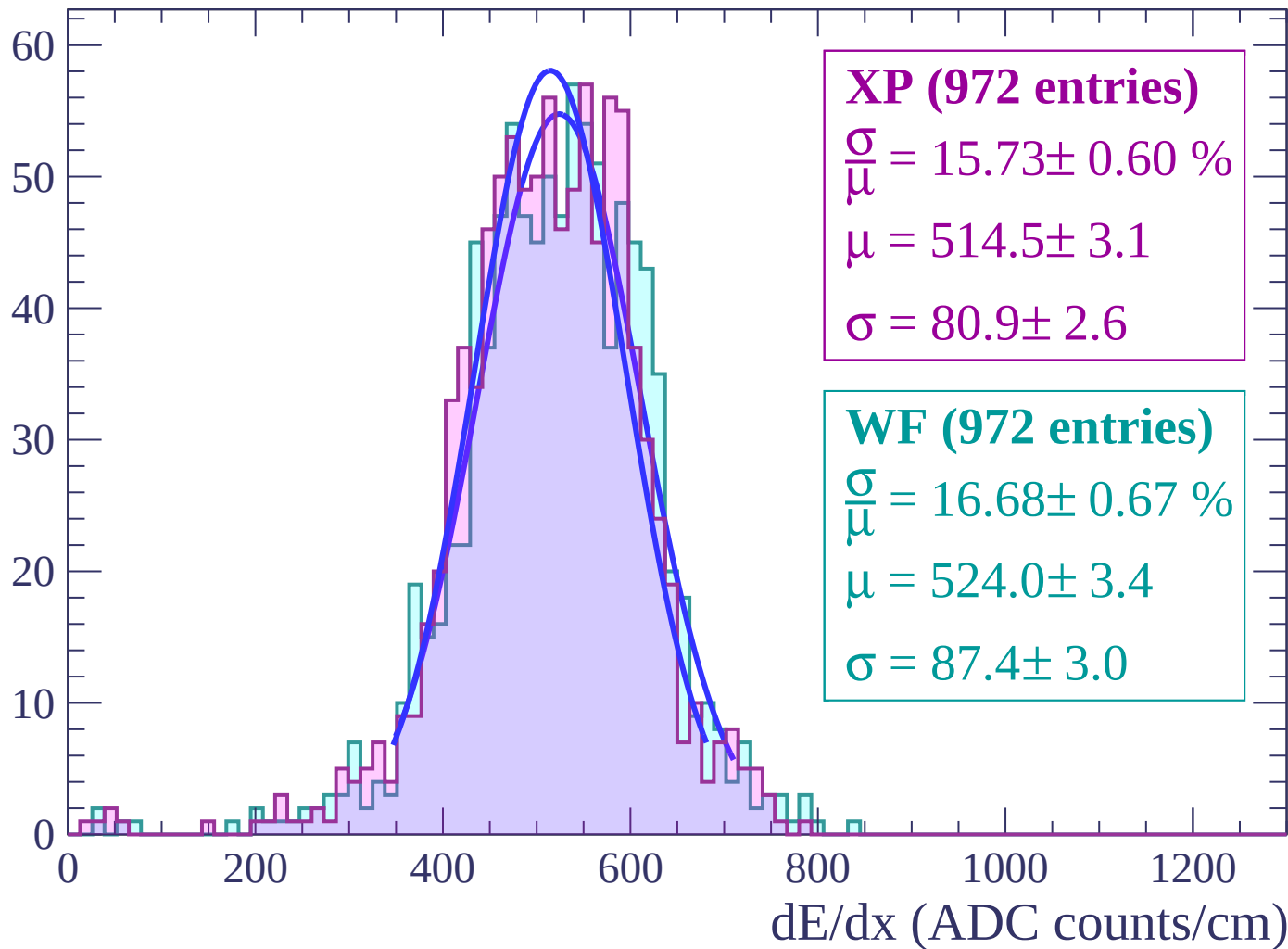
Energy loss | $-54 < \theta < -52$

Count



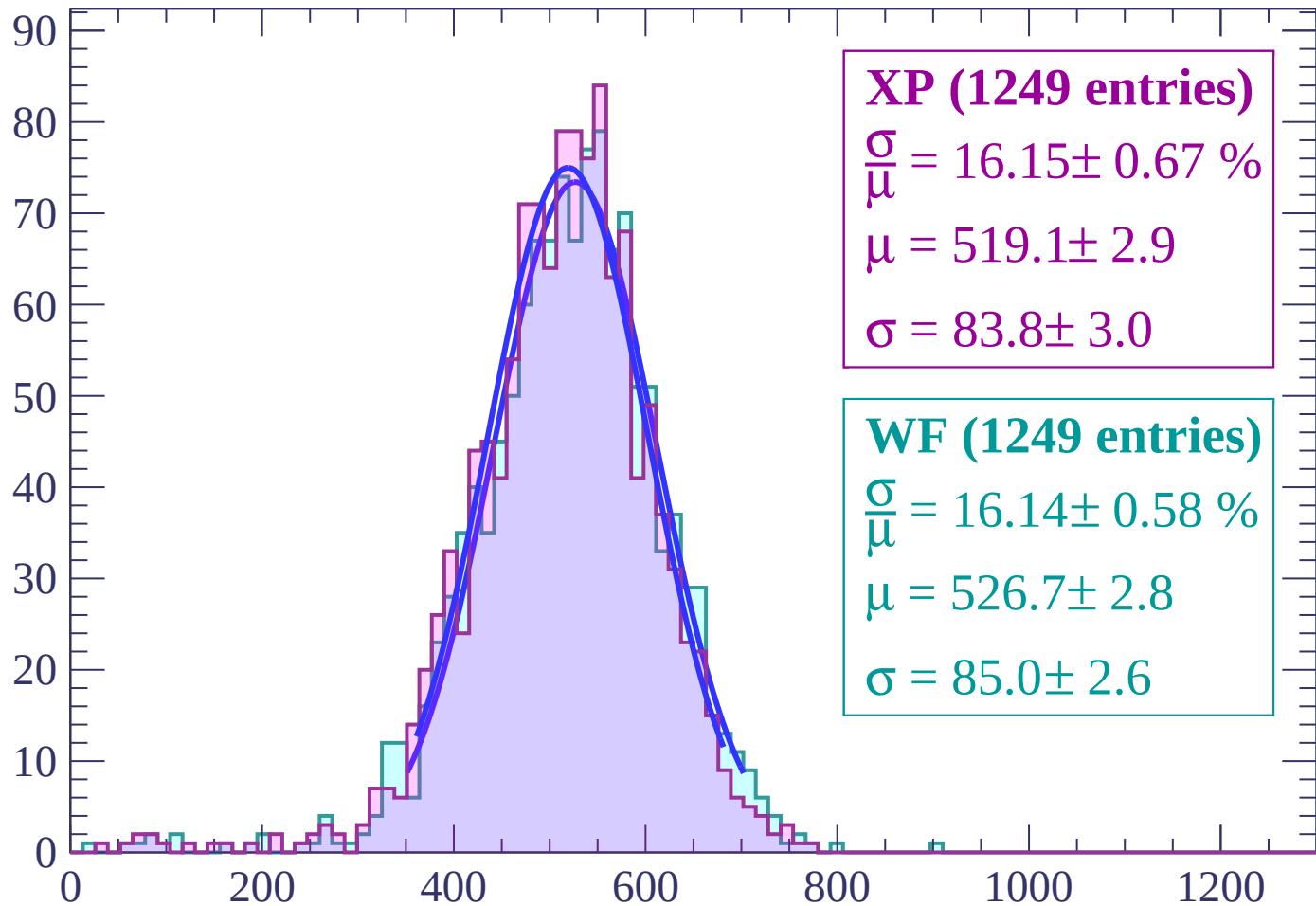
Energy loss | $-52 < \theta < -50$

Count



Energy loss | $-50 < \theta < -48$

Count



XP (1249 entries)

$$\frac{\sigma}{\mu} = 16.15 \pm 0.67 \%$$

$$\mu = 519.1 \pm 2.9$$

$$\sigma = 83.8 \pm 3.0$$

WF (1249 entries)

$$\frac{\sigma}{\mu} = 16.14 \pm 0.58 \%$$

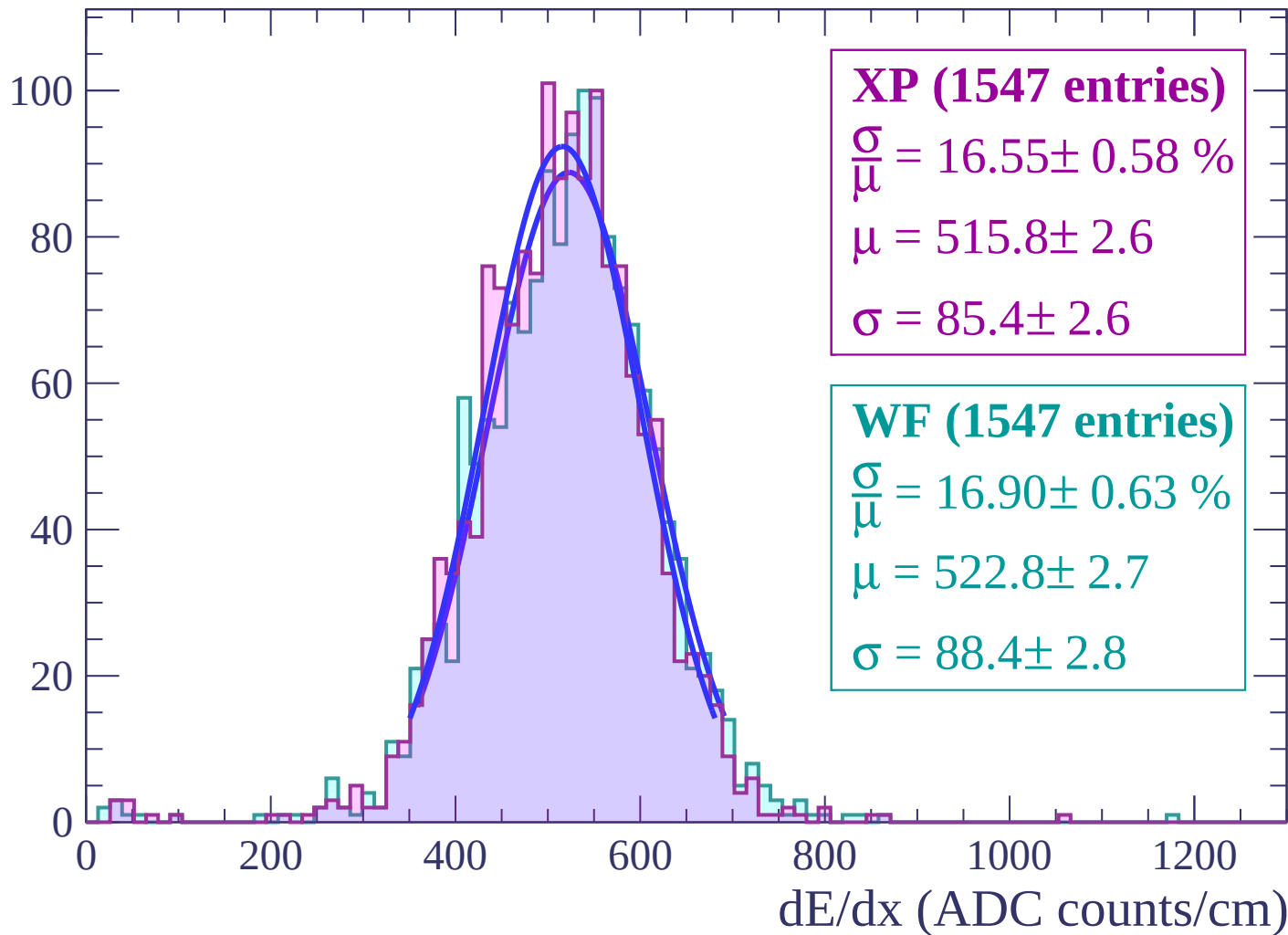
$$\mu = 526.7 \pm 2.8$$

$$\sigma = 85.0 \pm 2.6$$

dE/dx (ADC counts/cm)

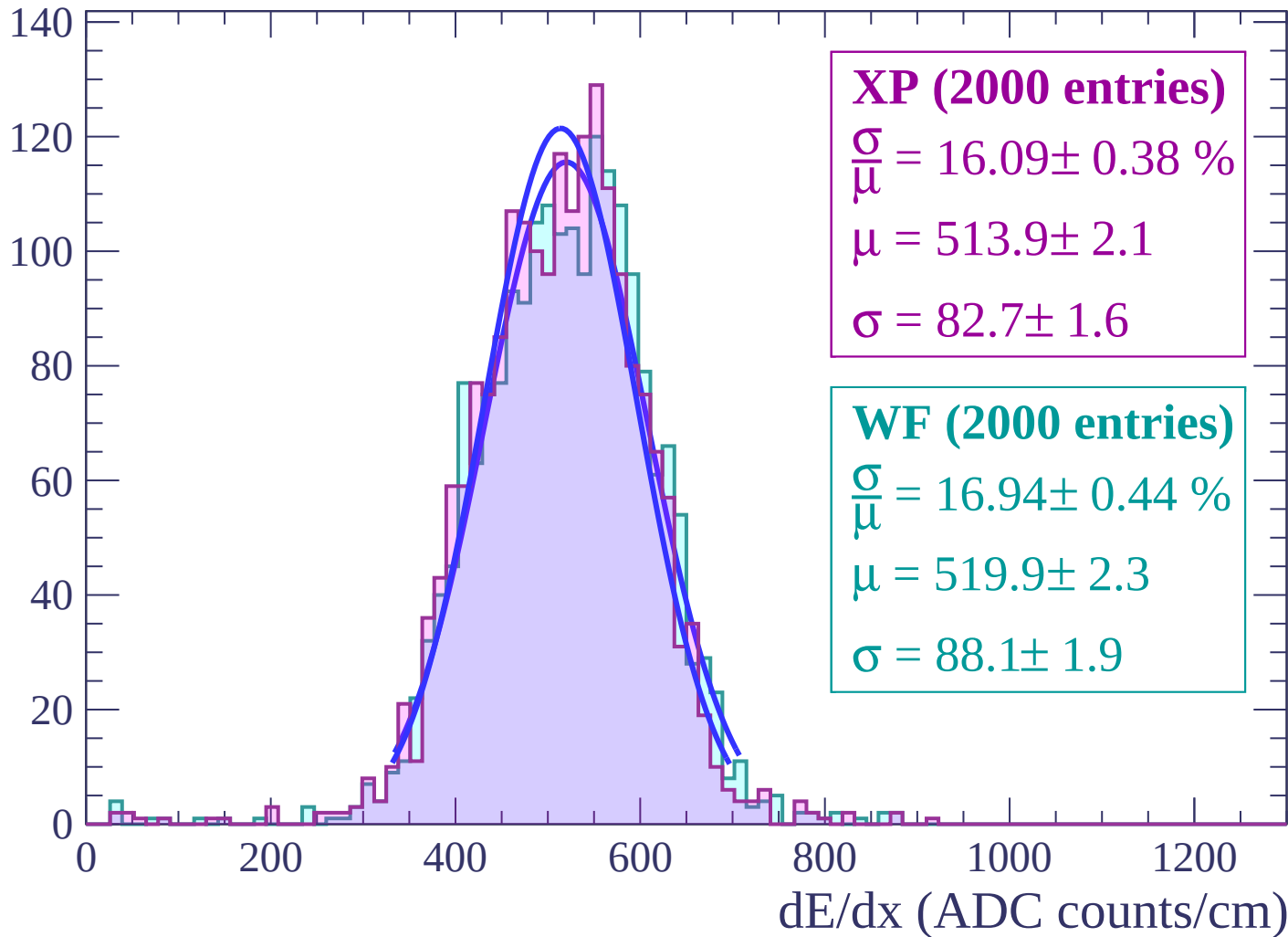
Energy loss | $-48 < \theta < -46$

Count



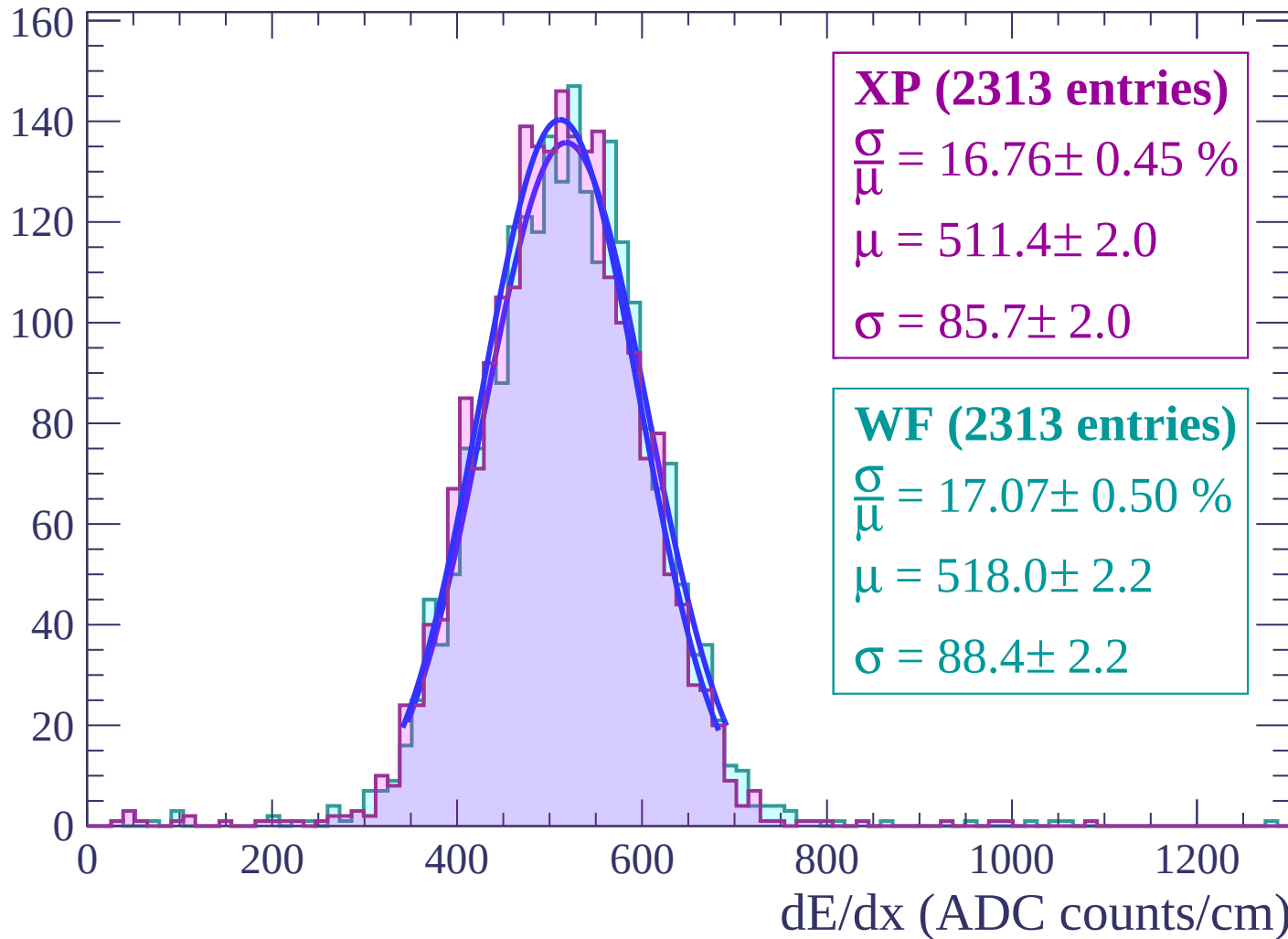
Energy loss | $-46 < \theta < -44$

Count



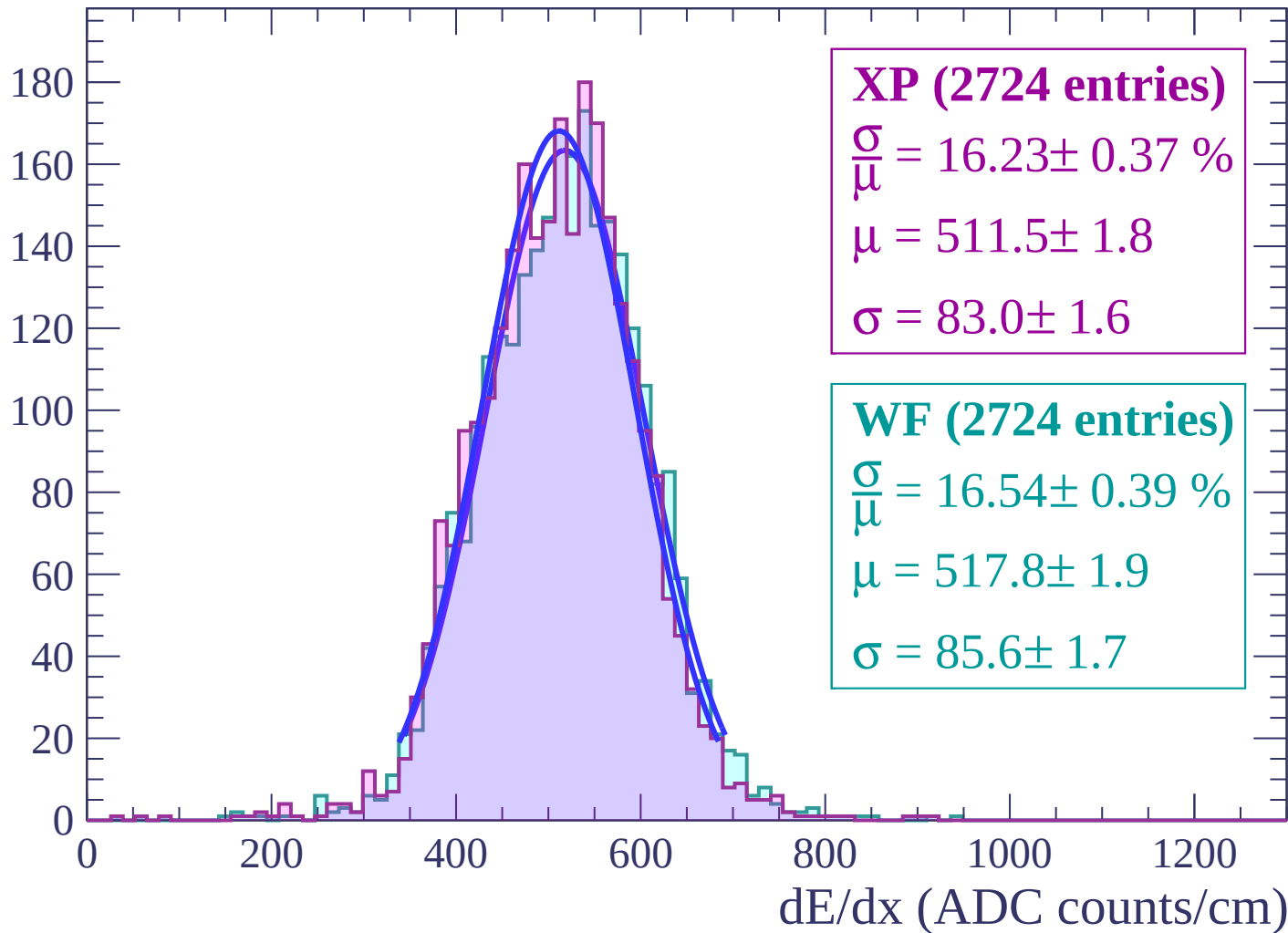
Energy loss | $-44 < \theta < -42$

Count



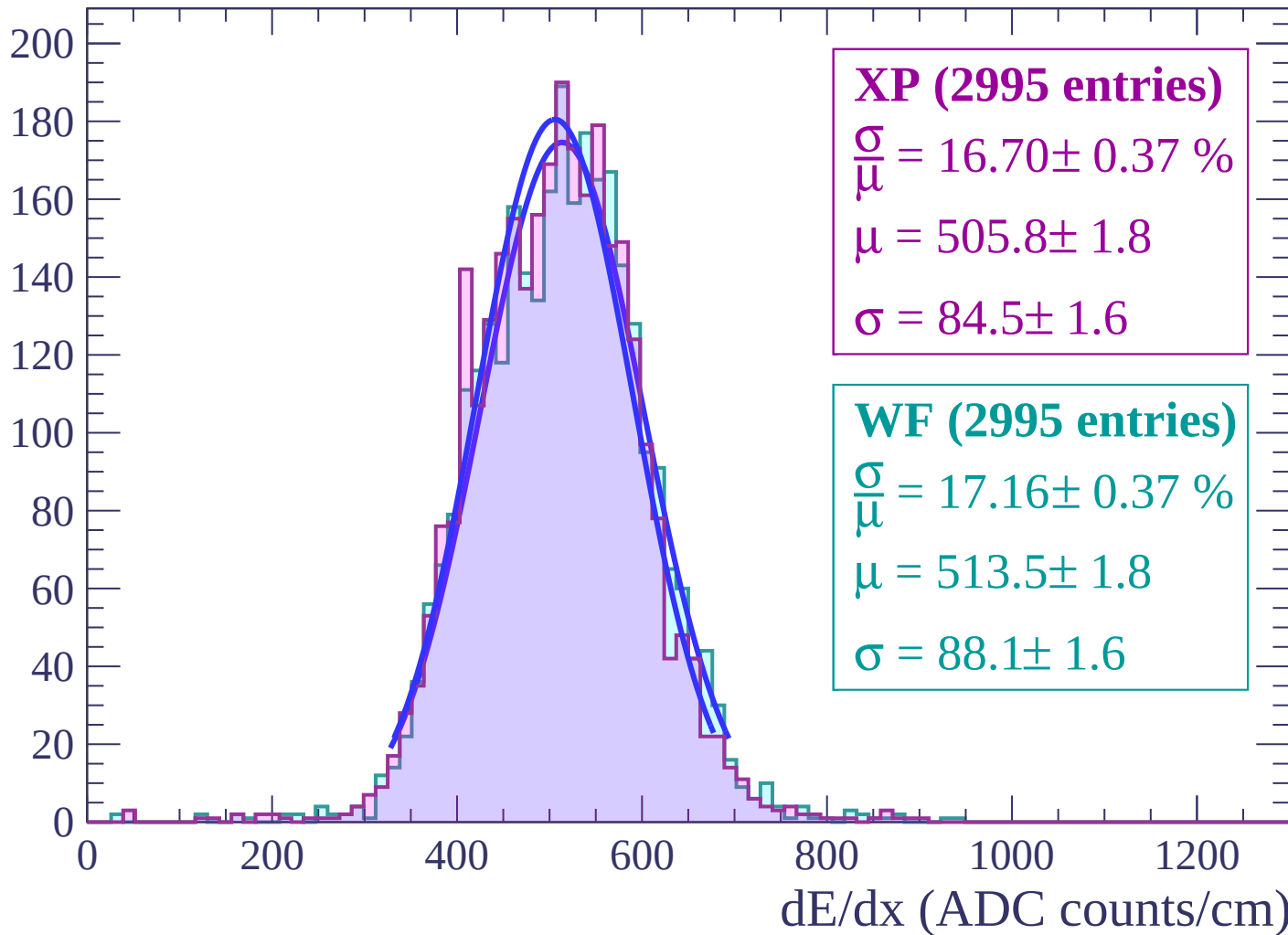
Energy loss | $-42 < \theta < -40$

Count



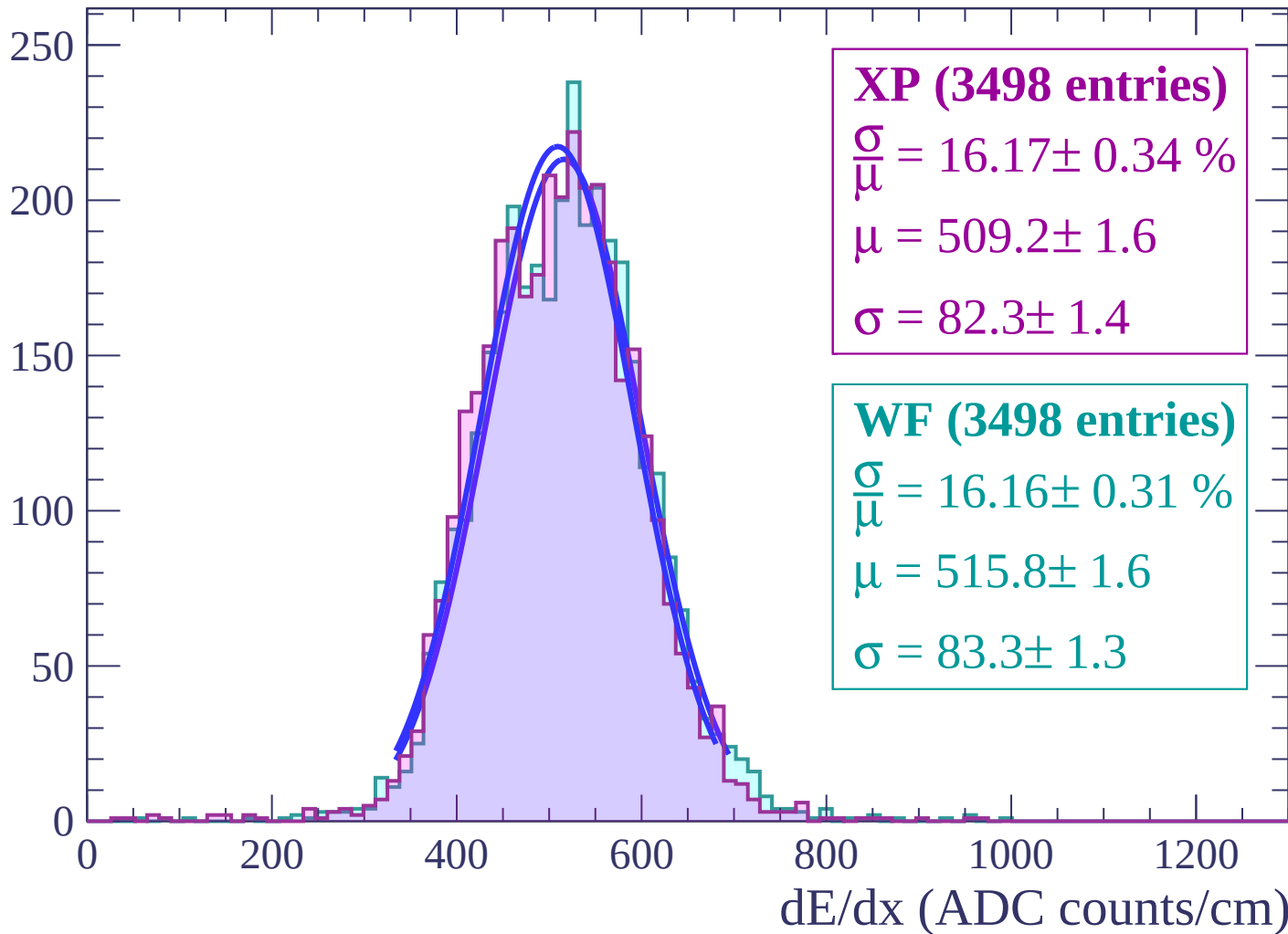
Energy loss | $-40 < \theta < -38$

Count



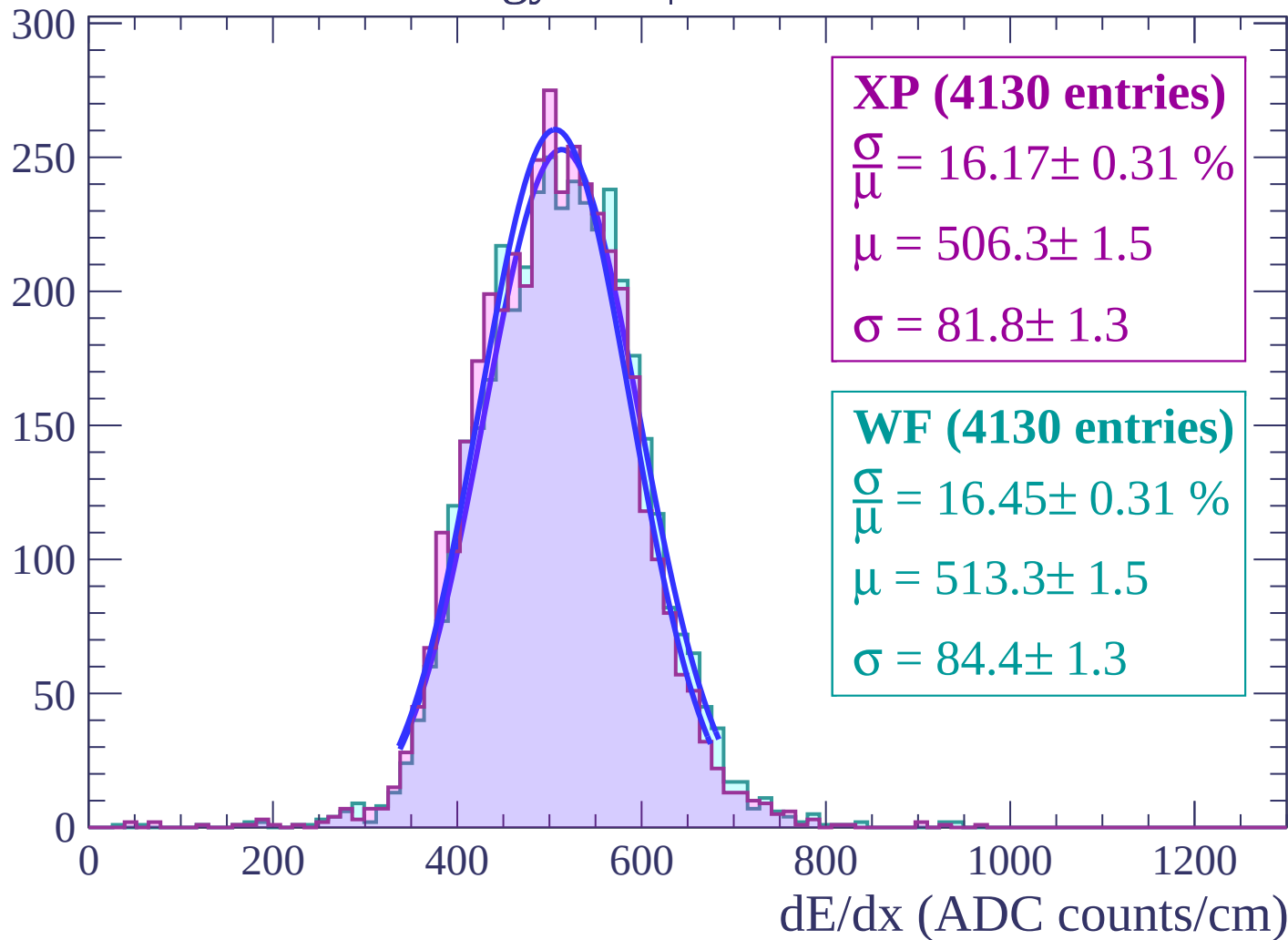
Energy loss | $-38 < \theta < -36$

Count



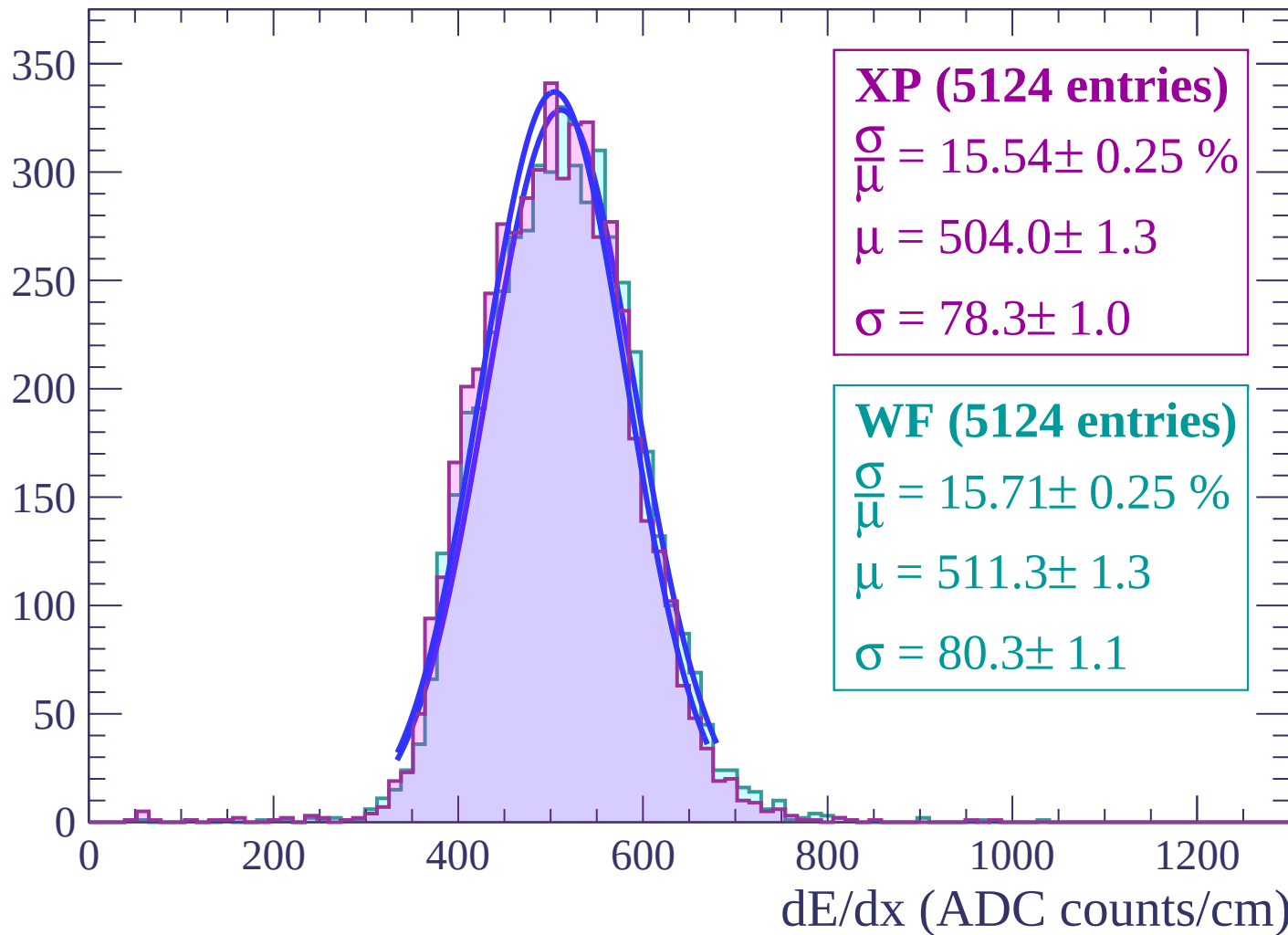
Energy loss | $-36 < \theta < -34$

Count



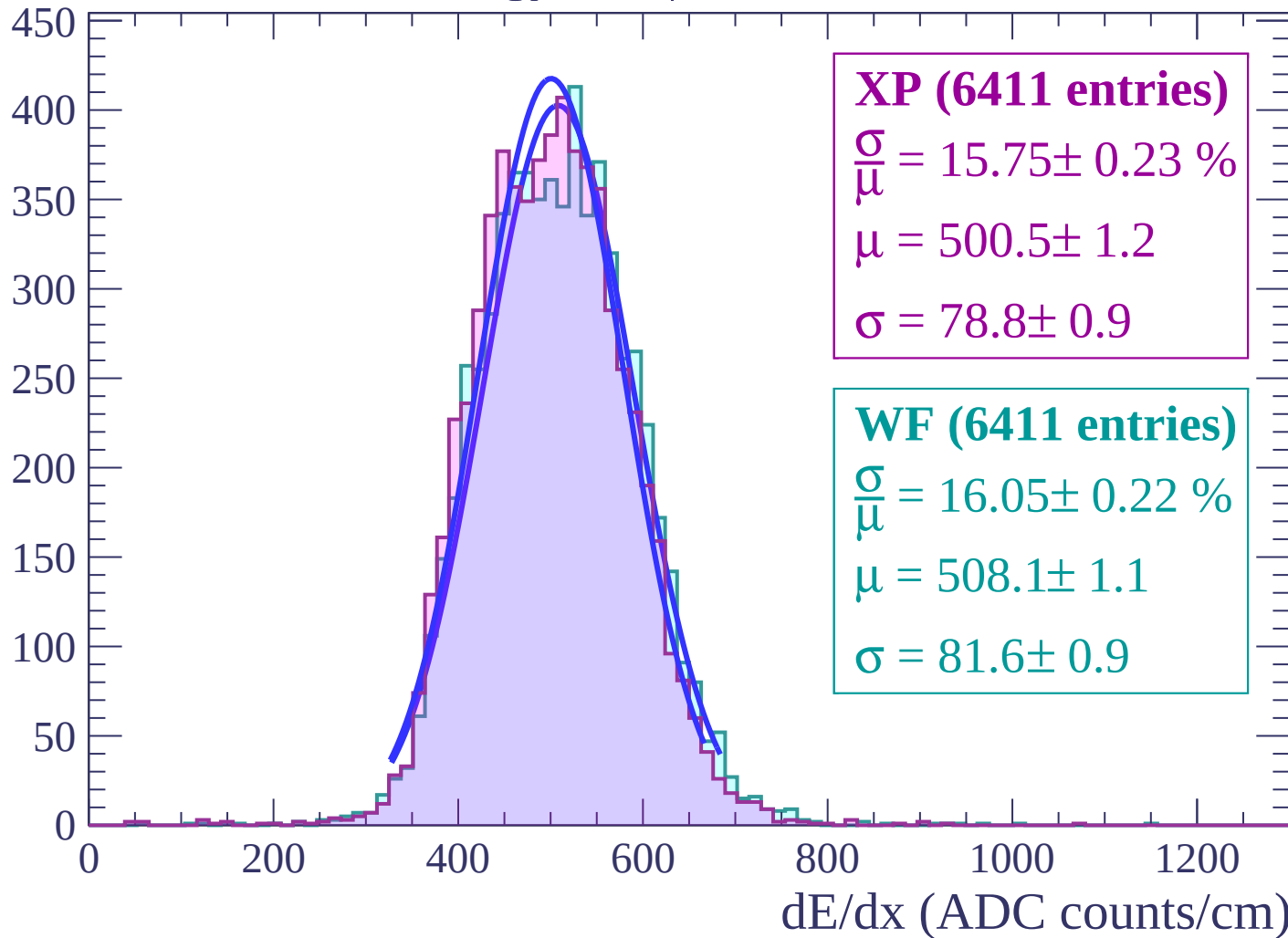
Energy loss | $-34 < \theta < -32$

Count

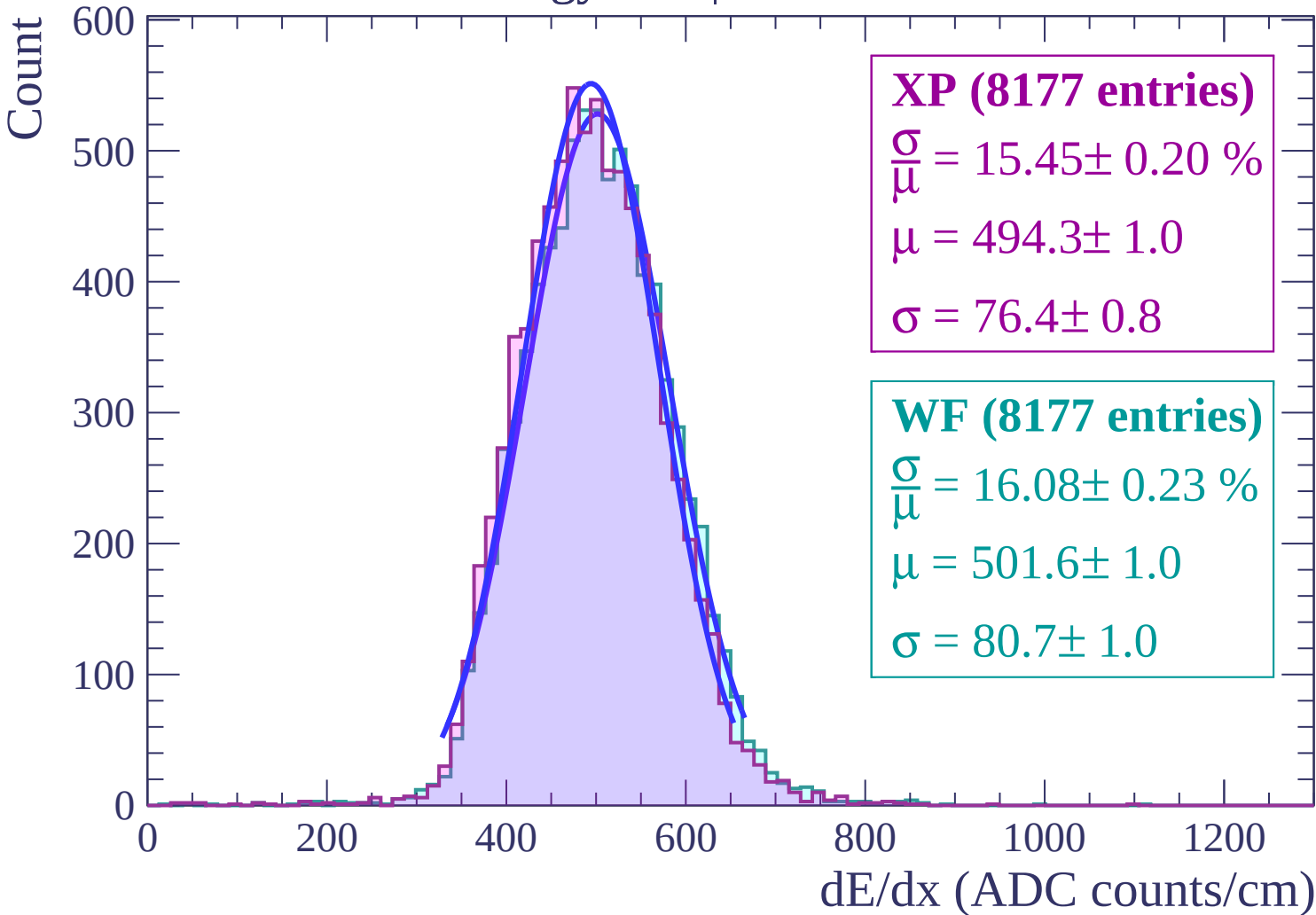


Energy loss | $-32 < \theta < -30$

Count

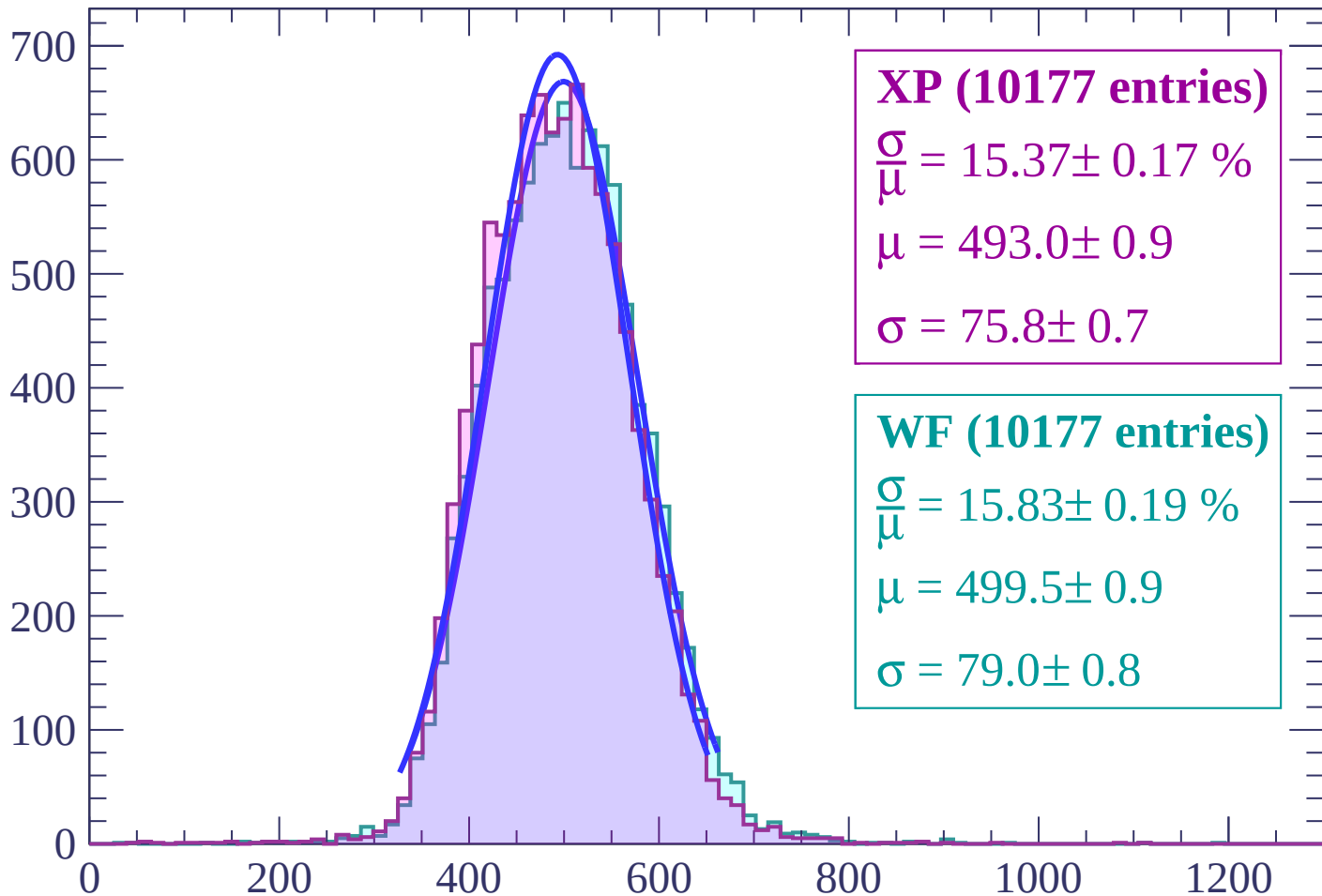


Energy loss | $-30 < \theta < -28$



Energy loss | $-28 < \theta < -26$

Count



XP (10177 entries)

$$\frac{\sigma}{\mu} = 15.37 \pm 0.17 \%$$

$$\mu = 493.0 \pm 0.9$$

$$\sigma = 75.8 \pm 0.7$$

WF (10177 entries)

$$\frac{\sigma}{\mu} = 15.83 \pm 0.19 \%$$

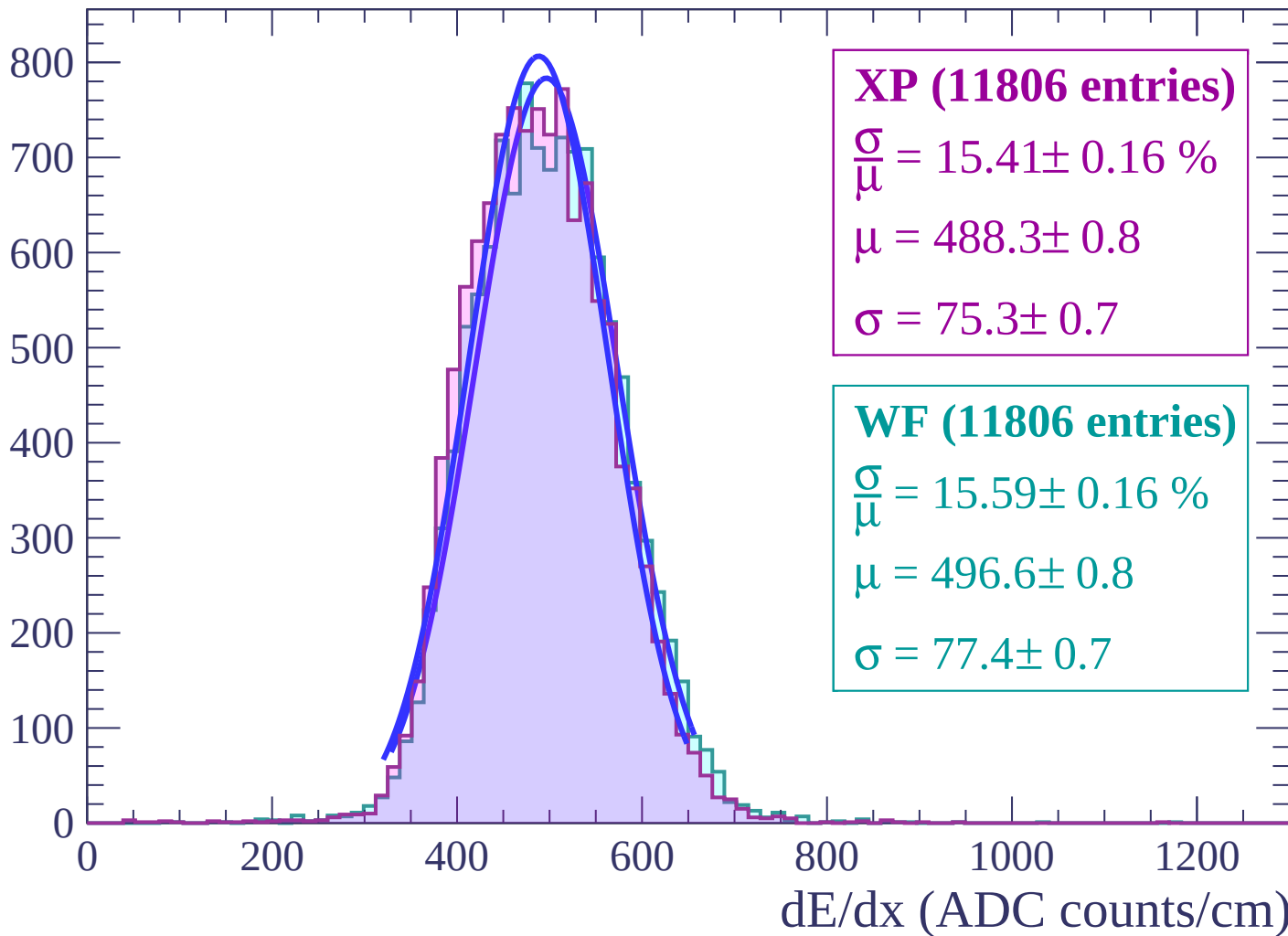
$$\mu = 499.5 \pm 0.9$$

$$\sigma = 79.0 \pm 0.8$$

dE/dx (ADC counts/cm)

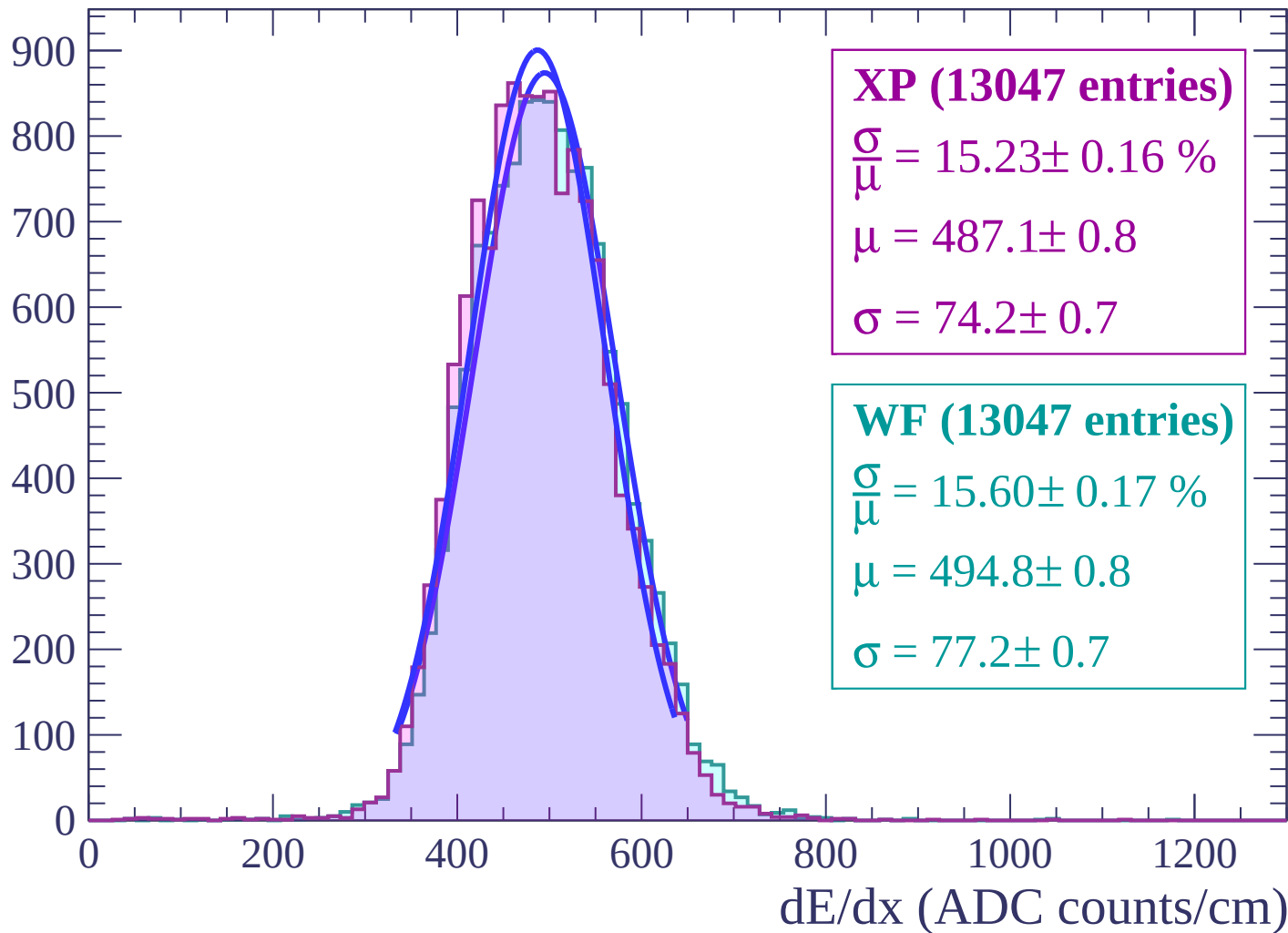
Energy loss | $-26 < \theta < -24$

Count



Energy loss | $-24 < \theta < -22$

Count



Energy loss | $-22 < \theta < -20$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (14495 entries)

$\frac{\sigma}{\mu} = 15.09 \pm 0.15 \%$

$\mu = 488.0 \pm 0.7$

$\sigma = 73.7 \pm 0.6$

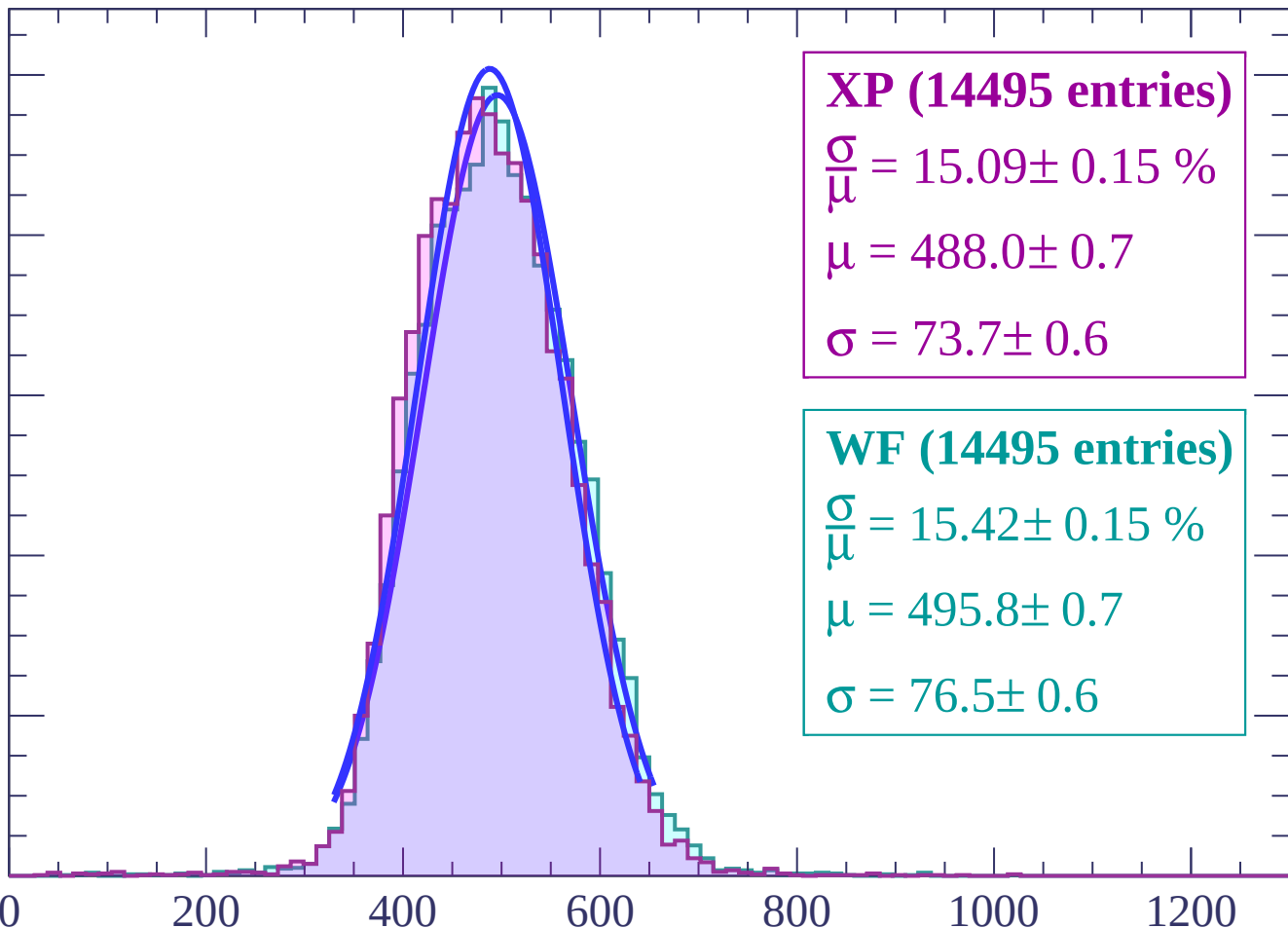
WF (14495 entries)

$\frac{\sigma}{\mu} = 15.42 \pm 0.15 \%$

$\mu = 495.8 \pm 0.7$

$\sigma = 76.5 \pm 0.6$

dE/dx (ADC counts/cm)



Energy loss | $-20 < \theta < -18$

Count

1000

800

600

400

200

0

200

400

600

800

1000

1200

XP (15959 entries)

$$\frac{\sigma}{\mu} = 15.28 \pm 0.13 \%$$

$$\mu = 486.7 \pm 0.7$$

$$\sigma = 74.4 \pm 0.5$$

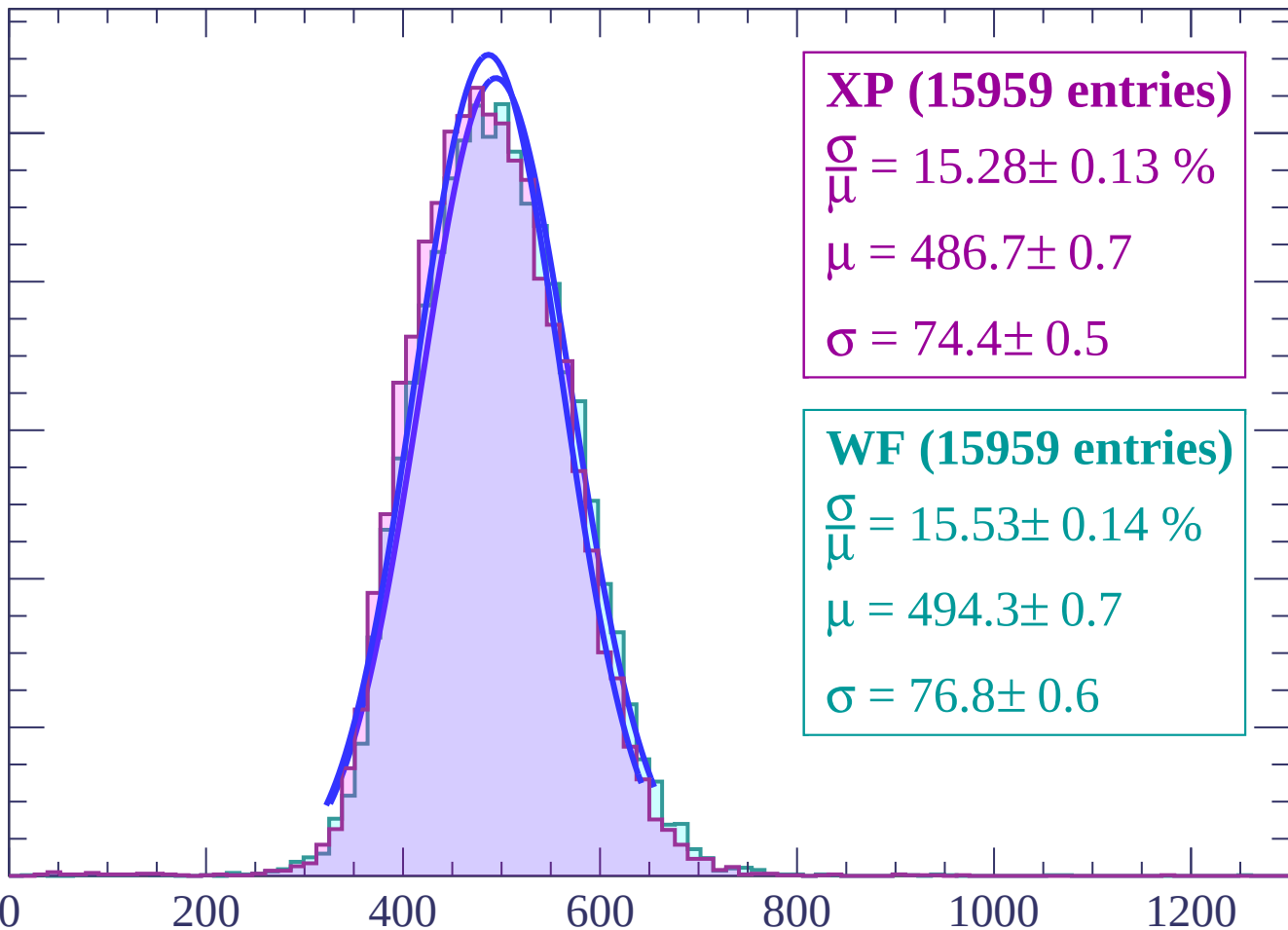
WF (15959 entries)

$$\frac{\sigma}{\mu} = 15.53 \pm 0.14 \%$$

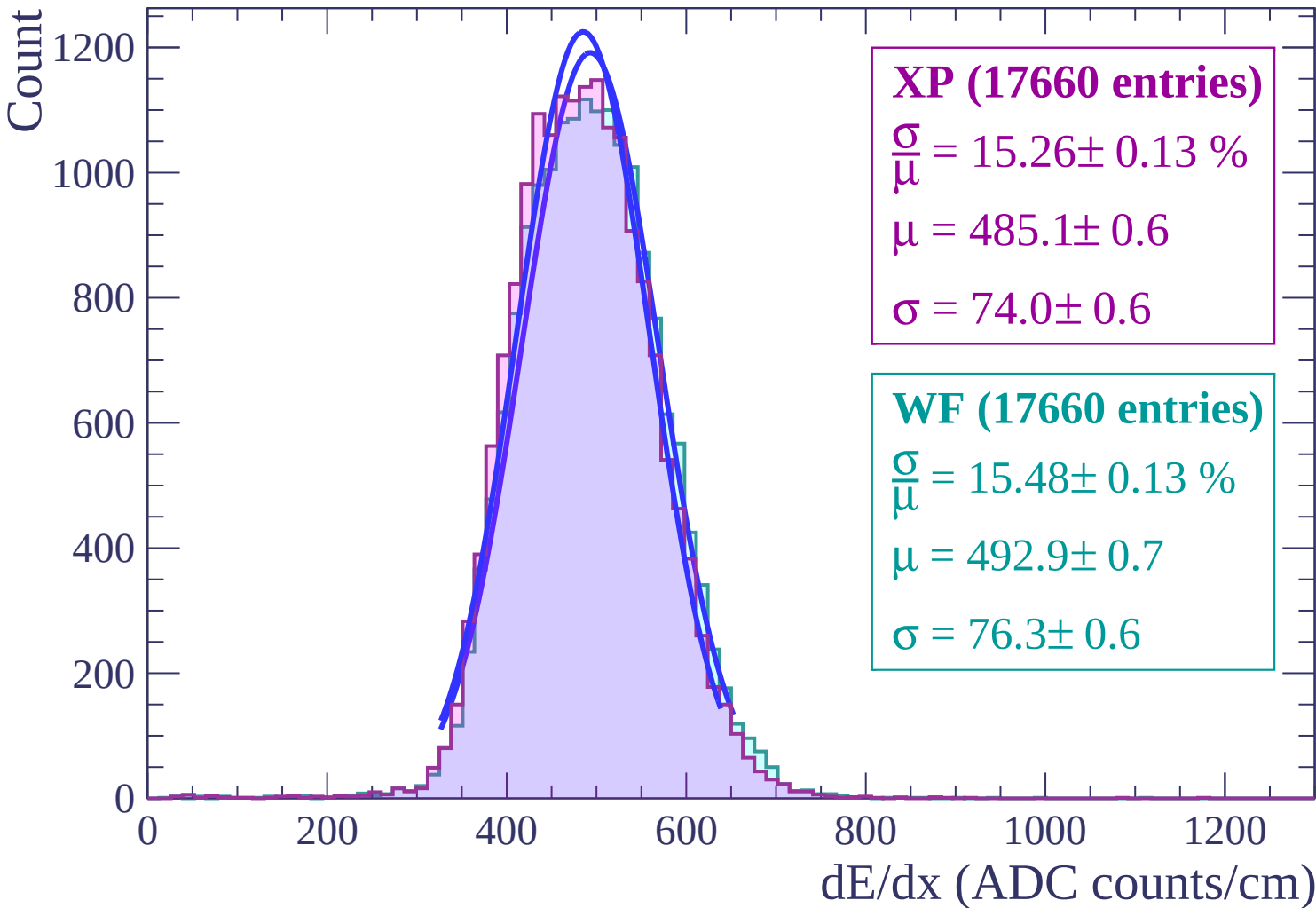
$$\mu = 494.3 \pm 0.7$$

$$\sigma = 76.8 \pm 0.6$$

dE/dx (ADC counts/cm)

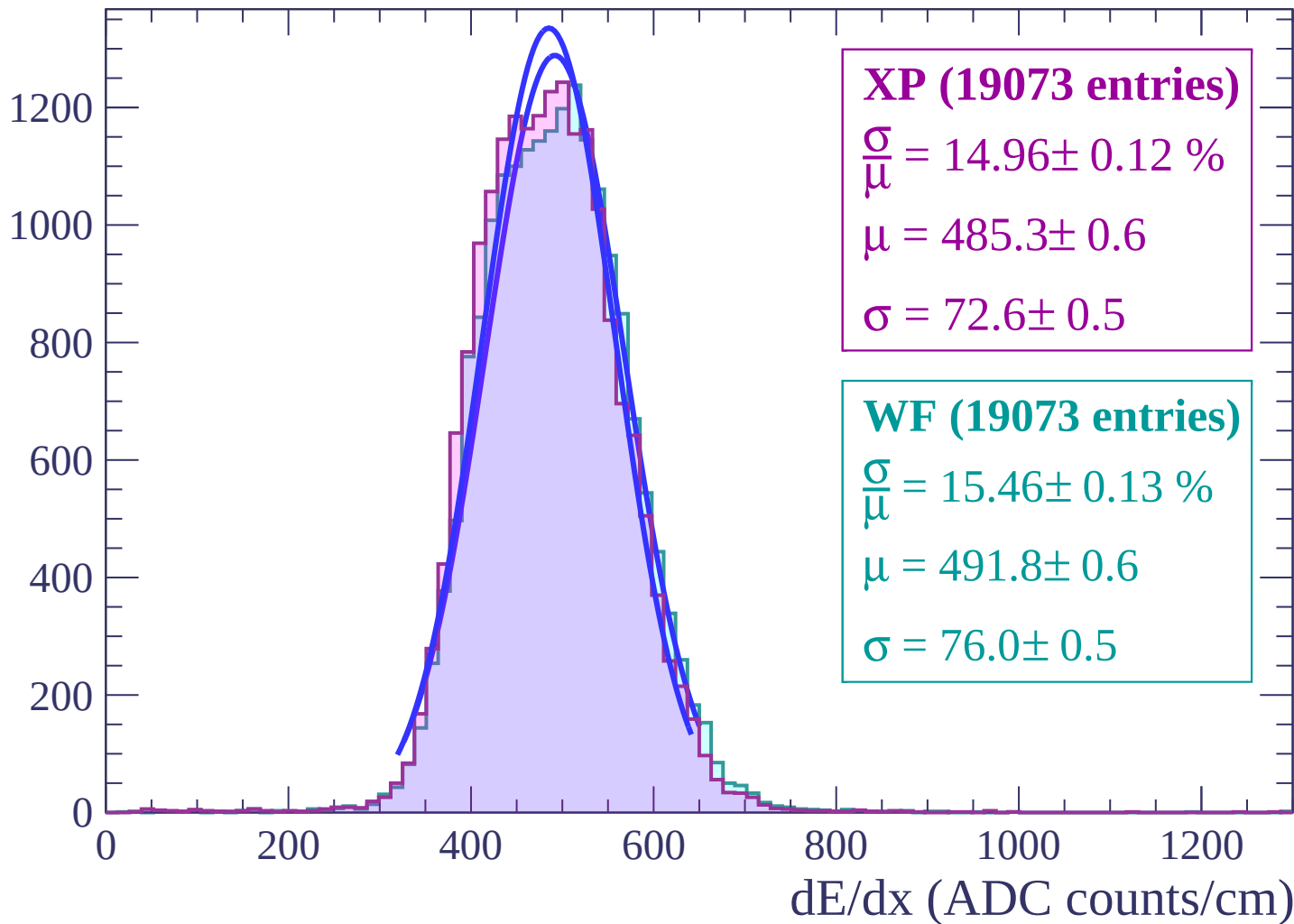


Energy loss | $-18 < \theta < -16$



Energy loss | $-16 < \theta < -14$

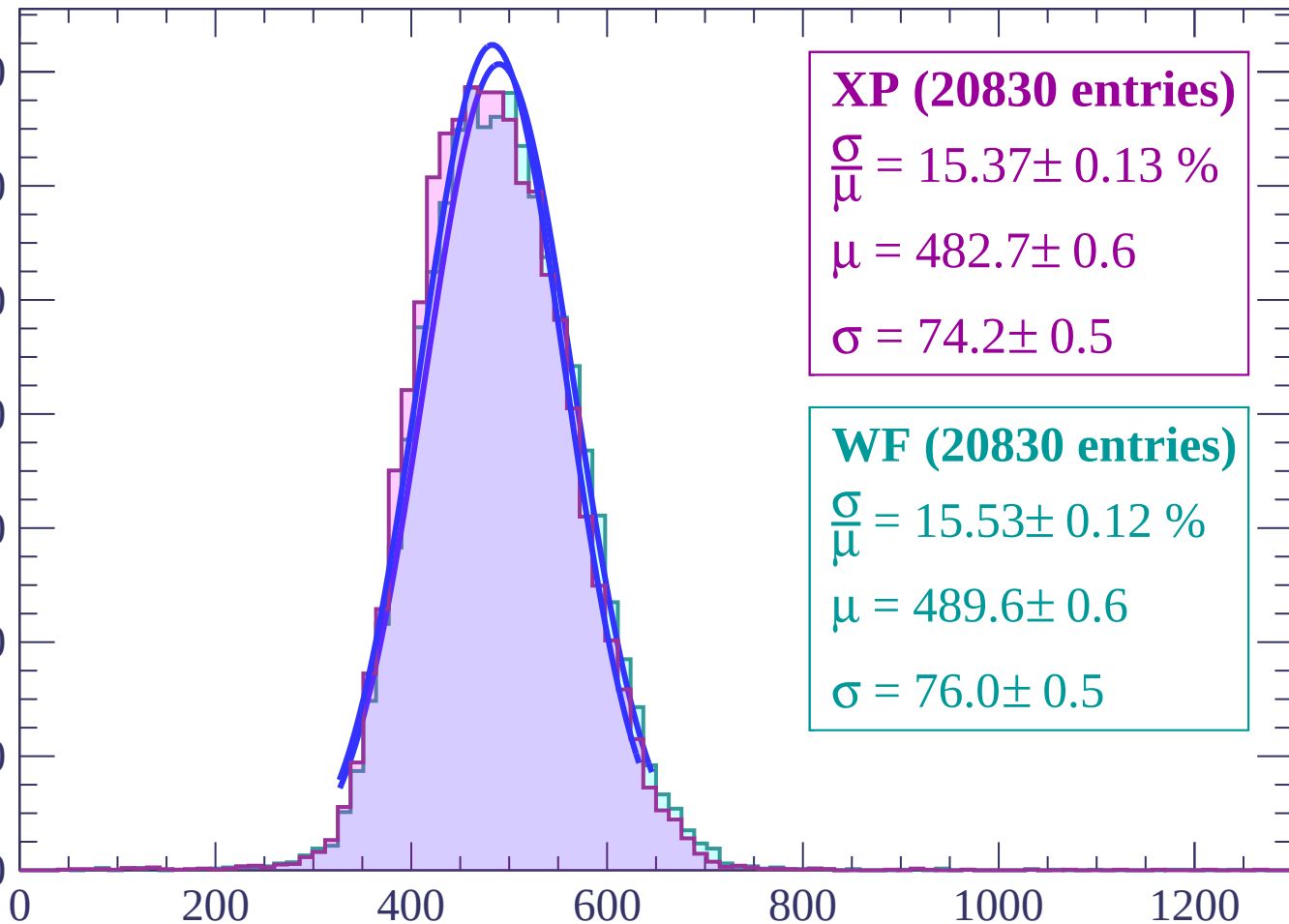
Count



Energy loss | $-14 < \theta < -12$

Count

1400
1200
1000
800
600
400
200
0



XP (20830 entries)

$\frac{\sigma}{\mu} = 15.37 \pm 0.13 \%$

$\mu = 482.7 \pm 0.6$

$\sigma = 74.2 \pm 0.5$

WF (20830 entries)

$\frac{\sigma}{\mu} = 15.53 \pm 0.12 \%$

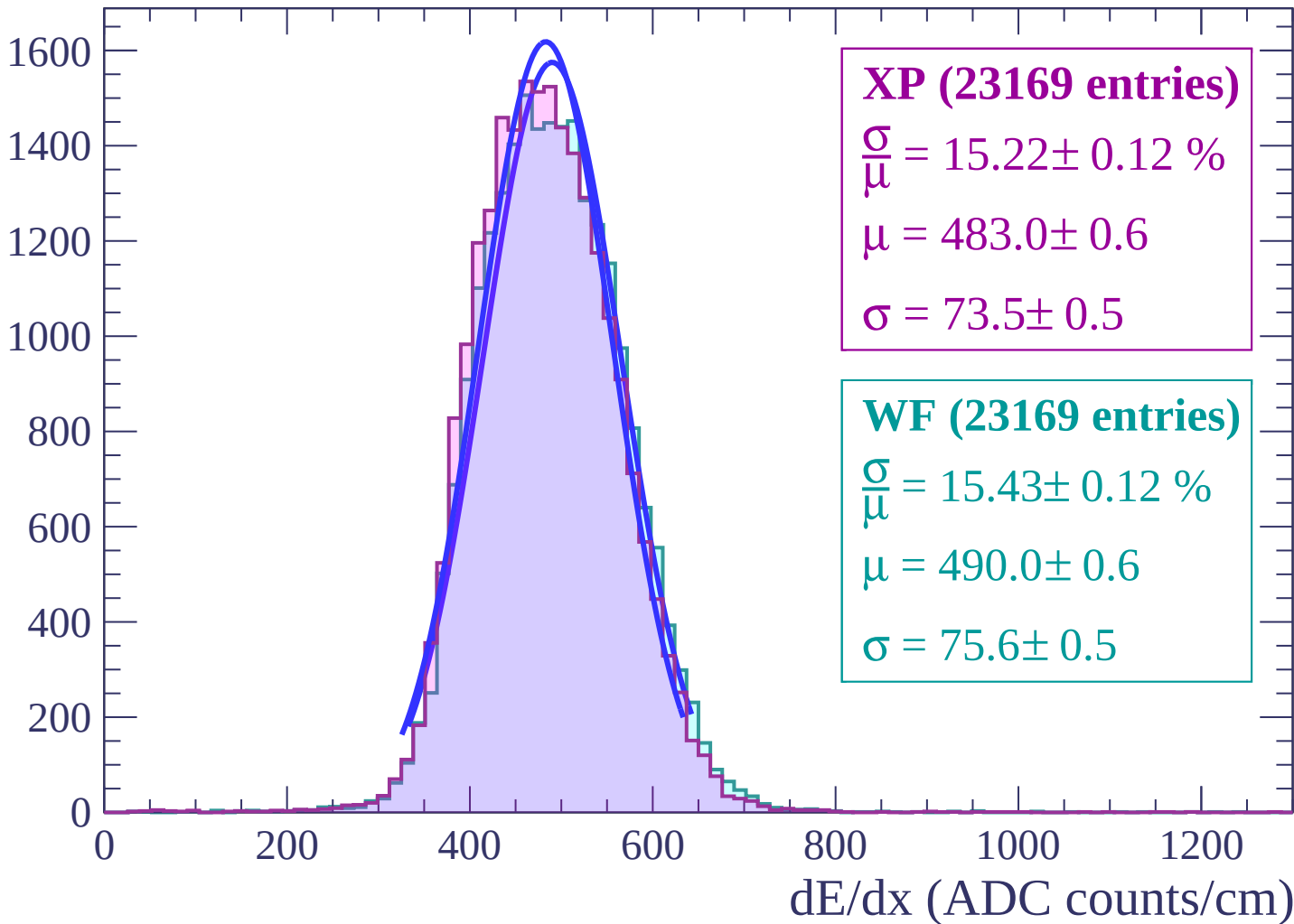
$\mu = 489.6 \pm 0.6$

$\sigma = 76.0 \pm 0.5$

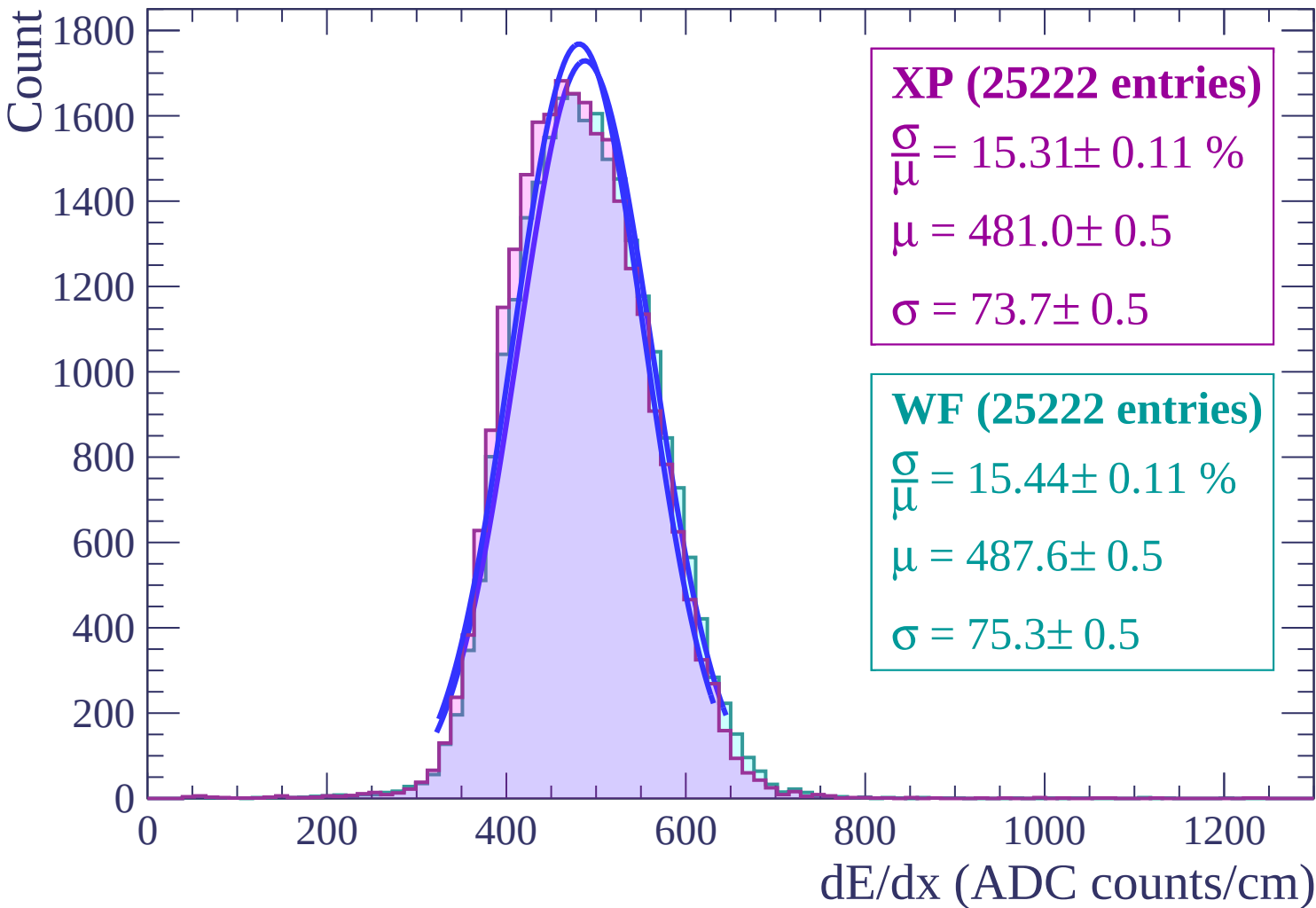
dE/dx (ADC counts/cm)

Energy loss | $-12 < \theta < -10$

Count



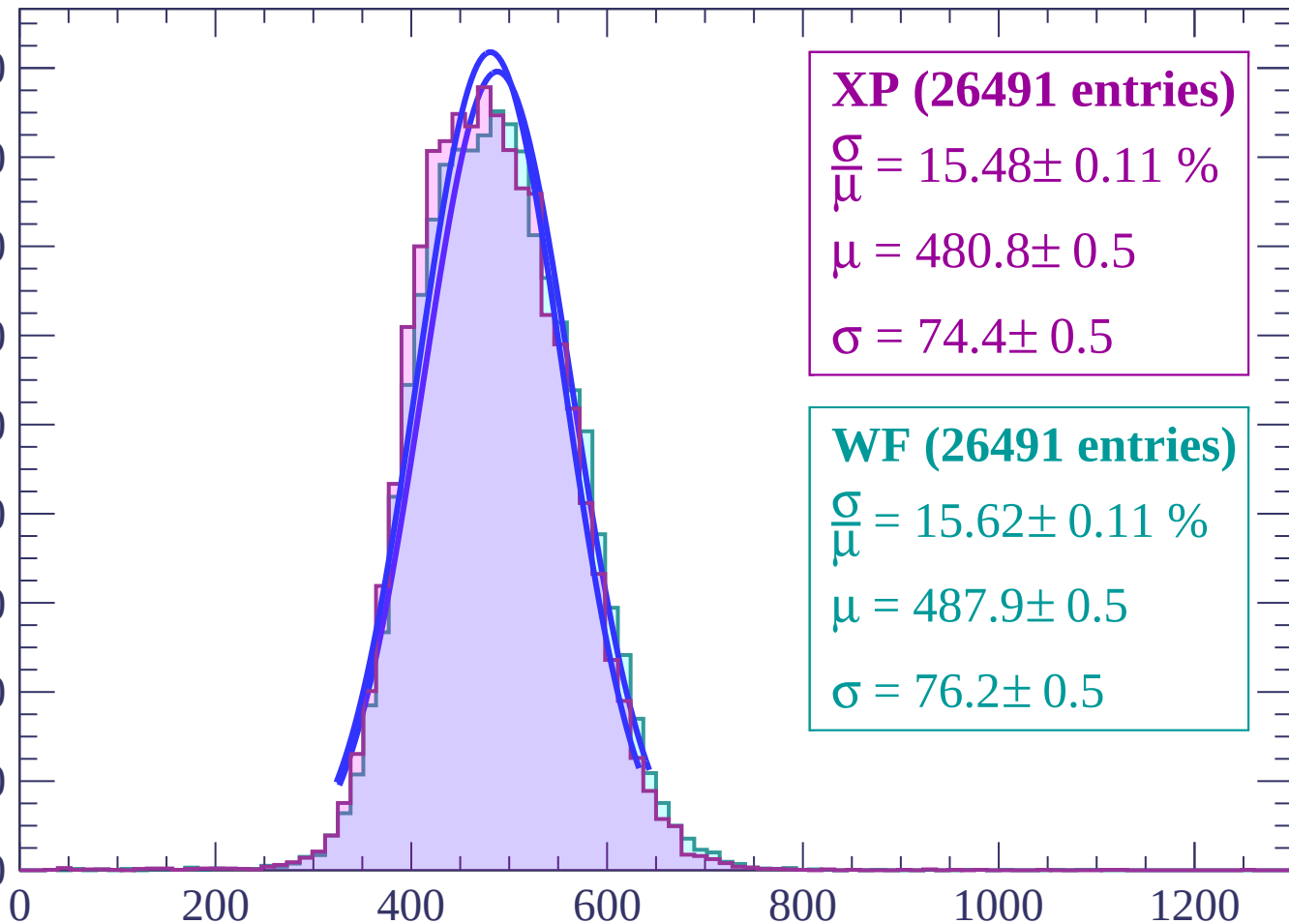
Energy loss | $-10 < \theta < -8$



Energy loss | $-8 < \theta < -6$

Count

1800
1600
1400
1200
1000
800
600
400
200
0



XP (26491 entries)

$\frac{\sigma}{\mu} = 15.48 \pm 0.11 \%$

$\mu = 480.8 \pm 0.5$

$\sigma = 74.4 \pm 0.5$

WF (26491 entries)

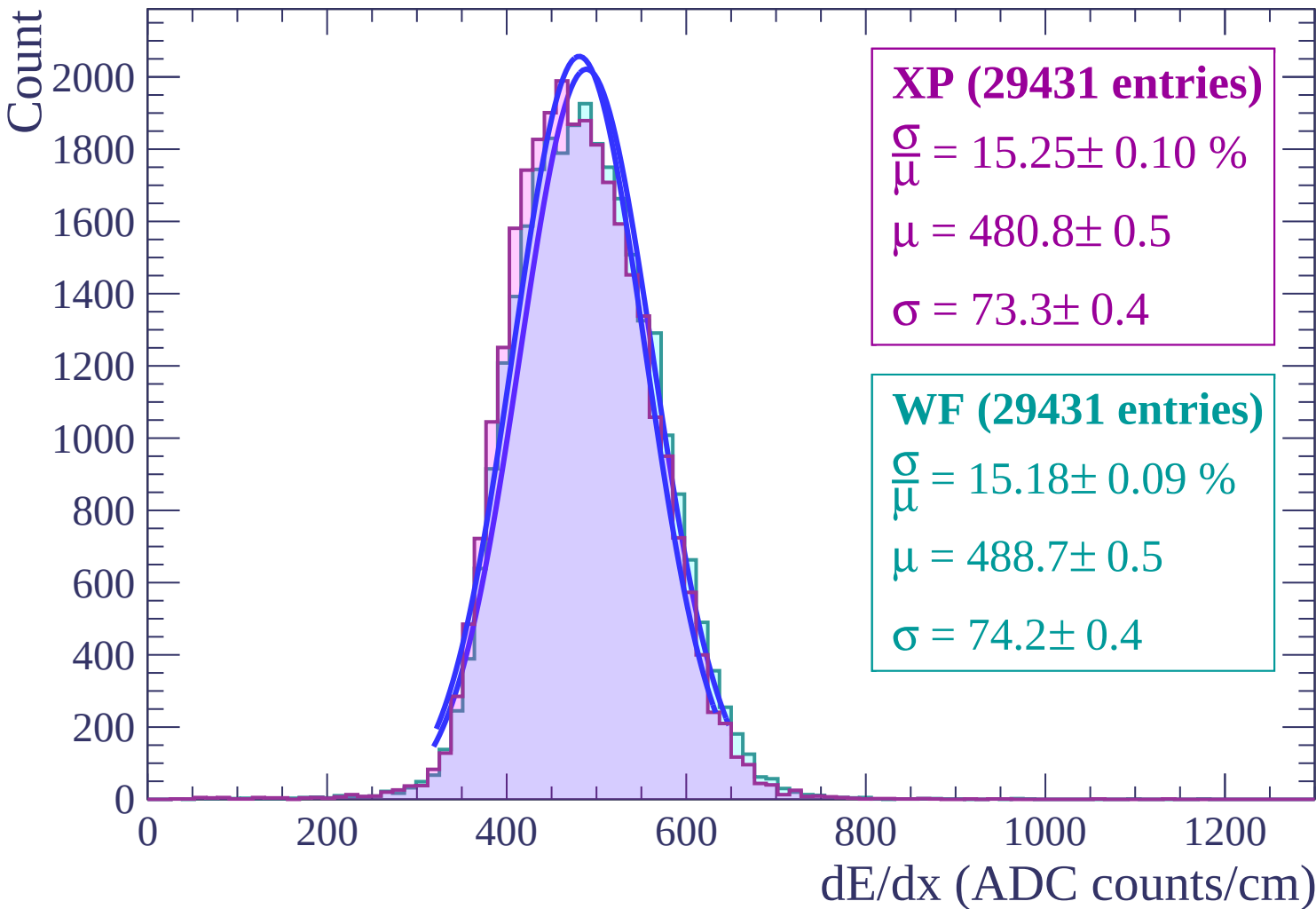
$\frac{\sigma}{\mu} = 15.62 \pm 0.11 \%$

$\mu = 487.9 \pm 0.5$

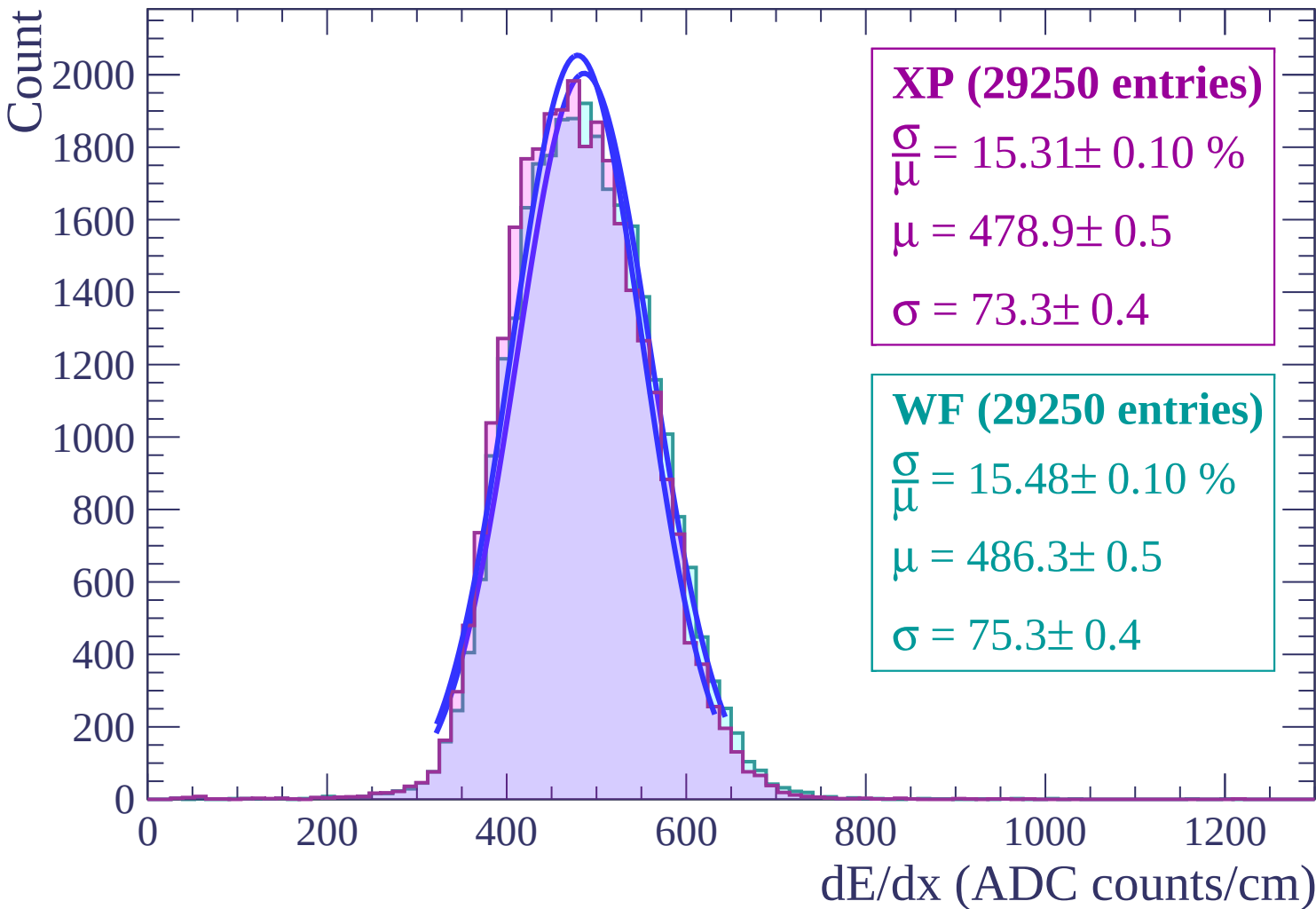
$\sigma = 76.2 \pm 0.5$

dE/dx (ADC counts/cm)

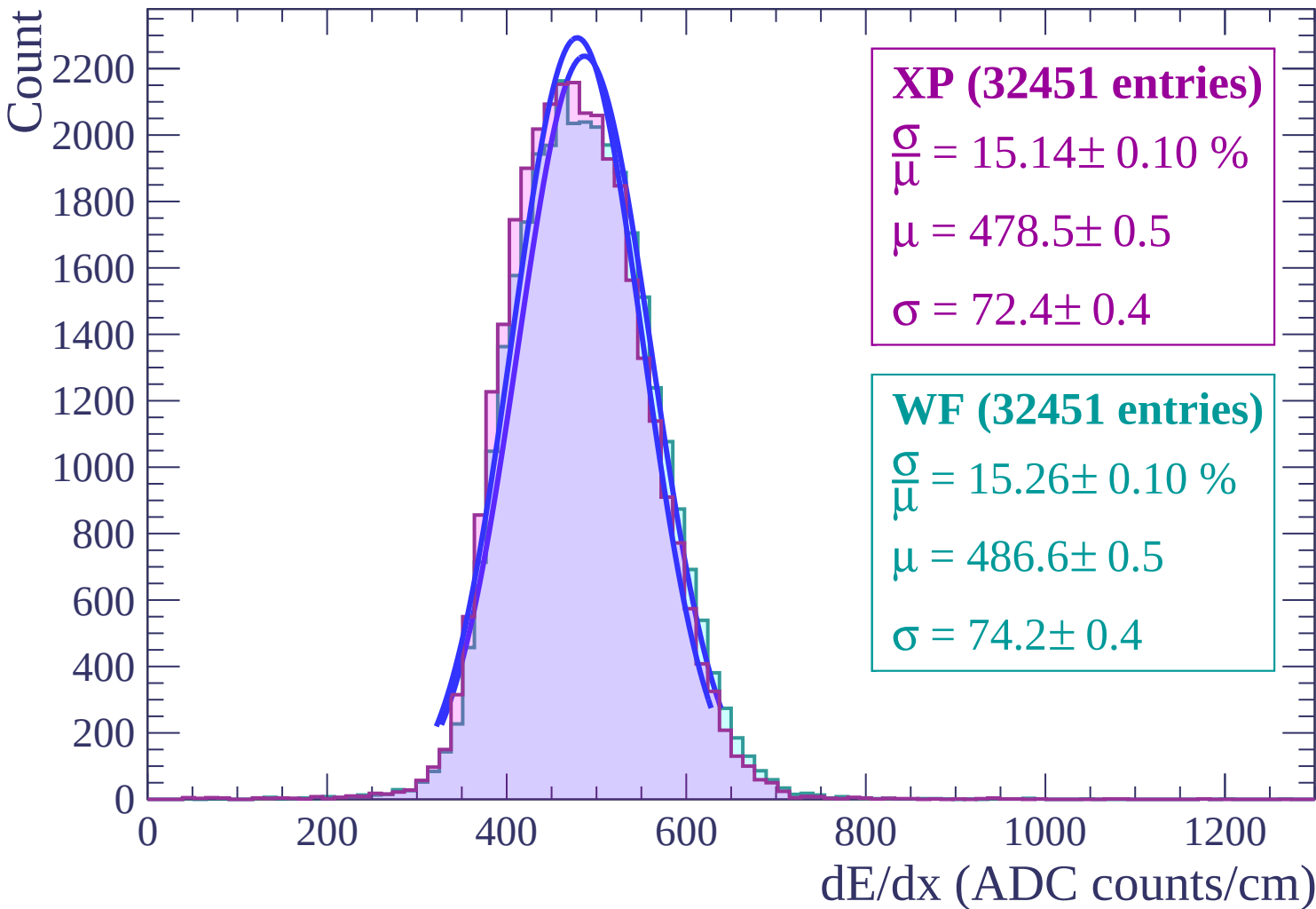
Energy loss | $-6 < \theta < -4$



Energy loss | $-4 < \theta < -2$



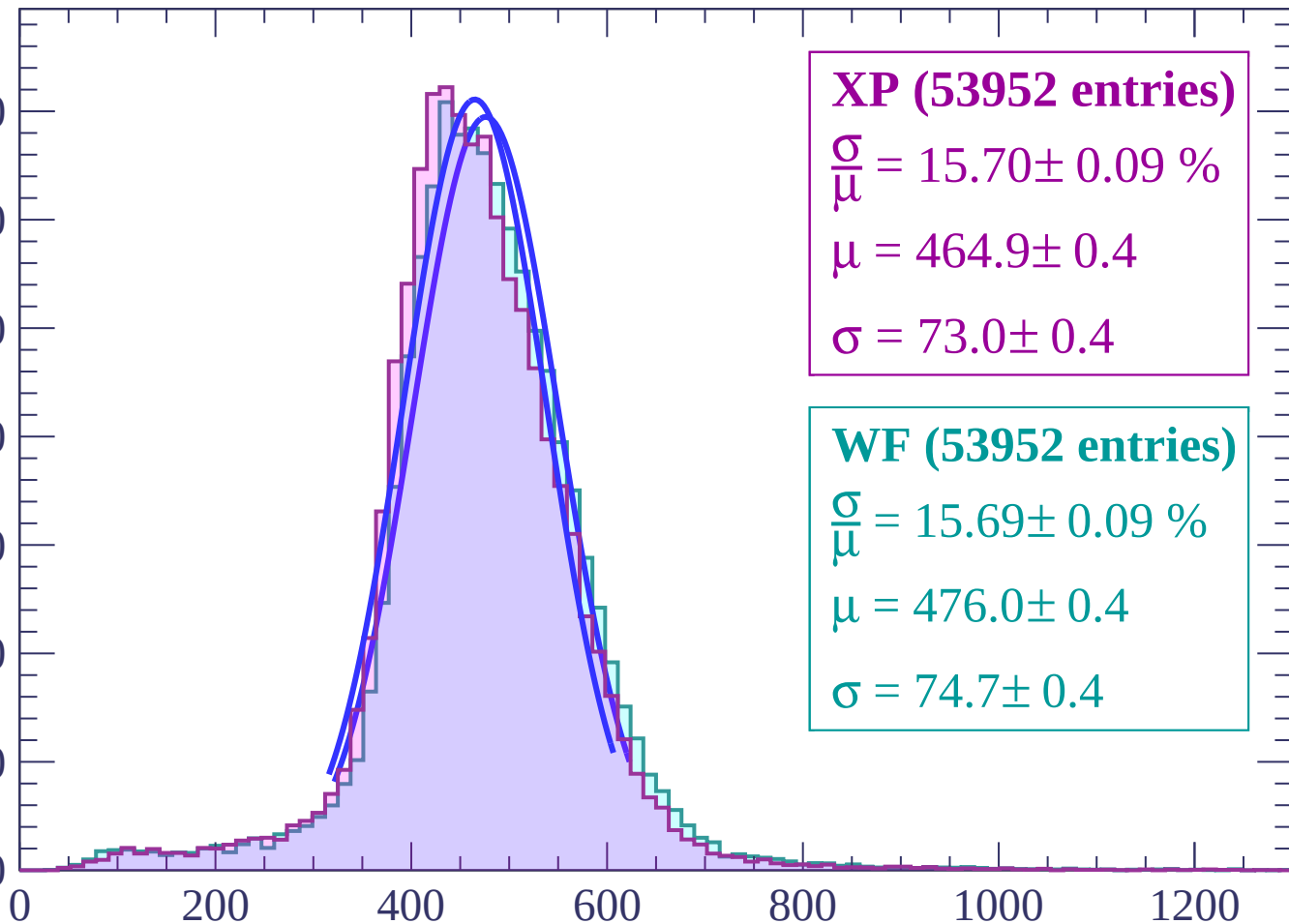
Energy loss | $-2 < \theta < 0$



Energy loss | $0 < \theta < 2$

Count

3500
3000
2500
2000
1500
1000
500
0



XP (53952 entries)

$\frac{\sigma}{\mu} = 15.70 \pm 0.09 \%$

$\mu = 464.9 \pm 0.4$

$\sigma = 73.0 \pm 0.4$

WF (53952 entries)

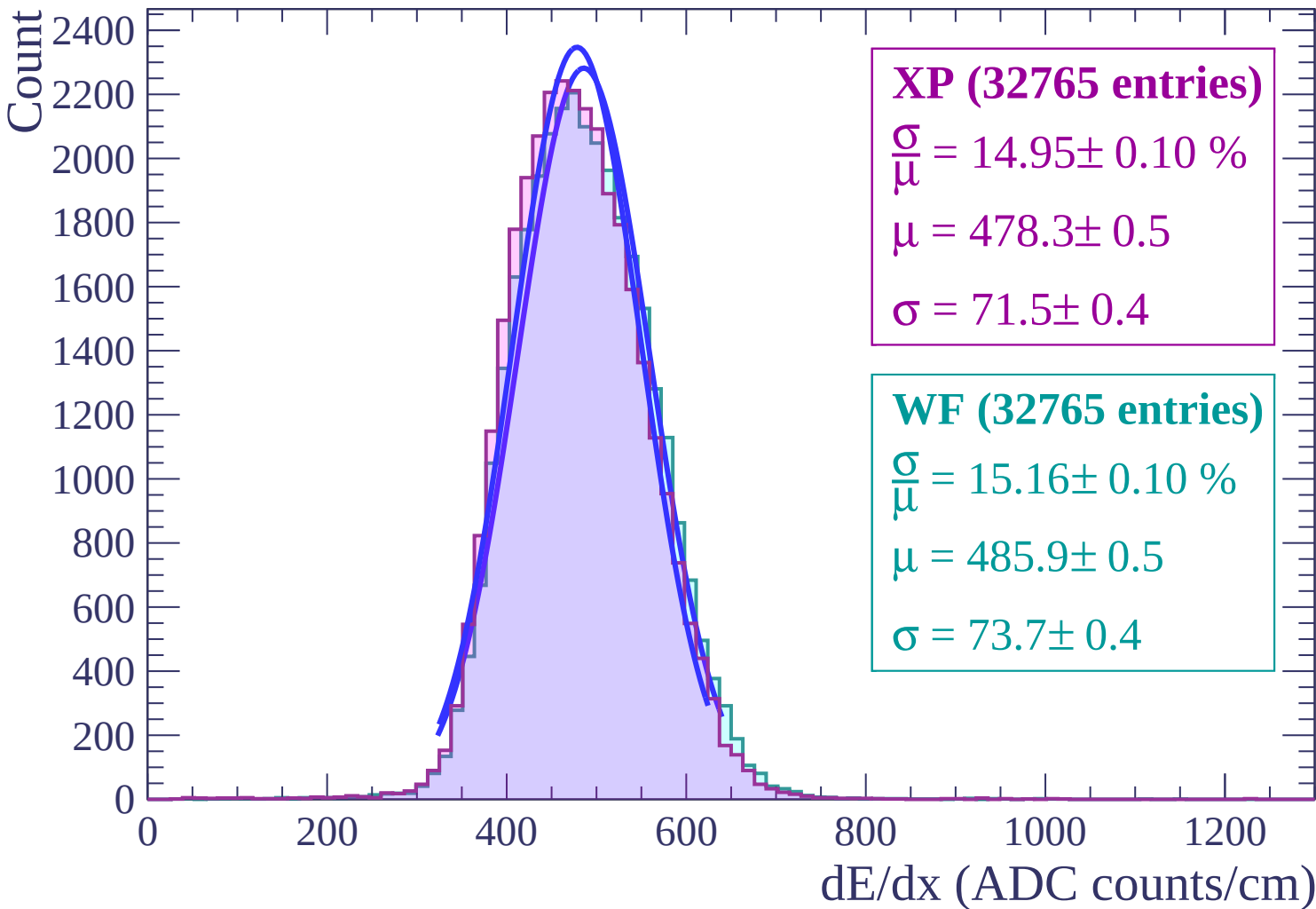
$\frac{\sigma}{\mu} = 15.69 \pm 0.09 \%$

$\mu = 476.0 \pm 0.4$

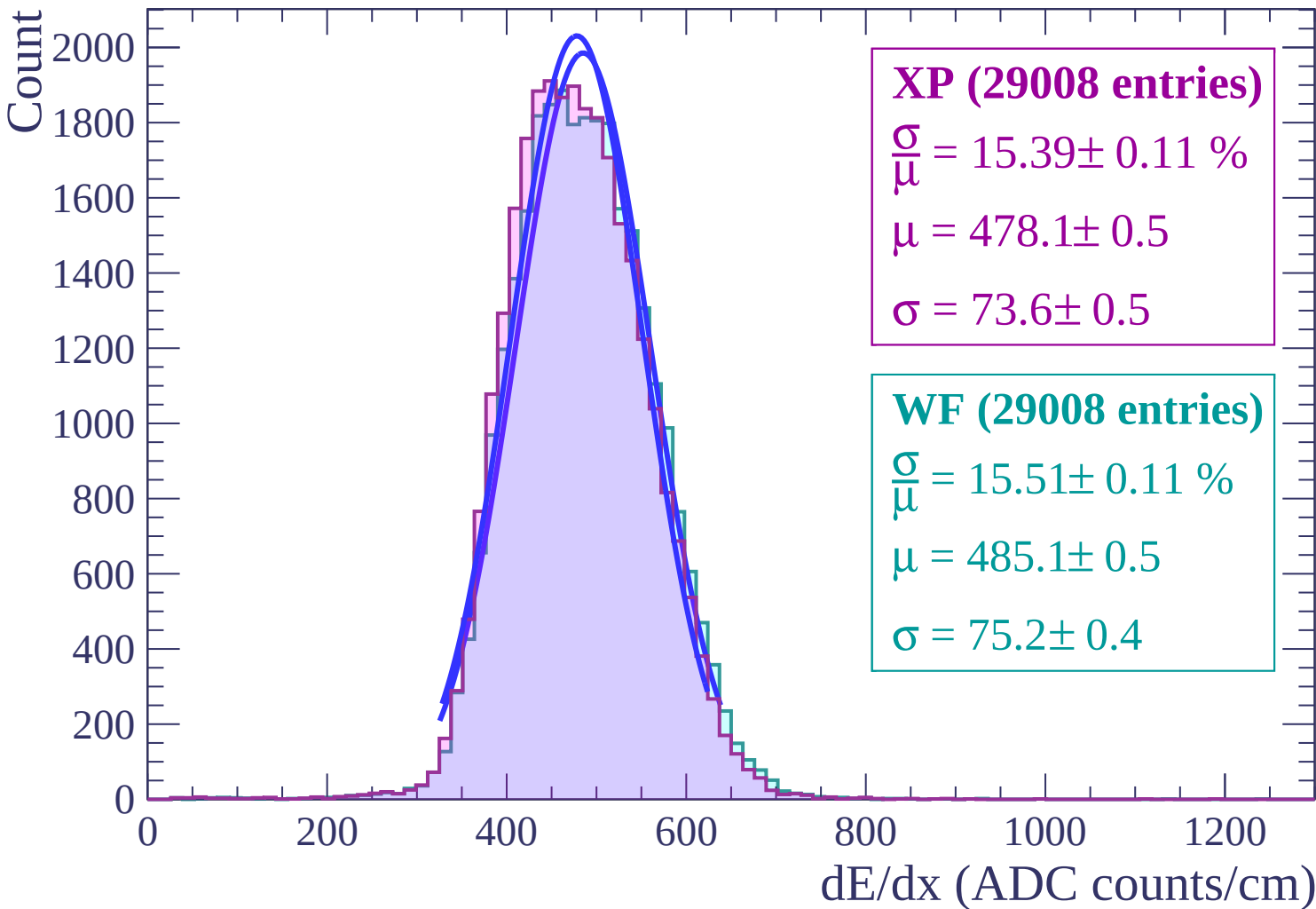
$\sigma = 74.7 \pm 0.4$

dE/dx (ADC counts/cm)

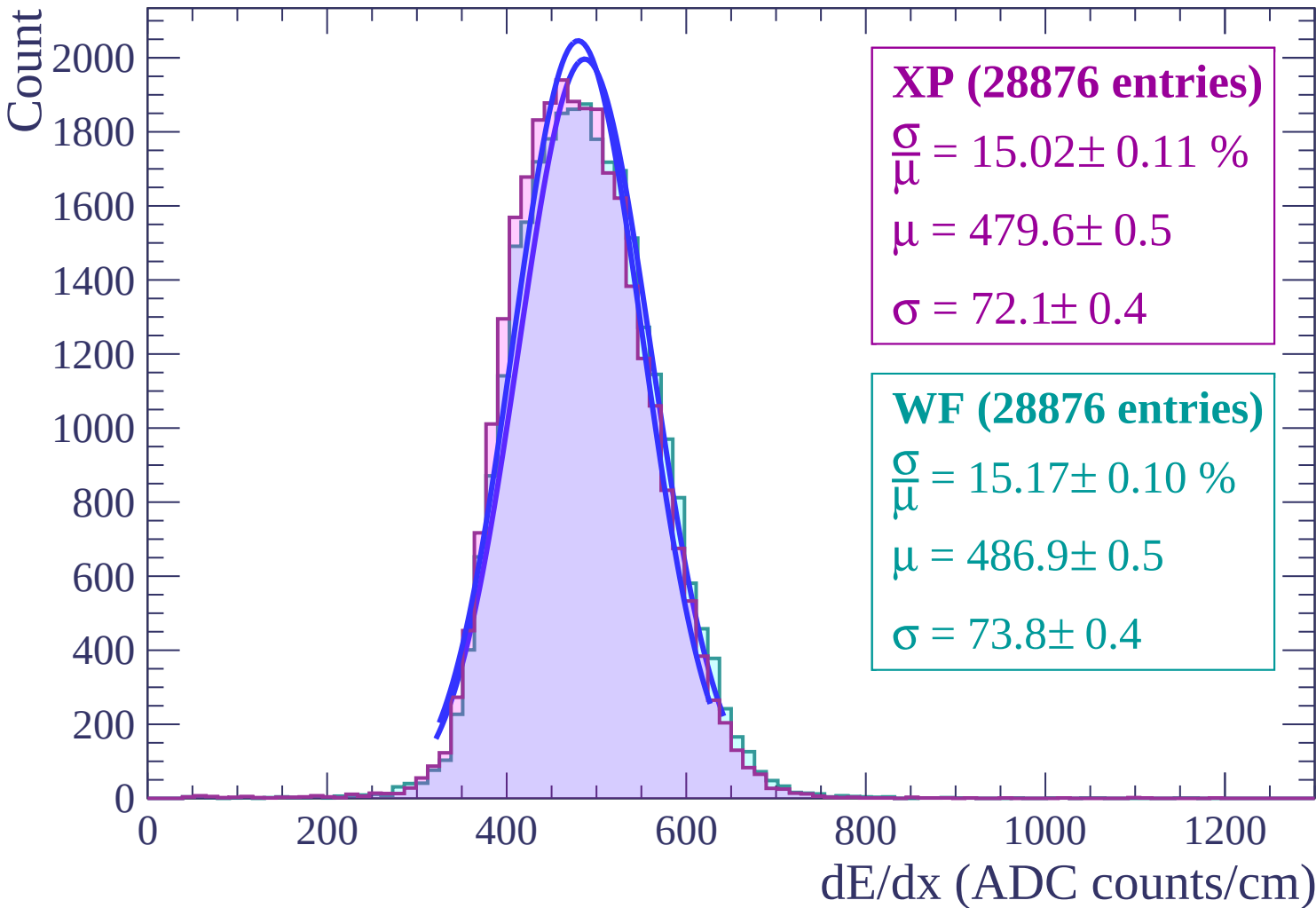
Energy loss | $2 < \theta < 4$



Energy loss | $4 < \theta < 6$



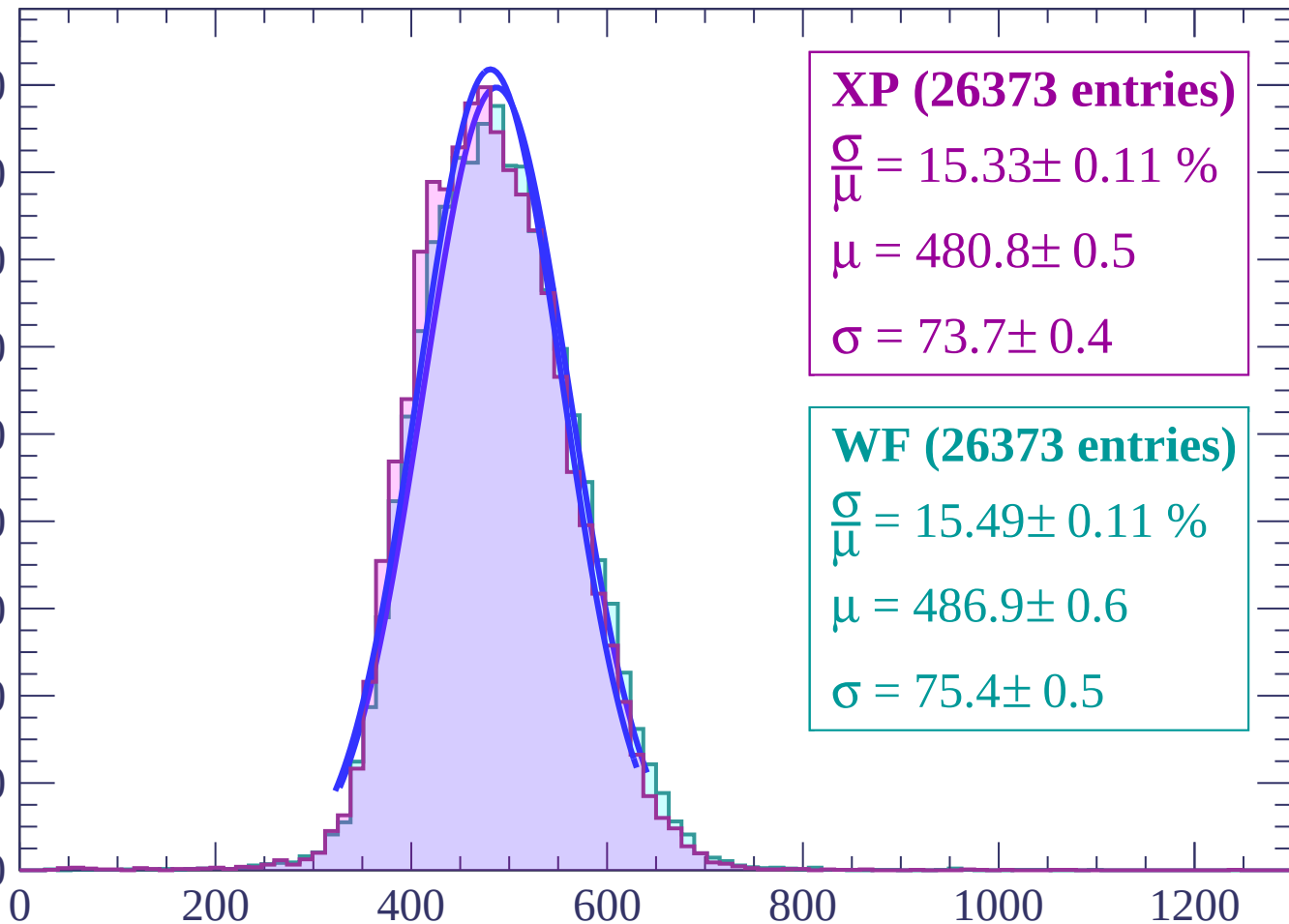
Energy loss | $6 < \theta < 8$



Energy loss | $8 < \theta < 10$

Count

1800
1600
1400
1200
1000
800
600
400
200
0



XP (26373 entries)

$\frac{\sigma}{\mu} = 15.33 \pm 0.11 \%$

$\mu = 480.8 \pm 0.5$

$\sigma = 73.7 \pm 0.4$

WF (26373 entries)

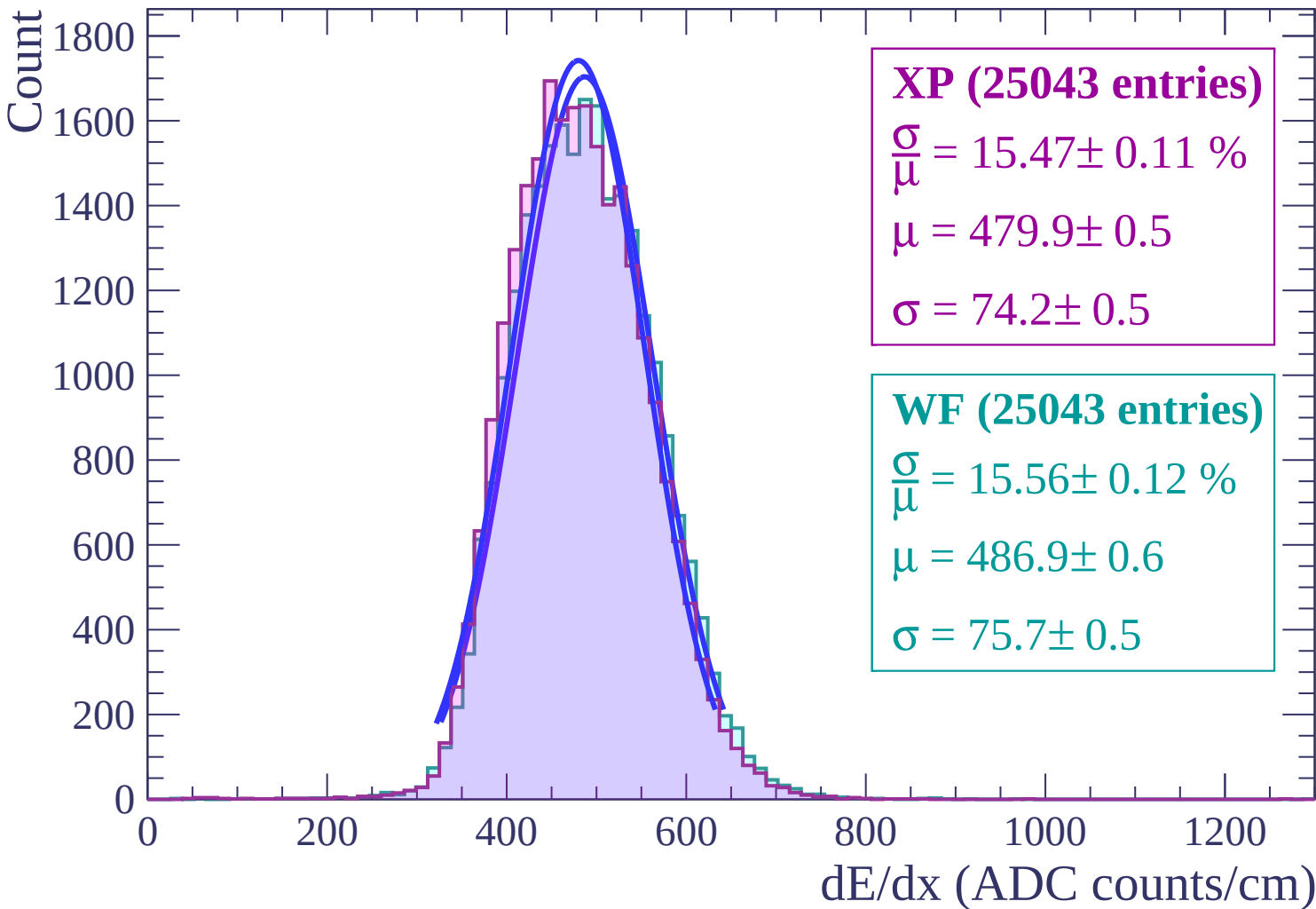
$\frac{\sigma}{\mu} = 15.49 \pm 0.11 \%$

$\mu = 486.9 \pm 0.6$

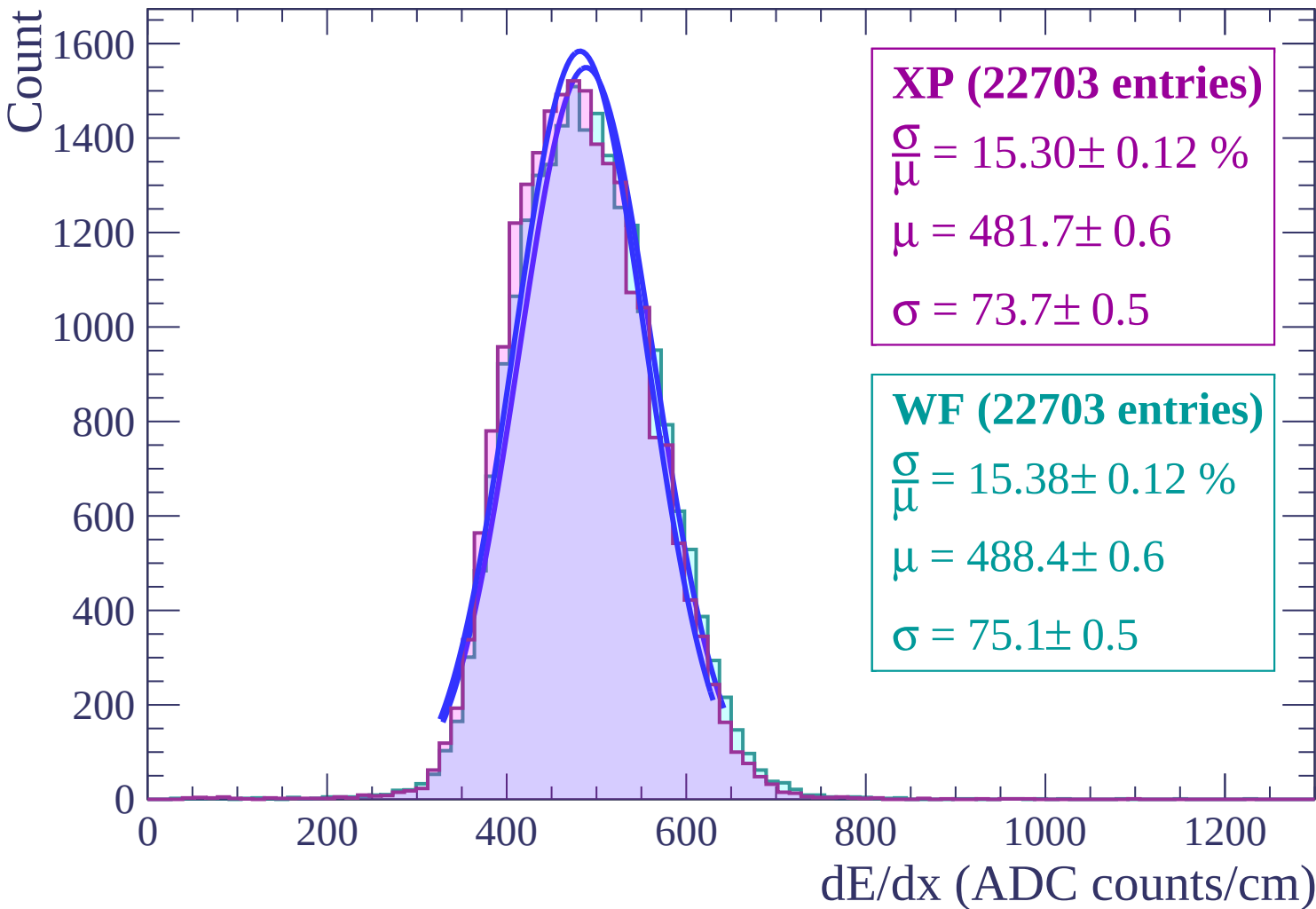
$\sigma = 75.4 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $10 < \theta < 12$



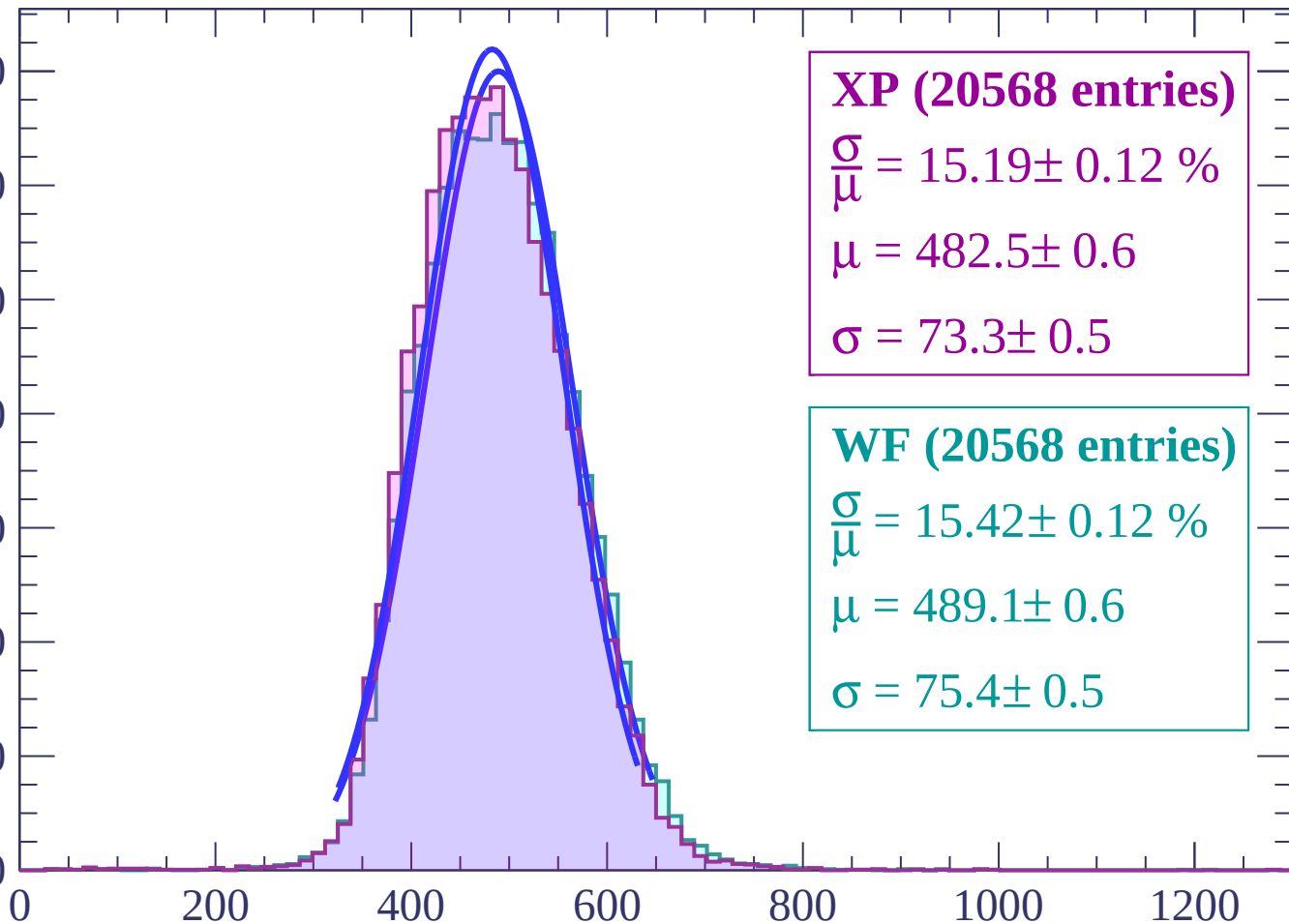
Energy loss | $12 < \theta < 14$



Energy loss | $14 < \theta < 16$

Count

1400
1200
1000
800
600
400
200
0



XP (20568 entries)

$\frac{\sigma}{\mu} = 15.19 \pm 0.12 \%$

$\mu = 482.5 \pm 0.6$

$\sigma = 73.3 \pm 0.5$

WF (20568 entries)

$\frac{\sigma}{\mu} = 15.42 \pm 0.12 \%$

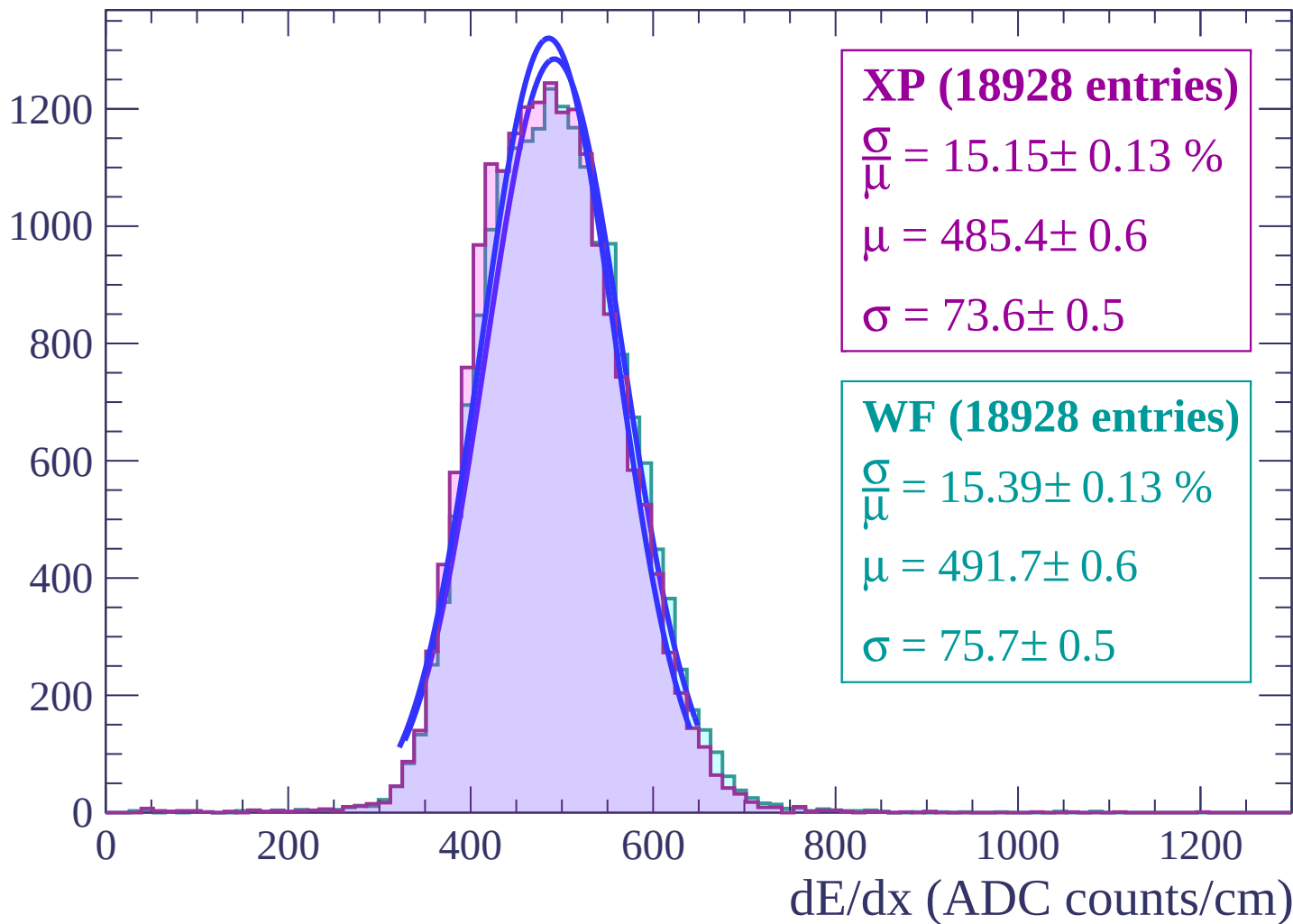
$\mu = 489.1 \pm 0.6$

$\sigma = 75.4 \pm 0.5$

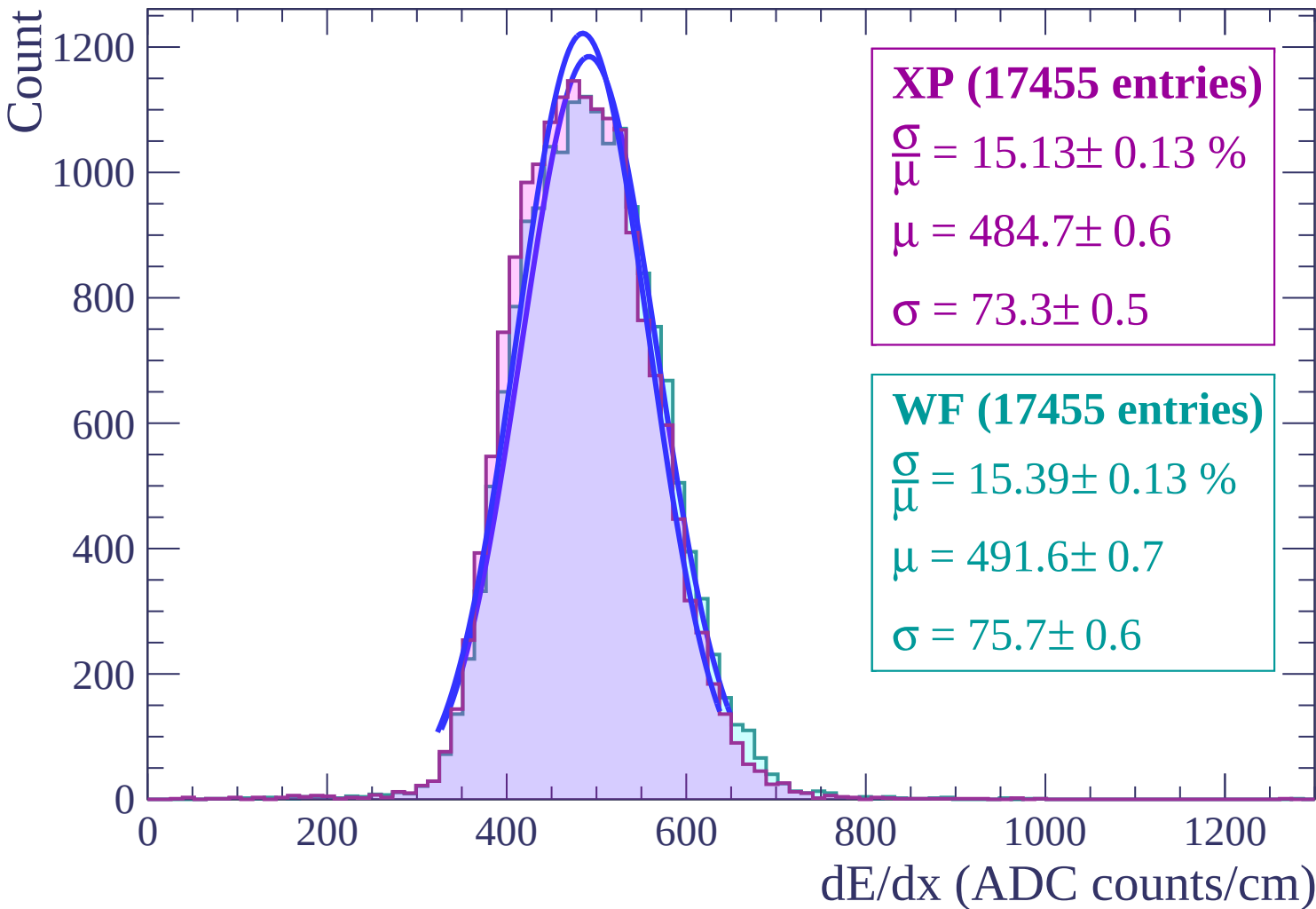
dE/dx (ADC counts/cm)

Energy loss | $16 < \theta < 18$

Count



Energy loss | $18 < \theta < 20$



Energy loss | $20 < \theta < 22$

Count

1000

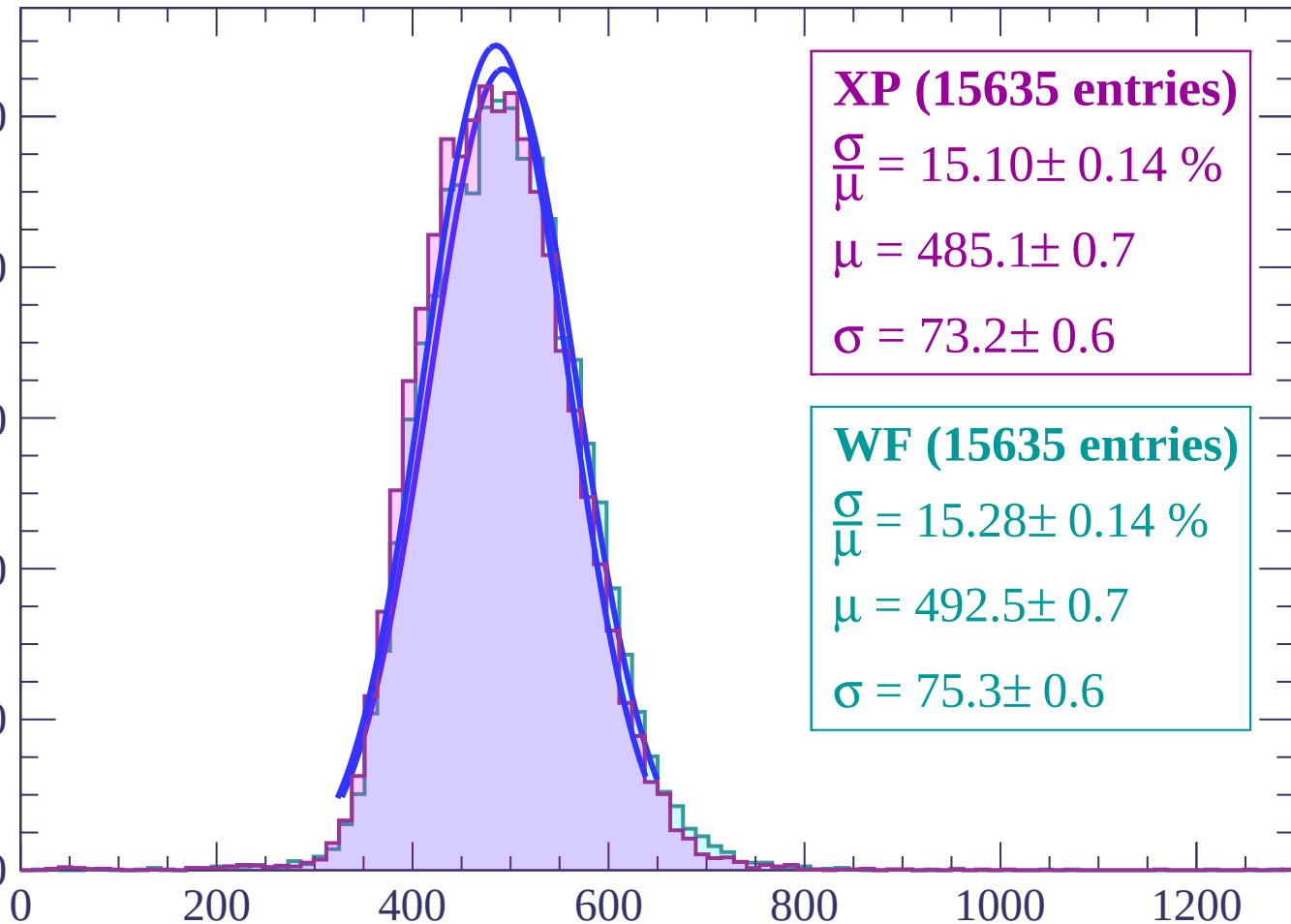
800

600

400

200

0



XP (15635 entries)

$\frac{\sigma}{\mu} = 15.10 \pm 0.14 \%$

$\mu = 485.1 \pm 0.7$

$\sigma = 73.2 \pm 0.6$

WF (15635 entries)

$\frac{\sigma}{\mu} = 15.28 \pm 0.14 \%$

$\mu = 492.5 \pm 0.7$

$\sigma = 75.3 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $22 < \theta < 24$

Count

1000

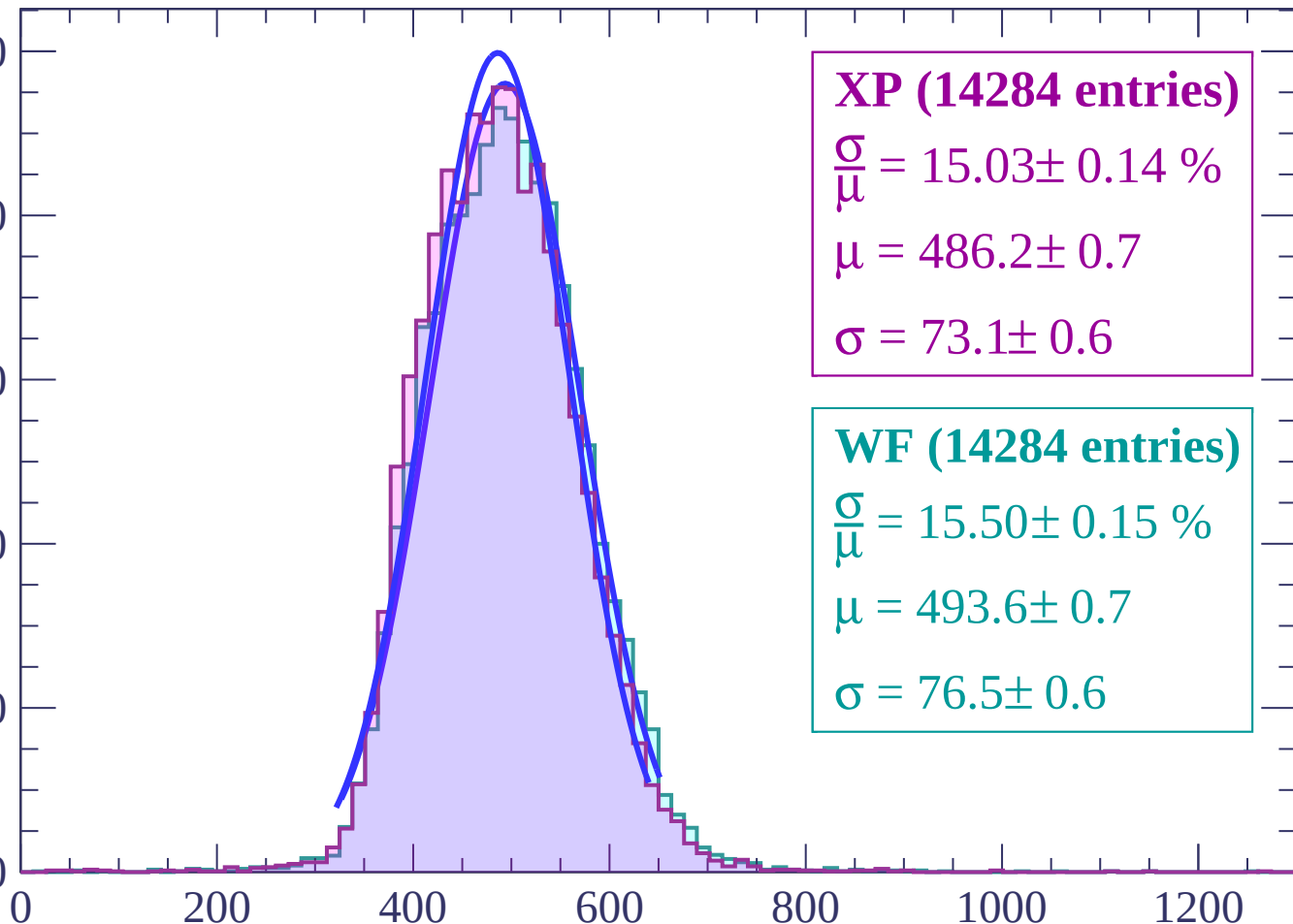
800

600

400

200

0



XP (14284 entries)

$\frac{\sigma}{\mu} = 15.03 \pm 0.14 \%$

$\mu = 486.2 \pm 0.7$

$\sigma = 73.1 \pm 0.6$

WF (14284 entries)

$\frac{\sigma}{\mu} = 15.50 \pm 0.15 \%$

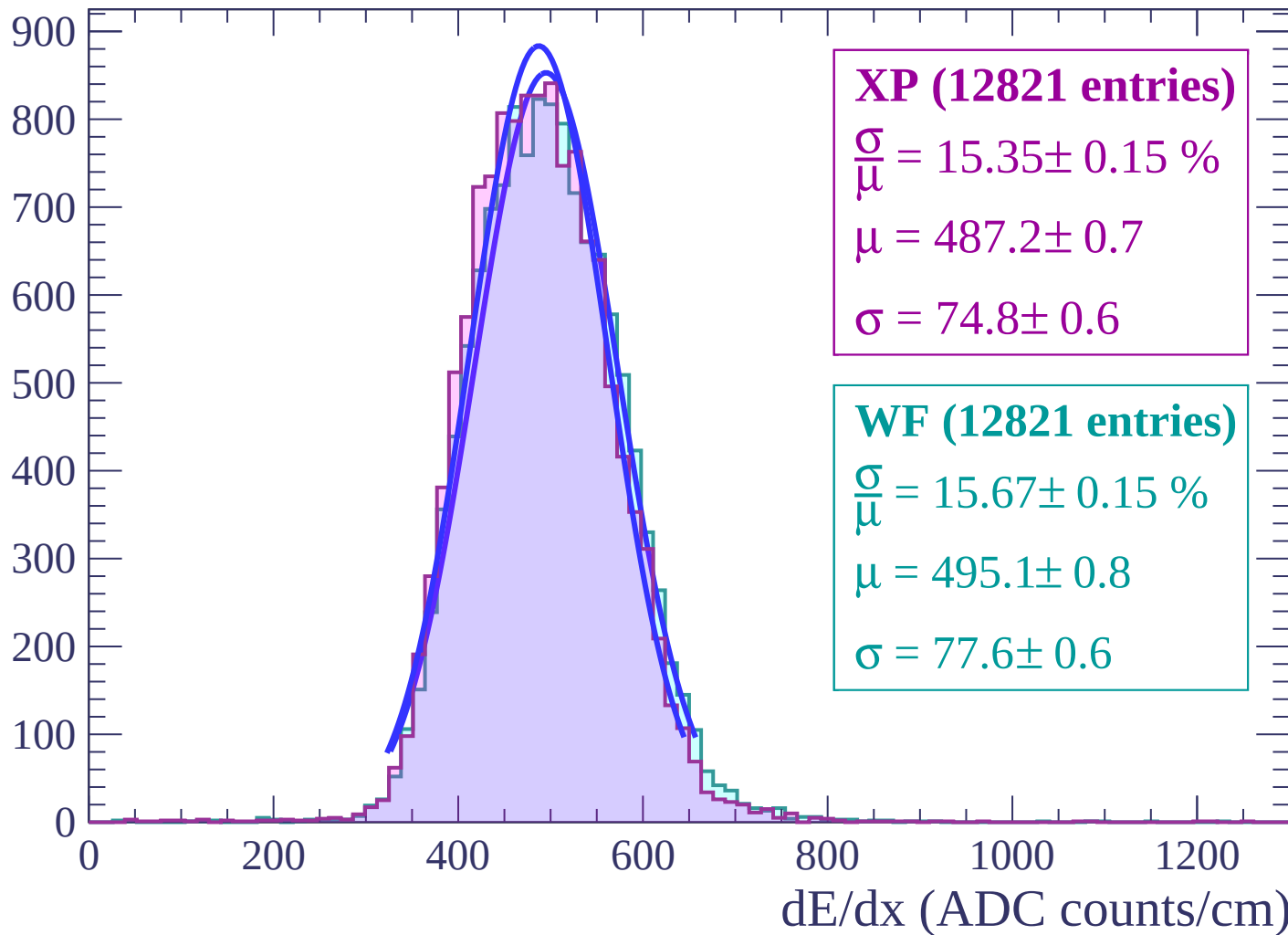
$\mu = 493.6 \pm 0.7$

$\sigma = 76.5 \pm 0.6$

dE/dx (ADC counts/cm)

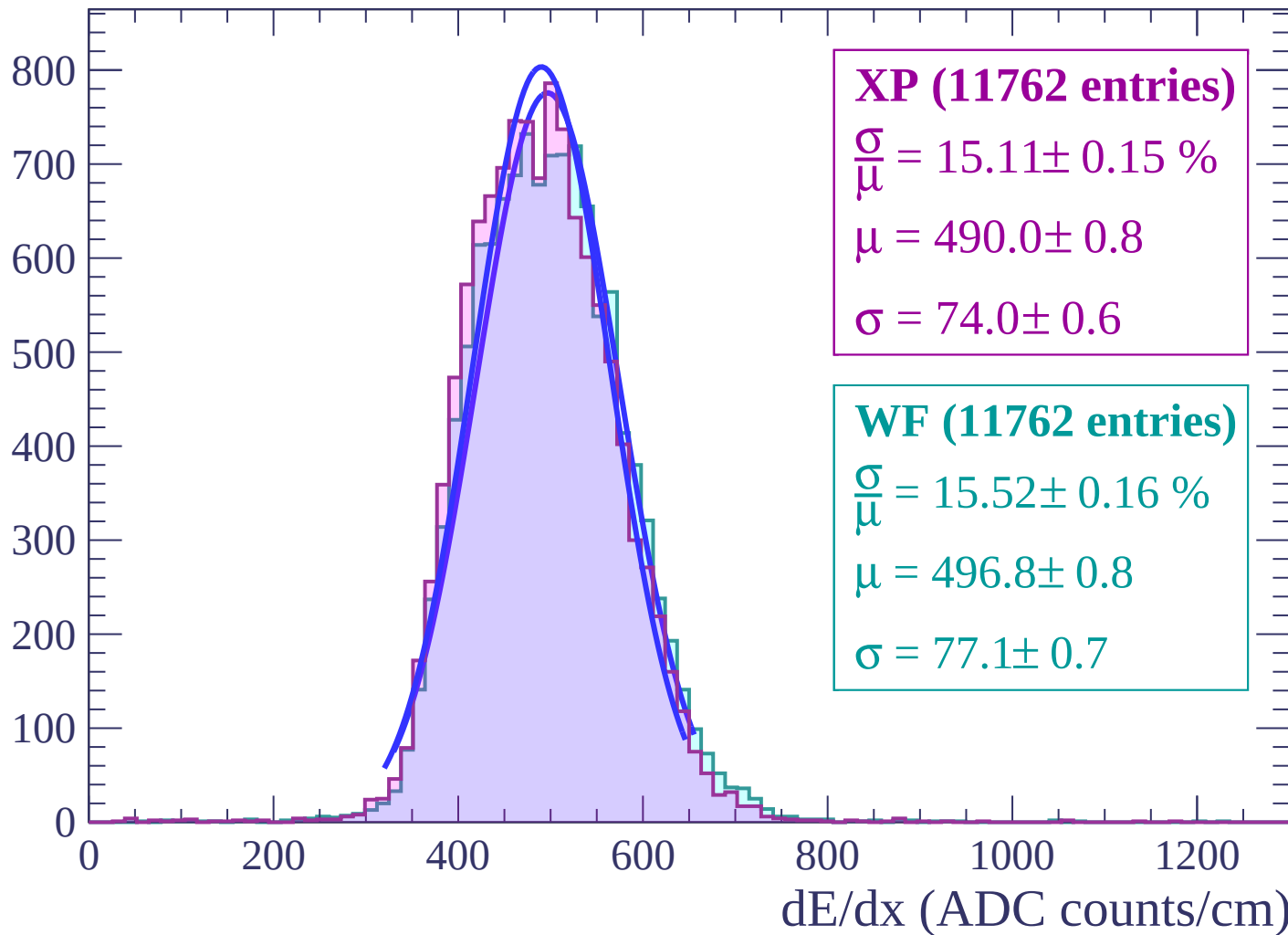
Energy loss | $24 < \theta < 26$

Count



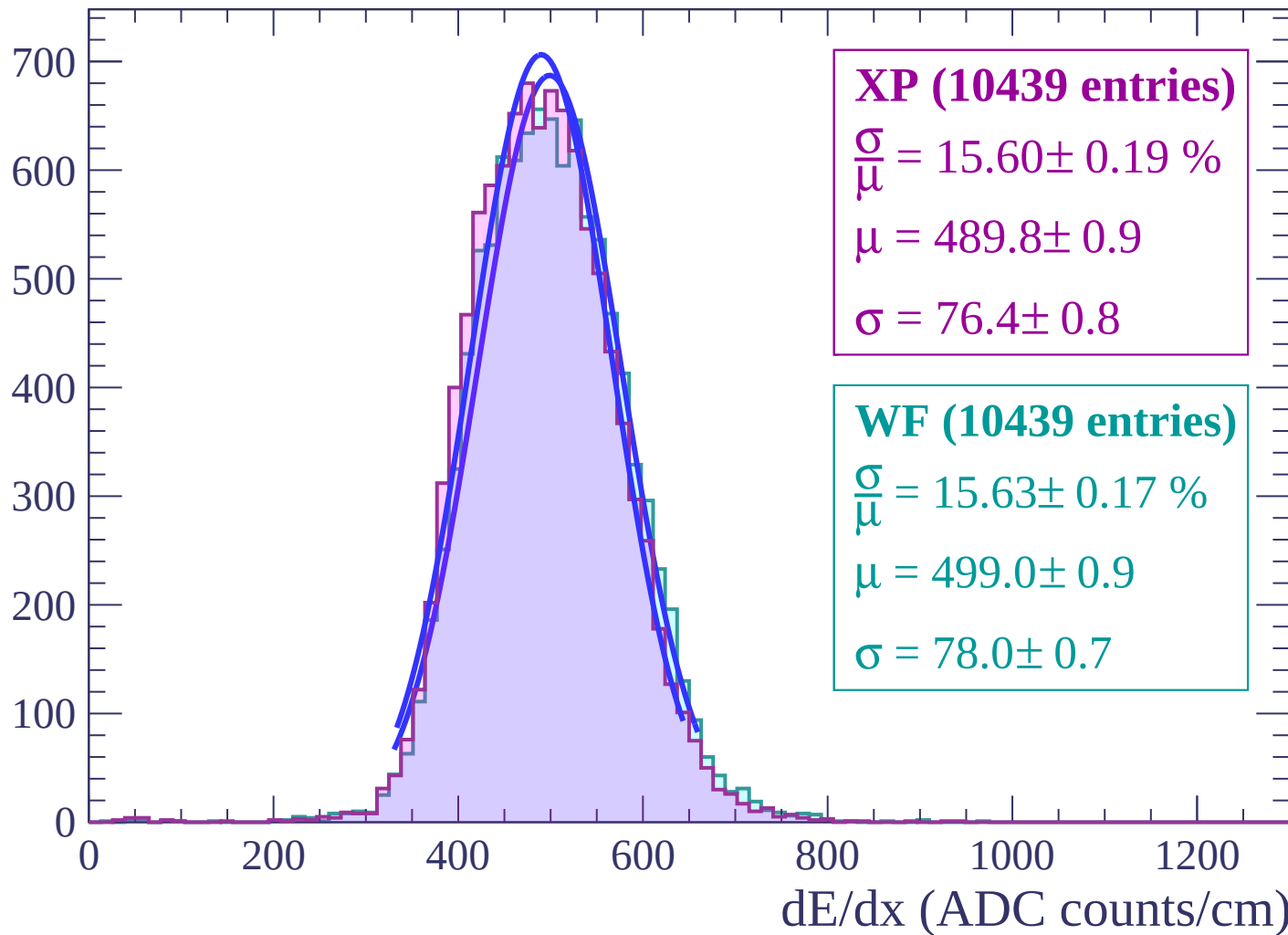
Energy loss | $26 < \theta < 28$

Count



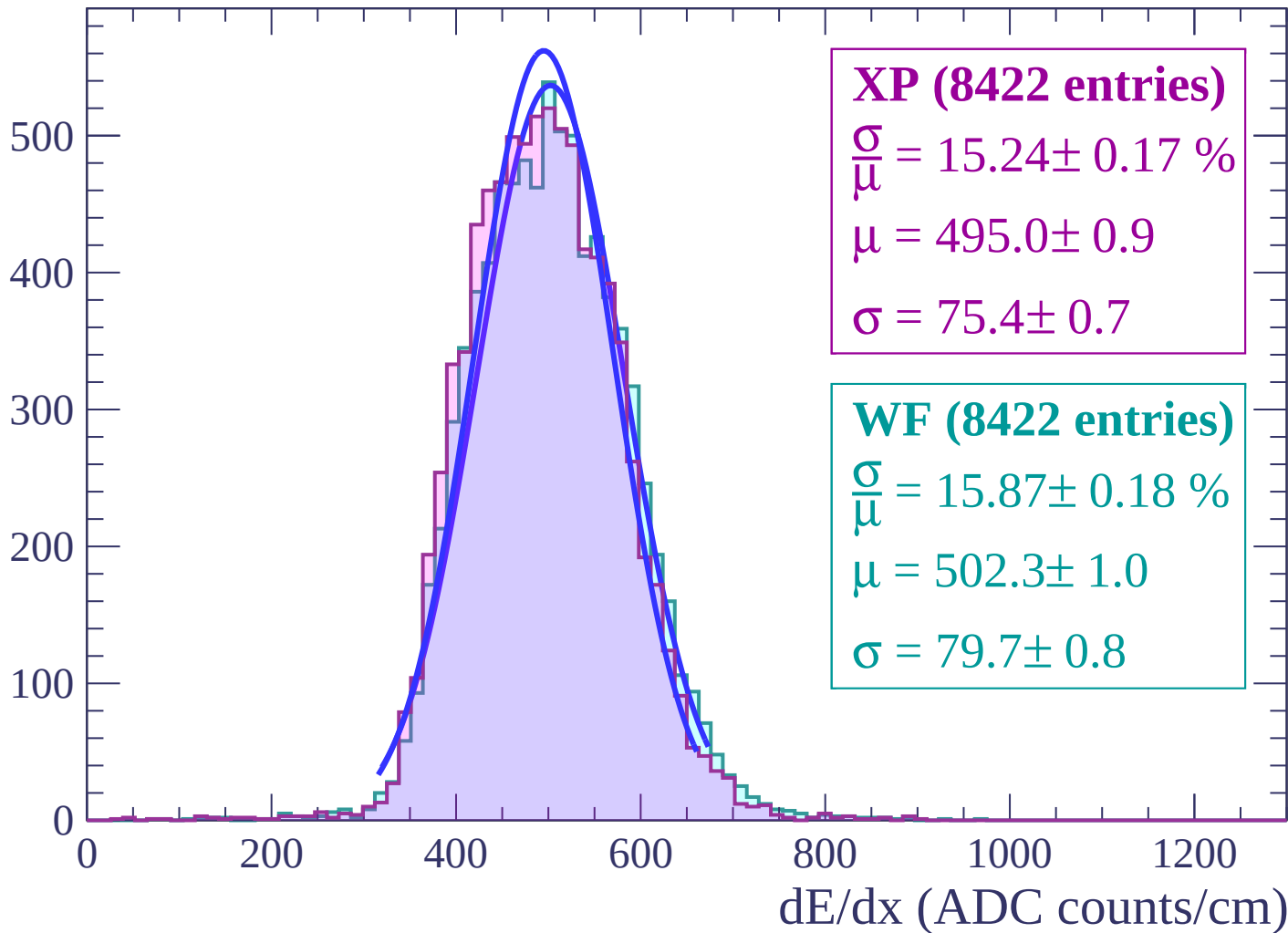
Energy loss | $28 < \theta < 30$

Count



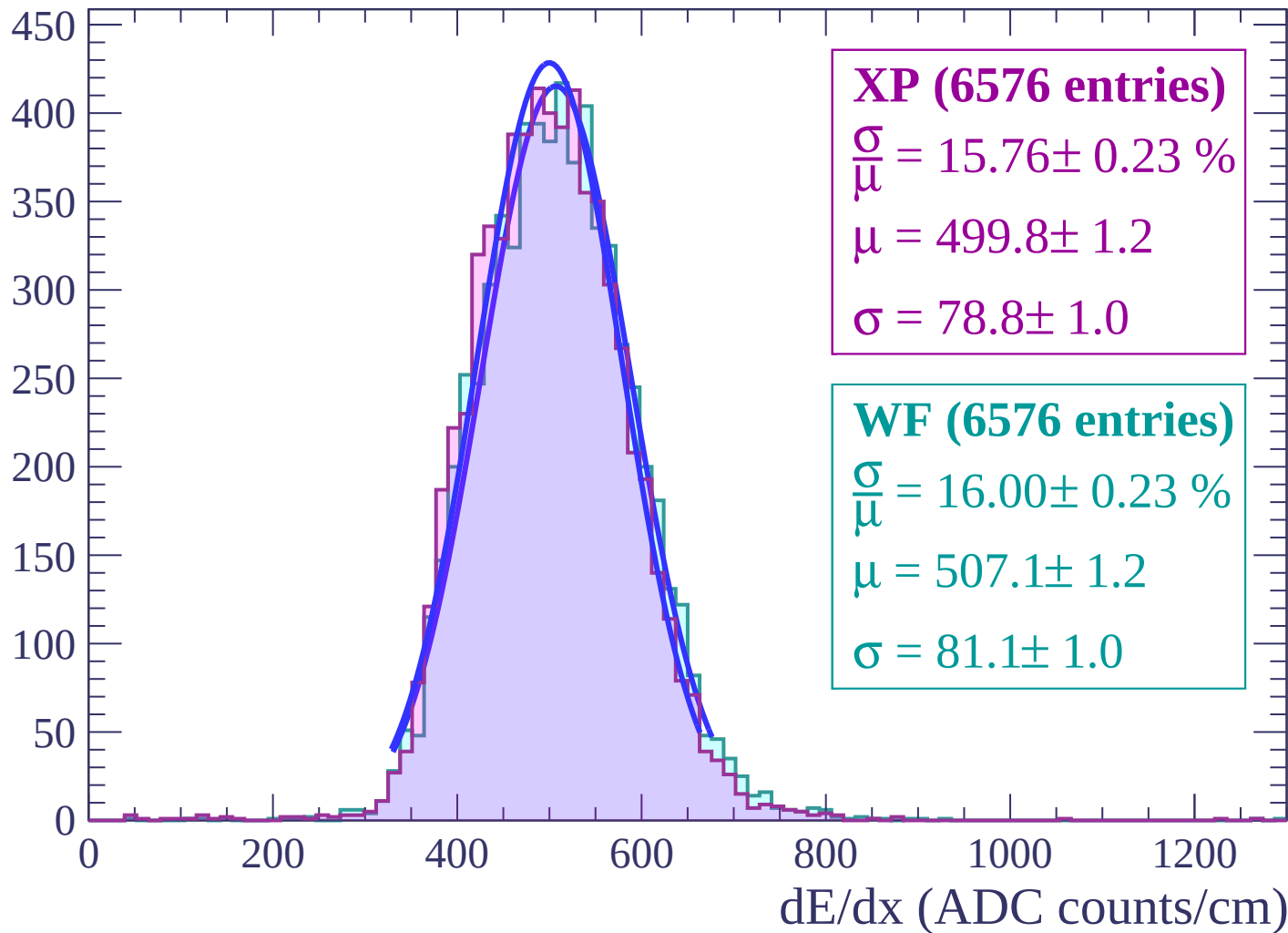
Energy loss | $30 < \theta < 32$

Count



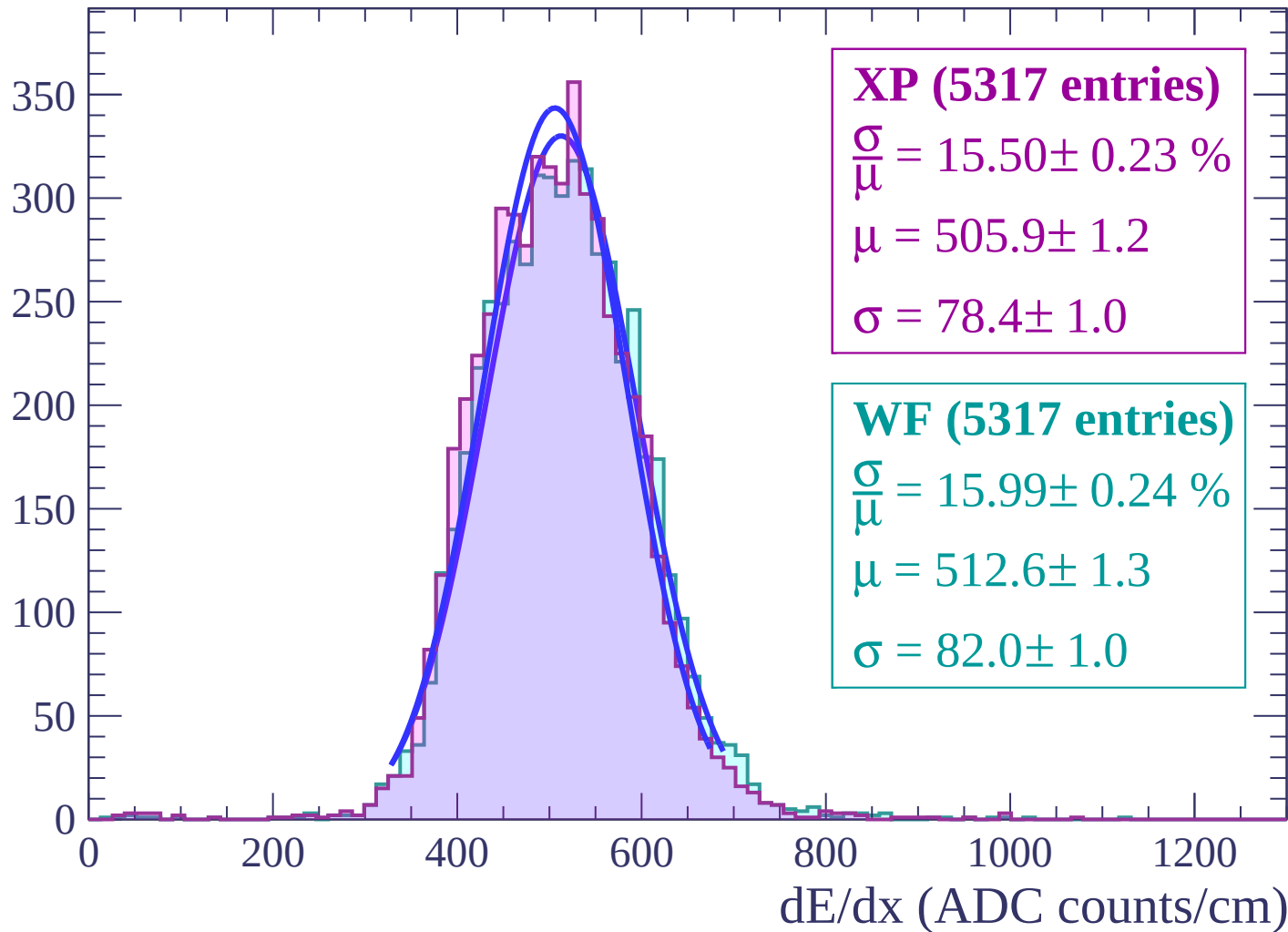
Energy loss | $32 < \theta < 34$

Count



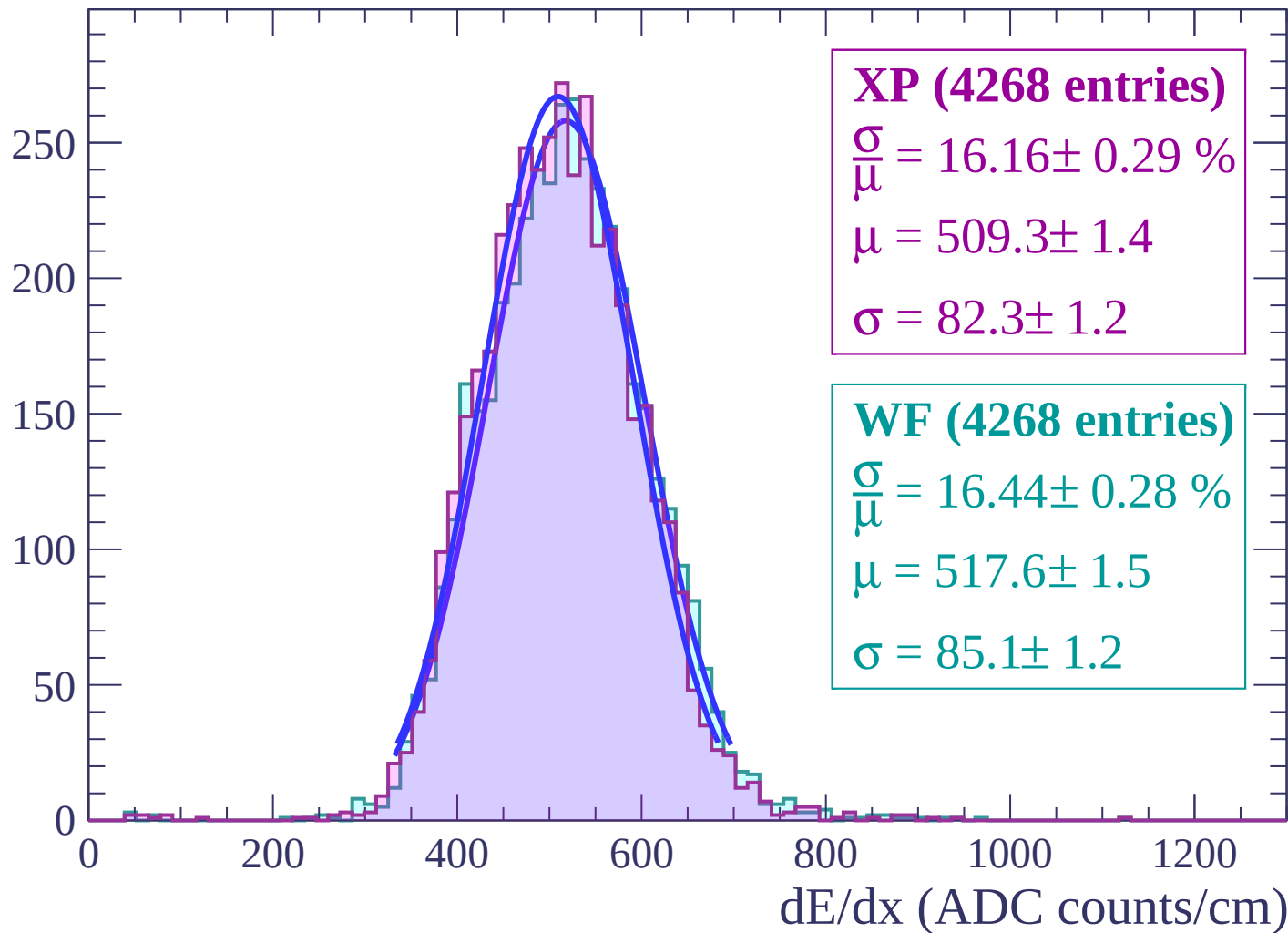
Energy loss | $34 < \theta < 36$

Count



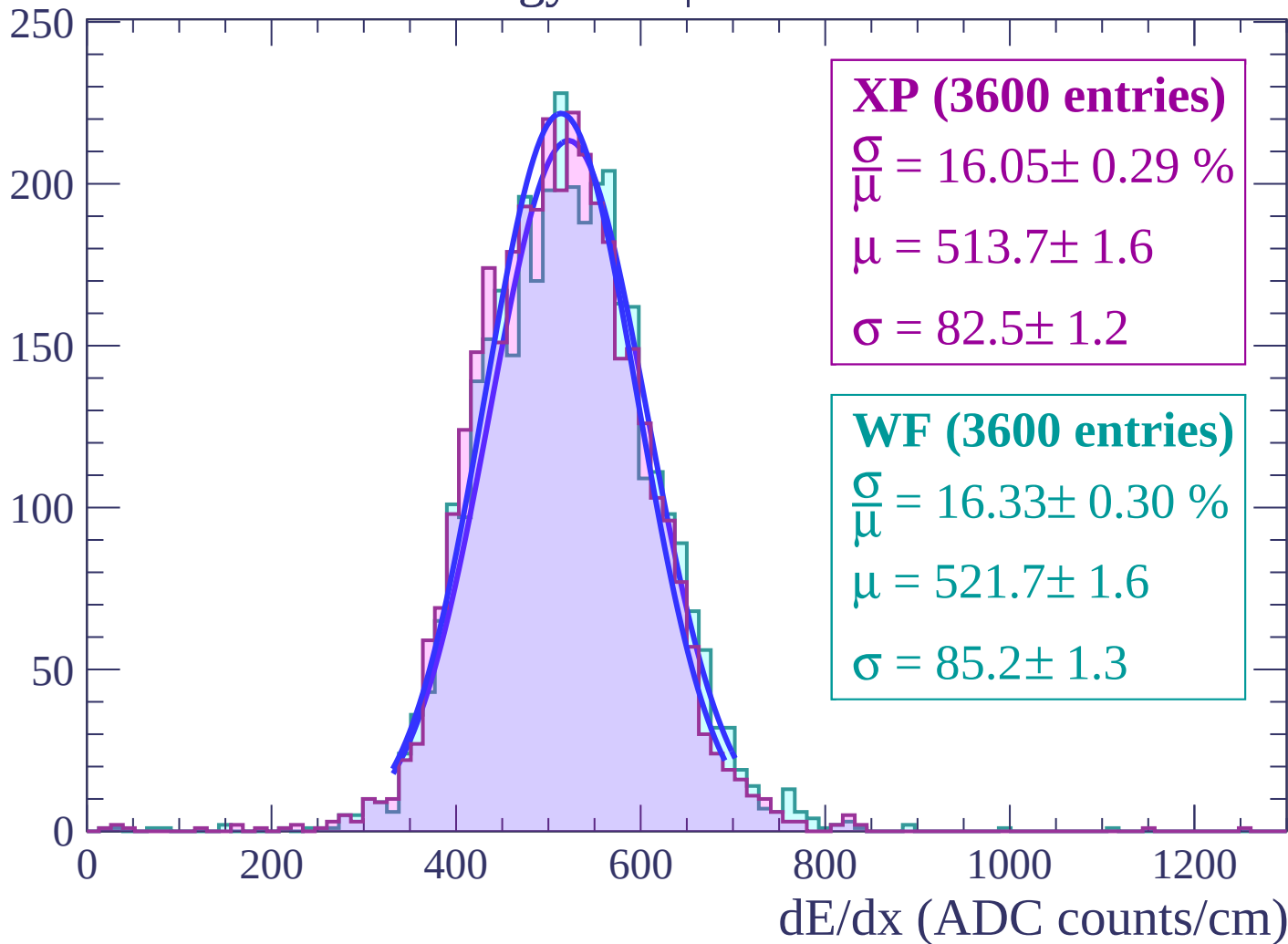
Energy loss | $36 < \theta < 38$

Count



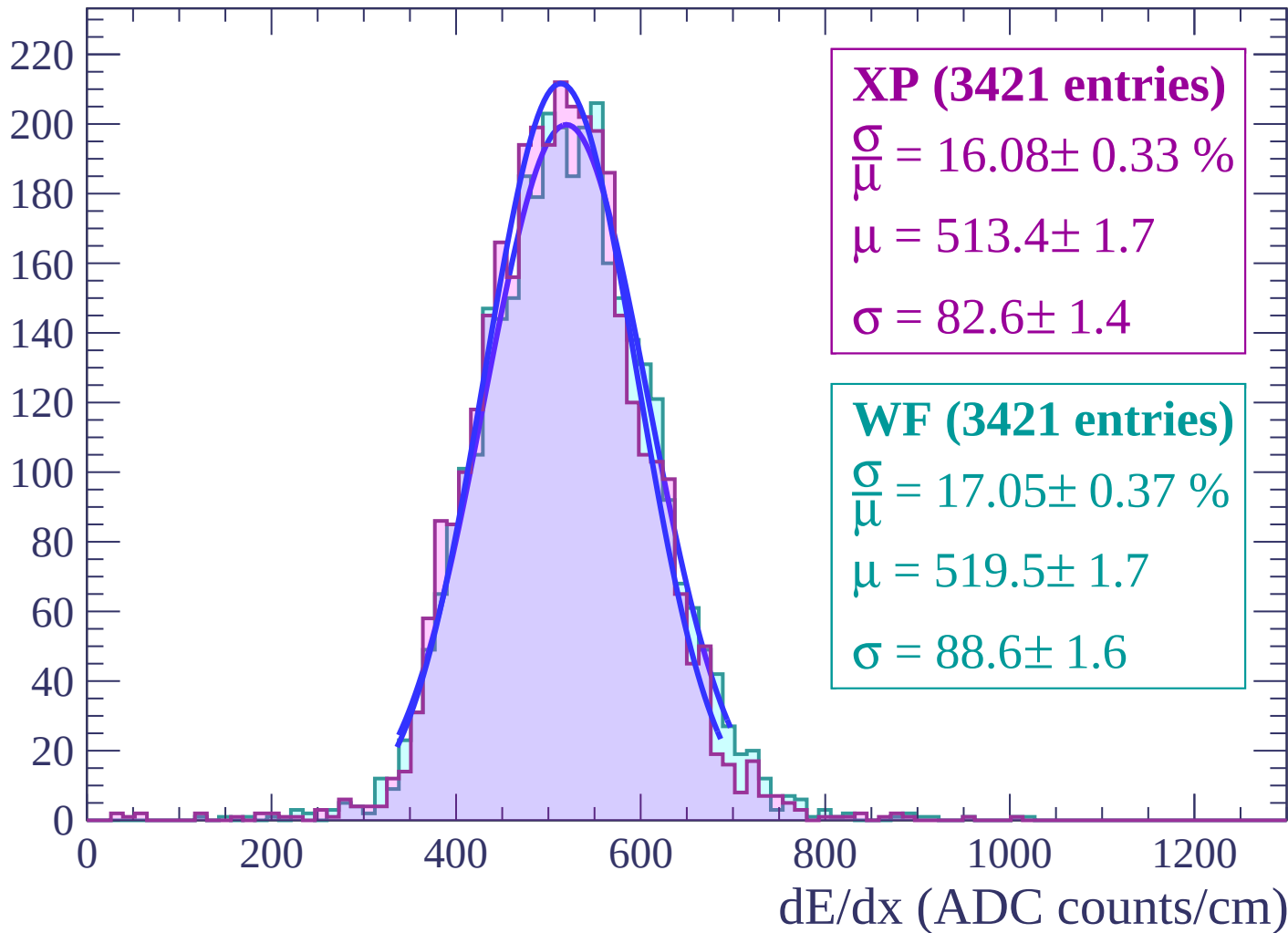
Energy loss | $38 < \theta < 40$

Count



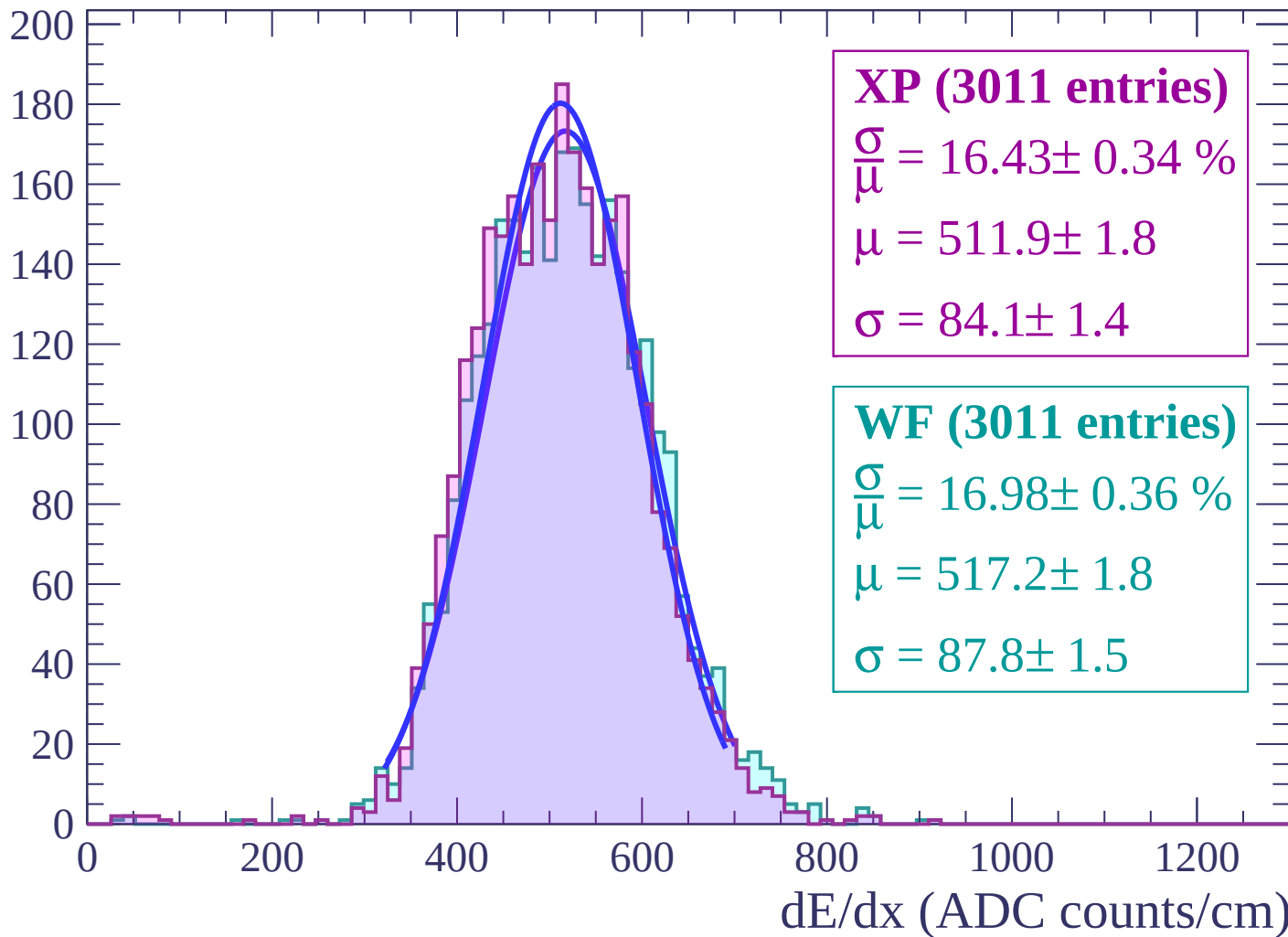
Energy loss | $40 < \theta < 42$

Count



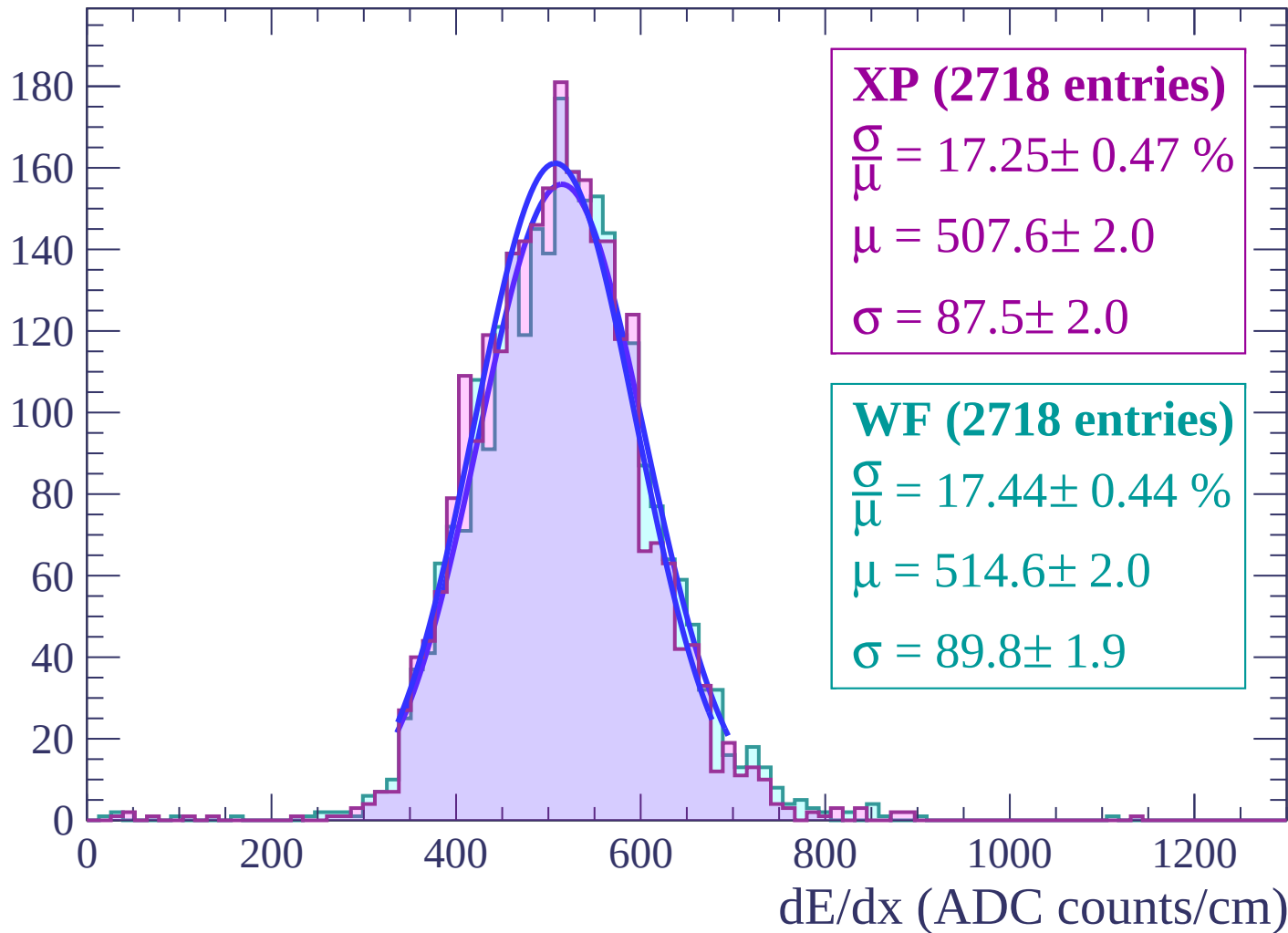
Energy loss | $42 < \theta < 44$

Count



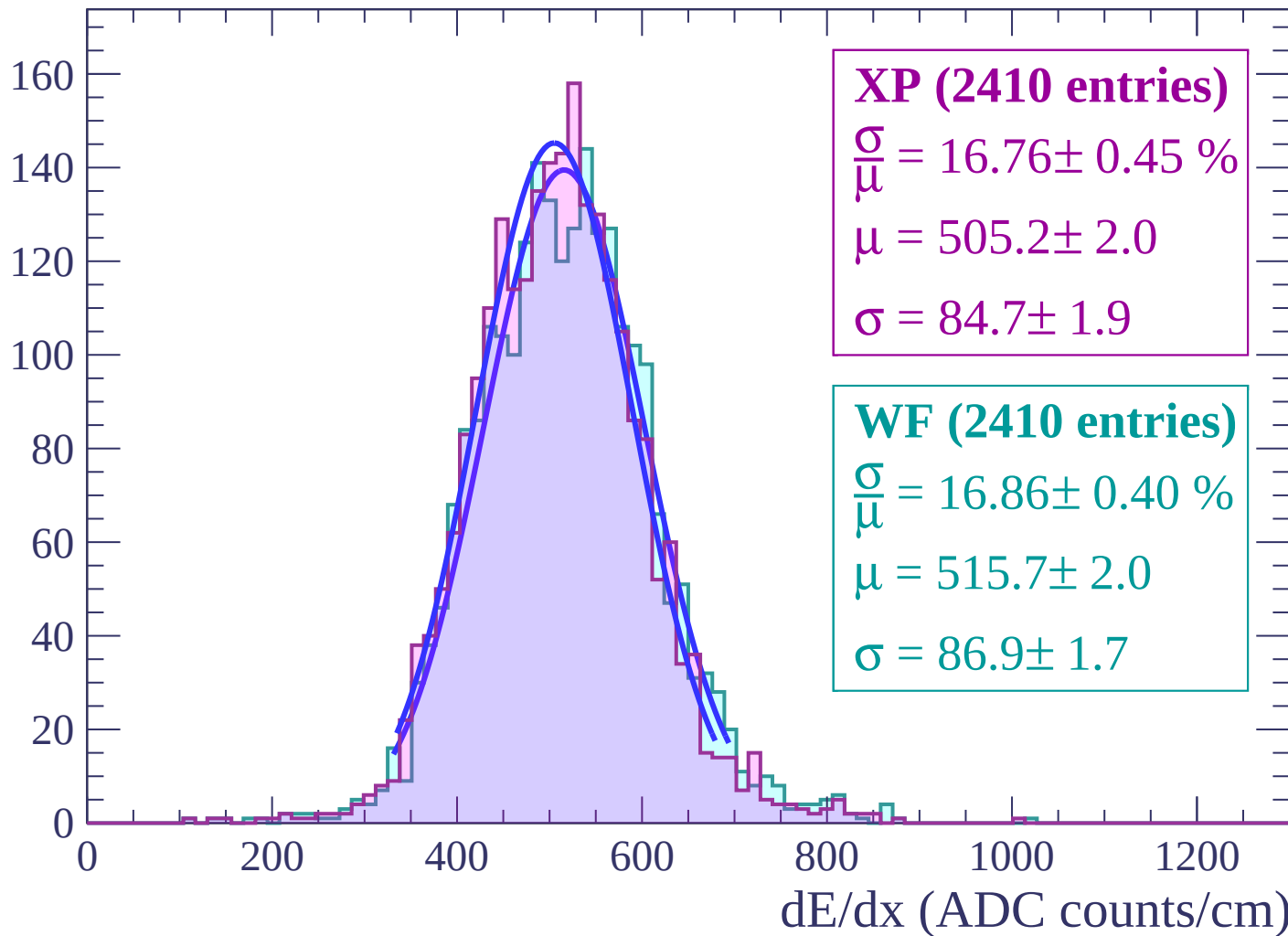
Energy loss | $44 < \theta < 46$

Count



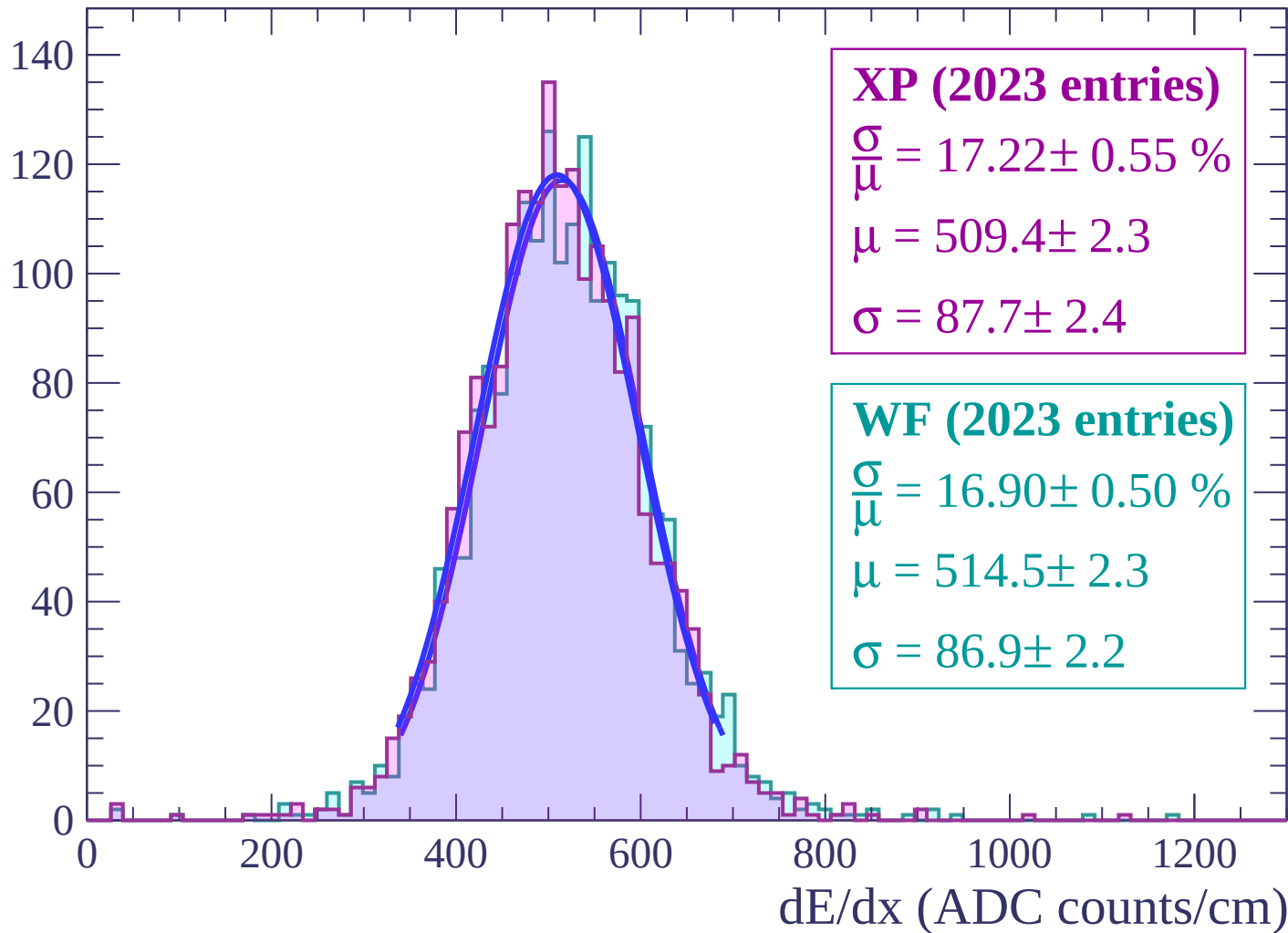
Energy loss | $46 < \theta < 48$

Count



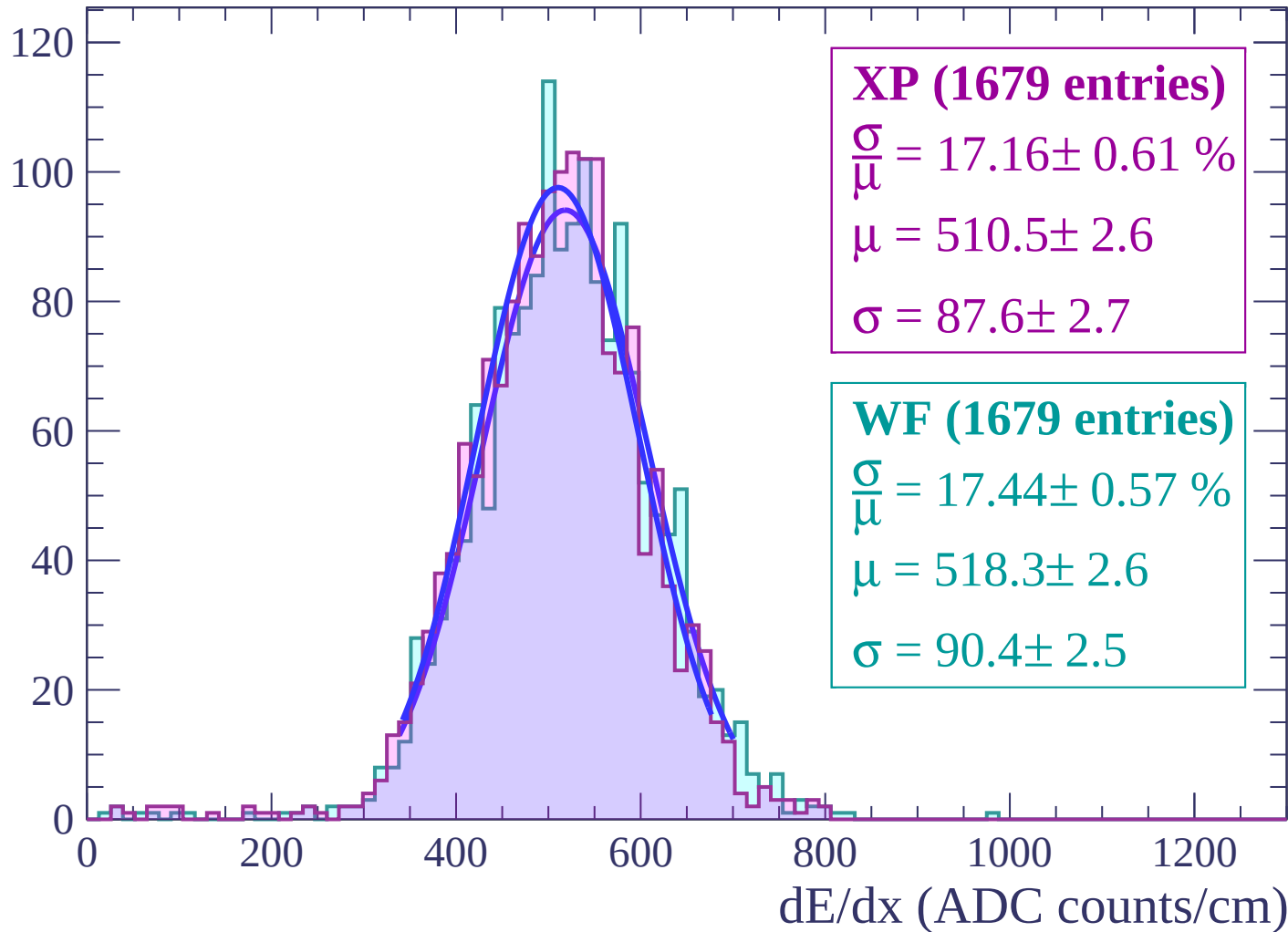
Energy loss | $48 < \theta < 50$

Count



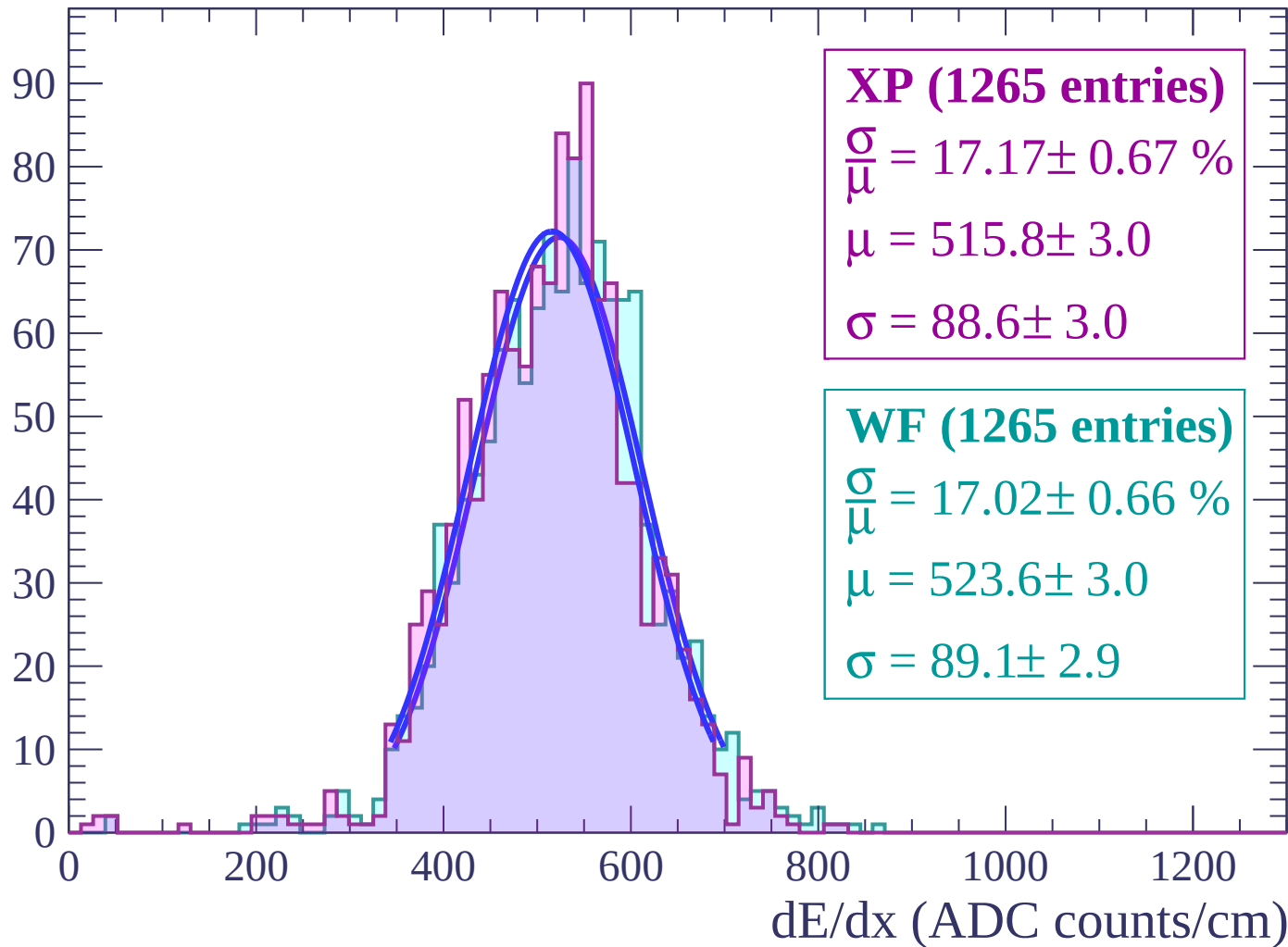
Energy loss | $50 < \theta < 52$

Count



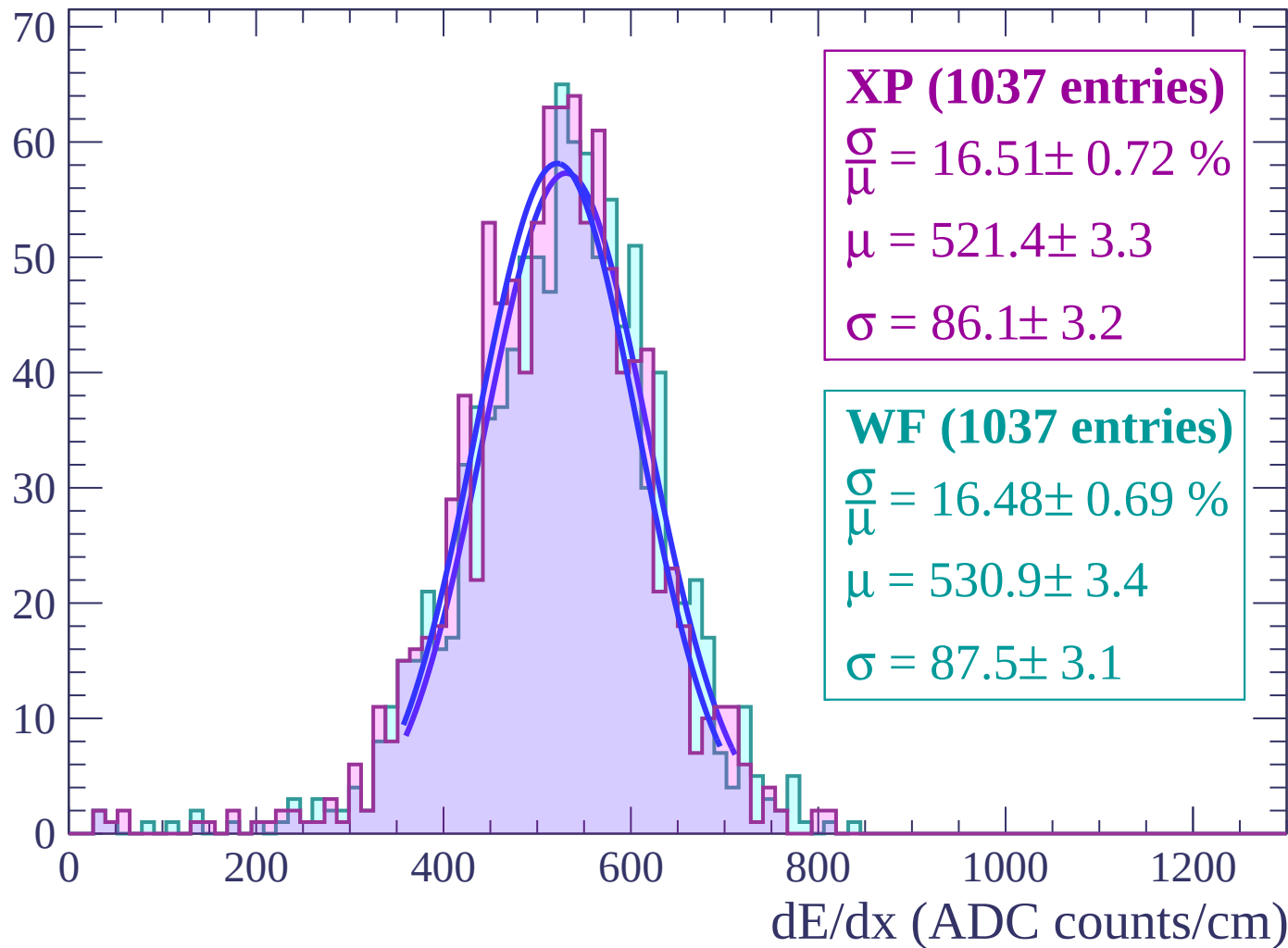
Energy loss | $52 < \theta < 54$

Count



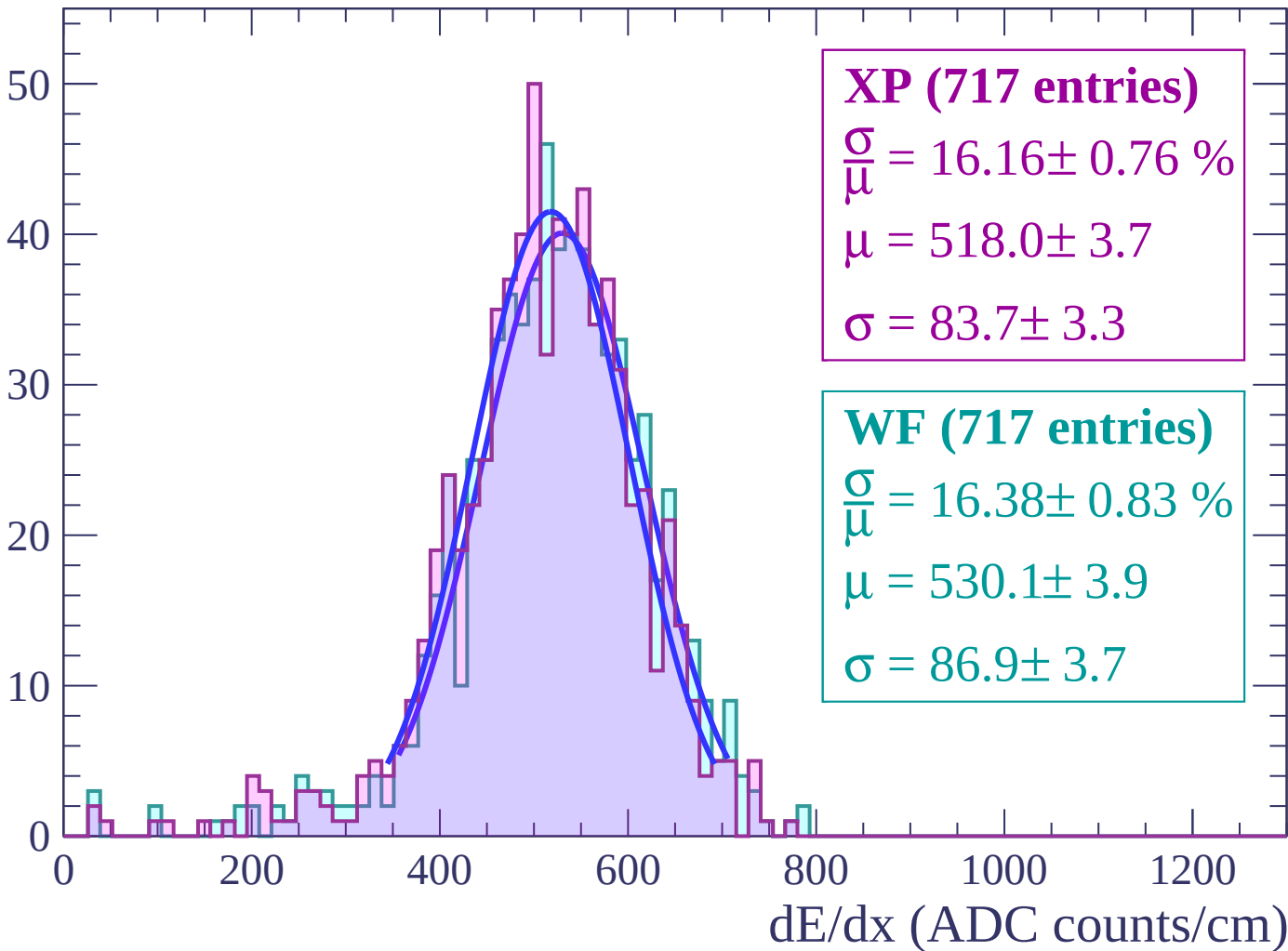
Energy loss | $54 < \theta < 56$

Count



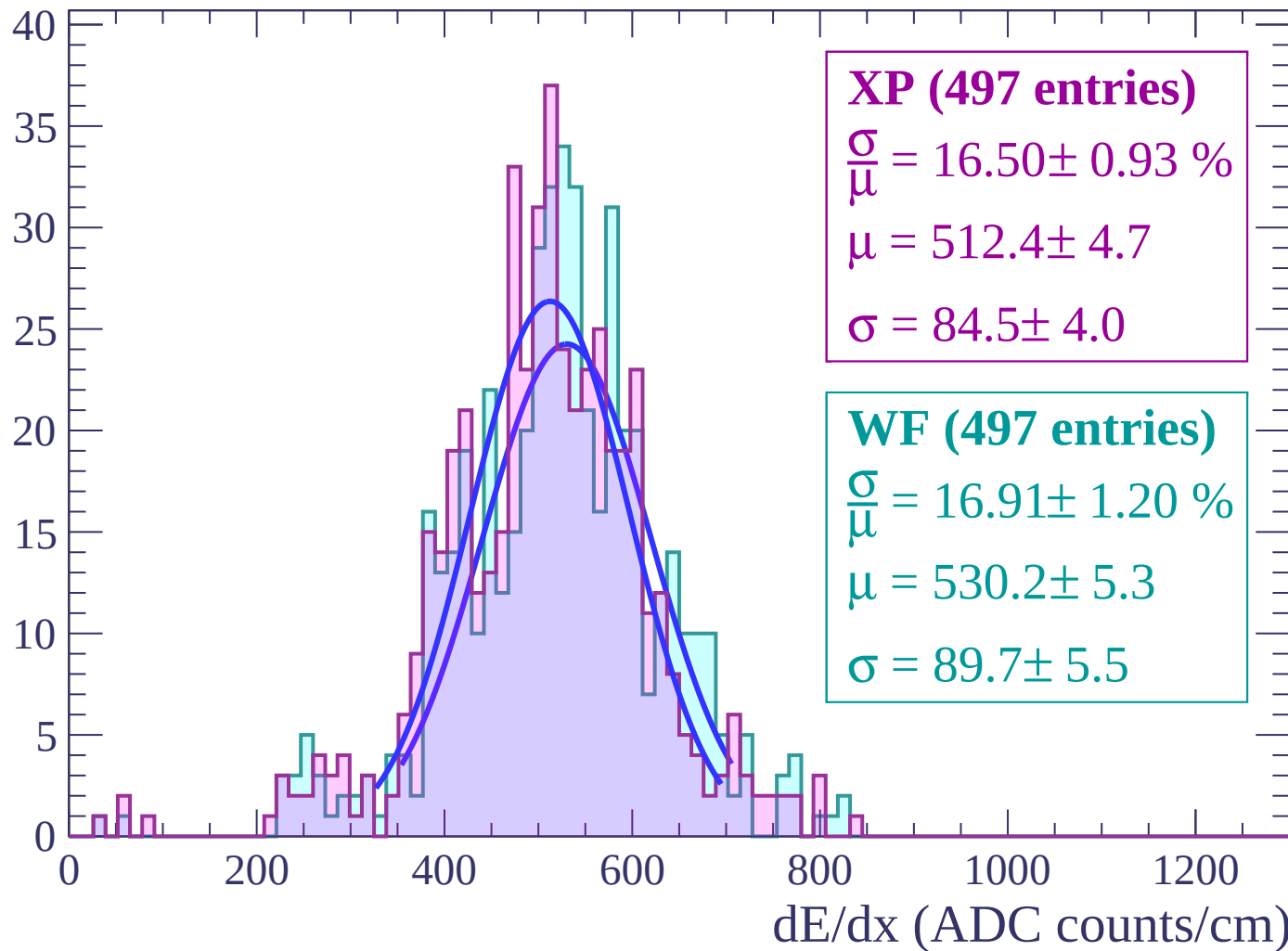
Energy loss | $56 < \theta < 58$

Count



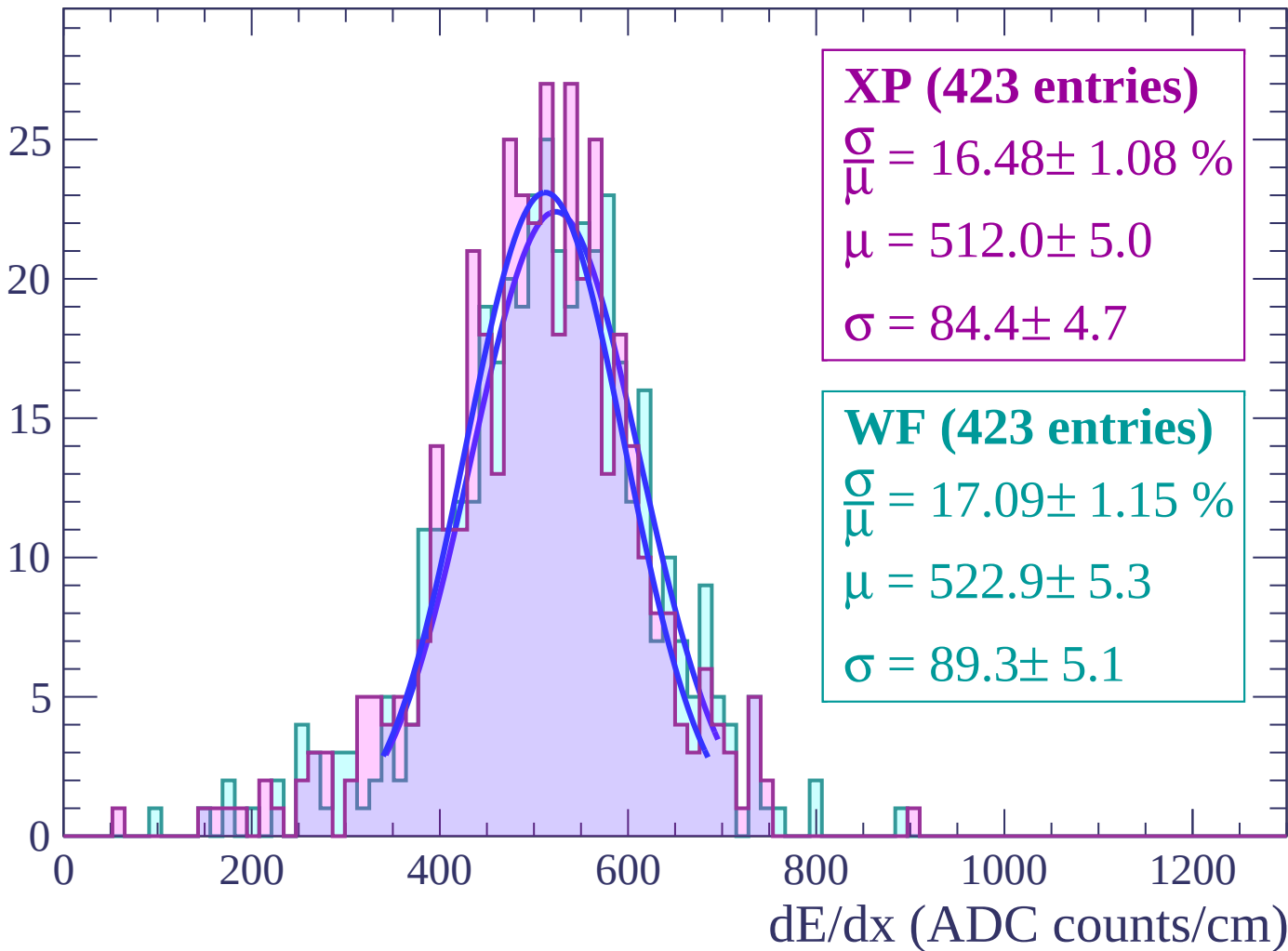
Energy loss | $58 < \theta < 60$

Count



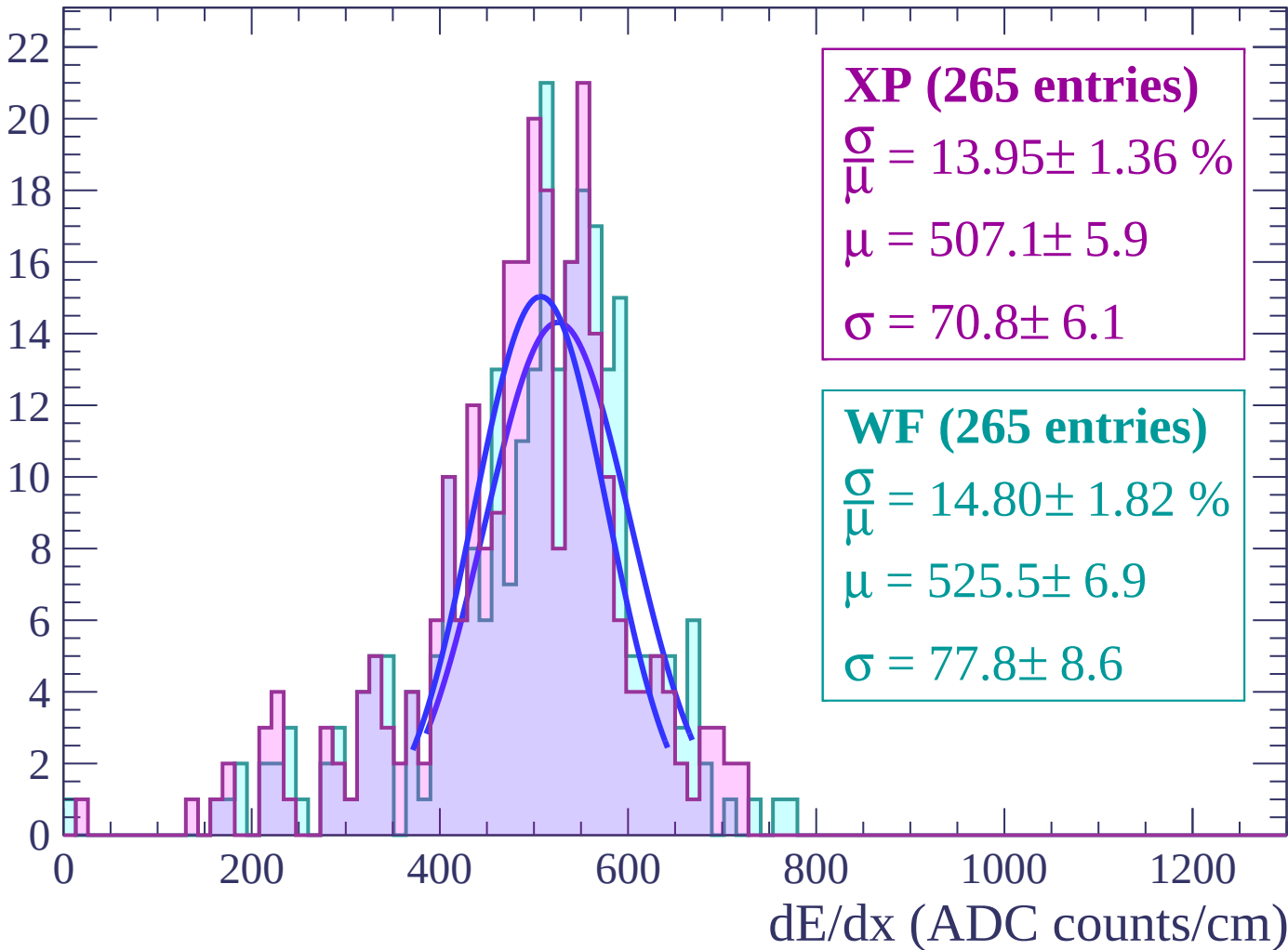
Energy loss | $60 < \theta < 62$

Count



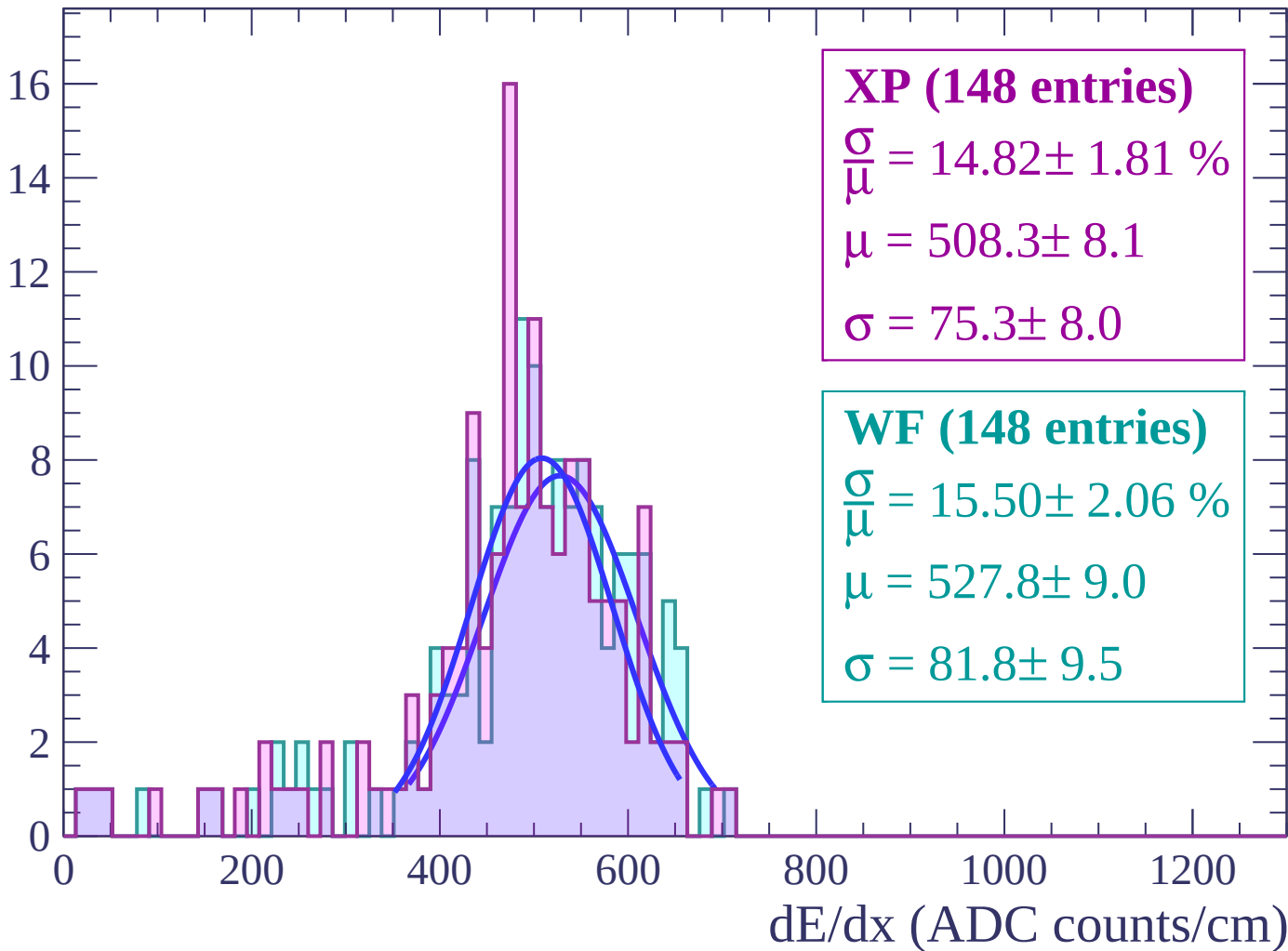
Energy loss | $62 < \theta < 64$

Count



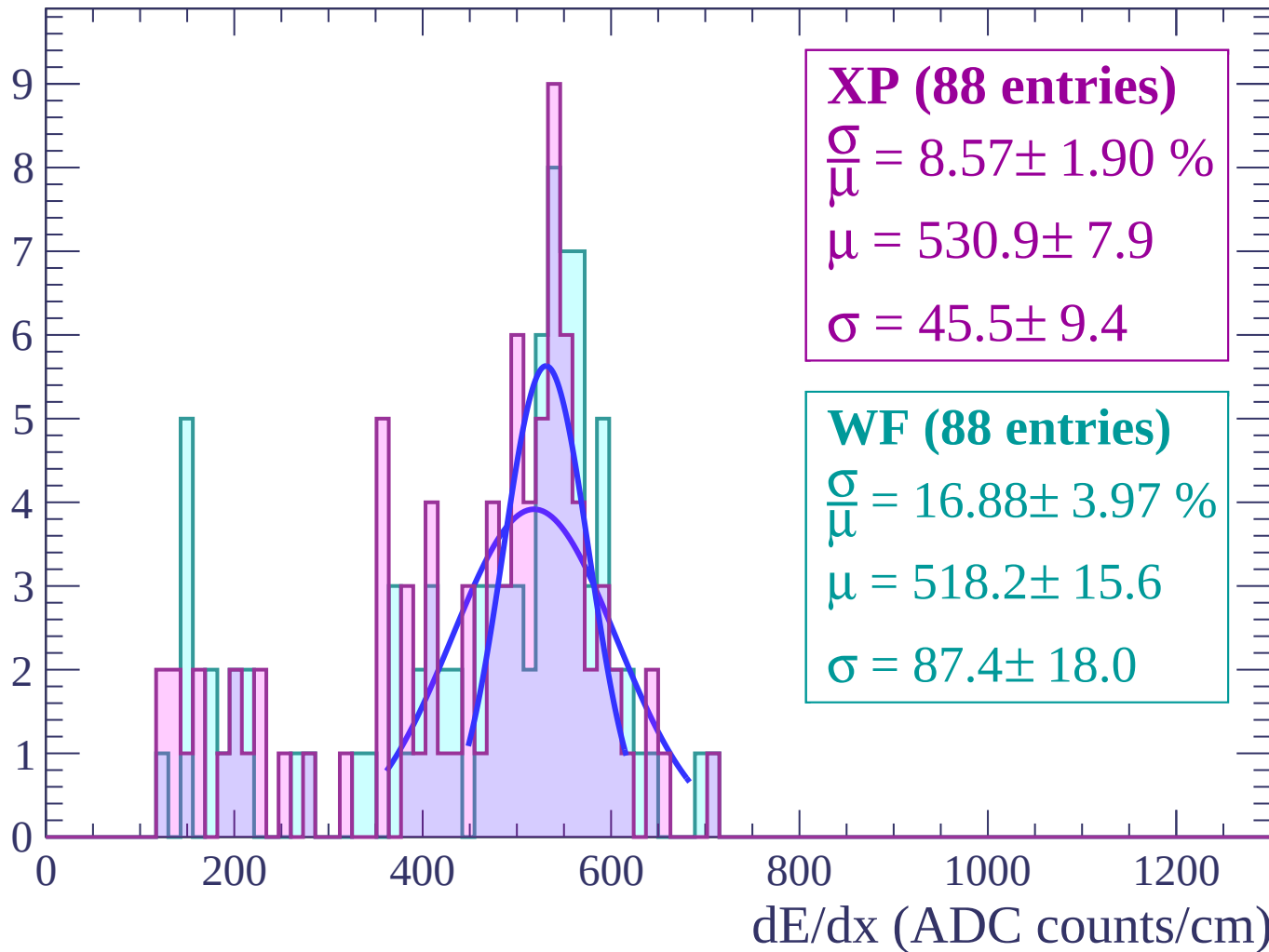
Energy loss | $64 < \theta < 66$

Count



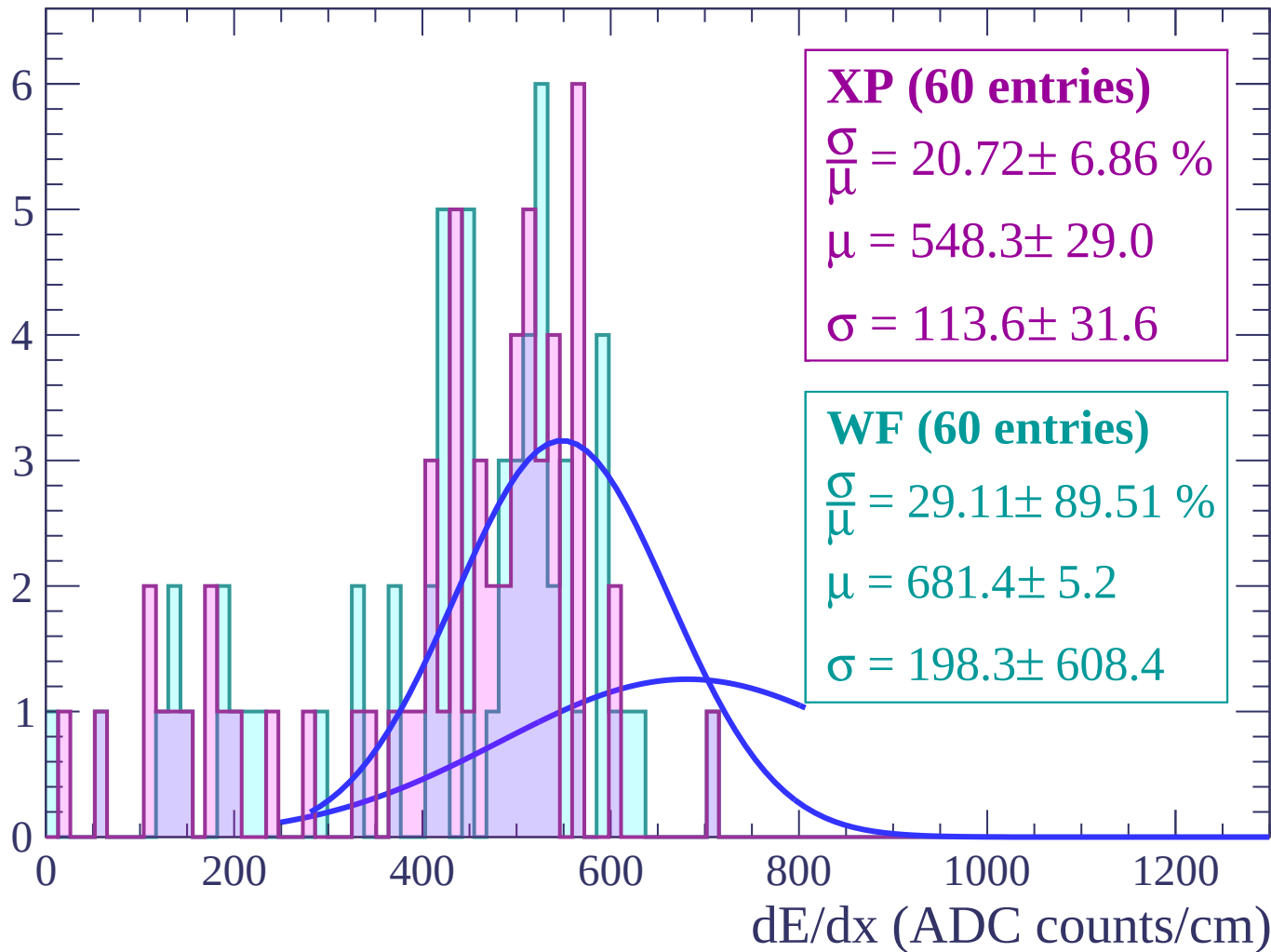
Energy loss | $66 < \theta < 68$

Count



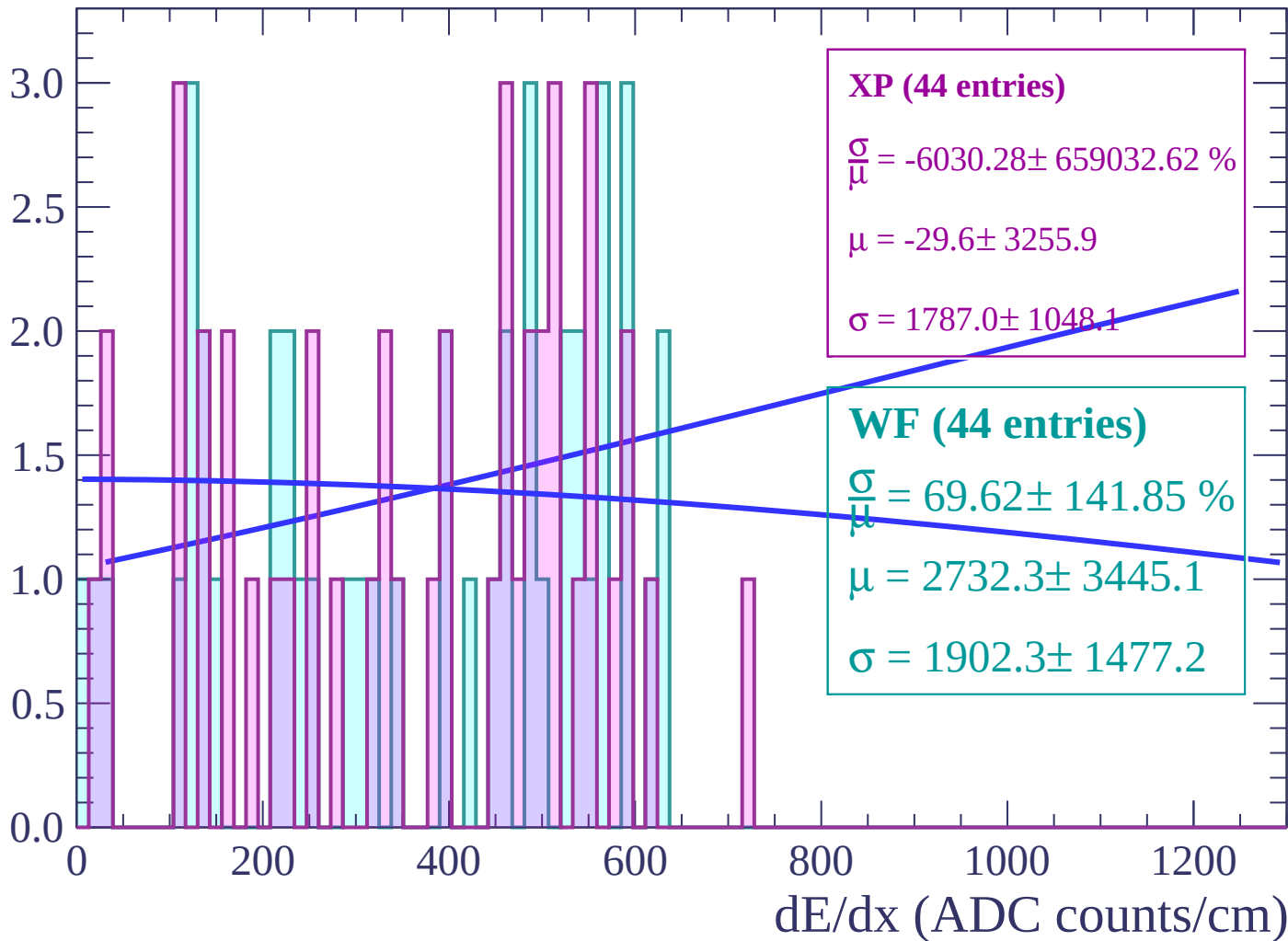
Energy loss | $68 < \theta < 70$

Count



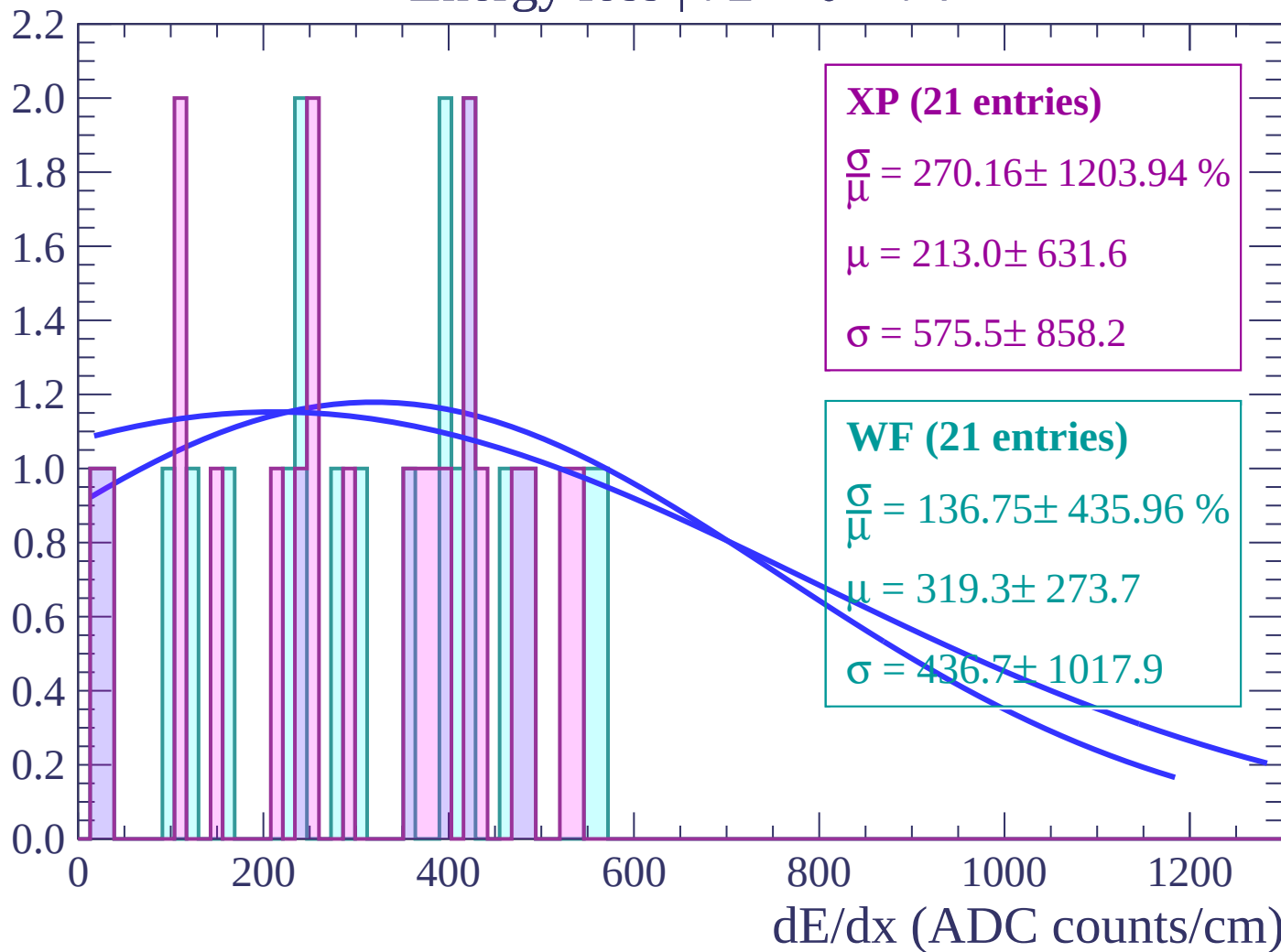
Energy loss | $70 < \theta < 72$

Count



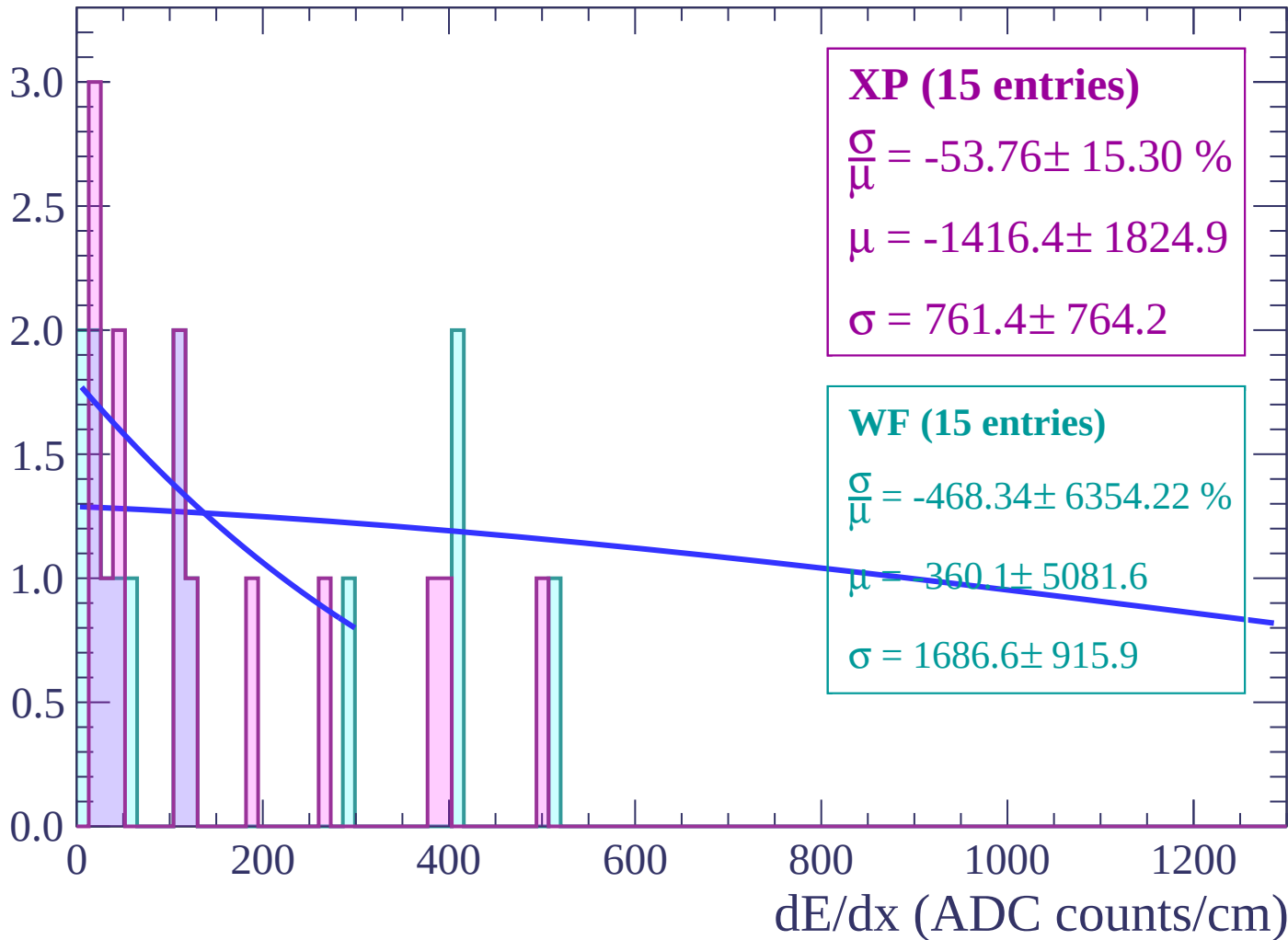
Energy loss | $72 < \theta < 74$

Count



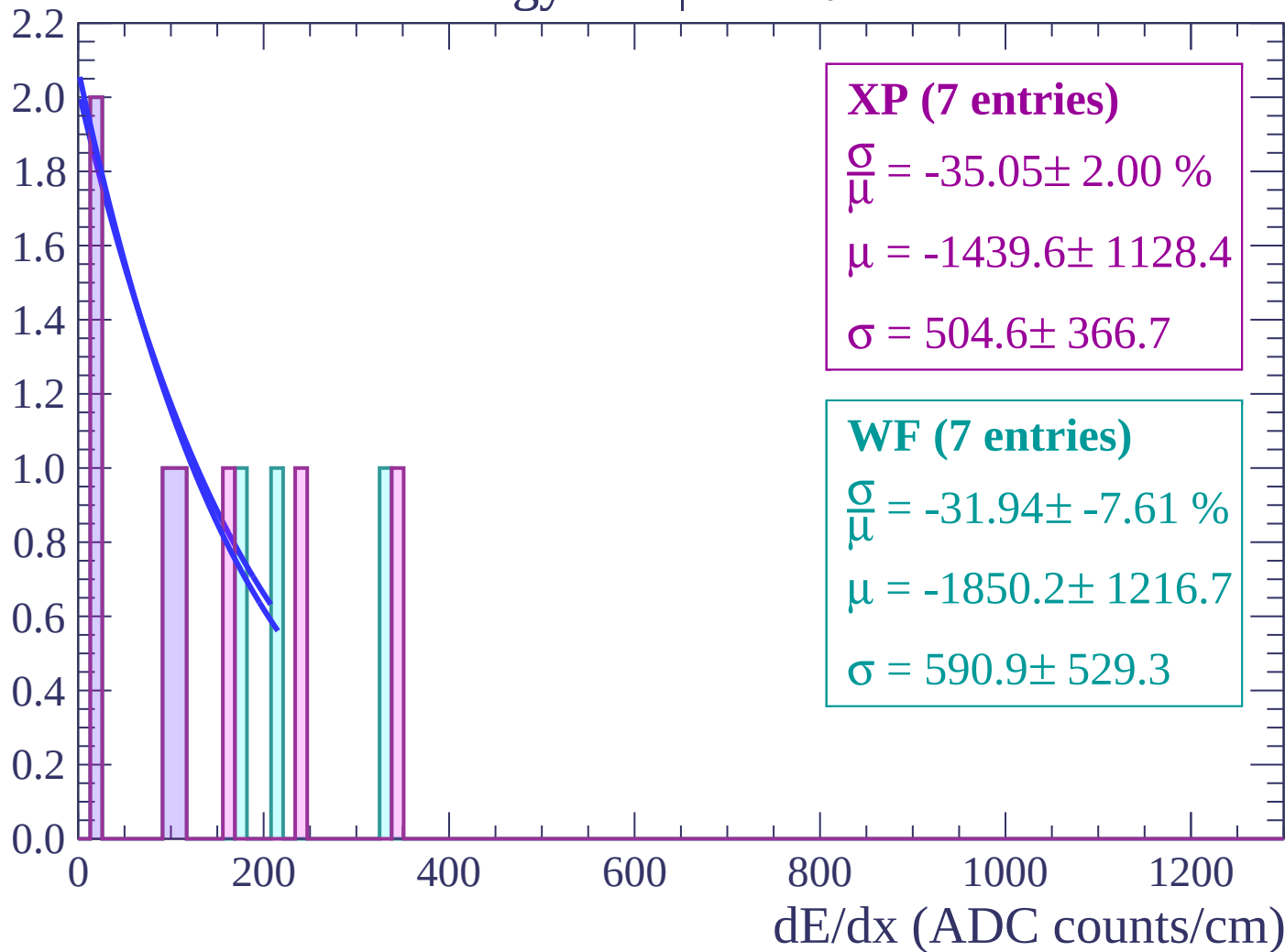
Energy loss | $74 < \theta < 76$

Count



Energy loss | $76 < \theta < 78$

Count



Energy loss | $78 < \theta < 80$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

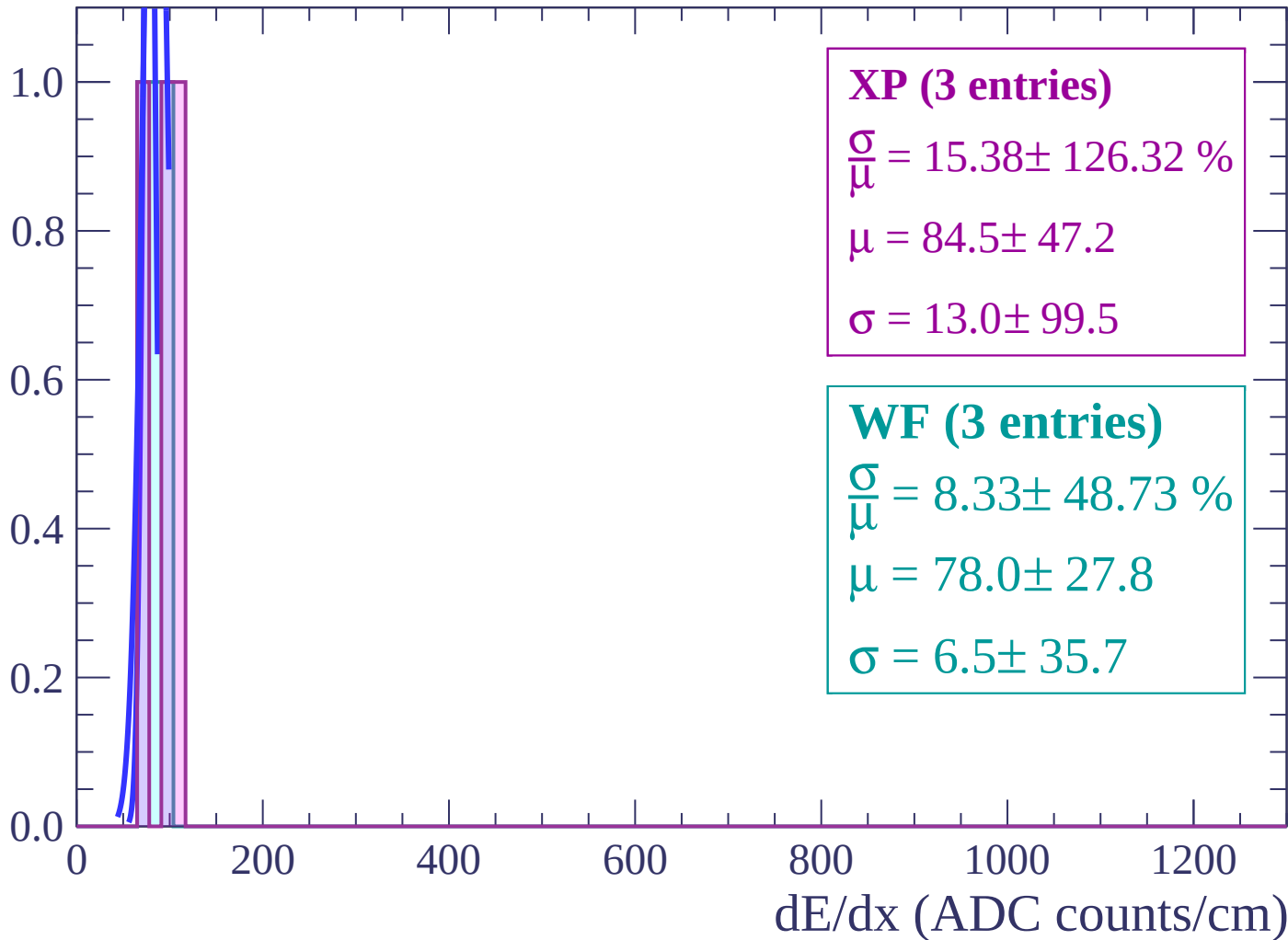
$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $80 < \theta < 82$

Count



Energy loss | $82 < \theta < 84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $84 < \theta < 86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $86 < \theta < 88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $88 < \theta < 90$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $90 < \theta < 92$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

